

The NEA Small Modular Reactor Dashboard: Third Edition



Nuclear Technology Development and Economics

The NEA Small Modular Reactor Dashboard: Third Edition

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Compilation of information for *The NEA Small Modular Reactor Dashboard*

When completing the assessments for *The NEA Small Modular Reactor Dashboard: Third Edition*, the OECD Nuclear Energy Agency (NEA) has exclusively used information from verifiable public sources. These sources are available on the NEA website (www.oecd-neo.org/dashboard-edition3-ref).

The majority of the sources are from third party references (e.g. governments, regulators, financiers, operators). None of the sources are from the SMR designers, except for some relating to fuel type, enrichment, reactor specifics and public announcements of financing. The NEA has made its best efforts to search available public sources which have been used to compile the assessment results.

Prior to publication, the SMR designers were consulted by the NEA and provided with a list of the sources used to compile the assessment. They were given the opportunity to comment on the draft assessments and submit additional information. If this further information was independently verified, it has been used in the final published assessment.

The criteria to obtain the assessment

The assessments are driven by objective criteria applied to information compiled from public sources. They are not the subjective judgement of analysts. The criteria used for the evaluation is in the “Track Progress of SMRs” section of this publication. The information used in the assessment was provided to the SMR designers for their awareness prior to publications. In this context, the SMR designers were provided with an opportunity to comment and provide further information that could be independently verified. The assessments in no way reflect the opinion of the OECD or the NEA.

Foreword



The global energy system is undergoing a profound transformation – driven by rapidly growing demand for electricity in both OECD countries and the Global South, the urgent need for clean energy, rapid digitalisation and a growing demand for resilient, dispatchable power sources. Against this backdrop, small modular reactors (SMRs) have advanced from a promising innovation to market-ready, low-carbon energy first-of-a-kind (FOAK) infrastructure projects.

The *NEA Small Modular Reactor Dashboard* (hereafter “*The NEA SMR Dashboard*”) has been the most cited reference tracking this progress since 2023. This third edition of the *NEA SMR Dashboard* offers not only a refreshed global overview but also in-depth analyses of financing, fuel supply and supply chain readiness. Drawing on the NEA’s proprietary, continuously updated SMR database, it provides policymakers and stakeholders with grounded, data-driven insights.

The overarching sector developments reflected in the assessments of the *NEA SMR Dashboard* are clear: the strategic drivers for SMR deployment – rising electricity demand (including from data centres and digital services), energy security imperatives and national goals set by many countries to reduce carbon emissions – are intensifying. SMRs are now a core part of low-carbon energy strategies in an increasing number of countries.

Private investment is surging, with an estimated USD₂₀₂₃ 5.4 billion of capital now flowing from venture and corporate sources. The United States leads in funds raised, although key European initiatives are gaining traction. Major global corporations such as Google, Amazon, Meta and Dow Chemical have recently bolstered this wave of investment to meet energy needs consistent with their environmental goals. And now with three SMR companies publicly traded, we are witnessing growing confidence in capital markets. The recent announcement that the World Bank will now consider financing nuclear energy projects also sends a clear and powerful signal to other financial institutions.

Equally notable is the evolution on the regulatory front. In 2024, more than 33 SMR designs began pre-licensing activities with nuclear regulators – a 65% increase from the 2023 edition. Major milestones include the United States granting construction licenses for two molten salt-cooled SMR designs – Kairos Power’s Hermes in Oak Ridge and Natura Resources’ MSR-1 in Texas. Ontario Power Generation (OPG) in Canada continues to lead in commercial deployment with the BWRX-300 at the Darlington site, with construction license approval. These developments illustrate a responsive and forward-leaning regulatory landscape.

The siting landscape has also evolved, with nearly 85 active discussions worldwide as of early 2025. As SMRs move from concept to construction, new markets are opening – not only for electricity, but also for non-electric applications like industrial heat and hydrogen production. The supply chain associated with SMR sector development is beginning to take shape, with universities, national laboratories and engineering firms contributing to capacity-building efforts.

Progress in SMR deployment is not limited to OECD countries. China approved the HTR-PM600S project’s environmental impact assessment, consisting of six high-temperature gas-cooled reactor (HTGR) modules at the Lianyungang Petrochemical Park in Jiangsu province. In Russia, two floating RITM-200S reactors are under construction.

Despite sector growth, potential constraints remain. High-assay low-enriched uranium (HALEU) fuel supply remains a key bottleneck for many technologies. This underscores the need for co-ordinated international action and support mechanisms to unlock fuel availability. Future editions of the *NEA SMR Dashboard* will also focus on the supply chain constraints of nuclear-specific components and high-demand non-nuclear infrastructure.

Importantly, since the data cutoff in February 2025, the pace of change continues with a steady drumbeat of announcements and investment plans. Some developments mentioned here occurred after that date and are not reflected in the analysis and charts of this edition.

Therefore, for the first time, this edition is accompanied by the launch of the NEA Small Modular Reactor Digital Dashboard (www.oecd-nea.org/smr-digital-dashboard) – an interactive platform offering real-time access to global SMR developments. This marks a significant leap forward in transparency, accessibility and utility for the international nuclear policy community.

The NEA thanks the many governments, companies and other partners who work with the Agency to make this accomplishment possible. The data and insights shared with us have made the NEA SMR Dashboard an increasingly comprehensive and valuable resource with each succeeding edition. Also, I thank the tremendously creative and dedicated team at the NEA that made this work possible.

We hope this third edition provides new clarity and insight in an increasingly complex and fast-moving energy landscape. As you will find in this Dashboard, SMRs are not a distant vision of the future – they are here today and provide competitive options to meet the energy security needs of countries all over the world.

William D. Magwood, IV
Director-General
Nuclear Energy Agency

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Executive summary

This third edition of the *NEA Small Modular Reactor (SMR) Dashboard* updates its assessment of the progress made in developing and deploying SMRs around the world and offers exclusive insights into key areas such as financing, supply chain development and the advancement of commercially available and qualified fuel. These new analyses draw on the NEA's proprietary SMR database, which offers a uniquely comprehensive view of SMR ecosystems around the world.

This report applies the same methodology and criteria as previous editions (2023, 2024) to assess progress in the areas of:

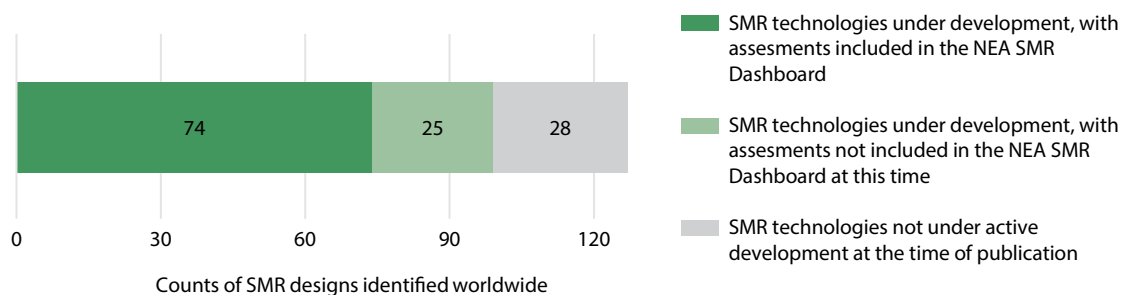
- **Licensing:** The criteria for assessing progress in licensing closely follow international licensing norms, including pre-licensing interactions with regulators, design approval, construction and the issuance of operating licences. SMRs with licensing activities in multiple jurisdictions at any level are accorded a bonus.
- **Siting:** The criteria for assessing progress in siting reflects decisions by site owners and considers licensing readiness of sites for SMR construction. A bonus is given to SMR technologies making progress at multiple sites at any level.
- **Financing:** The criteria for the financing assessments reflect both public announcements from reactor designers and financing reports from publicly available sources.
- **Supply chain:** The criteria for assessing progress in supply chain readiness consider increasing levels of commitment reflected in memoranda of understanding, binding contracts and formal partnerships, joint ventures or consortia with suppliers or engineering, procurement and construction companies.
- **Engagement:** The criteria for engagement reflect the number of engagements with people and communities associated with the SMR project, evidenced by memoranda of understanding, endorsements, town hall meetings or benefit-sharing agreements.
- **Fuel:** The criteria for assessing progress on fuel are based on progress towards the commercial supply of qualified fuel. Where there is a licensed and operating fuel fabrication facility, it is considered alongside others already being used in operating plants. The next stages of maturity include having contracts for fuel supply and a licence to operate the reactor with the specific fuel.

The *NEA SMR Dashboard* also includes information about each SMR design, such as technology concept, configuration, outlet temperature, size and fuel type, including requirements for enrichment.

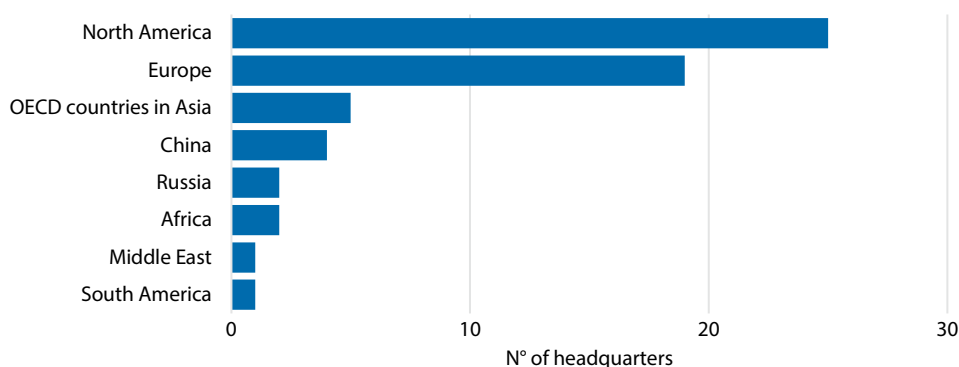
The *NEA SMR Dashboard* uses information exclusively from verifiable public sources. The assessments benefit from consultations with SMR designers to provide further information, subject to independent verification, and reflect developments as of 14 February 2025.

Growing momentum towards demonstration and commercialisation

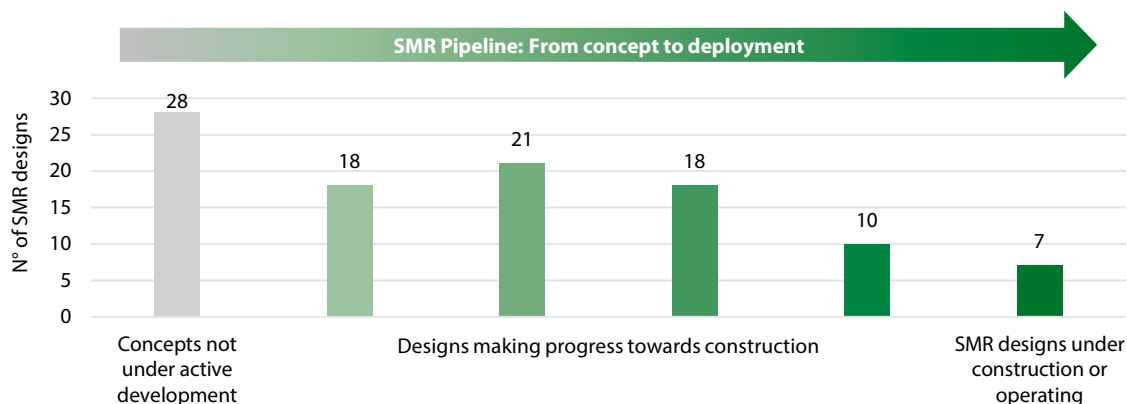
Interest in SMRs is growing worldwide, both in the private sector and among governments drawn to the technology's potential to support significant cuts to carbon emissions while providing energy security. For this edition of the *NEA SMR Dashboard*, 127 SMR designs were identified globally. Out of that total, 74 are analysed in this report; these are the designs for which there was enough publicly available information to assess and for which the designers were willing to participate. Of the other 53 designs, approximately 25 are under development but requested not to be included in the *NEA SMR Dashboard*, though they may be included in future editions; the others include SMR designs that are not under active development or have been cancelled or paused indefinitely (Figure ES1).

Figure ES1. **Count of SMRs identified worldwide**

In terms of geographic location, 30 of the designs covered in this edition are being pursued by 25 design organisations headquartered in North America, 20 designs by 19 organisations in Europe, 10 designs by 5 organisations in OECD countries in Asia, 5 designs by 4 organisations in China, 5 designs by 2 organisations in Russia, 2 designs by 2 organisations in Africa, 1 design by a single organisation in South America, and 1 design by an organisation in the Middle East (Figure ES2).

Figure ES2. **Locations of SMR designer headquarters by region**

Seven designs are either operating or under construction, and there is a strong pipeline of projects progressing toward first-of-a-kind deployment. While some SMR designs are at the conceptual stage, several may make progress in development in coming years (Figure ES3). The diversity of SMR designs presents an opportunity to streamline global supply chains and promote standardisation in order to enhance their economic viability.

Figure ES3. **SMR pipeline: Progress towards first-of-a-kind commercial deployment**

Of the SMR designs assessed in this publication, 51 are involved in pre-licensing or licensing processes across 15 countries, and there are approximately 85 active discussions between SMR developers and site-owners worldwide. Progress has also been made in financing and supply chain readiness, with an 81% rise in SMR designs confirming financing announcements since the *NEA SMR Dashboard: Second Edition* (2024). This influx of capital is catalysing progress in supply chain development and early-stage manufacturing capabilities.

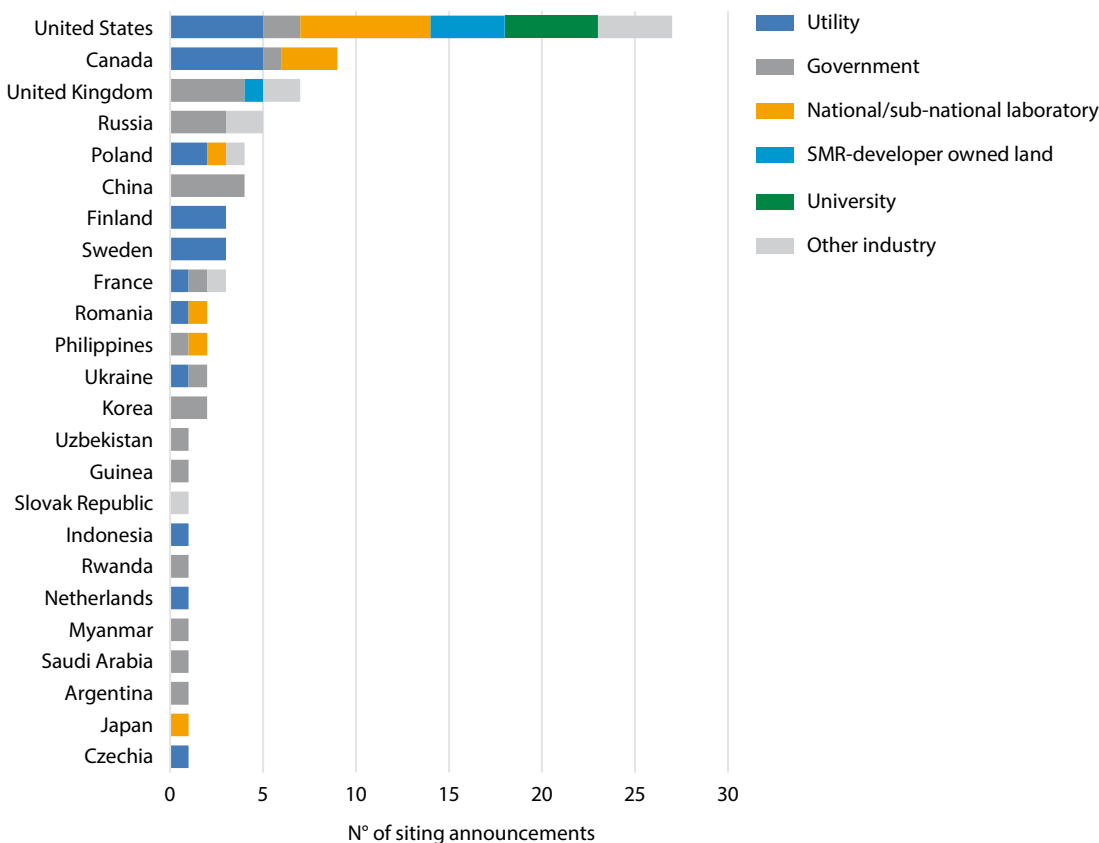
Emerging diversity in business models, site ownership, customers and financing

Most potential SMR sites (49) are associated with utilities and government-owned entities. Additionally, another 14 sites are linked to national or sub-national laboratories, often intended for demonstration purposes, while 11 are connected to private industrial players exploring SMRs as sources of clean, reliable power and heat. The United States is home to the greatest diversity of types of site owner, reflecting a broad spectrum of stakeholders now engaged in SMR deployment.

New types of customers and business models are poised to diversify the sector, moving beyond the historically utility-driven model. SMR projects are increasingly being considered for siting near industrial or commercial demand centres, or near retiring coal plant sites. Ownership structures are also evolving, with some vendors exploring Build-Own-Operate models, Power Purchase Agreements (PPAs), Virtual Power Purchase Agreements (VPPAs), or leasing arrangements, rather than relying solely on utilities as owners and operators.

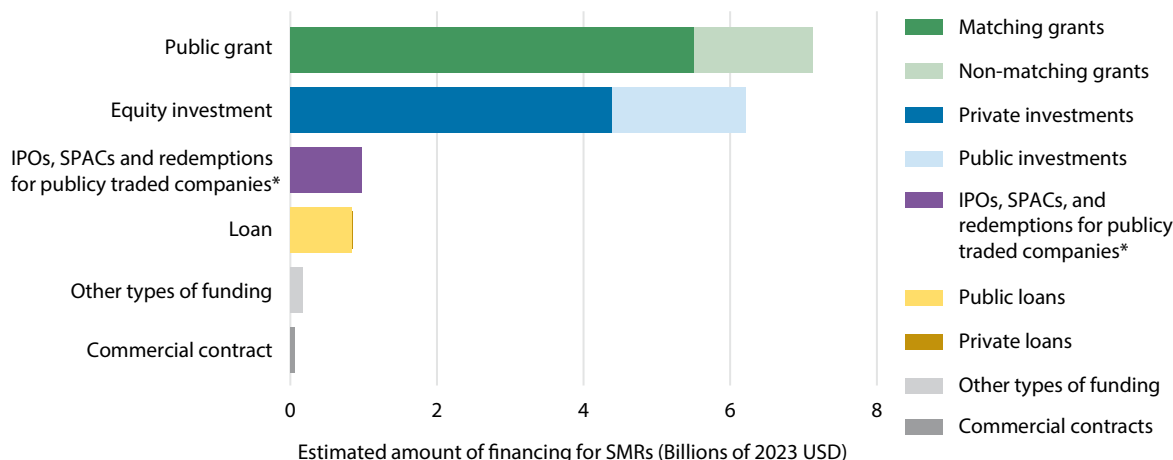
As part of its analysis of progress towards SMR deployment, the NEA is currently tracking 25 siting activities – involving specific SMR designers, site owners, and customers – in near-term markets beyond grid electricity generation. These are primarily driven by industrial partners rather than utilities. At the same time, some utilities are beginning to explore SMRs for emerging applications, such as hydrogen production, while legacy energy companies are examining SMRs as a means to decarbonise heat generation. These developments are giving rise to innovative, potentially transformative business models for the deployment of SMRs.

Figure ES4. **Siting activities, by country and site-owner type**



In terms of SMR financing, private capital is playing an increasingly important role, often complementing public matching grants (Figure ES5). The NEA is tracking approximately USD₂₀₂₃ 15.4 billion of financing towards SMRs worldwide. It estimates USD₂₀₂₃ 10 billion has come from public sources, alongside approximately USD₂₀₂₃ 5.4 billion comes private sources.

Figure ES5. **Sources of financing for SMR designs from OECD countries**



Note: "IPOs" stands for initial public offerings; "SPACs" stands for special-purpose acquisition companies.

Regional SMR supply chains are emerging

Analysis in this report reveals that countries are positioned to capture different types of benefits by participating differently in SMR global value chains. Figure ES6A shows the countries where supply chain activities support SMRs headquartered in that country. Figure ES6B identifies countries where supply chain activities support SMRs headquartered outside that country. In the United States, France, the People's Republic of China (China) and Russia, the majority of SMR-related supply chain activities support SMR designs that have headquarters within those countries. In contrast, in the United Kingdom, Canada and Italy, most SMR-related supply chain activities support SMR designs developed by design organisations headquartered abroad. In some cases, domestic supply chains support SMR projects with domestic sites, although the SMR design organisation has its headquarters abroad. For example, the GEVH BWRX-300 project at OPG's Darlington site in Canada is supported extensively by supply chain companies in Canada, although GEVH is headquartered in Wilmington, United States.

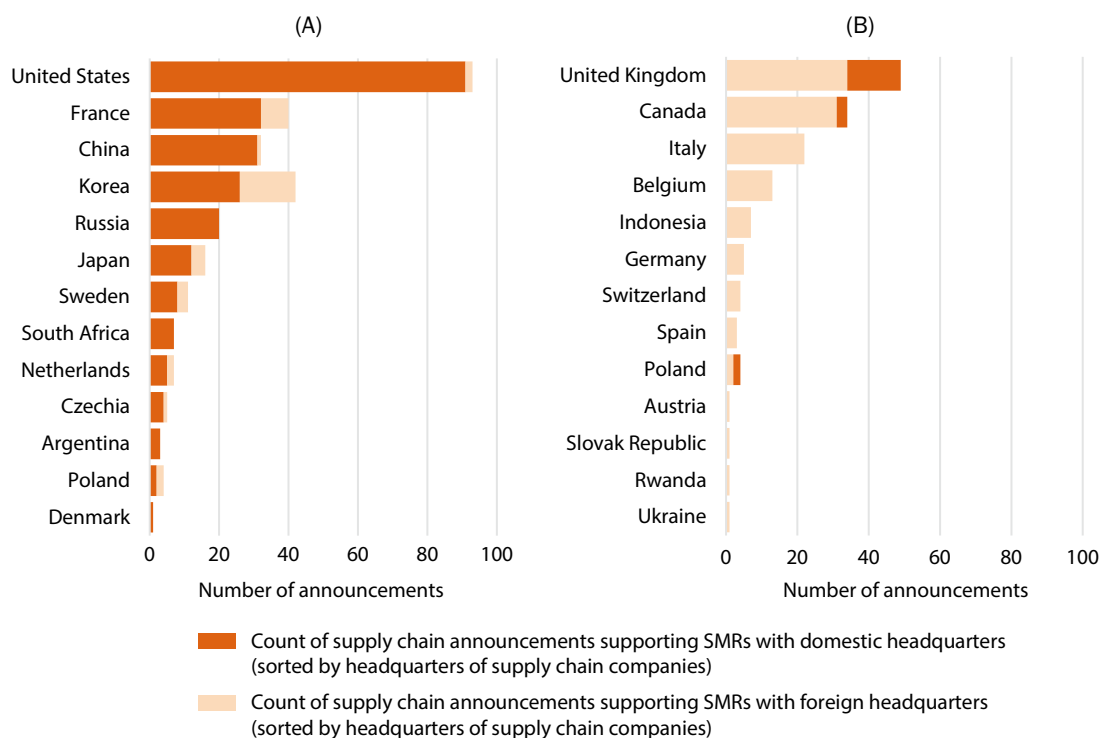
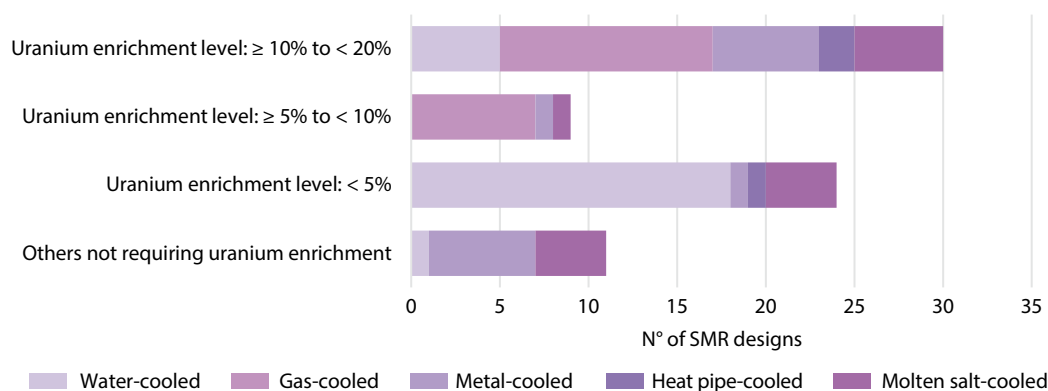
Additional analysis in the report highlights whether SMR design organisations are relying on domestic supply chains or imports. SMR developers in China, Korea, Russia, and Japan primarily rely on domestic supply chain companies.

In contrast, strong supply chain linkages are observed between the United States, Canada, the United Kingdom and Korea, as well as between France, Italy, the United Kingdom and Belgium.

Fuel supply and waste management

A majority of the SMR designs (39) reviewed in this report require high-assay low-enriched uranium (HALEU), which is defined as uranium enriched between 5% and 20% (Figure ES7). Of those, 30 plan to use uranium enriched at levels above 10% and below 20%, and nine plan to use uranium enriched at levels above 5% and below 10%.

The availability of HALEU remains a significant barrier to the deployment of many SMR designs, though some developers have engaged early to secure supplies for their first-of-a-kind reactors. The data collected shows that as of early 2025, more than half of the SMRs planning to use HALEU had not yet progressed beyond non-binding agreements or collaborative studies with national laboratories related to fuel supply.

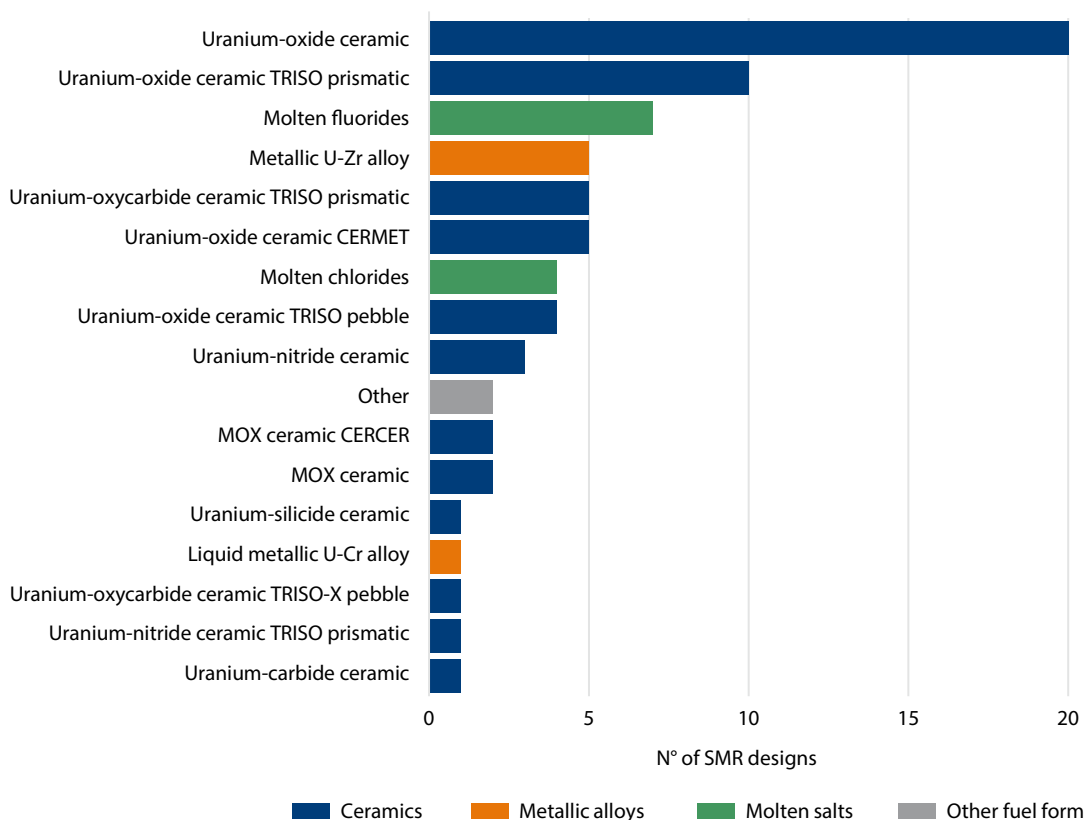
Figure ES6. **Supply chains supporting SMRs with (A) domestic headquarters; (B) foreign headquarters**Figure ES7. **Diverse SMR fuels: Uranium enrichment requirements**

SMR designs are based on an increasingly diverse range of fuel forms, most of which have not yet been licensed or qualified for use. Figure ES8 illustrates the diversity of fuel forms used in SMR designs. Uranium oxide ceramics are the most common fuel in commercial reactors today. However, innovative SMR designs are introducing a wide range of novel fuels that require different manufacturing processes and technologies.

Standard uranium oxide ceramic fuel is the most common fuel among the SMR technologies under active development, with 39 planning to use it as their fuel. Out of these 39 designs, 19 incorporate or plan to incorporate a composite fuel architecture, such as TRISO, distinguishing them from the conventional fuel used in today's large-scale light-water reactors and potentially altering significantly fuel fabrication requirements and performance characteristics. More broadly, 47 SMR designs, over 60% of those included in this edition of the *Dashboard*, rely on fuel forms that are not currently available at commercial scale.

The supply chains for uranium oxide ceramic fuel, and to some extent mixed oxide (MOX) ceramic fuel, are well established with multiple vendors and robust infrastructure for transport and regulation. There are currently no commercial-scale production facilities for many novel fuels, such as TRISO particles, metallic alloys, and molten salts, which will require new fabrication technologies.

Figure ES8. **Diverse SMR fuels: Aggregate physical-chemical forms and composite fuel architectures**



Waste management is another enabling condition that is critical to the successful deployment of SMRs. Some advanced SMR designs are being developed together with innovative strategies for the back end of the nuclear fuel cycle, where fuel could potentially be reused. Combined with recycling strategies, these reactors have the potential to reduce the volume of high-level waste to be managed by deep geological repositories and the uranium resource requirements for the front end of the nuclear fuel cycle. At the time of publication of this edition, there was insufficient available information from verifiable public sources to assess the progress of SMRs in terms of waste management planning and readiness for end-of-life cycle management. Work on future editions of the *NEA SMR Dashboard* is expected to include the development of a methodology and criteria for assessing progress in this area.

Future updates to the NEA SMR Dashboard

The NEA will continue to track SMR deployment progress and expand its analysis of SMR ecosystems. In 2025 the NEA is also launching the *NEA Small Modular Reactor Digital Dashboard* (www.oecd-nea.org/smr-digital-dashboard), a digital, interactive platform that will provide direct access to the NEA's SMR database. It offers stakeholders a real-time view of the status of SMR development around the world. Data for the *NEA SMR Digital Dashboard* will be collected on an ongoing basis, allowing for more frequent updates to the NEA's assessments and providing valuable analysis to policymakers and other interested readers.

The NEA SMR Dashboard: Third Edition

The *NEA SMR Dashboard: Third Edition* provides an authoritative global assessment of small modular reactor (SMR) development by going beyond technology readiness to assess progress, using objective criteria (see Annex 1), across six additional enabling conditions: licensing, siting, financing, supply chain readiness, engagement and fuel availability.

This third edition updates the NEA's global assessment of SMR progress and introduces new, in-depth analyses to support policymakers navigating this rapidly evolving field. In addition to tracking progress toward first-of-a-kind (FOAK) deployment, this edition offers exclusive insights into key areas such as SMR financing, supply chain development and the advancement of commercially available and qualified fuel. These new analyses draw on the NEA's proprietary SMR database, built and maintained over several years, which offers a uniquely comprehensive view of SMR ecosystems around the world.

Beyond updating deployment progress and deepening ecosystem analysis, the NEA is also launching the *NEA SMR Digital Dashboard* in 2025. This interactive platform (www.oecd-nea.org/smr-digital-dashboard) provides direct access to the NEA's SMR database and offers stakeholders a window into the status of SMR development around the world.

The NEA remains committed to maintaining and continuously improving these tools to help policymakers build confidence in SMR technologies and identify emerging challenges that may require targeted policy responses.

Energy policy priorities

Energy security and energy system resilience: Two emerging drivers for SMR development and deployment

Energy security and grid reliability have been significant drivers of renewed interest in nuclear power over the last few years. The disruption of Russian gas supplies in 2022 exposed the continued vulnerability of economies not just in Europe but around the world to imported sources of energy. This moment parallels the oil shocks of the 1970s, which created the first large-scale build-out of commercial nuclear power around the world (Toth and Rogner, 2006). Energy security and independence are emerging as key goals in national energy policies, and SMRs can play an important supporting role to achieve them.

Energy security concerns loom large for many countries. Unlike fossil fuels, which require near-continuous refuelling, nuclear energy is far less vulnerable to short-term fuel supply and price volatility. Plant owners often have contracts for nuclear fuels signed years in advance with fixed prices and delivery schedules. Traditional light water reactors (LWRs) typically refuel every 18-24 months, while SMRs could operate even longer. Some microreactor designs are planned for lifetime fuel, meaning they will be delivered fully-fuelled, operate for 10-30 years, and then be returned to a central facility for decommissioning and refuelling.

Many regions are meanwhile facing power capacity shortfalls during times of peak demand. Ensuring a reliable supply of electricity is getting more difficult for grid operators as they face a confluence of trends: Diminished thermal power plant capacity from fossil and nuclear phaseouts, reduced hydro generation from droughts and increased reliance on intermittent renewable energy. The threat of blackouts in the summer and winter of 2022 led several countries to reconsider their long-standing policies on nuclear energy. Some examples include:

- Sweden once had a plan to phase out nuclear power by 2010 and abandoned the policy in 2009 (Bergenäs, 2009). The current government has now called for two new large reactors to be completed by 2035 and an additional 10 large or small reactors by 2045 (Reuters, 2025a; WNA, 2025a).
- The Dutch government is looking to extend the operating life of its last operating nuclear power plant after reversing a decision to phase out nuclear power and has allocated funding for two SMRs with plans to have them operational by 2035 (IAEA, 2024; Proctor, 2023).
- In 2025, Belgium's parliament voted to reverse a 2003 phase-out of nuclear power. In 2022, they also extended the operating life of two of their older plants by 10 years (Le Monde, 2025).
- Italy, which closed its last nuclear power plant in 1990, is reconsidering its position on nuclear power in light of energy security issues (Landini, 2024).

The wide-scale grid blackout across Spain and Portugal in April 2025 drew renewed focus to the vulnerabilities of the power grid. Other recent grid crises like the 2021 Texas power grid failure or the 2022 summertime energy conservation campaign in Japan, highlight the importance of having adequate firm or dispatchable capacity, like nuclear energy, on the grid. A distributed web of SMRs could not only provide that critical inertia and load-following to keep the grid stable, but also black start capabilities if there is an outage.

Over 30 countries are in various stages of developing their first commercial nuclear power project. Many of these countries do not have the financial resources or the electricity demand to support a traditional large reactor and are thus more interested in SMRs. Almost every country that has nuclear power began by importing the technology from an existing nuclear supplier, and many of these aspiring nuclear countries are already signing bilateral nuclear co-operation agreements. For example:

- As Poland looks to start up its first nuclear power plant by 2036, the country has signed nuclear co-operation agreements with Canada (Reuters, 2025b), France (Pineau, 2025) and the United States (U.S. DOE, 2022).
- In May 2022, Japan and the United States announced an initiative to work collaboratively to develop SMRs along with Estonia, Ghana, Kazakhstan, Korea, Latvia, the Philippines, Romania, Ukraine and the United Kingdom (Hogan Lovells, 2023; UN, 2022).
- As part of a tour of southeast Asia in May 2025, French President Emmanuel Macron signed agreements or discussed collaboration on nuclear energy with Indonesia, Singapore and Viet Nam (Mai, 2025; Lin, 2025).
- Also, on the private sector side, Uganda has contracted Korea's KHNP to perform a two-year site study for the country's first potential SMR project (Barigaba, 2025).

From economies of scale to economies of multiples

While traditional, large reactors may still benefit from economies of scale, SMRs offer the opportunity to achieve economies of multiples (NEA, 2021). This approach does not depend on increasing the size of individual units but rather on the efficiencies gained through the repeated construction of standardised designs. SMRs are intended to be replicated across projects, leveraging accumulated experience to enhance cost and schedule performance. Serial construction, where the design remains stable across a defined number of units, is a key component of this model. Even at modest production volumes, repetition can allow for greater predictability, improved quality control, and reduced construction risk over time.

Experience from other industries highlights how learning can occur through repetition (Wright, 1936). In the aviation industry, aircraft programmes typically operate with well-defined designs that remain stable over many units, allowing manufacturers to identify and correct inefficiencies, transfer knowledge across teams, and improve co-ordination with suppliers over time. Factory fabrication has enabled a high degree of consistency in production, supporting this learning process by concentrating work in controlled environments where feedback loops are faster and execution is more predictable (Argote and Eppler, 1990).

The shipbuilding industry, particularly in naval and offshore segments, provides a comparable model. While production volumes are relatively low, progress has been made by applying modularisation and standardisation across repeated builds. Shipyards have developed methods to break down complex structures into standardised modules that can be fabricated in parallel, assembled more efficiently, and adapted for future designs with minimal disruption. Standardisation in design and construction methods has enabled shipyards to maintain skilled teams across multiple projects, thereby creating the continuity necessary for effective learning (Thompson, 2001).

Achieving economies of multiples depends on the ability to repeat standardised builds across a programme, supported by factory fabrication and delivery models structured around replication. Ongoing NEA work is reviewing the necessary conditions for applying this model to SMRs, drawing on multiple industries where limited-series production has demonstrated that consistency in design, modularisation and execution can drive cost and schedule improvements over time.

Reducing system costs and driving supply chain growth

Even with anticipated cost reductions, the levelised cost of electricity from an SMR may still be higher than alternatives, especially renewable energy (IEA/NEA, 2020). This begs the question: Why is nuclear energy needed when the cost of renewables has fallen?

Economy-wide energy modelling consistently finds that including clean firm power like nuclear and hydro in the grid reduces overall systems costs and improves performance. As renewables and grid-scale batteries become more cost-effective, the increasing deployment of SMRs could provide an optimal solution to reduce overall system costs while maintaining the reliability of a low-carbon grid (NEA, 2012; NEA, 2019; NEA, 2022a).

While there are risks for first movers, there are also unique benefits for those leading the way in building first-of-a-kind SMRs, whether they are utilities, universities or industrial campuses. First movers can expect positive spillover effects from innovations, such as new materials and processes, among others. Texas A&M University, for example, is planning to build at least four different SMR technologies on its campus, positioning the campus as an innovation hub (TAMU, 2025).

Many host communities are hoping to localise supply chains by hosting an early SMR demonstration, and many SMR vendors are equally interested in vertically integrating their supply chain into a regional network. Such localisation can create high-paying jobs beyond those involved in operating the SMR itself, which may be fewer in number than for a large nuclear power plant.

Despite their negative impact on the environment and public health, coal mines and power plants deliver various economic and social benefits to local communities. Consequently, utilities are pursuing projects that not only generate comparable levels of bulk electricity but also match or exceed the socio-economic benefits of the coal industry. In this context, coal-to-nuclear repowering presents a highly attractive option, especially when nuclear energy projects are situated near the same community and workforce that previously benefited economically from the now-retired coal power plant. SMRs are often the most suitable choice, as decommissioned coal plants typically have capacities under 600 MW electric. Even smaller units allow gradual additions of power as the coal plant is dismantled and the site is restored (NEA, forthcoming [a]).

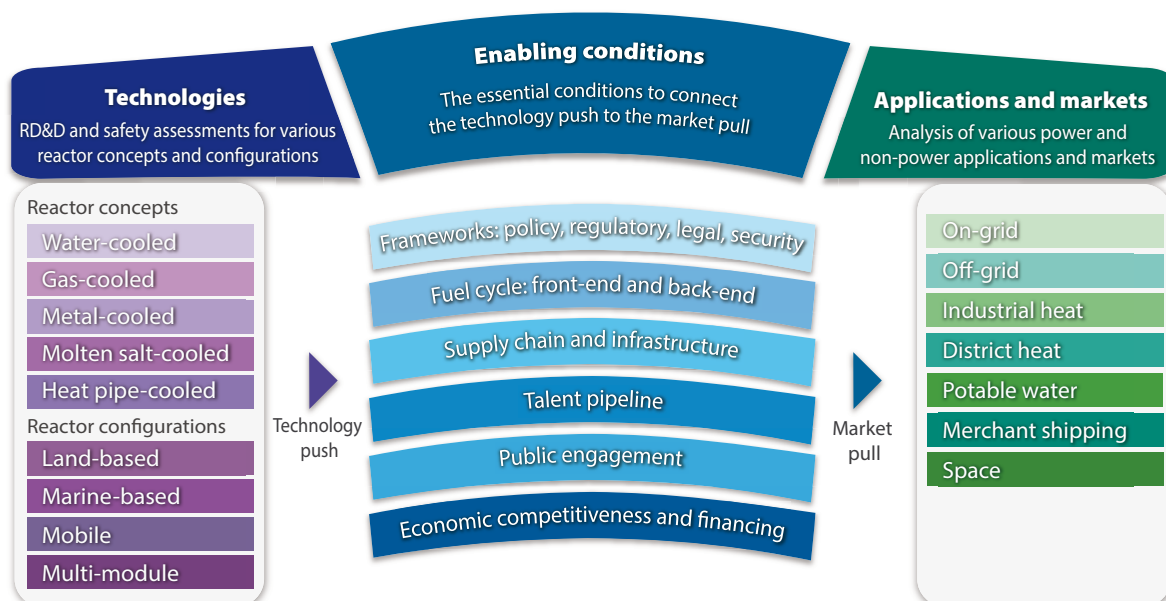
Enabling conditions for SMR deployment

Bridging technology push to market pull

In terms of applications and markets, SMRs are poised to serve multiple energy needs. While on-grid power generation – especially as an alternative to coal – remains a key global market, there is growing interest in diverse, disruptive applications. These include off-grid power and heat for remote areas such as mining operations; high-temperature heat for heavy industries like chemical and fertiliser production; and marine propulsion to replace carbon-intensive bunker fuels.

Several of these market-pull drivers are likely to increase demand for electricity broadly, but this will not directly translate into increased demand for nuclear energy or SMRs without the right enabling conditions. The following enabling conditions in Figure 1 below are critical to the successful commercial deployment of SMRs.

Figure 1. Enabling conditions



Source: NEA, 2023.

Governments, international organisations and industry organisations have a role in creating the enabling policy frameworks, along with enabling regulatory and legal frameworks. A reliable source of fuel for SMRs is also essential, including a plan for managing the back end of the fuel cycle. Similarly, a robust supply chain is required along with infrastructure and human resources. Public trust is essential for the successful implementation of any SMR project. Finally, SMRs will succeed only if they are cost-competitive and have access to economical sources of financing.

Box 1. **Security of small modular reactors**

provided by Lars van Dassen, Executive Director and Pierre Legoux,
Head of Programme Oversight and Implementation, the World Institute for Nuclear Security (WINS)

WINS is an international non-governmental organisation committed to building a global community of nuclear security professionals who are willing to work together to strengthen the security of nuclear and other radioactive materials. WINS' primary purpose is to improve the professionalism and competence of all those involved in nuclear security so nuclear and other radioactive materials are not used for terrorist or other criminal purposes by any actor, whether State or non-State. It promotes knowledge sharing and best practices through workshops and webinars, guidance documents, certified training programmes and benchmarking and evaluation services.

Security considerations for SMRs

SMRs present unique security considerations. Their small size and modularity can support surveillance and standardised security arrangements but may make it harder to implement layered physical protection. Many SMRs have passive safety systems, reducing reliance on human intervention, which can lower the security risks but not necessarily protect against all malicious acts.

SMRs may be deployed in remote and/or underground locations, offering protection from certain external threats, while potentially increasing cybersecurity concerns. Remote siting also poses co-ordination challenges for emergency response, requiring pre-established protocols and training including exercises and simulations.

Smaller staffing models, often tied to remote deployment, raise insider threat risks due to increased levels of access and authority in the hands of fewer staff. Remote operations make insider threat mitigation programmes, including enhanced personnel vetting, vital.

Security technologies evolve at a fast pace. SMRs will benefit from the most recent security technologies such as artificial intelligence, in surveillance, detection and assessment. However, it is expected that adversary capabilities will also increase, and they too will take advantage of the most advanced and sophisticated attack tools, including advanced weapons and unmanned aerial vehicles (UAVs). Design requirements, and particularly Design Basis Threat, will need to reflect this evolution.

Many SMRs rely heavily on sophisticated, highly integrated digital instrumentation and control systems, making them vulnerable to cyber threats. Furthermore, remote maintenance and monitoring creates vulnerabilities and potential entry points for cyberattacks.

The diverse and innovative nature of SMR designs often challenges existing regulatory frameworks, necessitating new, flexible and performance-based approaches to ensure compliance and effectiveness. Current regulatory frameworks may not adequately address cybersecurity threats specific to SMR designs, creating uncertainty and potential security gaps. It will require strong international collaboration on security expectations, including design certification and cybersecurity performance requirements.

Addressing security needs from the design phase

Addressing security challenges involves embedding security from the earliest stages of reactor design – a concept known as security-by-design. Integrating security considerations early enhances protection and reduces long-term operational costs and complexities associated with retrofitting security solutions after manufacture.

Security-by-design embeds security into the engineering process, addressing risks at the source rather than relying solely on external protective measures. Successfully implementing security-by-design requires more than just input from design engineers or security specialists – it demands a top-down commitment to treat security equally with nuclear safety. A collaborative approach among designers, operators, project managers and regulators is essential to ensure security is fully integrated throughout the facility lifecycle.

For example, security-by-design for SMRs could include underground siting, robust containment structures and passive safety systems that reduce vulnerabilities and mitigate the consequences of sabotage. Also included in security-by-design should be advanced surveillance and monitoring, including real-time remote surveillance and automated security response systems. Enhanced detection capabilities would also reduce reliance on human guards.

Optimised transport security protocols should also be considered in the design phase. Standardised, secure protocols for the transportation and handling of HALEU fuel should be developed, emphasising secure routing, containment measures and real-time tracking.

For digital assets, “security by default” principles, such as limited access, strict permissions, and encryption, would have to be applied from the start and users would have minimum extra work to protect their systems.

The role of stakeholder collaboration

Effective implementation of security for SMRs requires early and meaningful engagement with a broad range of stakeholders, including technology developers and designers, nuclear industry associations, regulators and government agencies. Collaboration at the initial stages of design helps identify and address security concerns proactively, ensuring that security measures are technically sound.

Stakeholder involvement not only fosters transparency and trust but also helps counter disinformation and misinformation, builds consensus and ensures compliance with legal and regulatory frameworks. Integrating stakeholder perspectives into the design process strengthens both the credibility and effectiveness of security measures from the outset.

International collaboration for SMR security

International collaboration is vital for SMR deployment. It can align regulatory requirements across countries, reduce duplication and cost, and strengthen safety and security through shared expertise. It can build public and political trust, enable faster international market access and support innovation and global supply chains. Working together ensures SMRs can be deployed efficiently, safely, securely and at scale worldwide. Examples of significant international activities include the NEA Working Group on New Technologies (WGNT), the SMR Regulators’ Forum hosted by the IAEA, and the IAEA Nuclear Harmonisation and Standardisation Initiative (NHSI), including the Security of SMRs Working Group.

WINS has undertaken work to support decision makers and other national stakeholders to become knowledgeable customers of reactor technology and related security matters and understand that early attention to security is critical to the successful implementation of a nuclear energy programme. WINS has implemented a series of workshops aiming at advancing the safe and secure deployment of SMRs and produced publications summarising security considerations and challenges of the various reactor designs and security by design principles that can be adopted by SMR developers.

Safety

SMR designs build on lessons learnt from over 60 years of experience in the nuclear energy sector to enhance safety and improve operational flexibility. Many SMR designs continue the process for safety improvements by incorporating more inherent safety features.

Some SMRs also leverage design features that have the potential to bring safety-related advantages such as:

- smaller reactor cores with smaller quantities of nuclear material;
- relatively large water inventories for water-cooled reactors;
- the use of accident-tolerant fuels and advanced fuels that can maintain their structural integrity even at higher temperatures; and
- operation at lower pressures and the use of passive safety systems that require fewer external sources of electricity to maintain safety, if well qualified. Some require no external sources of electricity at all.

The implementation of such design features could help reach safety objectives while reducing the reliance on human intervention and operational measures, enabling design simplifications and cost reductions.

The benefits of lower quantities of nuclear material in the reactor cores combined with passive safety features may also lead to smaller offsite emergency planning zones (EPZ). This may facilitate plant siting and potentially reduce grid requirements for on-grid SMRs. However, care must be taken as emergency planning may still be necessary due to security reasons.

The new technical features of SMRs present both opportunities and challenges in terms of safety (NEA, 2025a). While they potentially offer significant safety enhancements, the novelty and diversity of SMR designs mark a significant departure from established regulatory experience. Careful analyses backed by test data and validated codes and simulation tools will be required to establish that such systems are effective in the variety of circumstances in which there is dependence on them. Applying flexible,

performance-based and objective-oriented regulations could help accelerate the licensing process for new reactor designs and foster the international compatibility of national regulations (INSAG, 2021). Additionally, using SMRs in industries other than power generation will likely require more co-operation between nuclear safety authorities and other industry regulators.

Understanding the environmental impacts of SMRs

The Declaration to Triple Nuclear Energy Capacity, announced during the 2023 United Nations Climate Change Conference (COP28) and now signed by 31 countries, was also signed by 14 of the world's largest banks. An accompanying pledge was signed by 140 nuclear companies and 15 large energy users, including big technology companies like Amazon, Google and Meta (WNA, 2024). Since then, many countries have incorporated nuclear energy into their national decarbonisation strategies (Energy & Climate Intelligence Unit, n.d.). Some 30 countries have a net zero carbon commitment codified in law, including Germany (2045), Sweden (2045) and Canada (2050).

- Over 50 countries mention net zero targets in policy documents, including Japan (2050), Korea (2050) and Argentina (2050).
- Around a dozen countries have non-binding declarations or pledges to reach net zero emissions at some future date, including China (2060), India (2070), and Saudi Arabia (2060).

Climate change is not the only environmental issue impacted by the global energy system. Ecosystems can be degraded by air and water pollution from mining and fossil fuel combustion and disturbed by construction for large-scale development projects and extractive industries. SMRs can improve environmental outcomes in several regards:

- **Nuclear energy has a small physical footprint.** Unique among low-carbon energy sources, nuclear energy has one of the smallest land-use requirements of any electricity source, including the upstream and downstream footprints of mining and waste disposal (Lovering et al., 2022). More specifically, SMRs can be co-located near demand centres, thereby avoiding the need for high-voltage transmission lines.
- **SMRs could use significantly less water than large reactors.** One of the most significant environmental impacts of traditional light-water reactors is the water withdrawal and consumption required for primary and secondary cooling (Averyt et al., 2011). Many SMR designs use significantly less water (Thonney, 2020), and some models are designed to utilise the Brayton cycle to generate electricity, thereby consuming no water on site, making them attractive for use in arid regions (WNA, 2025b).
- **Critical minerals requirements are much lower.** Compared to other distributed, small-scale solutions like batteries and renewables, SMRs are not as dependent on critical minerals. The primary critical minerals needed for nuclear are copper, nickel and chromium (International Energy Agency, 2021).

Taking a life cycle approach to assessing environmental impacts requires considering the front end and back end of the fuel cycle.

The environmental impacts of uranium mining are well mitigated (NEA, 2014). Over the course of the last few decades, uranium mining activities and mine development processes have significantly improved with the overarching goal of minimising impacts on environment and health. Improvements have been accomplished through the development of increasingly stringent regulatory requirements driven by societal expectations regarding the protection of people and the environment, and through industry innovation. Although all types of mining have improved in this regard, the improvements in uranium mining have been more dramatic because of evolving societal expectations stemming from legacy health and environmental effects of the industry (NEA, 2023; NEA, 2025b). Rehabilitation and reclamation planning, the provision of financial guarantees and funding and proactive community engagement are all common practices in the leading global uranium mining jurisdictions.

SMRs introduce new fuels, moderators and coolants that will alter the characteristics of their used fuel and waste (NEA, 2024a). Changes in reactor size may also affect the volume and type of waste produced. Additionally, higher burn-up rates, increased end-of-cycle reactivity, elevated decay heat, and different fuel geometries may impact the transport and storage requirements for used fuel. These differences mean that current national waste management systems may not be suited for waste streams from future SMRs. These issues must be fully understood and integrated into long-term waste management strategies to maintain sustainable long-term waste management solutions.

Box 2. Ensuring operational readiness for SMRs in the United States

provided by Joe White, Director, New Nuclear, the Institute of Nuclear Power Operations (INPO)

In the United States, an unprecedented demand for new sources of energy generation, after decades of little or no growth, has placed a premium on its ability to recapitalise a large and high performing national nuclear industry. This includes extending the life of existing plants, increasing their power output, restarting previously shut down plants and building new ones, with a near-term focus on construction of small modular reactors (SMRs).

About INPO

For over four and a half decades, US utilities have continuously leveraged their association with the Institute of Nuclear Power Operations (INPO) to achieve common purpose and mutual support, such that, by many objective measures they are now among the highest performing national nuclear industries in the world. It will be vital to maintain that same focus and sense of community as we now enter into a decades-long effort to multiply nuclear generation capacity while sustaining the high performance of the existing US industry. Founded in 1979 in response to the Three Mile Island incident, INPO is a nonprofit organisation dedicated to promoting excellence in the safe and reliable operation of commercial nuclear power plants. Through adaptable programmes and expert services, INPO has consistently supported its members in managing risk and maintaining skilled teams in a high-stakes environment where the performance of one reflects on all.

Importantly, INPO also operates WANO-Atlanta Center (WANO-AC), one of four worldwide regional centres, with members from several countries outside the United States, including Canada, China, Mexico, Romania, South Africa and the United Arab Emirates. INPO has chosen to fully integrate the WANO-AC programme into its operations, including WANO New-Unit Assistance (NUA) and Pre-startup Reviews (PSURs). During its early years, INPO played a pivotal role in supporting the construction of new nuclear plants, and today, it is once again stepping forward to assist with the industry's ambitious efforts to expand nuclear generation both domestically and abroad – efforts that will require the same unity and focus that have long defined the US nuclear community.

SMR operational readiness

A vast array of large-scale project management and new construction operating experience and best practices exist within the databases at INPO/WANO-AC, and its sister organisations, the Electric Power Research Institute (EPRI) and Nuclear Energy Institute (NEI). An artificial intelligence (AI) project is currently underway to make this operating experience even more useful to our members. So long as some new entrants choose to remain outside of this well-established nuclear community, their large projects are at greater risk of repeating past mistakes.

Technology designers and developers are often heavily focused on meeting regulatory requirements and place less emphasis on defining what operations and activities are required to achieve high levels of station-wide safety and reliability. In the case of SMR developers, the primary focus tends to be on their reactor designs, to the exclusion of the operating models that must be developed to complete their concepts of station-level operations, including “balance of plant” considerations. Interestingly, INPO's more recent engagement within the legacy nuclear industry is dominated by plant secondary system reliability challenges, as primary side safety system issues have been largely eliminated and remain excellent nearly across the board.

In any new construction project, and particularly among new entrants who have not operated nuclear stations before, the creation of an effective station organisation that is ready to function at a high level is a major hurdle to operational readiness. The following are a few areas that challenge new nuclear operators:

- **Leadership and management roles and responsibilities** – Any ambiguities regarding functional and cross-functional responsibilities are greatly amplified in newly formed organisations, especially when personnel and/or equipment challenges emerge, which they are bound to do at a new nuclear station.
- **Culture** – Regardless of how newly formed a nuclear organisation is, its leaders must have engrained the values and behaviours of nuclear safety culture, technical conscience and a continuous improvement culture as its first line of defense in preventing decision-making errors.
- **Construction-to-operations transition** – Often, new nuclear projects have difficulty transitioning from a construction project mindset to an operations organisation. This is complicated by the breadth and timing of technical and organisational factors that are part of every transition, together with the ability of leaders to refocus the workforce away from construction and toward operations.
- **Lacking operational focus** – New nuclear organisations struggle, at times, to thoroughly address the breadth of equipment and systems conditions they are likely to confront leading up to fuel load, which can then manifest in latent problems during power ascension and plant operations.

As a result of these and many other challenges to aspiring new nuclear operators, most are tending to partner with seasoned nuclear utilities to leverage their vast operations expertise in effectively running a plant.

Lastly, the corporate organisation itself must be organised and ready to govern, oversee and support initial operations. This is especially important for new entrants into nuclear generation who often consider the corporate role in ensuring the success of the station an afterthought.

An unprecedented future

As the United States seeks to revitalise its position in new nuclear construction at the same time that its nuclear industry is also planning many concurrent major capital projects, the need to remaster and co-ordinate many competing skills and resources, such as major project management, licensing, permitting, contracting, supply chain management, fuel enrichment and accessing the future nuclear workforce, calls for a strategic approach by INPO and the industry.

INPO is currently leading a New Nuclear Advisory Panel, which serves as a platform for SMR and other developers, suppliers and operators to exchange views on INPO's strategies and role in overcoming these challenges and ensuring the successful recapitalisation of the US nuclear industry. As WANO-AC, the WANO New Unit Assistance Program will be tailored to meet the specific needs of each project, while new entrants also benefit from broadly shared lessons learnt.

2025 is only the beginning of an unprecedented future that promises to remake the US nuclear industry. SMR designs are at the forefront of this effort and their construction has begun.

For more information on INPO, see www.inpo.info

Innovative business models

New nuclear vendors are targeting a range of first customers and business models, diversifying away from the traditional, vertically integrated utility model. Projects may be sited close to demand at industrial or commercial sites, or near a retiring coal power plant. Rather than the utility being the owner and operator, some vendors are exploring a Build-Own-Operate model, power purchase agreements, or leasing arrangements.

This edition of the Dashboard tracks just under 85 SMR projects with active siting activity, defined as some form of engagement between a site owner and an SMR developer. Roughly half of these projects involve SMR developers and traditional actors such as governments or utilities. The other half are led by new types of site owners, including universities, national laboratories, industrial companies and SMR developers that have purchased or are seeking to purchase land for their first-of-a-kind deployments.

One third of siting activities are occurring in five near-term markets for SMRs:

- replacing coal for on-grid power;
- replacing fossil fuel cogeneration for industries, including hydrogen production;
- replacing diesel for remote mining operations;
- providing district heating; and
- delivering off-grid energy solutions.

Most of the siting activity in these emerging markets is driven by industrial partners rather than traditional utilities. However, there are also examples of utilities exploring SMRs for new energy sectors such as hydrogen production, as well as legacy energy producers looking to decarbonise heat generation. These new configurations of site owners and applications are giving rise to transformative business models for SMR deployment.

In 2024-2025, major technology companies, including Google, Amazon and Meta, announced direct support for SMRs, aiming to meet the clean electricity needs of their data centres on the short to medium terms. While these announcements suggest interest in deploying potentially several gigawatts of capacity from SMRs, no data centre operator has yet announced a specific site for hosting an SMR. This highlights the distinct nature of data centre demand, which differs from industrial heat and power users that often seek behind-the-meter solutions. In contrast, data centres are pursuing front-of-the-meter access to clean energy, typically through virtual power purchase agreements.

Tracking progress of SMRs: From concept to first commercial deployment

The *NEA SMR Dashboard* considers factors beyond technology readiness levels to provide a comprehensive assessment of progress towards commercialisation and deployment of SMR and Generation IV technologies. It augments data from other sources that focus on technical attributes and technology readiness levels. Taken together, assessments about technical readiness coupled with the *NEA SMR Dashboard's* assessments of licensing, siting, financing, supply chain, engagement and fuel reveal which SMR technologies and projects are moving fastest from concept to commercialisation in various markets around the world.

The *NEA SMR Dashboard* captures the progress of SMRs in six key areas:

i. **Licensing**; ii. **Siting**; iii. **Financing**; iv. **Supply chain**; v. **Engagement**; vi. **Fuel**.

In each area objective criteria have been defined to assess substantial progress towards first-of-a-kind deployment and commercialisation. The progress criteria are applied using verifiable public sources.

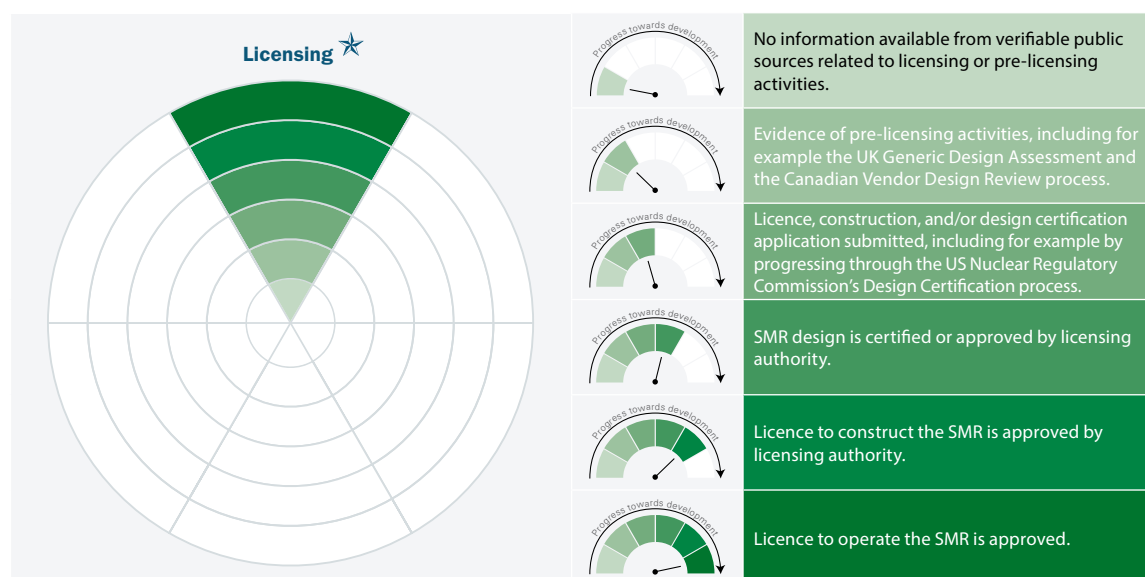
It is difficult to compare SMR technologies advancing in different contexts and jurisdictions, characterised by unique regulatory structures and approaches, siting requirements, financial models and policy environments, among other differences. Objective criteria have been defined to reflect substantial progress irrespective of differences across contexts and jurisdictions. The public information used to populate the *NEA SMR Dashboard* captures a “snapshot” in time based on the latest public information at the time of drafting.

The six areas and progress criteria are described below. The criteria are also summarised in Table 16 in Annex 1.

Licensing

The licensing process is critical for any SMR to reach market. Various nuclear safety regulators have been advancing efforts to get ready to regulate innovative SMRs. Many are working to develop new approaches and pathways for licensing, for example in some cases by collaborating to review an SMR design internationally from the earliest stages.

Figure 2. **Tracking progress in licensing**



Note: The star indicates a bonus given to SMRs with licensing activities in multiple jurisdictions at any level.

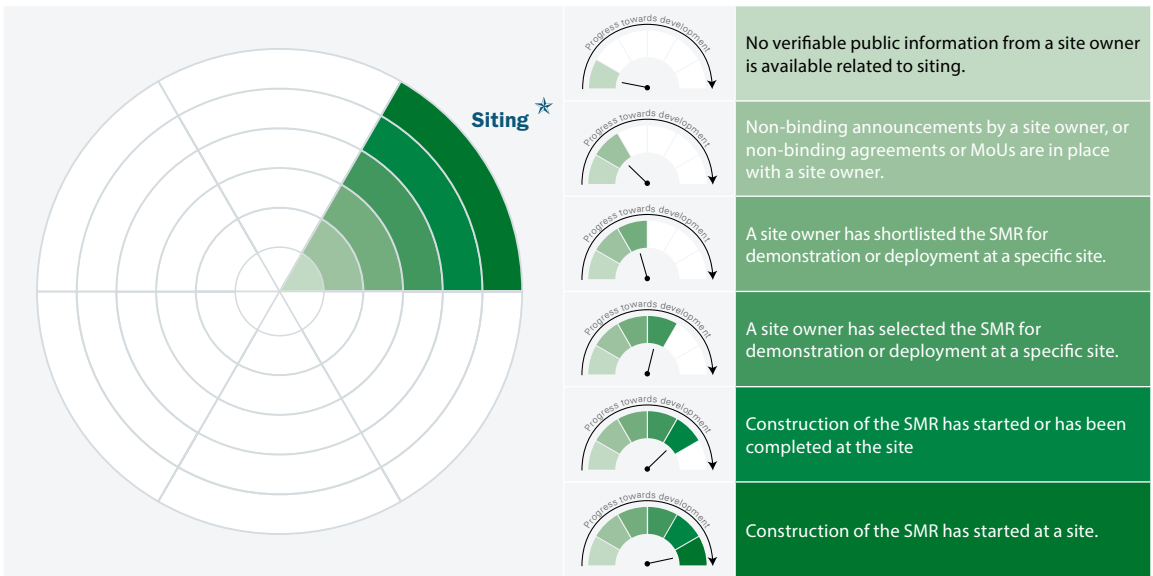
The *NEA SMR Dashboard* progress criteria for licensing closely follow international licensing norms, including pre-licensing interactions with regulators, design approval, construction and the issuance of operating licences. A bonus is given to SMRs with licensing activities in multiple jurisdictions at any level.

Siting

Siting requirements vary greatly between different SMR configurations: A multi-modular, land-based reactor will have a very different process of finalising a site than a floating reactor design.

The grading scale reflects one unifying factor against which all SMR designers could be assessed: Whether the site owner has publicly selected the SMR for their site. The next factor is whether the site has been licensed and is ready for that specific SMR to be constructed or deployed on the site. A bonus is accorded to SMR technologies making progress at multiple sites at any level.

Figure 3. Tracking progress in siting



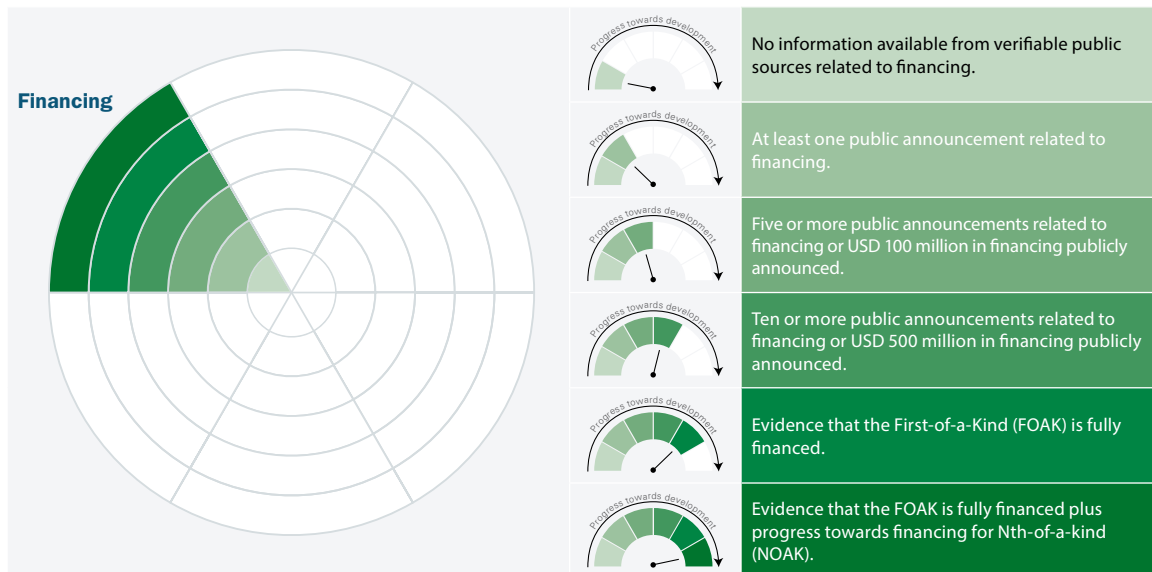
Note: The star indicates a bonus given to SMRs making progress at multiple sites at any level.

Financing

Securing financing is critical for any SMR technology to reach markets. However, transparency is understandably difficult to achieve as information may be considered commercially sensitive. Different SMR projects are also likely to require different levels of financing. Different business models are possible and different SMR designers will pursue different strategies. There are numerous paths to success with a combination of public and private financing, including a range of approaches to private financing, such as through debt, equity or bonds, or by becoming publicly traded.

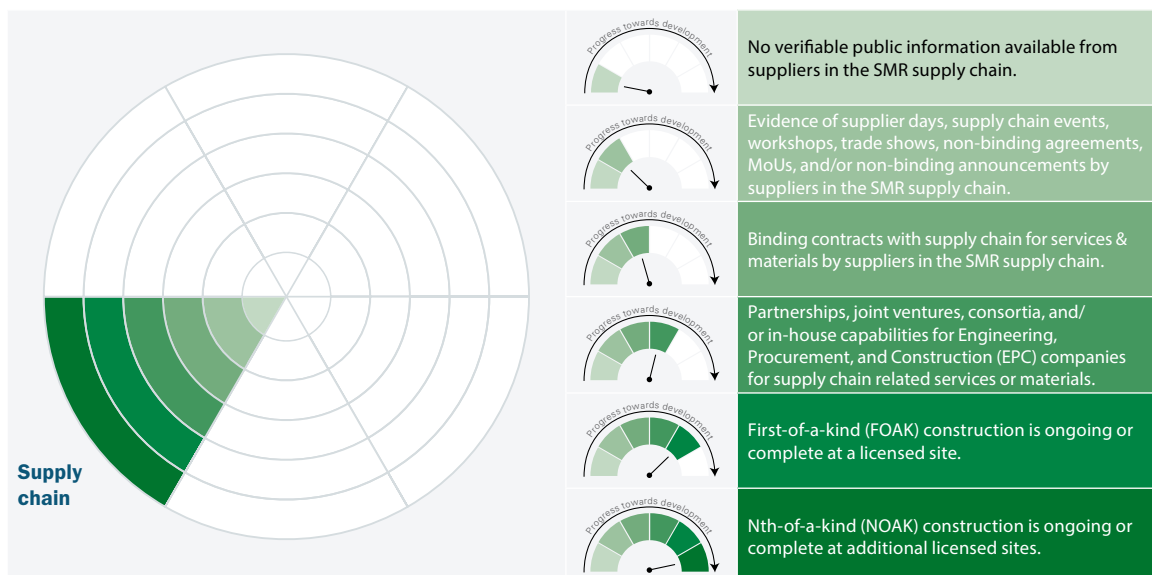
The *NEA SMR Dashboard* does not aim to audit any designer or validate the scale of financing required to bring a technology to market. Instead, public announcements on financing an SMR are utilised to indicate progress in this category. The progress criteria are based on the number or size of announcements in the public domain.

The financing category therefore has two pathways for progress: Either there is a significant number of financing announcements in the public realm for a particular SMR, or there are public announcements indicating the designer has raised significant funding. The pathways converge when it can be easily verified from public announcements that an SMR first-of-a-kind is considered fully financed, irrespective of the size of investments.

Figure 4. **Tracking progress in financing**

Supply chain

In most cases, a designer is not alone in developing an SMR, though there are exceptional cases where the designer can provide the full set of needed engineering, procurement and contracting. Contracts and partnerships must be in place to develop a supply chain for designing, constructing and operating the reactor. This category maps the maturity of the supply chain for each reactor through public announcements by suppliers and partners. Announcements from suppliers, engineering, procurement and construction organisations are all considered, as are announcements from universities, labs and research institutions when they are supplying research and development services to an SMR project.

Figure 5. **Tracking progress in supply chain readiness**

The progress criteria are based on the increasing level of commitment of memoranda of understandings, binding contracts, formal partnerships, joint ventures or consortia. A supply chain is considered mature when construction of the first-of-a-kind SMR is ongoing, and a well-developed supply chain is indicated by the construction of an ‘nth-of-a-kind’ SMR.

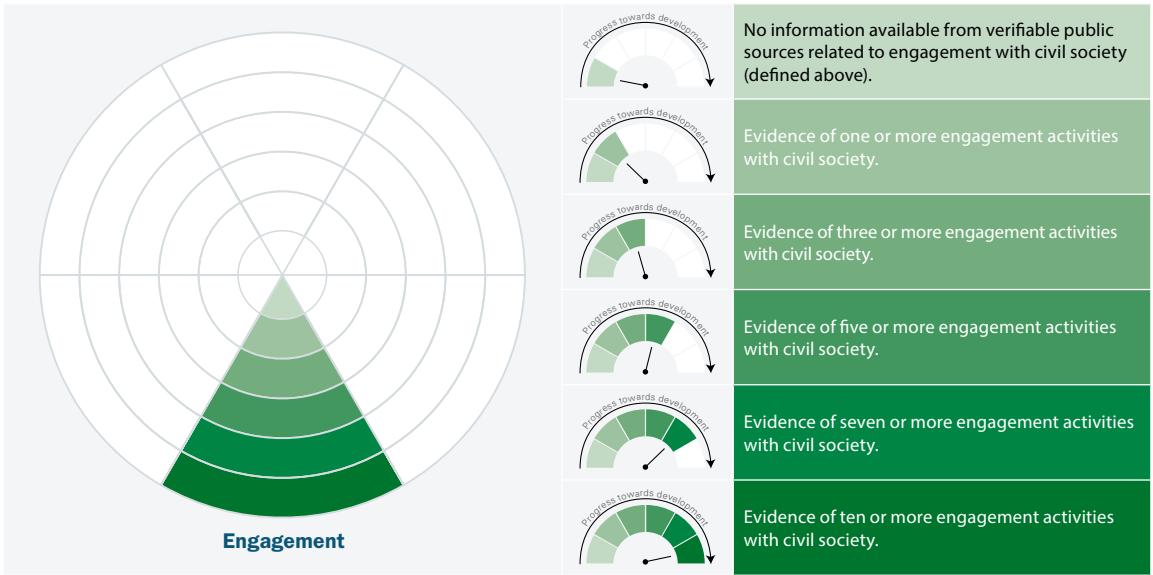
Engagement

Engagement with communities is vital to the success of an SMR. In some regions, a project cannot advance without building strong engagement with Indigenous communities from the onset.

The progress criteria will reflect the number of engagements with people and communities associated with the particular SMR project. Engagement activities in mainstream, non-specialist media through videos, podcasts or interviews will be considered as well as memoranda of understandings, endorsements, town hall meetings and benefit-sharing agreements from the following stakeholder groups:

- National governments
- Subnational governments
- Indigenous governments
- Labour unions
- Non-governmental organisations
- Civil society organisations
- Community organisations
- Universities
- End users and customers
- Advisory boards

Figure 6. Tracking progress in engagement

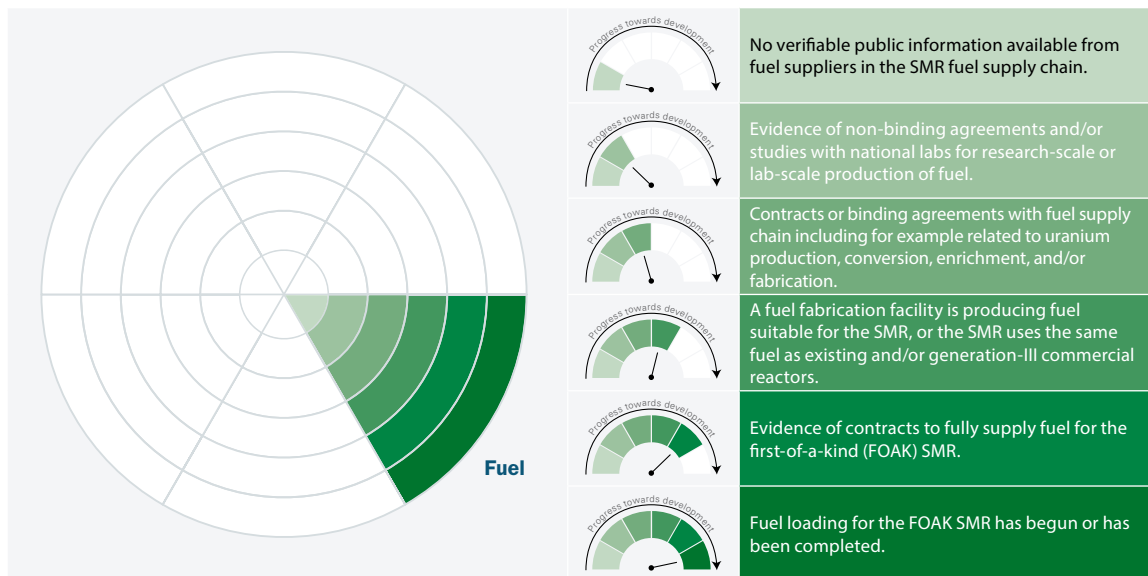


Fuel

The majority of light water reactor SMR designers plan to use low-enriched uranium fuel in the same form that is currently commercially available. Many other SMRs plan to use fuels that span a range of compositions and levels of enrichment, with enormous variation within those categories. Most such fuels have never been licensed to operate in a reactor or are not commercially available. There are multiple steps to achieving fuel qualification, and they differ between regulators. There are nevertheless clear indicators of the progress of an SMR towards a safe and secure supply of licensed and qualified fuel for operation.

The SMR progress criteria are based on the progress made towards commercial supply of qualified fuel. Once a licensed and operating fuel fabrication facility exists for a fuel, it is considered alongside others already being used in operating plants. For SMRs at this level of maturity, the next stages include contracts for fuel supply and a licence to operate the reactor with the specific fuel.

Figure 7. **Tracking progress in fuel availability**



Key findings from the third edition of the NEA SMR Dashboard

Over the course of preparing various editions of the *NEA SMR Dashboard*, a substantial amount of data and market intelligence has been collected. This has enabled the NEA to evaluate actual deployment progress and offer insights into the factors driving the commercialisation of SMR designs. This section highlights key findings from the data and analyses gathered so far, along with recent policy developments and market trends. The assessments in this edition of the *NEA SMR Dashboard* are based on progress up to a cutoff date of 14 February 2025.

Global momentum for SMRs

SMR technology is gaining momentum worldwide, driven not only by growing private sector interest but also by increasing policy-level interest in its potential to support deep decarbonisation. This edition of the *NEA SMR Dashboard* includes a total of 74 SMR designs currently under active development. Of these, 30 designs are being pursued by 25 design organisations headquartered in North America, 20 designs by 19 organisations in Europe, 10 designs by 5 organisations in OECD countries in Asia, 5 designs by 4 organisations in China, 5 designs by 2 organisations in Russia, 2 designs by 2 organisations in Africa, 1 design by a single organisation in South America, and 1 design by an organisation in the Middle East (see Figure 8).

Figure 8. Locations of SMR designer headquarters by region

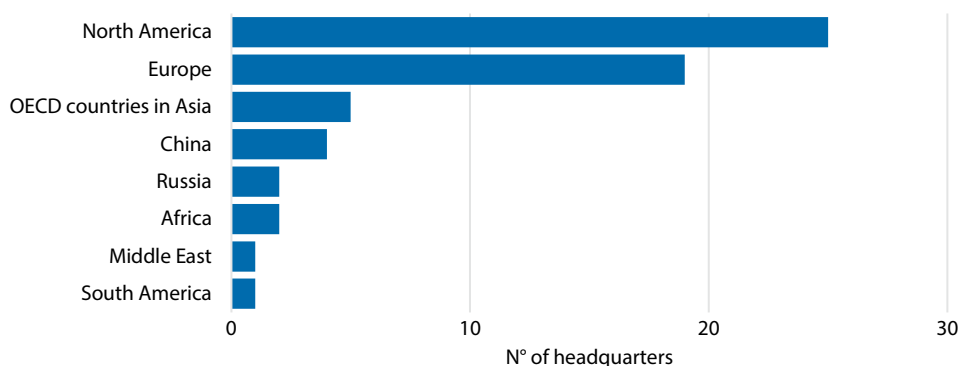


Figure 9 highlights the geographical distribution of the industry, showing where SMR designer headquarters are located around the world. Figure 10 illustrates global progress in siting SMR designs for research, demonstration or commercial deployment.

Figure 9. Map of SMR designer headquarters

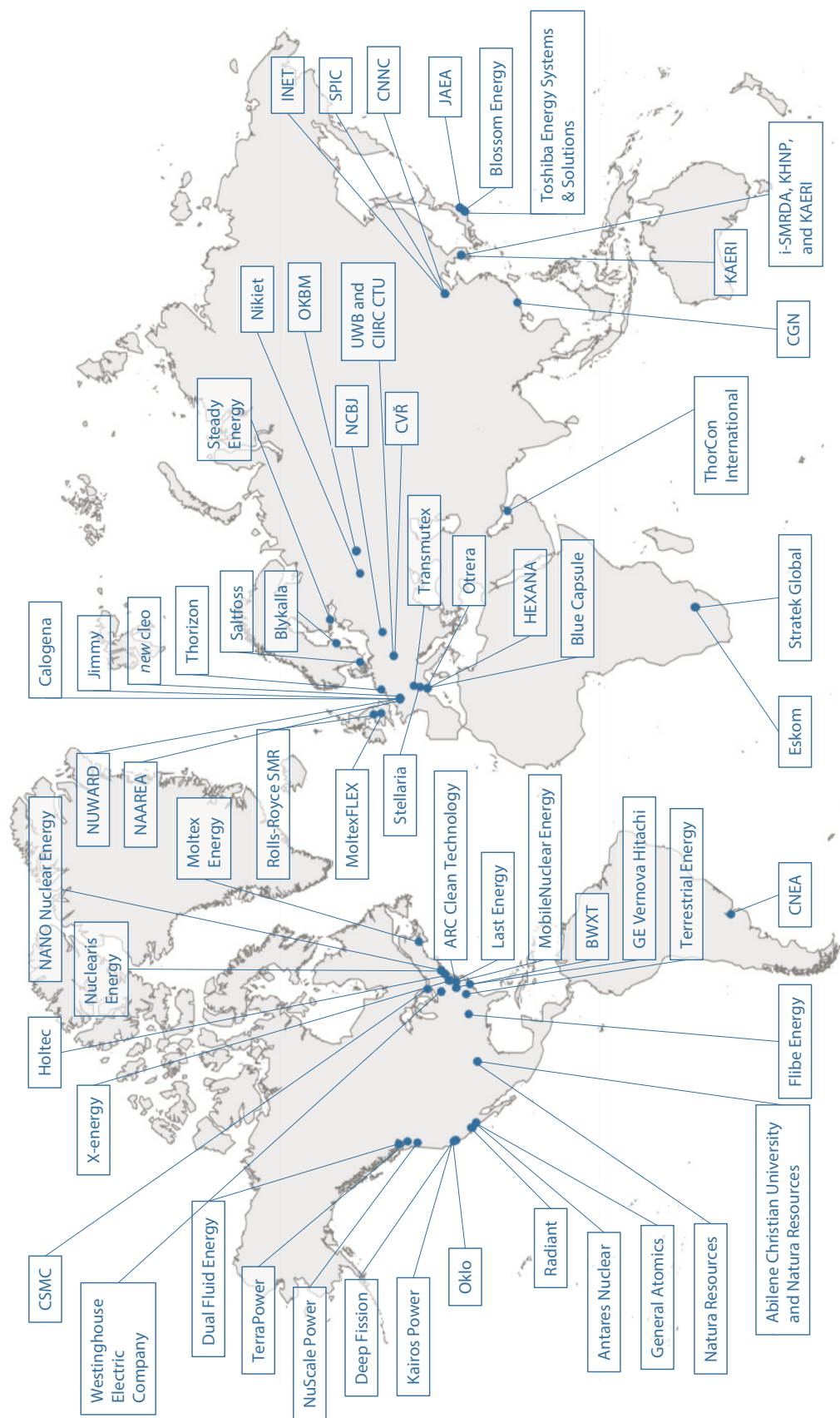
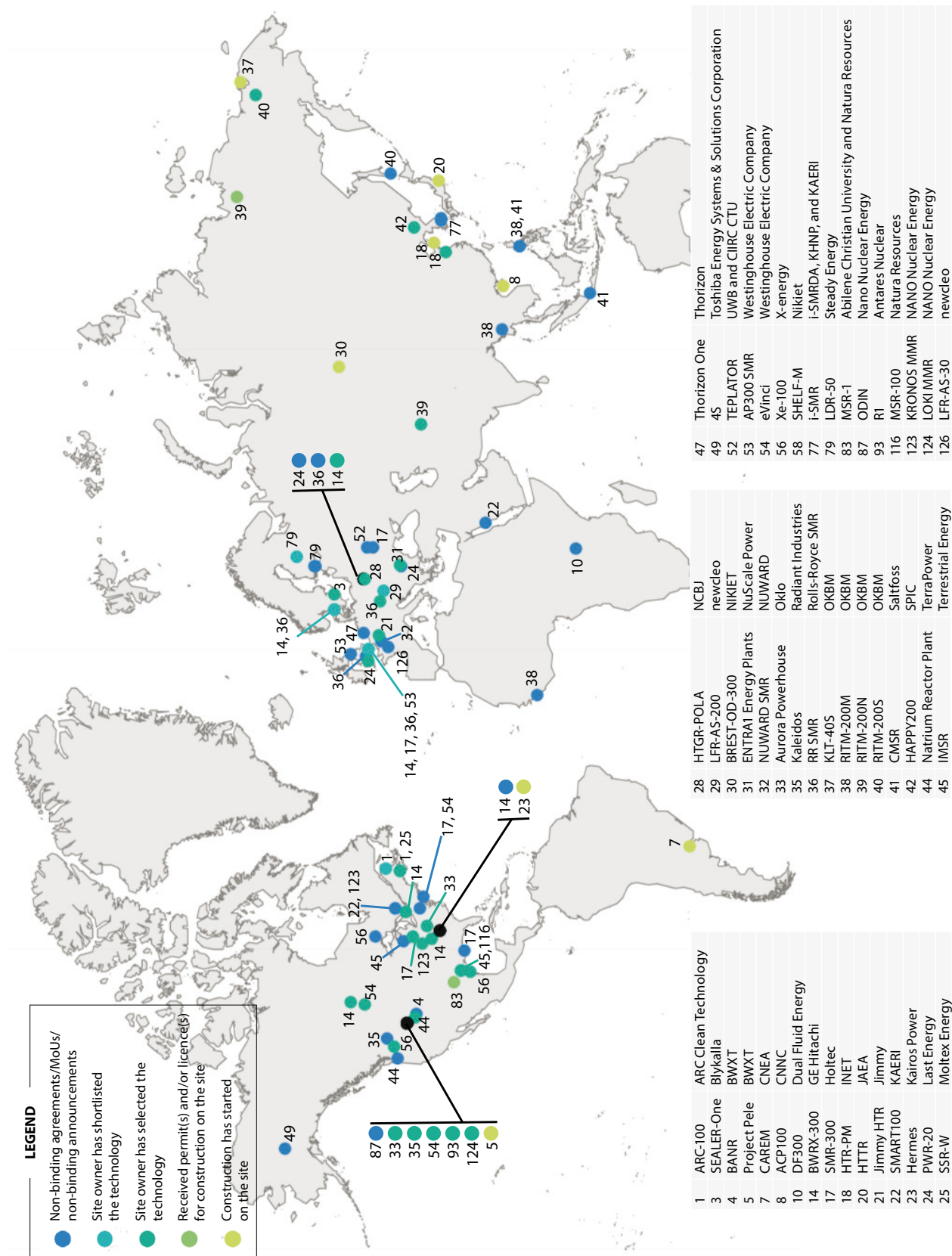


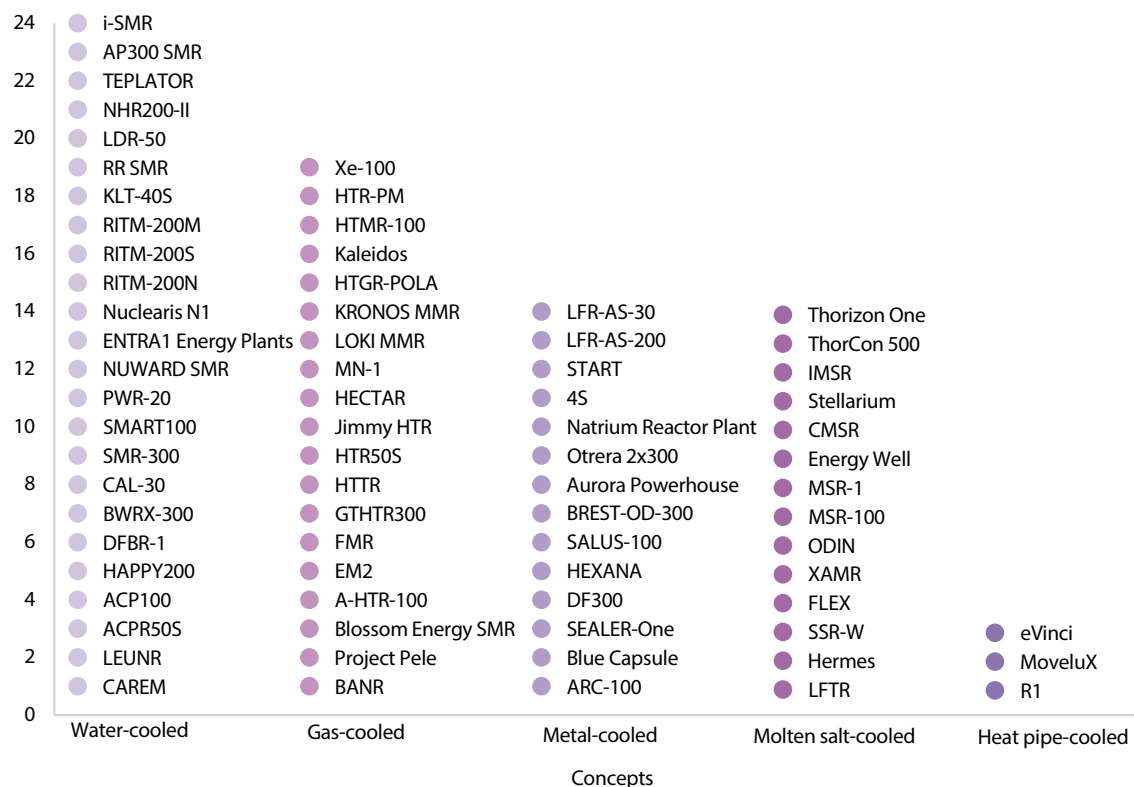
Figure 10. Map of SMR sites



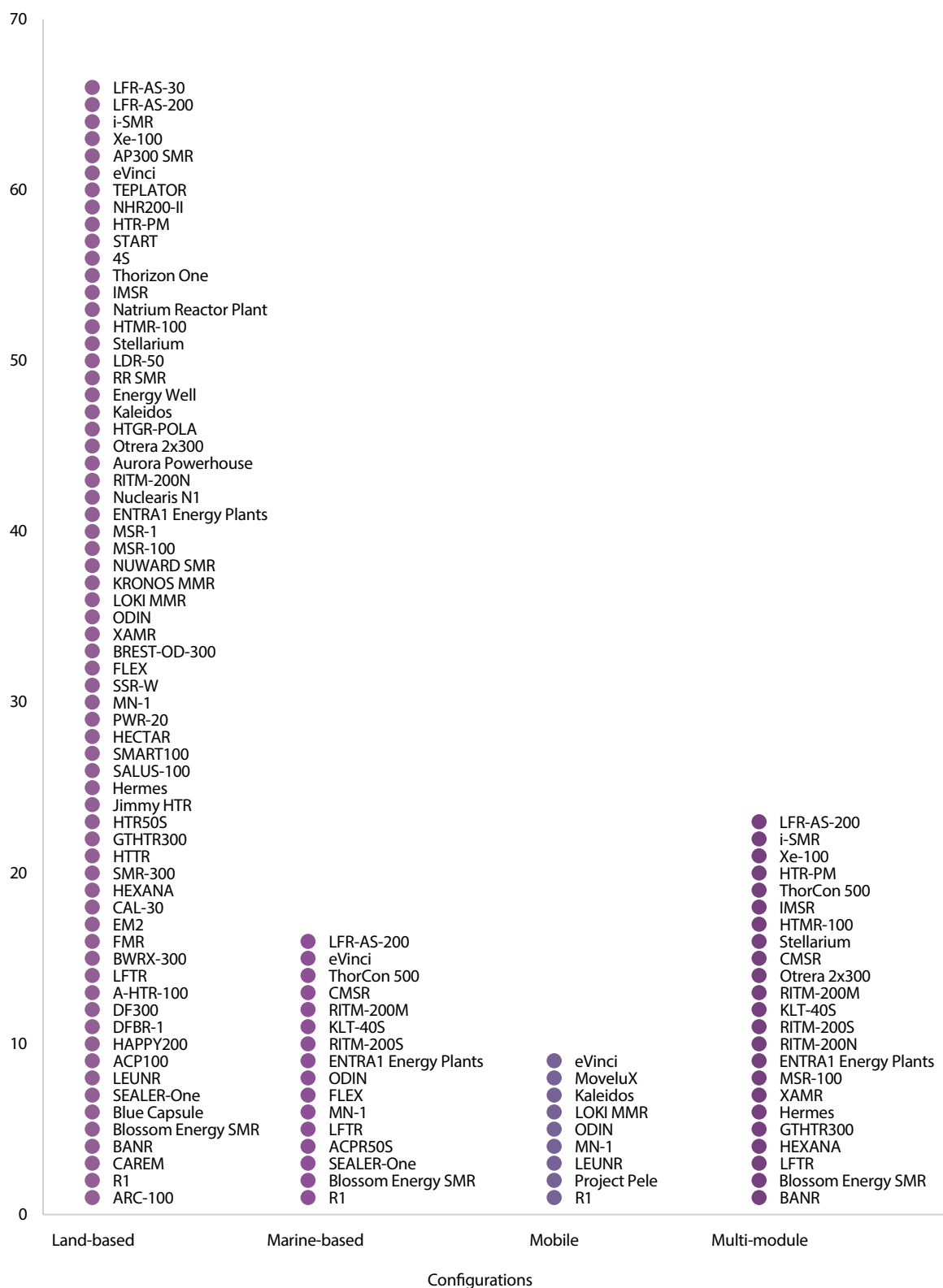
Different SMRs for different applications

A wide variety of SMR designs are being developed globally to serve different purposes. These designs vary in reactor concepts (Figure 11), configurations (Figure 12), and neutron spectrums (Figure 13). The range of reactor concepts extends beyond traditional water- and gas-cooled designs to include metal-cooled, heat pipe-cooled, and molten salt-cooled SMRs.

Figure 11. SMR concepts

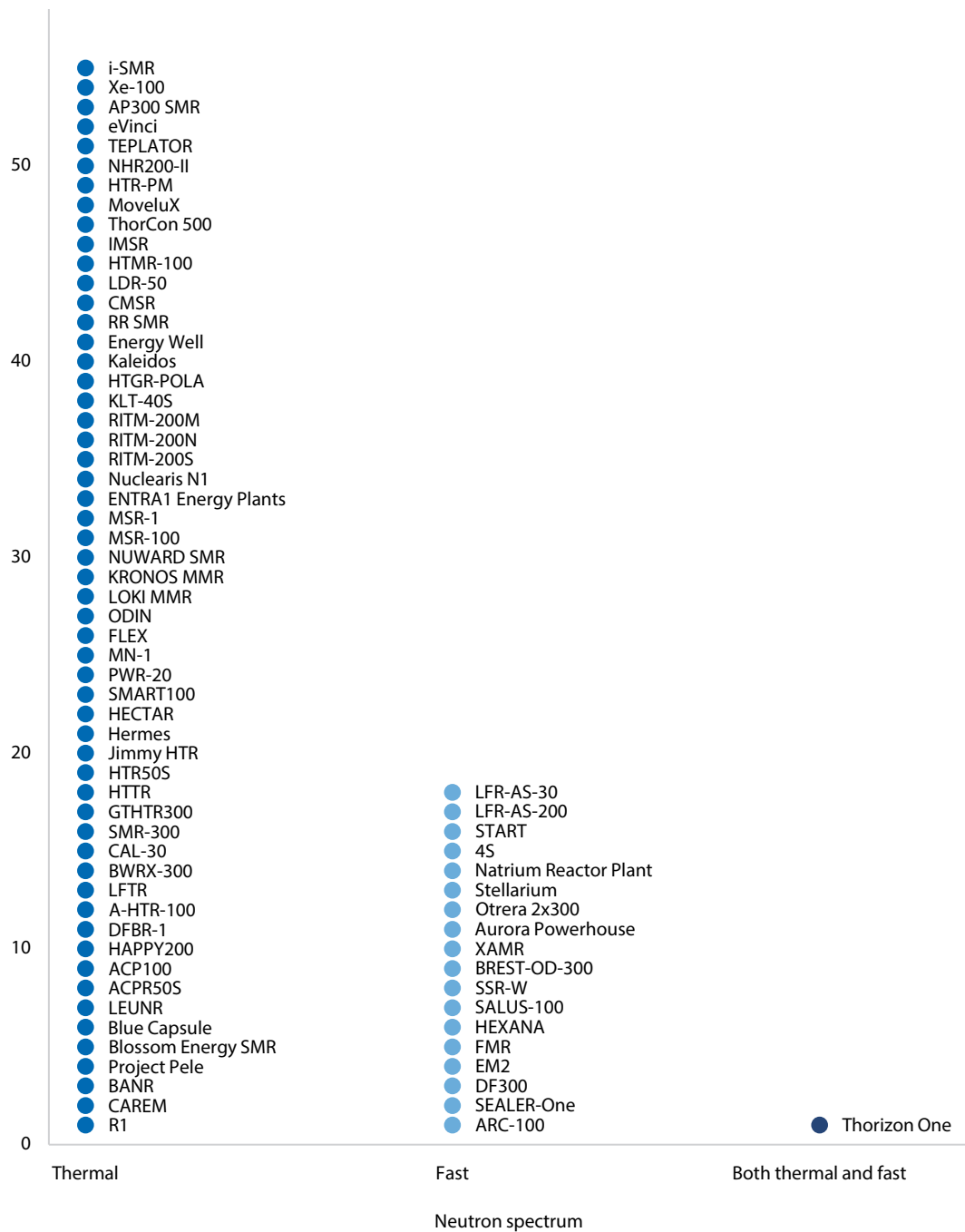


SMR configurations are not limited to land-based installations; marine-based reactors are already in operation. Additional innovative deployment pathways are also being planned, including multi-module reactor systems and mobile reactors designed for portability.

Figure 12. **SMR configurations**

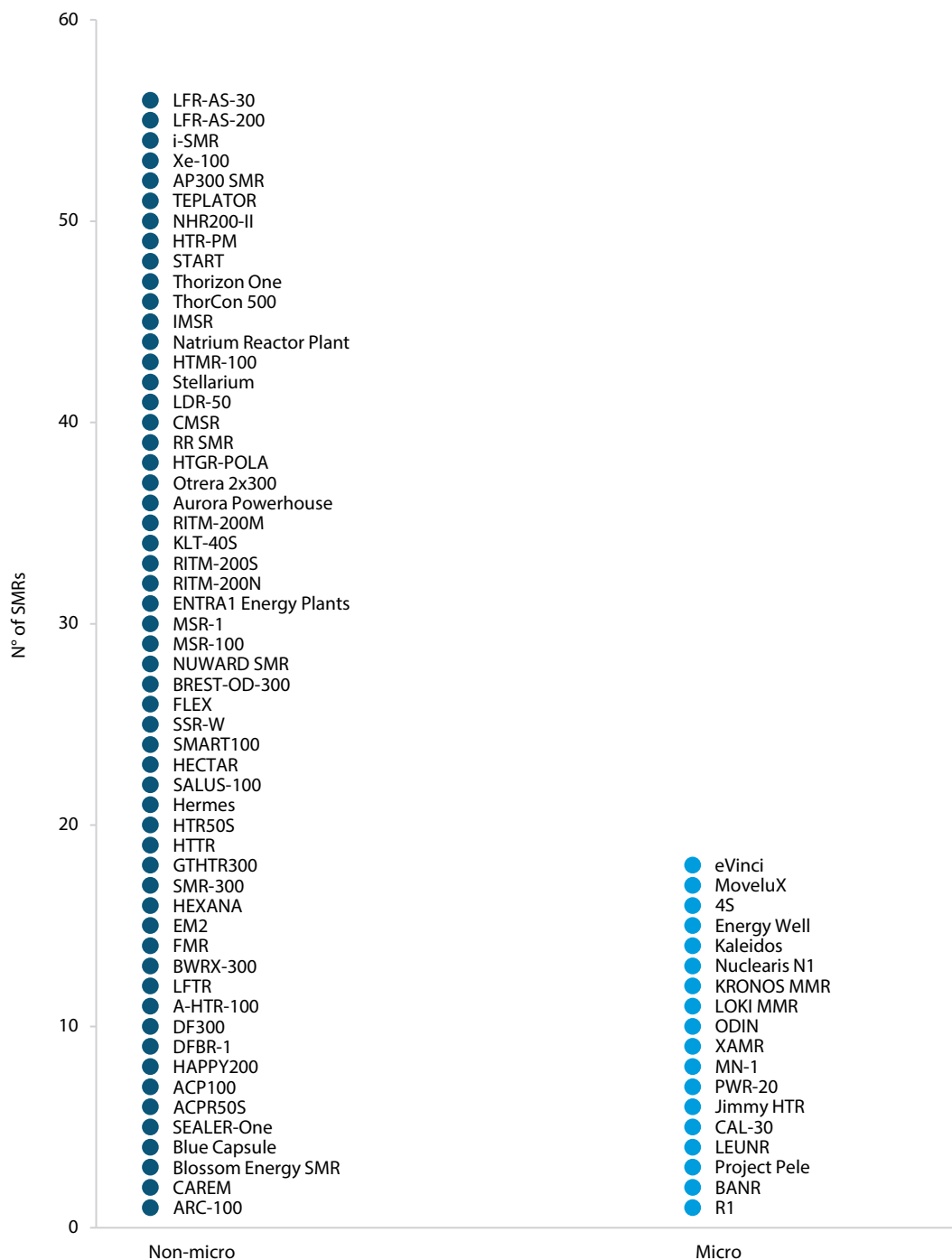
While most SMRs under development are designed to operate within the thermal spectrum, several are fast spectrum designs, and one can operate in both thermal and fast spectrums.

Figure 13. **SMR neutron spectrums**



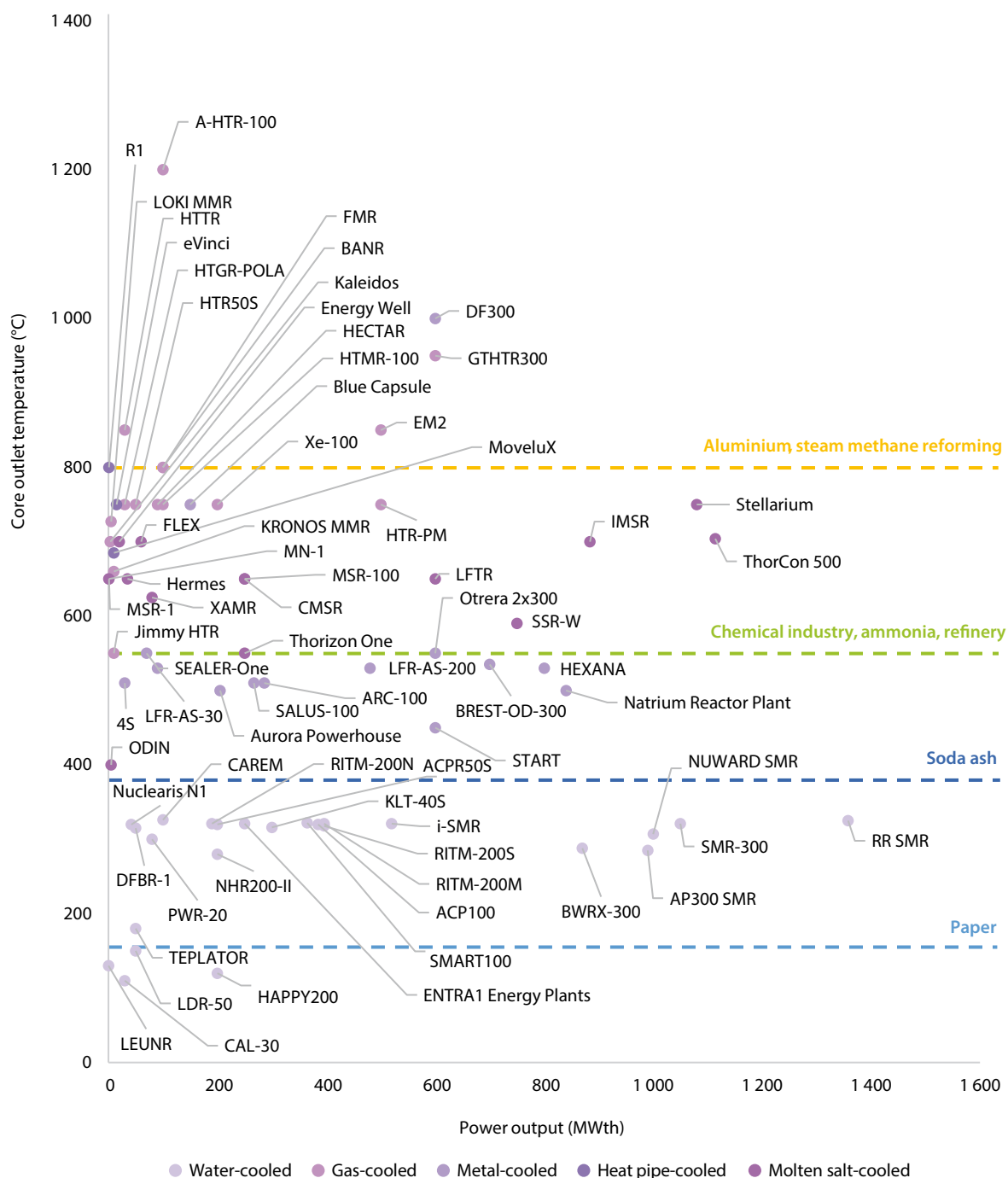
Many of the innovative designs featured in this edition of the *NEA SMR Dashboard* are micro-SMRs (or micro-reactors). A total of 18 designs are micro-SMRs, with an average electrical output of less than 10 MWe, significantly lower than the 180 MWe average for non-micro SMRs.

Figure 14. **Micro SMR designs**



While small and micro modular reactors are collectively defined by their compact size, there is significant diversity within this category. These reactors differ in power output, outlet temperature and fuel cycle. They range in size from less than 1 MWe to over 300 MWe, and in temperature from low core outlet temperatures around 110°C to more than 1 200°C. This wide range allows them to meet the specific temperature requirements of hard-to-abate industrial sectors (Figure 15).

Figure 15. **Range of sizes and temperatures for heat applications**



SMR designs currently under development incorporate a variety of fuel cycles, each involving different fuel types and enrichment levels (Figure 16 and Figure 17). Enrichment levels range from 5% or lower, similar to existing commercial light-water reactors, up to just under 20% for some advanced designs. These higher enrichment requirements could significantly impact the fuel supply chain, potentially requiring new infrastructure and investments to ensure adequate supply.

Figure 16. Range of sizes and uranium enrichment requirements

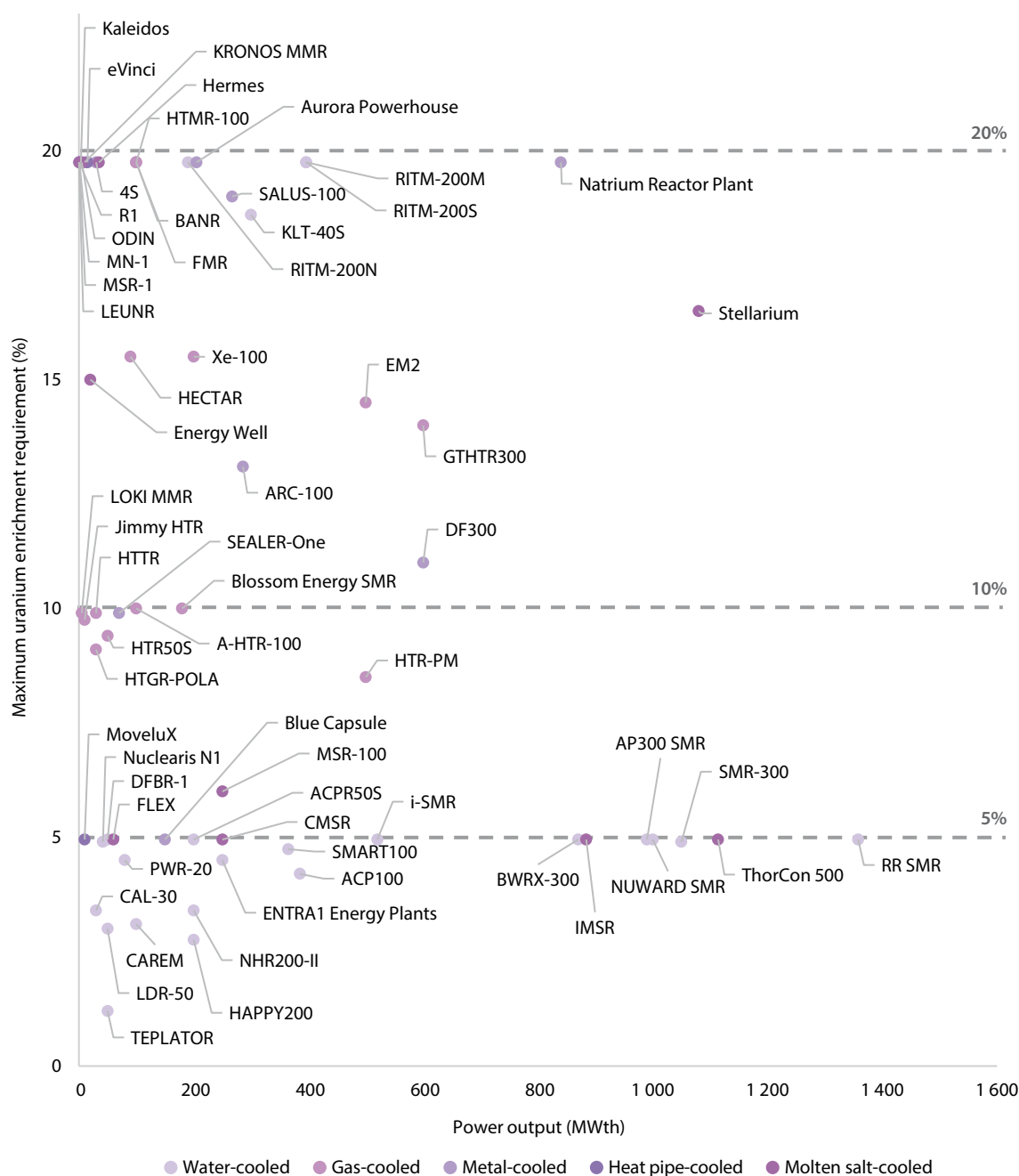
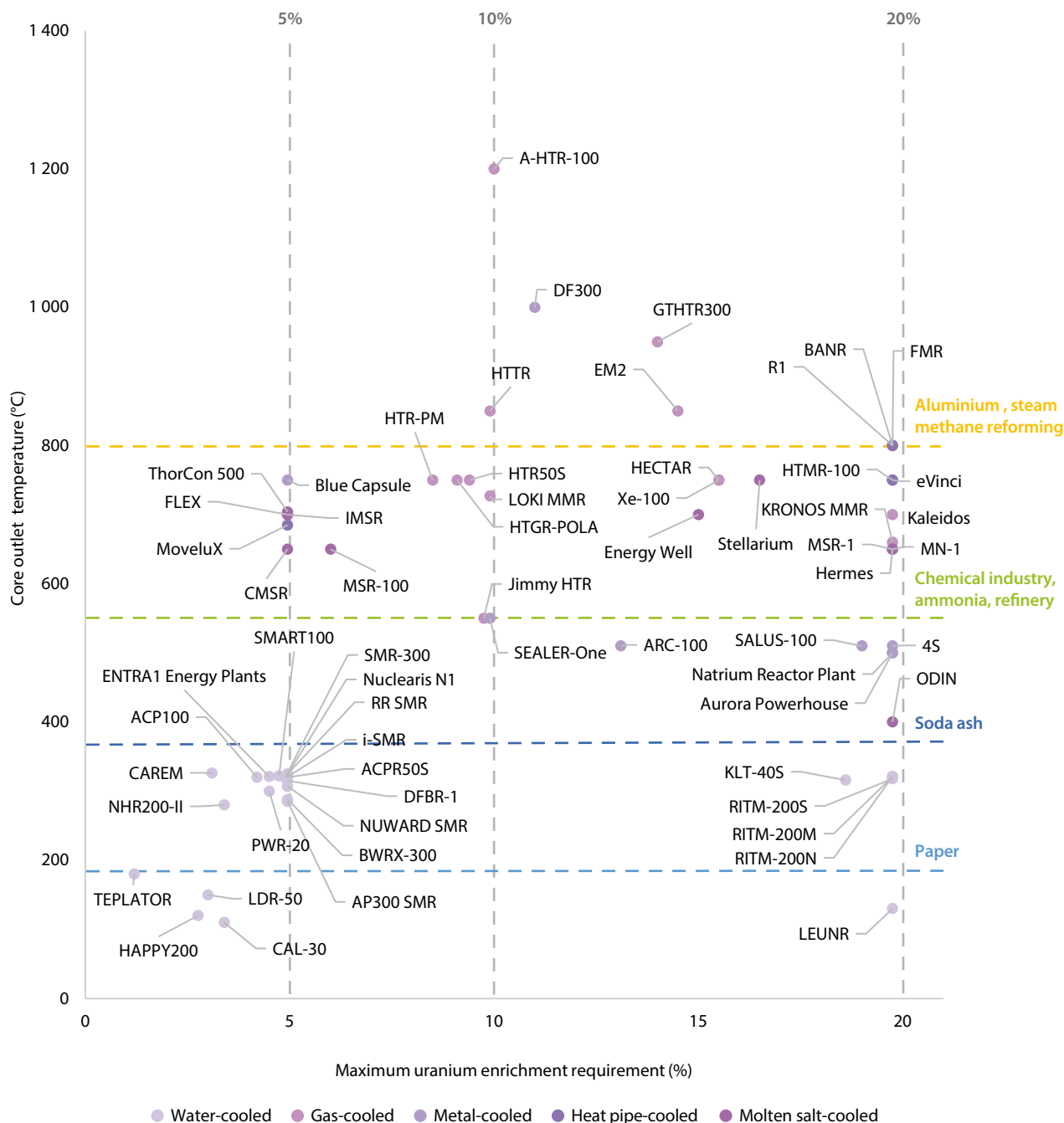


Figure 17. Range of temperatures and uranium enrichment requirements



The range of technical characteristics – concepts, configurations, neutron spectrums, sizes, and temperatures – enables SMRs to broaden the traditional market of nuclear energy. Some SMR technologies may be particularly suited to provide heat to hard-to-abate industrial sectors such as chemical production or oil and gas extraction, while others may be better suited for reliable electricity production in remote or offshore locations. SMR technologies may also be optimised for non-power applications including to produce medical isotopes or to reduce or recycle radioactive waste.

Data collected for the *NEA SMR Dashboard* reveals real progress in six near-term markets for SMRs:

- fossil fuel cogeneration replacement for industries, including hydrogen production;
- diesel replacement for mining;
- off-grid power;
- coal replacement for on-grid power;
- district heating; and
- on-grid reliable power for data centres.

This edition of the *Dashboard* identifies 25 projects making real progress at specific sites with commercial customers in these six near-term markets. Figure 18 shows the locations of these projects around the world. This does not include projects for data centres, which are not site-specific at this time. More information about announcements in all seven near-term markets, including data centres, is set out in Table 1.

Figure 18. **Map of near-term markets for SMRs**



1	ARC-100	ARC Clean Technology	40	RITM-2005	OKBM
4	BANR	BWXT	42	HAPPY200	SPIC
14	BWRX-300	GE Vernova Hitachi	44	Natrium Reactor Plant	TerraPower
17	SMR-300	Holtec	49	4S	Toshiba Energy Systems & Solutions Corporation
18	HTR-PM	INET	52	TEPLATOR	UWB and CIIRC CTU
20	HTTR	JAEA	53	AP300 SMR	Westinghouse Electric Company
21	Jimmy HTR	Jimmy	54	eVinci	Westinghouse Electric Company
31	ENTRA1 Energy Plants	NuScale Power	56	Xe-100	X-energy
35	Kaleidos	Radiant	79	LDR-50	Steady Energy
36	RR SMR	Rolls-Royce SMR			

A brief description of each is presented in Table 1.

Table 1. **Examples of demand for SMRs in near-term markets**

Fossil fuel cogeneration replacement for industries, including hydrogen production
• Canada's Belledune Port Authority shortlisted ARC-100 to generate a minimum of 1 gigawatt of firm heat and electricity for industrial users at the port's announced Green Energy Hub. • Orlen Synthos Green Energy is exploring the BWRX-300 design for hydrogen production and decarbonising chemical activities in Poland. • The Japan Atomic Energy Agency (JAEA) has been engaged in discussions with the Nuclear Regulatory Authority (NRA) of Japan about a hydrogen production facility utilising heat generated by the HTTR. • Kommersant, a Russian news outlet, reported that natural gas company Gazprom is considering RITM-200S units for the Kirinskoye and Yuzhno-Kirinskoye gas condensate fields in Sakhalin region. • Dow Chemical has selected X-energy to provide power and steam using four Xe-100 reactors to its manufacturing facility in Seadrift, Texas. • Westinghouse signed a Memorandum of Understanding (MOU) with the United Kingdom's Community Nuclear Power (CNP), an SMR development facilitator, for the construction of four AP300 SMRs at the North Tees Group Estate site in North Tees, United Kingdom. The site is planned to be in the heart of the Teesside regional hydrogen hub. • In Canada, Ontario Power Generation (OPG) has signed a Framework Agreement with X-energy to explore potential deployment of Xe-100 in Canada at remote and industrial sites. • Holtec is a member of the Mid-Atlantic Regional Clean Hydrogen Hub (MACH2) that includes Holtec's Oyster Creek nuclear site in New Jersey, United States, where Holtec is considering deploying its SMR-300. • WEC is working with the Saskatchewan Research Council (SRC) to develop a project to deploy the eVinci microreactor in Saskatchewan, Canada, where it would provide heat and electricity and support the testing of industrial, research and energy use applications. • Jimmy has submitted a licence application (Demande d'Autorisation de Création) to the French regulator to deploy its SMR at the Bazancourt site in France to supply heat to Cristal Union's food processing plant. • BWXT is working with Tata Chemicals Soda Ash Partners (TCSAP) to explore the feasibility of, and to develop the commercial terms of, deploying a series of BANR SMRs at the TCSAP's Green River site in Wyoming, United States. The BANR SMR would supply electricity and steam for trona ore mining and production of soda ash, used in the manufacture of glass and soap. • The Chinese government approved the deployment of the HTR-600S as part of Phase 1 of the Jiangsu Xuwei Nuclear Energy Heating Power Plant, which aims to supply a large volume of steam to the Lianyungang Petrochemical Park for a range of processes by 2030.
Diesel replacement for mining
• The chairman of Baimskaya copper mine and mineral processing facility in Cape Nagleynyn, Russia, signed an agreement to purchase heat and electricity from four floating nuclear power plants, with two RITM200S each, to be deployed and operated by ROSATOM. • Radiant has signed an MoU with the US mining company Idaho Strategic Resources to jointly study the suitability and licensing process to deploy a Kaleidos SMR at a site owned by Idaho Strategic Resources.
Off-grid power
• The Alaska Electric Light and Power Company and the City of Nome both wrote to the NRC indicating their interest in the 4S SMR for replacing diesel generators in remote areas of Alaska, United States.
Coal replacement for on-grid power
• PacifiCorp has selected the Natrium Reactor Plant to replace the retiring coal-fired Naughton Power Plant in Kemmerer, Wyoming, United States. • The Maryland Energy Administration partnered with X-energy to investigate the feasibility of repowering a current coal site in Maryland, United States, with a Xe-100 SMR. • Holtec has two co-operation agreements with NAEK Energoatom to plan for the deployment of SMRs in Ukraine to replace outdated or damaged units at thermal power plants. • In Romania, RoPower, a joint venture of Nuclearelectrica and Nova Power & Gas, has contracted NuScale for engineering and site evaluation of an ENTRA1 Energy Plant with a 6-NPM configuration to replace coal at the at Doicești plant.

District heating

- In 2021, the HTR-PM at the Shidaowan Nuclear Power Plant site in Shandong province, China, was connected to the electrical grid and, in March 2024, it was connected to a local district heating system.
- A tripartite agreement was signed between World ThermoExport, Teploatom Service, and the Ukrainian city of Slavutych to explore the possible deployment of a TEPLATOR reactor in Slavutych, in northern Ukraine, for district heating.
- The HAPPY200 SMR was selected for the Baishan Nuclear Energy Heating Project in China to provide district heating to an area of over 80 square kilometres.
- Steady Energy signed letters of intent and agreements with Finnish utilities Helen, Kuopion Energia, and Keravan Energia about the potential deployment of LDR-50 SMRs for district heating in the Helsinki, Kuopio, and Kerava regions, respectively.

Data centres

- While announcements by Google, Amazon and Meta cannot be depicted on the map in Figure 18, as they are not tied to specific sites, it is important to note their impact in creating a new source of near-term demand.
- Kairos Power signed an agreement with Google in October 2024 to construct and operate a series of SMRs to sell electricity to Google under Power Purchase Agreements (PPAs) to power data centres.
- Amazon has taken an equity stake in X-Energy, targeting deployment of 5 GW of SMR capacity by 2039. Amazon has also funded a feasibility study for at least four Xe-100 SMR modules near Columbia Generating Station in the United States, with Amazon having first-of-take rights for the generated electricity from this project.
- In December 2024, Meta issued a Request for Proposals for both SMRs and large-scale reactors to support up to four GW of capacity beginning in the early 2030s.

Figure 19 shows that, as of 2025, gas-cooled and water-cooled SMR concepts are preferred options for near-term markets, including beyond grid electricity generation. Figure 20 also provides insight into the types of site-owners progressing toward SMR deployment in near-term markets. Notably, heavy industry decarbonisation initiatives involve a diverse range of site-owners, whereas other markets tend to be dominated by specific site-owner types: government and utility-owned sites for district heating; utility-owned sites for coal replacement; government and industry-owned sites for off-grid power; and industry-owned sites for mining.

The figure 21 presents progress on siting for each of the 25 projects where site owners and customers are already engaged with specific SMR designers for deployment in near-term markets.

Figure 19. Near-term markets for SMRs, by SMR concept

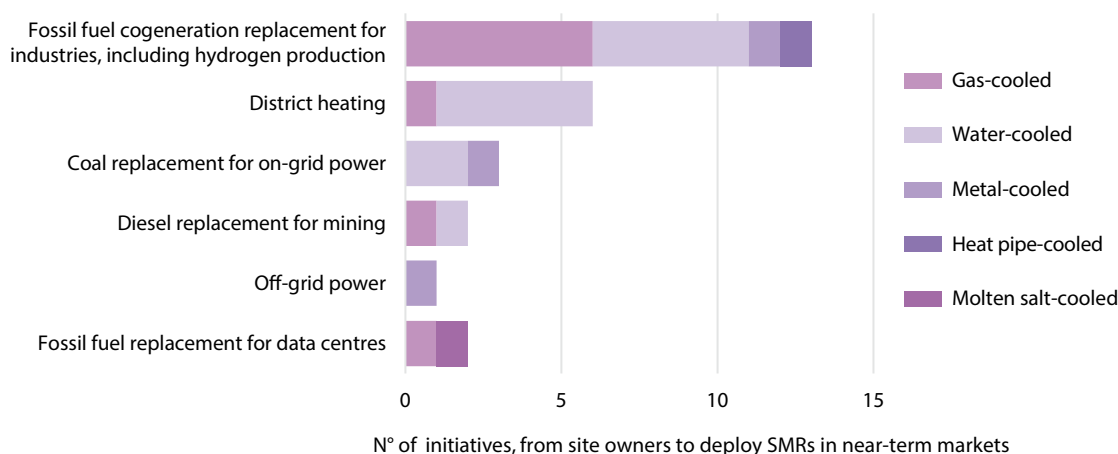
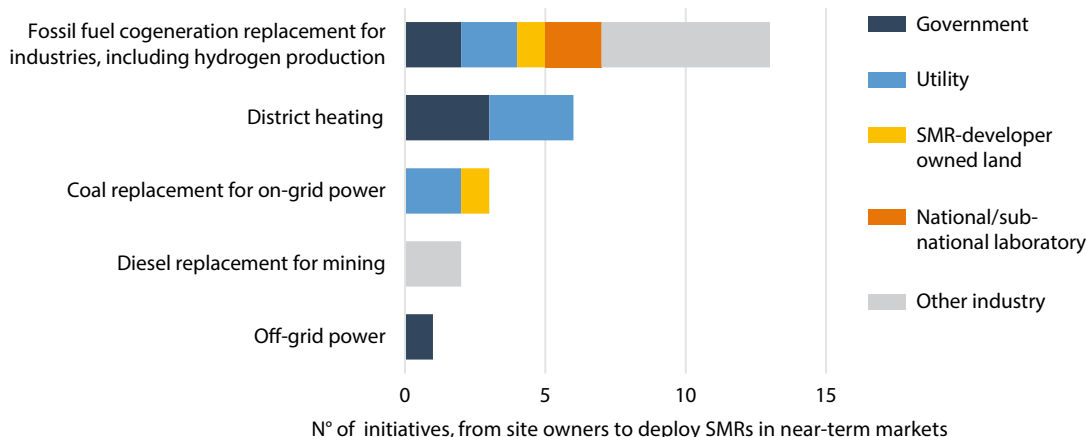
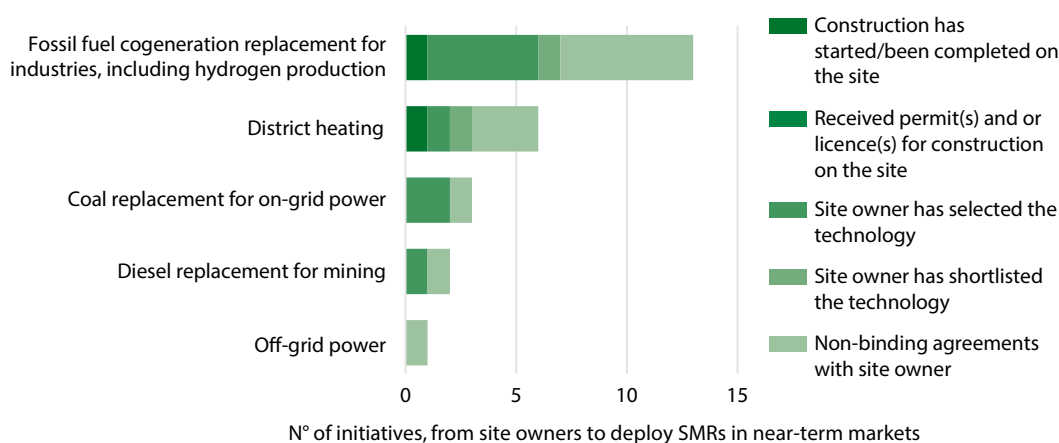


Figure 20. Near-term markets for SMRs, by site-owner type*



* Data centres are not included in this figure as announcements by data centres to date are not site-specific.

Figure 21. Progress toward deployment in near-term markets for SMRs*



* Data centres are not included in this figure as announcements by data centres to date are not site-specific.

Box 3. Industrial applications of SMRs

Industrial applications of SMRs include the near-term markets set out in Table 1, as well as several medium-term markets. Together, these industrial applications represent a transformation of the market for nuclear energy, from a historically utility-driven market to a new landscape characterised by diversified customers with different requirements and expectations (NEA, 2022). Some new industrial customers may also become strategic partners, creating opportunities for sector coupling and cross-sectoral learning and innovation.

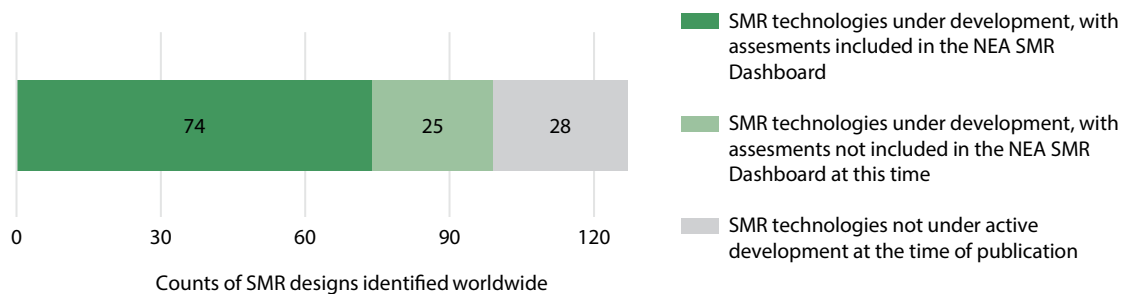
- Mining:** The NEA report *SMRs for Mining: Opportunities and Challenges for Small Modular Reactors* (NEA, 2024b) found that proximity to the electricity grid is a key driver in the feasibility of a mining project and identified a clear role for SMRs to provide reliable electricity and heat to power the mining industry. The analysis noted a particular opportunity for micro-SMRs, which were found to already be competitive with diesel generation at off-grid mines in remote locations representing a current market of 2 GWe for off-grid brownfield mines to replace fossil generation. It also noted that critical minerals essential to the clean energy transition are more often located in remote regions compared to primary metals, suggesting a growing need for remote mining and an energy source to power these mines.

- **Replacing coal generation:** Energy generation from coal power has been steadily declining since the early 2000s, generating positive environmental benefits but removing reliable baseload energy source. NEA analysis identifies 2.2 TWe of coal power globally, with 505 GWe of this generation considered to be in high priority regions where nuclear energy could be used to supplement this generation by 2050. SMRs to replace coal may also benefit from existing transmission infrastructure and a transitioning workforce from the coal sector in localised areas (NEA, forthcoming [a]).
- **District energy:** 25% of energy produced globally is used to heat and cool homes and commercial buildings, where over 90% of this heat is produced from fossil fuels. While large nuclear power plants have been used for district heating in the past, SMRs are being explored to provide a reliable source of non-emitting heat to district energy networks with projects moving forward in Europe and at university campuses in North America. In Finland, the LDR-50 SMR is being explored to provide heat to existing district energy networks in the cities of Kuopio, Helsinki and Kerava. (NEA, forthcoming [b])
- **Chemicals:** There is a history of nuclear energy being used for chemical production, including at the Bruce Power site in Canada prior to the 1990s, and history is repeating itself with growing interest in China, Europe, Korea, Russia and the United States (NEA, 2022). In the United States, Dow Chemicals has applied for a permit to construct the Xe-100 SMRs as a source of steam to replace existing assets approaching end of life. In Europe, Orlen Synthos has begun environmental and siting work for their planned deployment of the BWRX-300. In China and Russia, construction has started to deploy SMRs in the chemicals sector, including an 800 MW version of the HTR-PM SMR in China and the BREST-300 lead-cooled fast reactor in Russia.
- **Remote communities:** The compact size and reliable energy production of SMRs and micro-SMRs make them an option for areas lacking grid access, such as remote communities. These remote communities represent many different physical settings with unique challenges, such as extreme heat or cold and permafrost. Remote communities in the Arctic, Africa and island states have vastly different requirements, energy options, and technical and logistical challenges. Integrating these diverse needs and energy options with nuclear energy will require important design considerations. There are currently several such efforts, including exploratory work in Alaska, United States, and Canada, and efforts to deploy the RITM-200N in Yakutia in Russia.
- **Merchant shipping:** Maritime shipping accounts for approximately 80%, by volume, of global trade and contributes roughly 3% of global emissions annually (UNCTAD, 2024). Micro-SMRs ranging from 1 to 100 MWth may provide an opportunity to power the shipping industry, creating an alternative to fossil fuels. Nuclear reactors for maritime applications have been deployed on icebreakers and submarines. There is some limited experience with nuclear-powered merchant shipping as well. The first was the US NS Savannah, commissioned from 1959 to 1971 (Savannah Association, n.d.). The most recent was the Russian NS Sevmorput, commissioned in 1988, refueled in 2003, and still operating today in Russian waters (WNA, 2025c; Marine Traffic, n.d.). Some changes to legal, regulatory and policy frameworks may be required to enable deployment of SMRs more broadly in this market. Blykalla, Core Power, LeadCold and Kairos are among the companies exploring this potential market for SMRs.
- **Data centres:** Global data centre electricity demand is surging due to the growing use of artificial intelligence, blockchain technologies and other cloud-based technologies. To secure reliable power for this growth, data centre operators are increasingly showing interest in nuclear energy and SMRs. In the United States, technology companies have announced plans to support the deployment of SMRs. Google has partnered with Kairos Power to deploy 500 MW of SMRs for its data centres by the mid-2030s. Amazon selected X-Energy for an immediate 320 MW SMR project in Washington State and to enable over 5 GW of new nuclear capacity by 2039.
- **Steel:** The production of steel accounts for 7% of greenhouse gas emissions globally and requires process heat of more than 1 000°C (IEA, 2020). Nuclear energy is being considered to produce clean hydrogen that can be used for the direct reduction of iron to produce steel, with processes being actively explored in Korea by POSCO, in Europe by HYBRIT, and in North America.
- **Oil and gas extraction:** Thermal energy from SMRs can be used directly in many industrial processes, with an immediate opportunity to use steam in the extraction of oil. In Canada, the Pathways Alliance representing 95% of oil sands production is working together to explore SMRs among other sources of energy to provide large volumes of steam for oil recovery through in-situ steam-assisted gravity drainage (Pathways Alliance, 2023). Off-shore oil rigs may also benefit from a compact supply of clean heat and electricity to replace on-site fossil fuel combustion.
- **Space:** Space nuclear power systems have been utilised for the past 60 years to advance space exploration and improve the knowledge of the solar system (ISECG, 2023). Recent global efforts suggest that the expanded use of micro-SMRs is critical for future planned activities in space, with space SMR concepts being designed by Westinghouse, BWX Technologies, Nano Nuclear Energy, Antares Nuclear and Canadian Space Mining Corporation.

The above applications reflect growing markets for nuclear energy beyond traditional electricity generation. SMRs have the potential to become a replacement for current fossil energy sources to address many of these markets, driving a real near-term need for technology demonstrations and commercialisation.

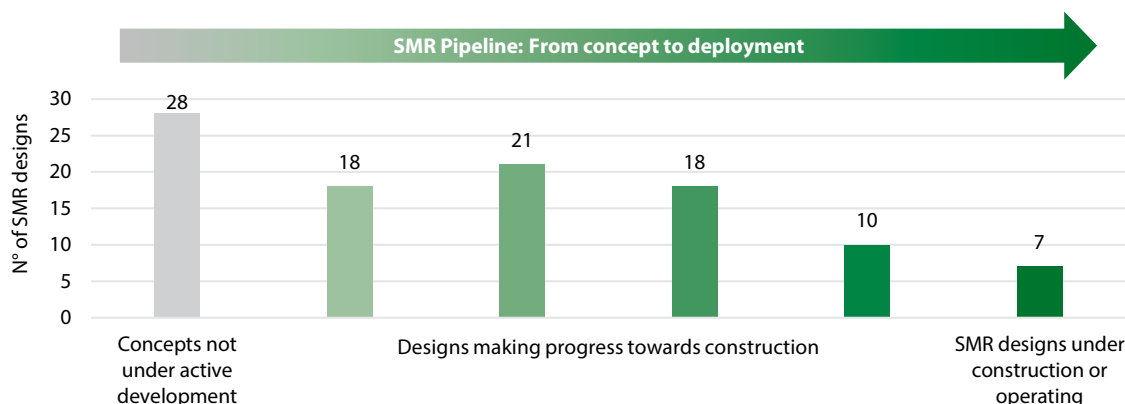
Progress towards demonstration and commercialisation

Figure 22. **Count of SMRs identified worldwide**



The NEA has identified 127 SMR designs around the world. Of those, 74 are included in this third edition of the *NEA SMR Dashboard*; these are the SMRs for which the requisite publicly available information was assessable and for which the relevant designers were willing to participate. The other 53 include approximately 25 that are under development but requested not to be included in the *NEA SMR Dashboard* at this time but may be included in the future; the others include SMR designs that are not under active development, may be without human or financial resources, or have been cancelled or paused indefinitely (Figure 22).

Figure 23. **SMR pipeline: Progress towards first-of-a-kind commercial deployment**

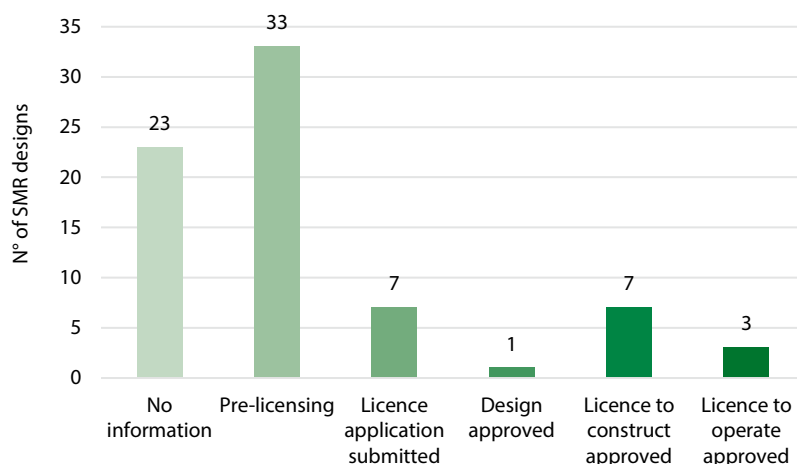


The *NEA SMR Dashboard* offers the most comprehensive assessment to date of progress toward SMR commercialisation, distinguishing between designs making tangible advancements and those still in early development (Figure 23). Seven designs are already either operating or under construction, with a strong pipeline of projects progressing toward first-of-a-kind deployment. While some SMR designs remain at the conceptual stage, several may accelerate development in the coming years. The diversity of designs presents challenges and opportunities to streamline global supply chains, promote standardisation, and enhance the economic viability of SMRs.

The following sections explore global trends and progress in key areas critical to SMR deployment: licensing, siting, financing, supply chains, stakeholder engagement and fuel.

Licensing

Figure 24. **Licensing progress worldwide**



Most licensing progress for SMRs is concentrated in early-stage engagement with nuclear safety regulators and non-commercial designs

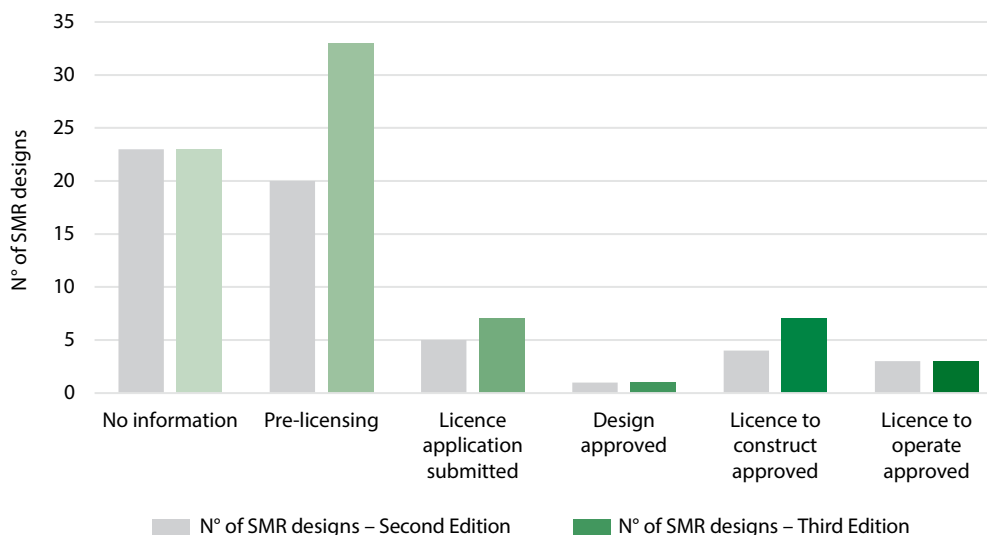
Figure 24 shows the pipeline of SMRs progressing through licensing around the world. Three SMR designs are licensed to operate, seven have been granted a licence to construct, one has been granted design approval, seven have submitted a licence application, and 33 SMR designs are engaged in pre-licensing activities. For 23 SMR designs there is no information available from verifiable public sources related to licensing or pre-licensing activities.

Table 2. **Licensing highlights**

- As of 2025, the NEA is tracking 51 SMR designs involved in pre-licensing or licensing activities with nuclear safety regulators across 15 countries worldwide.
- Since 2023, global pre-licensing activity has increased by 65%, reflecting accelerated progress in many regions. Notably, France and the United States have seen significant growth in the number of SMR designs entering the licensing process.
- In 2024, three construction licences in the United States were approved:
 - The Department of Defense’s Project Pele
 - Kairos Power’s demonstration molten salt-cooled reactor, Hermes
 - Natura Resources’ molten salt-cooled research reactor, MSR-1
- These are the first licences to construct Generation IV SMRs in the United States in decades. Although these specific examples are not themselves commercial projects – one is a defence project, the other two are demonstration and research projects, respectively – they signal a wave of innovation in the nuclear sector. While the United States is home to the largest number of pre-licensing and licensing activities, China and Russia are further ahead in deployment of novel Generation IV SMRs. For example, China has operated its gas-cooled HTR-PM since 2021, and Russia has begun construction of the fast-spectrum BREST-OD-300 reactor, demonstrating progress across a variety of SMR concepts.

Significant global progress in the licensing of SMRs has occurred since the release of the *NEA SMR Dashboard: Second Edition* (NEA, 2024c). The number of SMR designs undergoing pre-licensing has increased by 65%, reflecting strong engagement between SMR developers and nuclear regulators, particularly in France and the United States. Progress towards formal licensing is also notable, with five applications submitted by SMR developers in Canada, France, the United Kingdom and the United States. Additionally, three construction licences have been approved, all in the United States: the Department of Defense’s Project Pele, Kairos Power’s Hermes, and Natura Resources’ MSR-1. More information is presented in Table 3.

Figure 25. **Comparison of licensing pipelines between the second (2024) and third (2025) editions of the NEA SMR Dashboard***



* The NEA SMR Dashboard: Second Edition is based on data as of 10 November 2023. The NEA SMR Dashboard: Third Edition is based on data as of 14 February 2025.

Table 3. **Licensing progress worldwide, detailed**

Licence to operate approved
As of February 2025, there are three SMRs deployed and operating: The HTR-PM in China, the floating KLT-40S in Russia and the High Temperature Test Reactor (HTTR) in Japan.
Licence to construct approved
- Since the publication of the <i>NEA SMR Dashboard: Second Edition (2024)</i>, licences to construct have been approved for three SMR designs: the Department of Defense Project Pele, Kairos Power's demonstration molten salt-cooled reactor Hermes, and Natura Resources' molten salt-cooled research reactor MSR-1. - Those three SMR designs add to the previously approved licences to construct for four SMR designs: The CAREM SMR in Argentina, the ACP100 in China, as well as the RITM-200N and the lead-cooled, fast-spectrum BREST-OD-300, both in Russia.
Design approved
Korea's Nuclear Safety and Security Commission (NSSC) approved the SMART100 SMR design in 2024. SMART100 is an updated version of the SMART SMR that was originally designed by KAERI and received a Standard Design Approval from NSSC in 2012.
Licence/construction/design certification application submitted
A total of seven SMR designs have submitted an application to regulators: GE Vernova Nuclear Energy's BWRX-300 in Canada; CGN's floating ACPR50S in China; Jimmy's High Temperature Reactor SMR in France; OKBM Afrikantov's floating RITM-200S in Russia; Rolls-Royce's SMR in the United Kingdom; and NuScale Power's ENTRA1 Energy Plant SMR, and TerraPower's Natrium Reactor Plant in the United States.
Pre-licensing
- Thirty-three SMR designs have started pre-licensing activities. - This third edition of the <i>NEA SMR Dashboard</i> records a significant increase in pre-licensing activities worldwide, up by 65% since the previous edition (2024). - France stands out with seven SMR designs having entered pre-licensing during 2024.
No information available
As of February 2025, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities for 23 SMRs.

Figure 26 presents the number of SMRs in pre-licensing or licensing activities with nuclear safety regulators around the world. China, Japan and Russia have SMR designs already licensed to operate. Various SMR designs have been granted licenses to construct or design approval in the United States, Russia, China, Argentina and Korea. License applications are under review by regulators in the United States, France, Canada, the United Kingdom, China and Russia. Most pre-licensing engagement is taking place in the United States, France, and Canada.

Figure 26. **Number of SMRs in pre-licensing or licensing activities with nuclear safety regulators, by country**

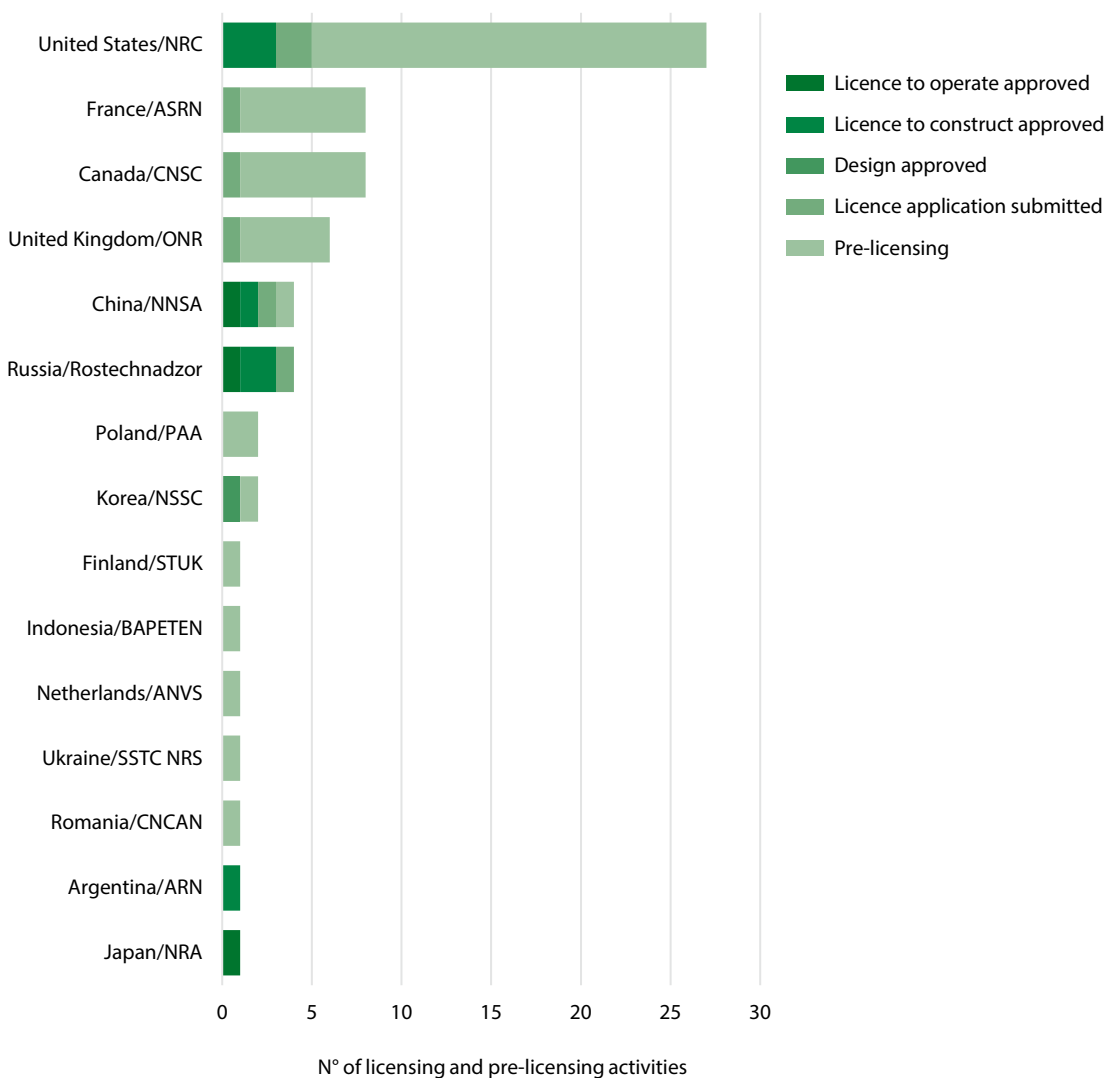
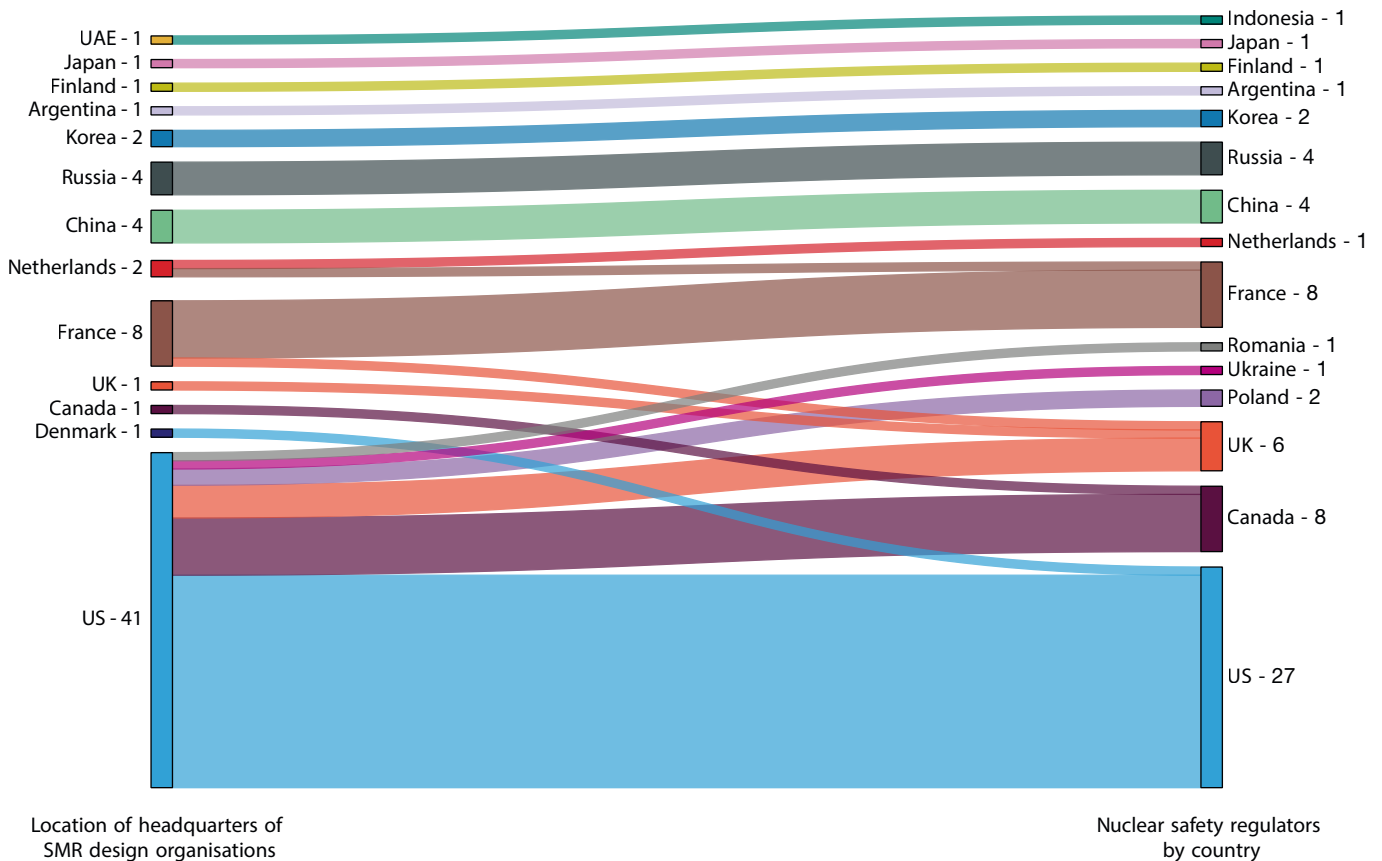


Figure 27 presents another view of licensing activities. The countries that are home to the headquarters of the SMR design organisations appear on the left. The countries that are home to the nuclear safety regulators appear on the right. The bars indicate pre-licensing and licensing activities with domestic and foreign nuclear safety regulators. While most SMR designers are advancing pre-licensing and licensing activities with their domestic nuclear safety regulators, 18 projects involve SMR designers working with multiple regulators or regulators abroad.

Figure 27. **SMR pre-licensing and licensing activities with domestic and foreign nuclear safety regulators**



While most non-water-cooled and non-land-based reactor designs remain in the pre-licensing phase, progress across all SMR concepts and configurations (Figures 28 and 29) can be observed. Notably, the United States has approved construction licences for two molten salt-cooled reactors, Hermes, a demonstration reactor, and MSR-1, a research reactor. This progress reflects the commitment of implicated nuclear safety regulators to develop the necessary expertise for licensing innovative Generation IV designs.

Figure 28. Licensing progress, by SMR concept

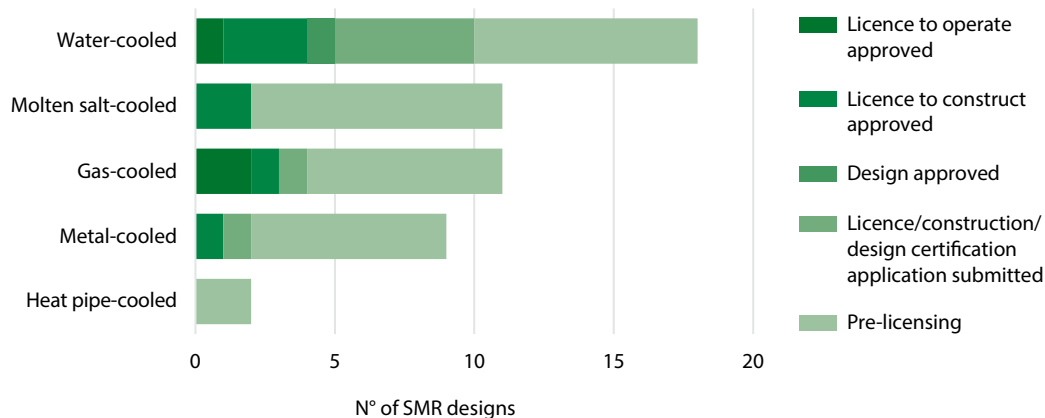
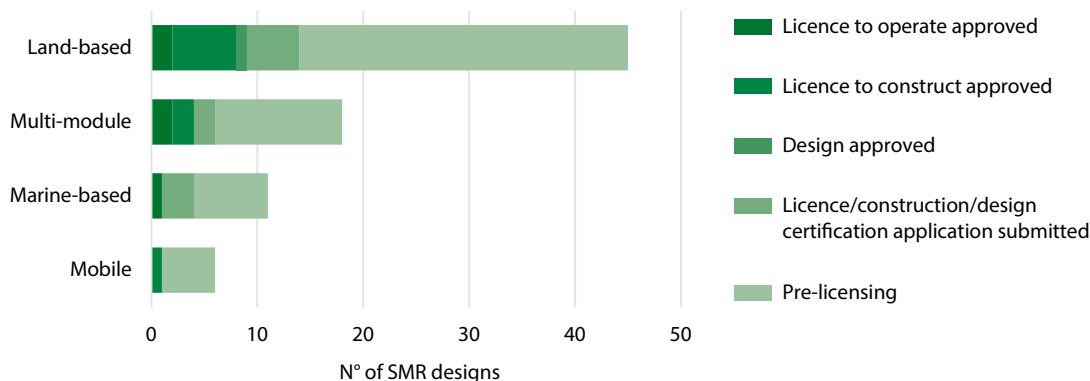
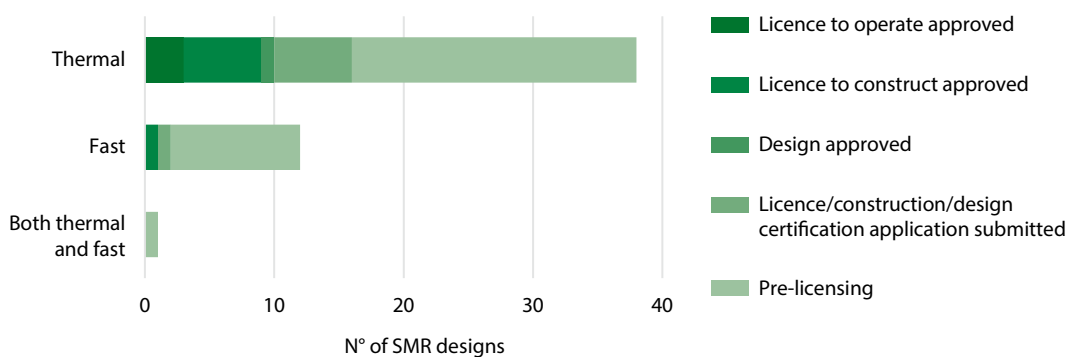


Figure 29. Licensing progress, by SMR configuration



There has been limited global progress in the licensing of fast-spectrum SMR designs (Figure 30). To date, only one such design, the BREST-OD-300 in Russia, has received a construction licence. Within OECD countries, TerraPower's Sodium Reactor Plant stands out, having submitted a construction permit application to the US Nuclear Regulatory Commission in 2024 for deployment on its proposed site in Kemmerer, Wyoming.

Figure 30. Licensing progress, by SMR neutron spectrum



Box 4. **AI opportunities to accelerate new energy technologies**

provided by Mikaela Winter, Senior Manager, Energy Policy, Conor Kelly, Power & Utilities Decarbonization Software Lead, Mansour Belhadj, General Manager, Energy & Resources Industries, Microsoft

Throughout history, each wave of societal transformation has been catalysed by a general-purpose technology, one capable of reshaping entire economies, redefining work, and unlocking new frontiers of human potential. From the steam engine to electricity, from the printing press to the internet, these breakthroughs have marked inflection points in how the world functions.

Artificial intelligence (AI) now stands at the threshold of becoming the most consequential of them all. With its capacity to learn, reason and scale across sectors, AI is not merely another tool in the digital toolbox; it is a force multiplier for innovation, problem-solving and system-wide transformation.

In the context of the global energy transition, AI arrives at a critical juncture. It offers the ability to expand access to clean energy, accelerate the deployment of advanced technologies, and decarbonise complex systems. Specifically, AI's strength in data synthesis, predictive analytics, and intelligent automation positions it to unlock breakthroughs in how energy is produced, regulated and consumed. For the nuclear sector and SMRs in particular, this could be the key to moving faster, safer, and more cost-effectively into a clean energy future.

o understand how AI can deliver on this promise, it is essential to examine the specific capabilities that make it uniquely suited to transform complex, safety-critical sectors like energy and in particular, nuclear power.

Game-changing capabilities of AI:

Measure, predict and optimise complex systems

AI can enable people to discern patterns, predict outcomes and optimise performance in systems that are too complex for traditional analytic methods.

Accelerate the development of technologies and solutions

AI can accelerate the discovery and development of innovations and new discoveries.

Empower the workforce

AI can empower the workforce by enabling targeted training and assistance and enabling the workforce to access and distil vast archives of data, regulations, science and policy documents so that the workforce can easily find the information they need to understand and manage complex challenges.

How to unlock AI's full potential – The five plays:

- **Invest in AI to accelerate solutions:** AI has numerous applications that can enhance efficiency, optimise business operations and provide game-changing breakthroughs to challenges and bottlenecks. AI can help to expedite the integration of clean energy onto electric grids, develop energy storage solutions, reduce waste, optimise supply chains and enable accurate grid forecasting in advance of current capabilities.
- **Develop digital and data infrastructure for the inclusive use of AI:** Data is the foundation on which AI operates, shaping its insights, predictions and decision-making capabilities. Yet, there are major gaps and accessibility challenges that constrain the development of accurate and representative AI models. Even when data exists, it can be inaccessible or difficult to use because it is locked in institutional silos, not digitalised, or in incompatible formats. Data standards, sharing mechanisms and platforms are needed to increase the usability of data in AI models.
- **Minimise resource use in AI infrastructure design and operations, expand carbon-free energy, and support local communities:** AI operations require resources such as energy and water. To expand the power and applications of AI, AI operations must align with global sustainability and decarbonisation goals to achieve a net zero and nature-positive future. The continued development of AI will increase the demand for resources, but how much more electric power capacity is needed will depend on how AI models are designed and operated, how technologies advance and the pace of AI adoption. In response to these challenges, it is important to focus on minimising resource use, accelerating and expanding access to carbon-free electricity, including emerging nuclear energy technologies, and supporting local communities in achieving their social, economic and sustainability goals.

- **Advance AI policy principles and governance:** AI technologies can have a positive impact on both the environment and society by accelerating sustainable business practices and the energy transition. The infrastructure that hosts the computing power needed to yield these benefits may affect resource use too, such as by increasing power needs while reducing water reliance. Governments have an opportunity to enable the positive impacts of AI by crafting policies that harness its capabilities to benefit and ensure alignment with sustainability outcomes while also mitigating the resource impact that will result from the increased demand for AI.
- **Build workforce capacity:** To harness the transformative power of AI requires a solid foundation of human capacity to use AI tools. Building a workforce prepared to use AI requires holistic learning pathways that cultivate AI fluency within the context of the relevant industry(ies) and market(s) (Microsoft, 2025)

Opportunities for AI in power systems and clean energy technologies

The applications of AI in the energy industry span data management and optimisation, predictive analytics and maintenance, workforce management, forecasting, grid and resource optimisation, health and safety, and more. AI is already being utilised to improve grid reliability and capacity while reducing costs (Microsoft, 2024), enhancing the security of energy supply through comprehensive virtual inspections of transmission and distribution lines (Microsoft, 2022), and advancing the permitting process for new energy resources (Lloyd's Register, 2025).

Permitting has long been a critical bottleneck for the rollout of energy infrastructure worldwide. Small modular reactors (SMRs) have the potential to provide clean, firm power in a variety of sizes and configurations. The time-intensive, complex and cost-intensive permitting and licensing processes introduce unpredictability and risk to the development and deployment of SMRs globally. Microsoft's AI capabilities can enhance the productivity of energy infrastructure developer and regulator permitting operations, reducing the cost and time to permit new energy infrastructure.

- **Draft complex regulatory documents** – AI can greatly enhance the speed of drafting permitting documents with the relevant data and in the required format for human review and refinement. The solution can provide answers on whether the document meets the regulation or if additional information is needed. The capabilities also provide references to the relevant regulation(s) or requirement(s).
- **Query large, complex datasets, past applications, regulations, specifications, requirements and reports** – AI-powered assistants can help engineers and regulatory professionals navigate complex nuclear licensing rules and regulations. These AI copilots provide real-time responses to queries such as, What additional safety documentation is required for my SMR permit? These capabilities reduce human workload while ensuring compliance.
- **Accelerate review of draft documents** – AI agents can reviews draft documents against the regulations, specifications and requirements, allowing permitting teams to spot-check and resolve missing information, issues or inconsistencies before submission to the regulator. AI-powered compliance validation ensures that licensing applications meet all necessary requirements, reducing regulatory back-and-forth and accelerating approvals.

Featured use case: Lloyd's Register to use generative AI to advance the application of nuclear technology in maritime.

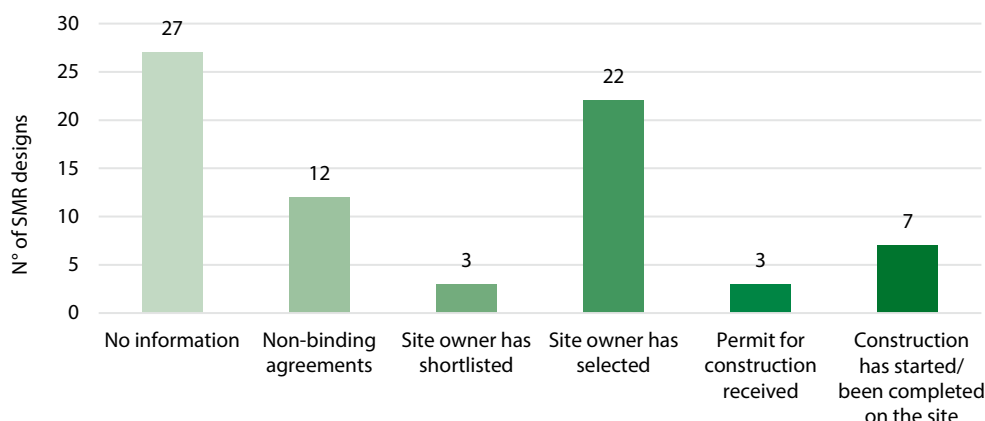
"Together, we're tackling one of the biggest challenges in deploying nuclear technology, which is navigating complex, slow, and costly licensing processes."

Mark Tipping – Global Offshore Power To X Director, Lloyd's Register

The intersection of artificial intelligence and nuclear innovation marks a decisive turning point in the global energy transition. As AI reshapes industries and infrastructure, its deployment must be guided toward solutions that unlock climate resilience, energy equity and regulatory agility. Nuclear, particularly SMRs, offers a pathway to scalable, carbon-free baseload power, but only if deployment barriers like licensing and permitting are fundamentally transformed. Microsoft's Generative AI for Permitting Solution Accelerator is an example of how advanced digital tools can move from concept to critical infrastructure enabler, equipping governments, regulators and industry with the speed and precision this decade demands. By aligning policy, technology and talent, we can convert today's AI revolution into a foundation for clean energy security, delivered faster, governed transparently and shared equitably across the globe.

Siting

Figure 31. **Siting progress worldwide**



SMR siting is advancing globally, with non-commercial projects and a growing diversity of types of site owners

Figure 31 shows the pipeline of SMRs progressing through siting around the world. In Russia, one SMR design received its permit for construction, one started construction on site and one is already operating. In China, one SMR is under construction, and one is already operating. Construction started on site for one SMR in Argentina and one SMR is also already operating in Japan. In the United States, two SMR designs received permits for construction and one started construction. Twenty-two SMR designs have been selected by site owners for deployment on their sites, while three have been shortlisted. Twelve SMR designs have entered into non-binding agreements with site owners that are considering possible deployment. For 27 SMR designs there is no information related to siting that was readily available from any site owners.

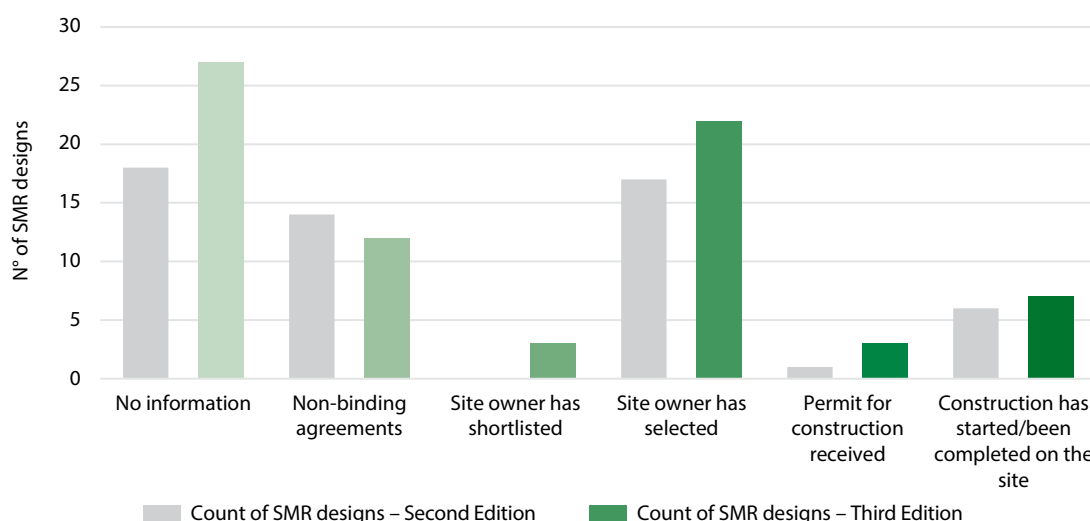
Table 4. **Siting highlights**

- As of 2025, the NEA is tracking 84 siting activities involving SMR developers and site owners, a 20% increase since 2024, covering 47 SMR designs worldwide.
- The United States leads in the volume of siting activities, while China and Russia are further ahead with construction and operations.
 - In China, the HTR-PM gas-cooled reactor entered commercial operation in 2023, and construction has begun on the ACP100 water-cooled SMR.
 - In Russia, the KLT-40S floating, water-cooled SMR has been in operation since 2022, and construction is underway for the BREST-OD-300, a fast-spectrum, metal-cooled reactor.
- Since 2023, notable progress in siting activities has been observed in Canada, Czechia and the United States:
 - In Canada, Ontario Power Generation (OPG) completed early site preparation for its first BWRX-300 SMR at the Darlington site and initiated work for three additional BWRX-300 SMRs. Among OECD countries, Canada is the first mover on commercial SMR deployment.
 - In Czechia, the utility CEZ Group concluded its SMR design selection process, choosing the Rolls-Royce SMR for up to 3 GWe of capacity at the Temelín site, with early site work scheduled to begin in 2025.
 - In the United States, construction has started on Kairos Power's Hermes, a demonstration molten salt-cooled reactor, and construction permits were granted for Natura Resources' MSR-1 and the Department of Defense's Project Pele.
- Approximately half of global siting activities involve Generation IV SMR designs, reflecting the progress of these advanced reactor concepts.
- Nearly half of siting activities are now associated with non-utility site owners, such as national and sub-national laboratories, heavy industry customers, and SMR developers themselves. These new ownership models reflect a level of innovation in the business models for SMR deployment.
- Russia is also expanding its international engagement, actively pursuing agreements with newcomer countries such as Guinea, Myanmar and the Philippines for potential deployment of floating SMRs.

As with licensing, there has been significant progress in siting since the *NEA SMR Dashboard: Second Edition* (NEA, 2024). In several cases, SMR developers are planning to deploy their first-of-a-kind SMR on land they own: Holtec International's SMR-300, Kairos Power's Hermes, and TerraPower's Natrium Reactor Plant in the United States; and Last Energy's PWR in the United Kingdom.

The siting of SMRs with the express goal to foster innovation ecosystems and supply chain development is also observed. For instance, the Texas A&M University System, a network of universities and state agencies in Texas, United States, has selected Terrestrial Energy's IMSR, Kairos Power's KP-FHR, Natura Resources' MSR-100, and Aalo Atomics' Aalo-1 for deployment at its RELLIS campus. More information is presented in Table 5.

Figure 32. **Comparison of siting pipelines between the second (2024) and third (2025) editions of the NEA SMR Dashboard***



* The *NEA SMR Dashboard: Second Edition* is based on data as of 10 November 2023. The *NEA SMR Dashboard: Third Edition* is based on data as of 14 February 2025.

Table 5. **Siting progress worldwide, detailed**

Construction has started or been completed on-site
Seven SMR designs have begun or completed construction on-site, including two in China (ACP100 and HTR-PM), two in Russia (KLT-40S and BREST-OD-300), one research reactor in Japan (HTTR), one in Argentina (CAREM), and one in the United States, a non-grid connected demonstration reactor Hermes.
Permits for construction received
Permits for construction have been received for the RITM-200N in Russia, as well as for the research molten salt-cooled reactor MSR-1 and the US Department of Defense Project Pele in the United States.
Site owner has selected the technology
- Twenty-two SMR designs have been selected by site owners for deployment around the world across 27 different sites. - GE Vernova Nuclear Energy's BWRX-300 stands out with having been selected for deployment at four different sites: at OPG's Darlington site in Ontario and with SaskPower in Saskatchewan; in Indiana, United States, for American Electric Power; and in Poland for Orlen Synthos Green Energy. - Three other SMR designs have each been selected at two different sites: In the United States, the Aurora Powerhouse has been selected for deployment at both Idaho National Laboratory and for the Southern Ohio Diversification Initiative in Piketon, Ohio. The Xe-100 design has been selected for deployment for Dow Chemical in both Seadrift, Texas, and for Energy Northwest in Richland, Washington. The eVinci microreactor has been selected for deployment at both Idaho National Laboratory, and for the Saskatchewan Research Council, in Saskatoon, Canada. - Progress on siting is also observed in Czechia where the utility CEZ Group selected Rolls-Royce SMR for the potential deployment of up to 3 GWe at the Temelín site, with early preparatory work scheduled to begin in 2025.

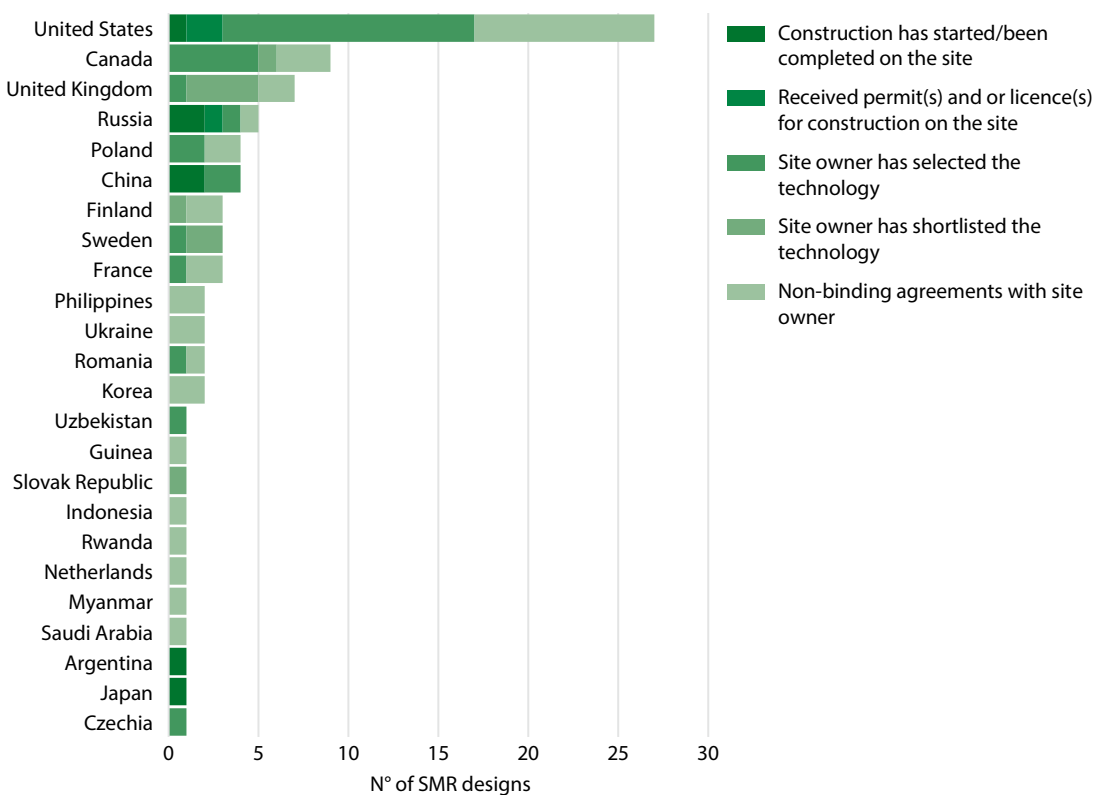
Site owner shortlisted the technology
As of 14 February 2025, Westinghouse's AP300 is included in the shortlist for the Great British Nuclear (GBN) competition in the United Kingdom and Steady Energy's LDR-50 has been shortlisted for potential deployment in Kuopio, Finland, specifically for district heating.
Non-binding agreements with site-owners
Twelve SMR designs have entered into non-binding agreements with site owners.
No information
For 27 SMR designs, corresponding to more than a third of all SMR designs included in the <i>NEA SMR Dashboard: Third Edition</i>, there is no information related to siting that was readily available from any site owners at the time of assessments.

SMR deployment expands globally with regional trends emerging

Figure 33 presents the number of site owners involved in SMR deployment projects worldwide, grouped by country. Since the previous edition of the *Dashboard*, there has been an increase in global siting activities. Notably, construction has begun on a new SMR design in the United States, construction permits have been granted for two additional US designs, and site selection processes are progressing in both Czechia and the United Kingdom.

Three SMR designs are currently in operation: the high-temperature gas-cooled HTR-PM in China; the research-focused high-temperature gas-cooled HTTR in Japan; and the floating, light water-cooled KLT-40S in Russia. and. Several other SMR designs are under construction, including the light water-cooled CAREM in Argentina, the ACP100 in China, the lead-cooled demonstration reactor BREST-OD-300 in Russia, and the molten salt-cooled demonstration reactor Hermes in the United States.

Figure 33. Siting progress, by country



The United States is home to the largest volume of siting activities, while Russia and China are further ahead with development. Canada is farthest ahead among OECD countries with the BWRX-300 project by GEV at Ontario Power Generation's Darlington site. Although the construction permit was approved after the cut-off date for this edition of the *Dashboard* (14 February 2025) and is therefore not reflected in the current analysis, it marks a major milestone. Among the seven siting activities in the United Kingdom, four are linked to the Great British Nuclear (GBN) competition. The selection of Rolls-Royce's RR SMR through GBN was also announced after the cut-off date for this edition of the *Dashboard* and is not reflected in the current analysis.

In Czechia, ČEZ concluded its SMR selection process for the Temelín site with the choice of the Rolls-Royce SMR, marking the country's first siting activity and a significant development since the second edition of the *Dashboard*.

Russia continues to stand out for its early engagement with countries that have limited or no experience operating nuclear power plants. This includes non-binding expressions of interest from the governments of Guinea, Myanmar and the Philippines for the potential deployment of the floating, light water-cooled RITM-200M. Additionally, Uzbekistan has selected the land-based RITM-200N design.

About half of the siting activities worldwide involve Generation IV SMR designs, reflecting progress with and interest in these advanced reactor concepts

Figure 34 illustrates siting activities across different SMR design concepts in countries around the world. About half of all siting activities are to water-cooled SMR designs. In many countries, including the United Kingdom, Finland, Romania, Ukraine, and Korea, siting activities are exclusively related to water-cooled designs, suggesting near-term deployment goals. In contrast, three countries – the United States, Canada, and France – are home to siting activities for three or more different SMR design concepts, suggesting a more diverse set of interests. Finally, some countries, such as Indonesia, Japan, the Netherlands, Rwanda and the Slovak Republic, are focused solely on Generation IV SMR concepts, suggesting longer-term deployment objectives.

Figure 34. **Siting activities, by country and SMR concept**

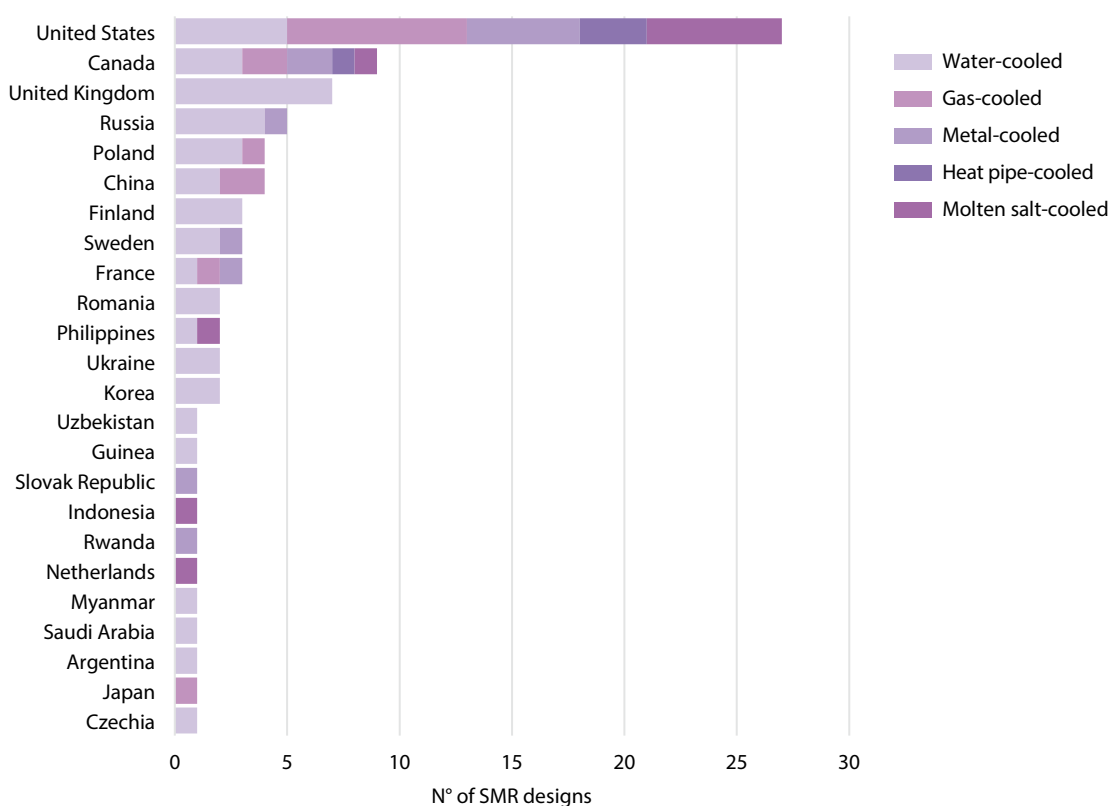
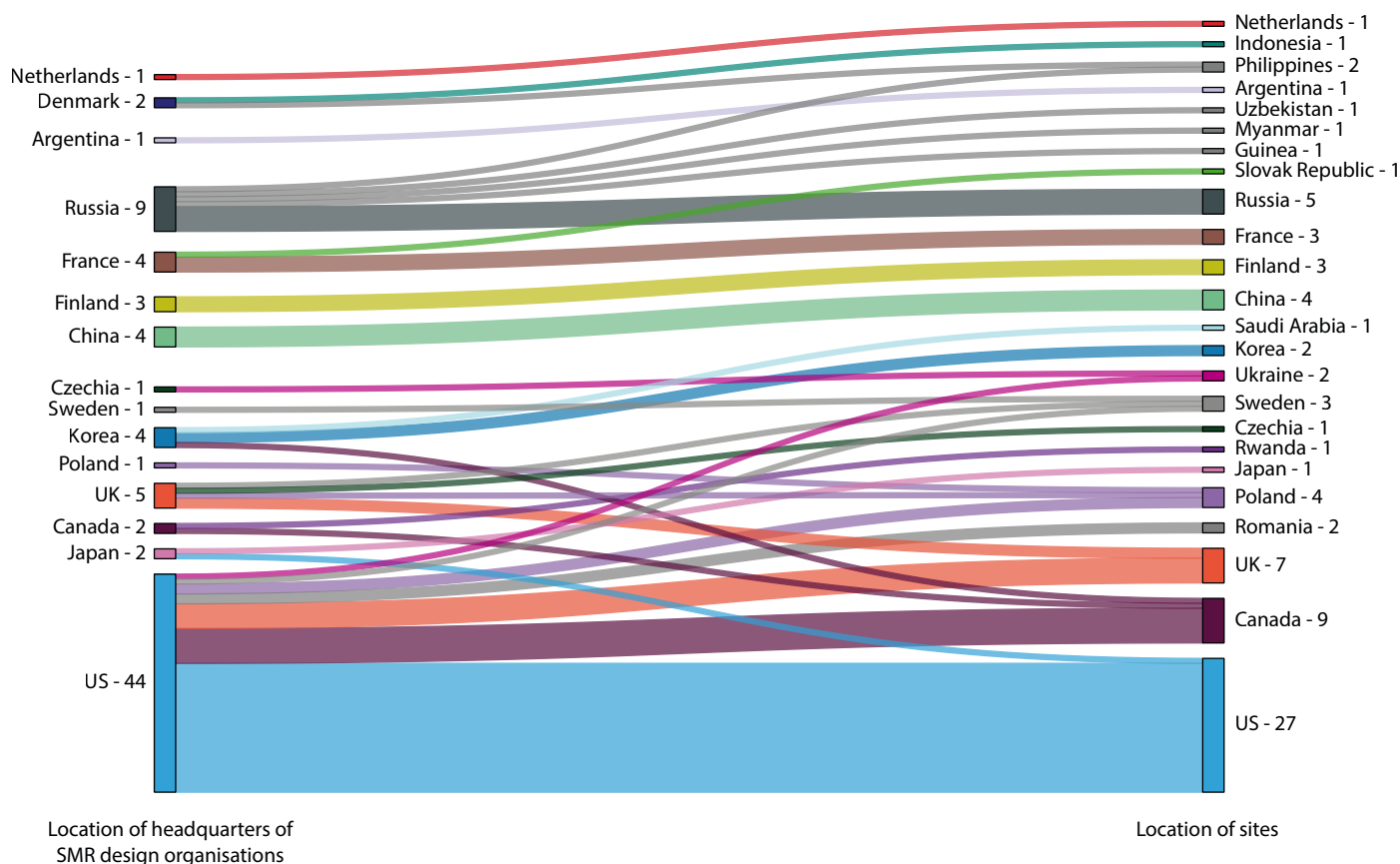


Figure 35 presents another view of siting activities. The countries that are home to the headquarters of the SMR design organisations appear on the left. Site locations appear on the right. The bars indicate siting activities in domestic and foreign markets.

Figure 35. **SMR siting in domestic and foreign markets**



SMRs with headquarters in the United States and Russia are the most active in siting activities in export markets, with 18 and 4 siting activities abroad, respectively, albeit in different markets. US-origin SMRs are active on siting in the United States, Canada, the United Kingdom, Poland, Romania, Sweden, and Ukraine. Russian-origin SMRs are active on siting in Russia, Guinea, Myanmar, the Philippines and Uzbekistan.

While most SMRs are sited on government or utility-owned land, new site owners are reshaping business models in the nuclear energy sector

Figure 36 presents the types of site owners involved in SMR siting activities across different countries. The majority of potential SMR sites (49) are associated with utilities and government-owned entities. However, 14 siting activities are linked to national or sub-national laboratories, often intended for demonstration purposes, while 11 are tied to private industries exploring SMRs as a source of clean, reliable power and heat. The United States is home to the greatest diversity of types of site owners, reflecting a wide range of stakeholders engaged in potential SMR deployment.

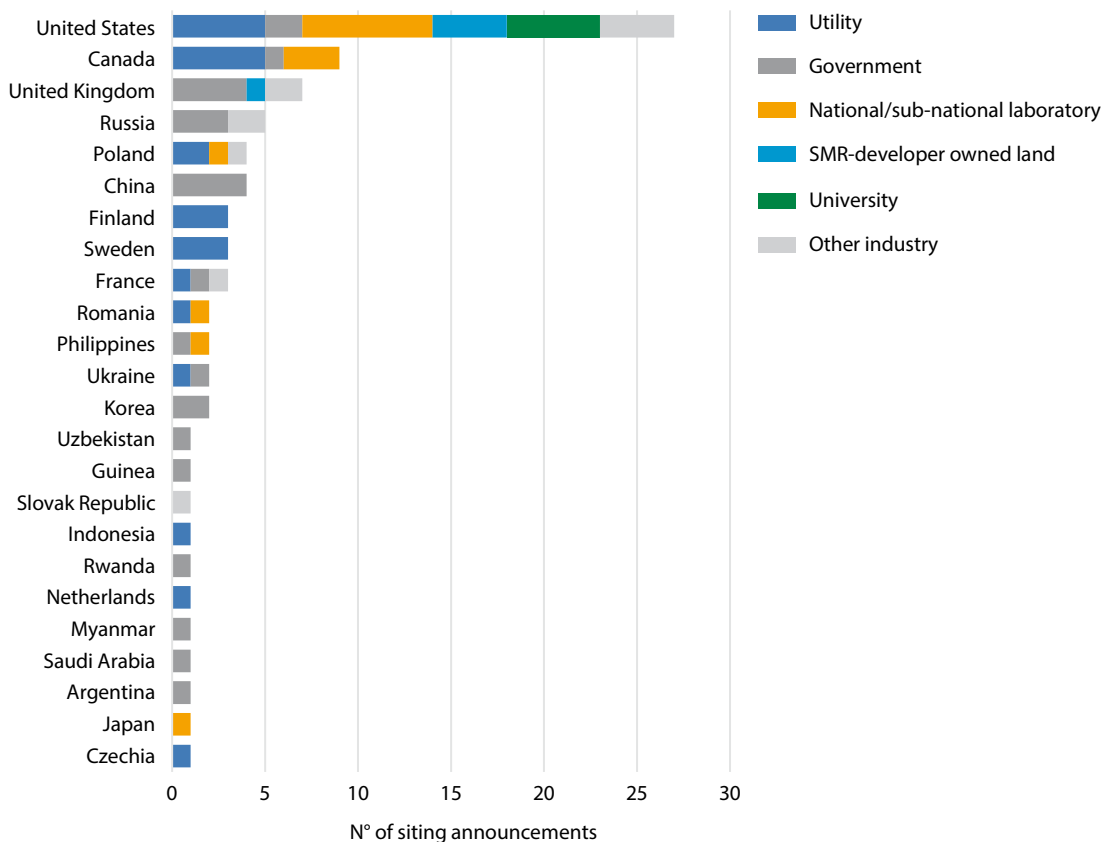
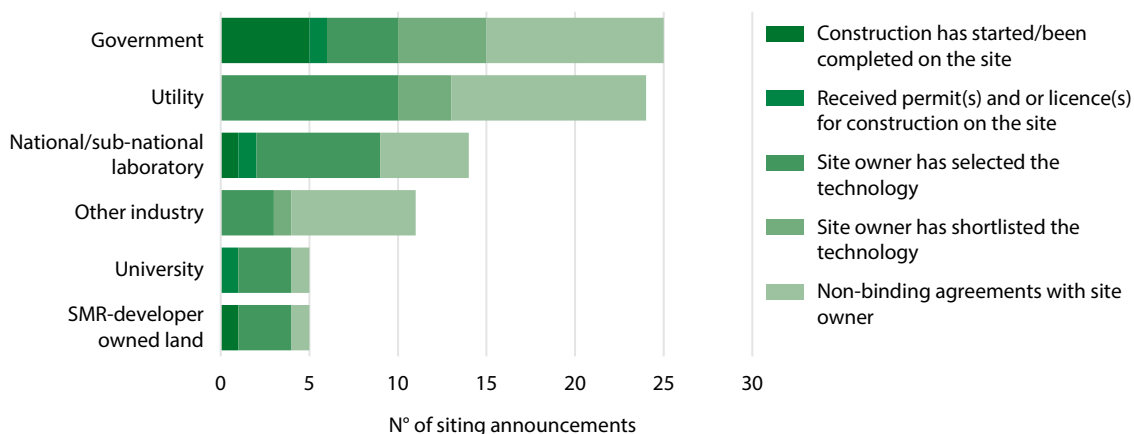
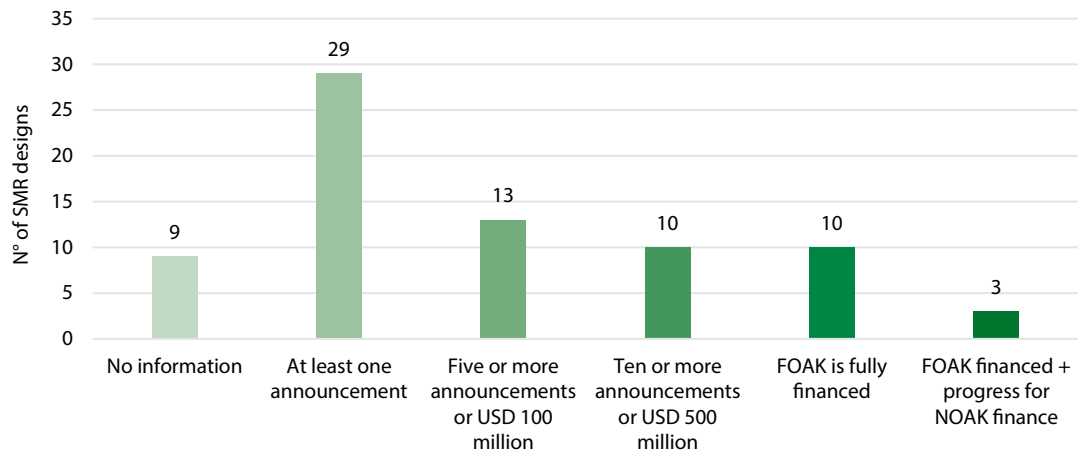
Figure 36. **Siting activities, by country and site-owner type**

Figure 37 illustrates the advancement of SMR siting activities across various types of site owners. SMR deployment is most advanced on government-owned land, particularly in China and Russia. In contrast, within OECD countries, significant progress is being made on sites owned by other types of organisations, such as national or sub-national laboratories, heavy industry customers, and SMR developers who own the land designated for their first-of-a-kind deployments.

Figure 37. **Siting progress, by site-owner type**

Financing

Figure 38. Financing progress worldwide



Global progress in SMR financing shows notable growth in early private and public funding initiatives worldwide

Figure 38 shows the pipeline of SMRs progressing through financing around the world. Three SMR designs deployed in Canada, China and Russia have made progress towards financing for fleet deployment and 10 SMR first-of-a-kind projects that are already fully financed in Argentina, China, Japan, Russia and the United States. Ten SMR designs have ten or more announcements or have secured at least USD 500 million in funding and financing. Thirteen have at least five announcements or have secured at least USD 100 million. Twenty-nine SMR designs have at least one announcement related to funding and financing. There is no information available from verifiable public sources related to financing activities for nine SMR designs.

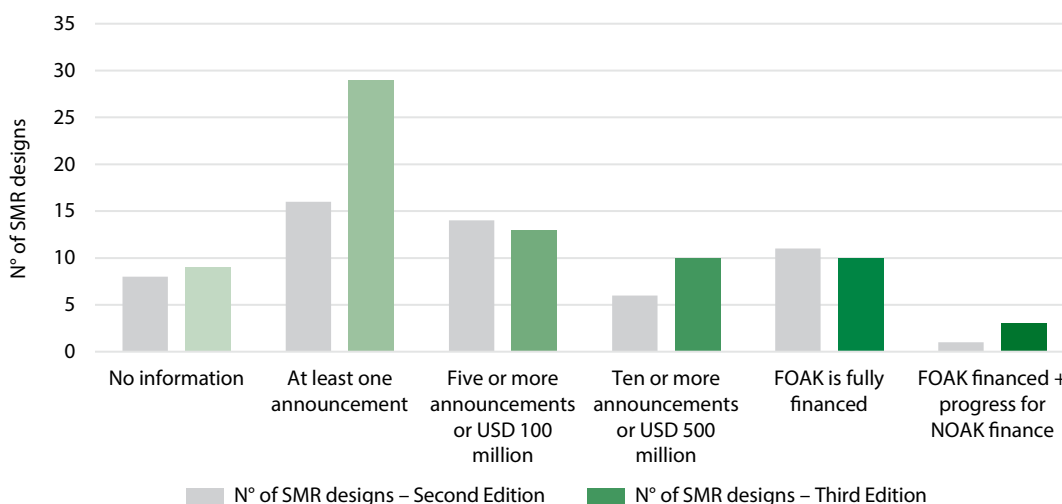
Table 6. Financing highlights

• According to the data compiled in this edition of the <i>NEA SMR Dashboard</i>, it is estimated that a total of USD₂₀₂₃ 15.4 billion of funding has been secured for SMR development and deployment. Of this, around USD₂₀₂₃ 10 billion is sourced from public funding, while USD₂₀₂₃ 5.4 billion comes from private sources. • Three SMR designs have secured full funding for their first-of-a-kind deployment and also begun making progress to secure funds for fleet deployment: – HTR-PM, a high-temperature gas-cooled reactor in China – RITM-200S, a floating pressurised water-cooled reactor in Russia – BWRX-300, a boiling water-cooled reactor in Canada • Private capital is playing an increasingly important role in SMR financing, often complementing public matching grants. • Since the second (2024) edition of the <i>NEA SMR Dashboard</i>, there has been an 81% increase in SMR designs that have announced at least one funding or financing deal. Seed and early-stage private fundraising has surged globally, expanding well beyond the United States. • However, large-scale fundraising efforts remain concentrated in the United States, with several notable examples: – Radiant Industries completed a USD₂₀₂₃ 100 million Series C round to support testing of its Kaleidos microreactor at the INL DOME facility in Idaho, United States. – Between 2024 and 2025, X-energy raised USD₂₀₂₃ 700 million in a Series C-1 round, led by Amazon, to support the deployment of its Xe-100 reactor. – In 2022, TerraPower secured over USD₂₀₂₃ 864 million. • The only major exception outside the United States is newcleo, an SMR developer headquartered in France, which had raised USD₂₀₂₃ 570 million, primarily from Italian investors, by the end of 2024. • While many SMR developers have succeeded in securing early-stage funding for research and development, most have yet to secure the full financing needed for first-of-a-kind deployment.

- Analysis of SMR financing and supply chain ecosystems reveals two trends: Financiers in five countries account for up to 95% of global financing for SMRs – United States, Korea, Canada, France, and the United Kingdom; and 90% of total financing worldwide is redirected to contracts with supply chain companies in the same country.
- Although water-cooled and thermal-spectrum designs have received most of the financing, investments are flowing to a broad range of technologies, including diverse concepts, configurations and neutron spectrums.

There was an 81% increase in SMR designs that have secured at least one funding or financing announcement between publication of the second (2024) and third (2025) editions of the *NEA SMR Dashboard* (Figure 39). Private seed and early-stage fundraising efforts have seen a global surge, extending beyond the United States. While such activity is evident worldwide, in Europe it is more commonly tied to public grants and matching grants, where private capital is raised to complement public investment, unlike the more independently driven private fundraising in the United States. Several private funding rounds in the United States have raised amounts in the hundreds of thousands of dollars or more, levels that remain rare globally. One exception outside the United States is newcleo, an SMR developer headquartered in France, that has raised over USD₂₀₂₃ 570 million since 2022, primarily from Italian investors.

Figure 39. **Comparison of financing pipelines between the second (2024) and third (2025) editions of the NEA SMR Dashboard***



* The *NEA SMR Dashboard: Second Edition* is based on data as of 10 November 2023. The *NEA SMR Dashboard: Third Edition* is based on data as of 14 February 2025.

Table 7. **Financing progress worldwide, detailed**

First-of-a-kind financed and progress for fleet deployment

- Three SMR designs have fully financed their first-of-a-kind and are already making progress on financing for fleet deployment:
 - HTR-PM, a high-temperature gas-cooled reactor in China
 - RITM-200S, a floating pressurised water-cooled reactor in Russia
 - BWRX-300, a boiling water-cooled reactor in Canada
- In August 2024, the Chinese government approved Phase One of the Jiangsu Xuwei Nuclear Heating Power Plant project. This phase is to include the first HTR-600S, comprising three HTR-PM modules.
- In the United States, the Tennessee Valley Authority (TVA) has submitted an application for up to USD₂₀₂₃ 800 million in funding from the US Department of Energy's Generation III+ Small Modular Reactor Program to support the deployment of the BWRX-300. This complements the broader portfolio of funding and financing mechanisms, such as loans, equity investments and grants, that the BWRX-300 project is leveraging to support the planned deployment of up to four units at the Darlington site in Canada.

First-of-a-kind fully financed

- Ten additional SMR designs worldwide that have secured financing for their first-of-a-kind projects. These designs originate from Argentina, China, Japan, Russia and the United States.
- Of these ten, one is a research reactor (HTTR in Japan), seven are prototype or demonstration projects. Two are commercial projects:
 - RITM-200N in Russia.
 - Kaleidos, a gas-cooled microreactor developed by Radiant Industries in the United States.
- In November 2024, Radiant Industries completed a Series C fundraising round of USD₂₀₂₃ 100 million to support testing of the Kaleidos microreactor at the INL DOME facility in Idaho, United States.

Ten or more announcements or USD 500 million

- Ten SMR designs have either received ten or more public funding announcements or secured over USD 500 million in financing. These include three designs from the United States and seven from Europe.
- In 2024-2025, X-energy secured a USD₂₀₂₃ 700 million Series C-1 capital raise, led by Amazon, to advance the deployment of its Xe-100 reactor.
- newcleo has raised over USD₂₀₂₃ 570 million between 2020 and 2024, marking the largest private capital fundraising effort for SMR development in Europe to date.

Five or more announcements or USD 100 million

- Thirteen SMR designs from Canada, Denmark, France, Korea and the United States have made five or more announcements or secured USD 100 million in financing.

At least one announcement

- Twenty-nine SMR designs have made at least one announcement in financing of which 13 are newly included in this edition of the *Dashboard*.

No information available

- For nine SMR designs, there is no available information from verifiable public sources related to financing activities.

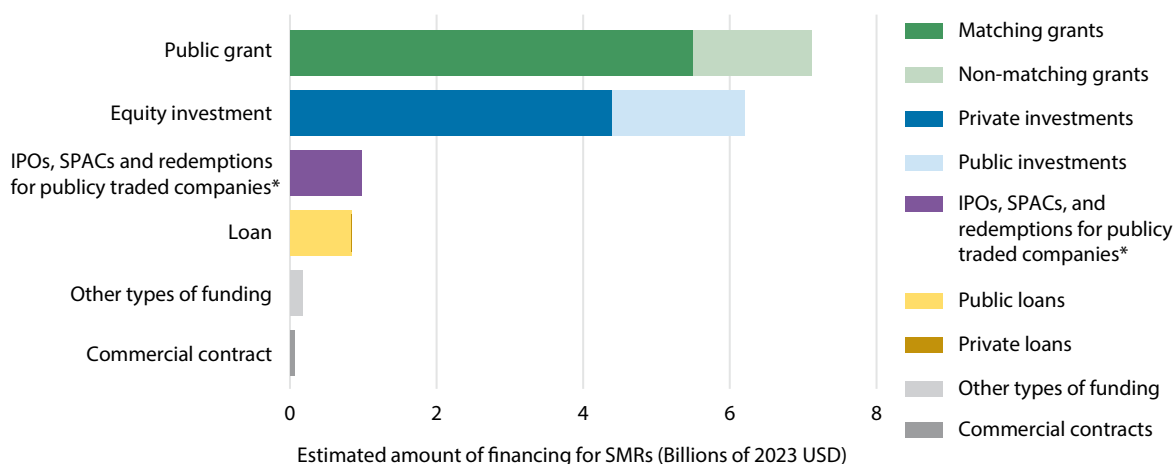
Public funding remains essential for SMR development, but private capital is now contributing nearly as much to support the sector

More than 380 verified financing announcements outside of China and Russia, totalling approximately USD₂₀₂₃ 15.4 billion, have been made. An estimated USD₂₀₂₃ 10 billion has come from public sources, alongside approximately USD₂₀₂₃ 5.4 billion comes private sources.

Public grants and equity investments contribute nearly equal amounts to total SMR funding (Figure 40). Public grants, estimated at USD₂₀₂₃ 7.1 billion, are primarily matching grants that require private co-funding. Equity investments, estimated at USD₂₀₂₃ 6.2 billion, are predominantly from private sources. Together, these two funding types account for over 87% of global SMR funding, totalling approximately USD₂₀₂₃ 13.3 billion.

Currently, three SMR companies are publicly traded, all in the United States: NuScale, Oklo and NANO Nuclear Energy. Combined, they have raised approximately USD₂₀₂₃ 975 million through special-purpose acquisition company (SPAC) mergers, initial public offerings (IPOs), and warrant redemptions. Loans represent a slightly smaller share, totalling USD₂₀₂₃ 846 million, including USD₂₀₂₃ 747 million from the Canada Infrastructure Bank to support GE Vernova Hitachi's BWRX-300 project at OPG's site in Darlington, Canada, and USD₂₀₂₃ 95 million from the US Export-Import Bank to support NuScale's deployment of the Entra1 Energy plant in Romania.

Finally four commercial contracts have been awarded to two SMR developers: Radiant Industries for Kaleidos and Westinghouse Electric Company for eVinci. These contracts, totalling USD₂₀₂₃ 25 million, are all with US government entities, including the US Department of Defense, the US Department of Energy, the Air Force Research Laboratory, and the Idaho National Laboratory.

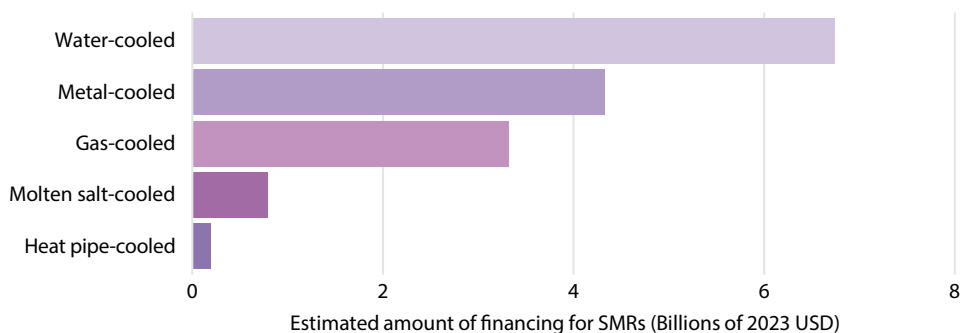
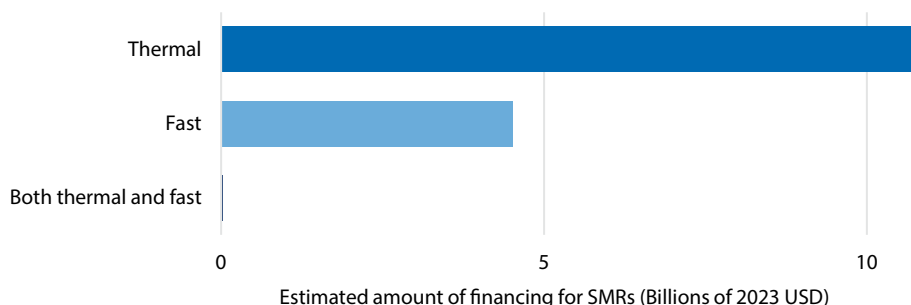
Figure 40. **Sources of financing for SMR designs from OECD countries**

*IPOs: initial public offerings; SPACs: special-purpose acquisition companies.

Over half of total SMR financing supports non-water-cooled concepts

Figure 41 shows that more than half of all financing has been allocated to Generation IV reactor concepts, including USD₂₀₂₃ 4.3 billion and USD₂₀₂₃ 3.3 billion for metal-cooled and gas-cooled reactor concepts, respectively. Water-cooled reactor concepts have secured USD₂₀₂₃ 6.7 billion in financing, while molten salt-cooled and heat pipe concepts have secured USD₂₀₂₃ 789 million and USD₂₀₂₃ 197 million, respectively.

Figure 42 shows that approximately USD₂₀₂₃ 10.8 billion has been invested in thermal-spectrum SMR designs, while USD₂₀₂₃ 4.5 billion has been directed toward fast-spectrum designs.

Figure 41. **Financing for SMRs by configuration**Figure 42. **Financing for SMRs, by neutron spectrum**

Figures 43, 44A, 44B and 45 represent flows of financing internationally. For financiers based in countries around the world, Figure 43 reveals the share of financing allocations to SMRs with domestic versus foreign headquarters.

Figure 43. **Share of financing allocated to SMRs with domestic versus foreign headquarters**

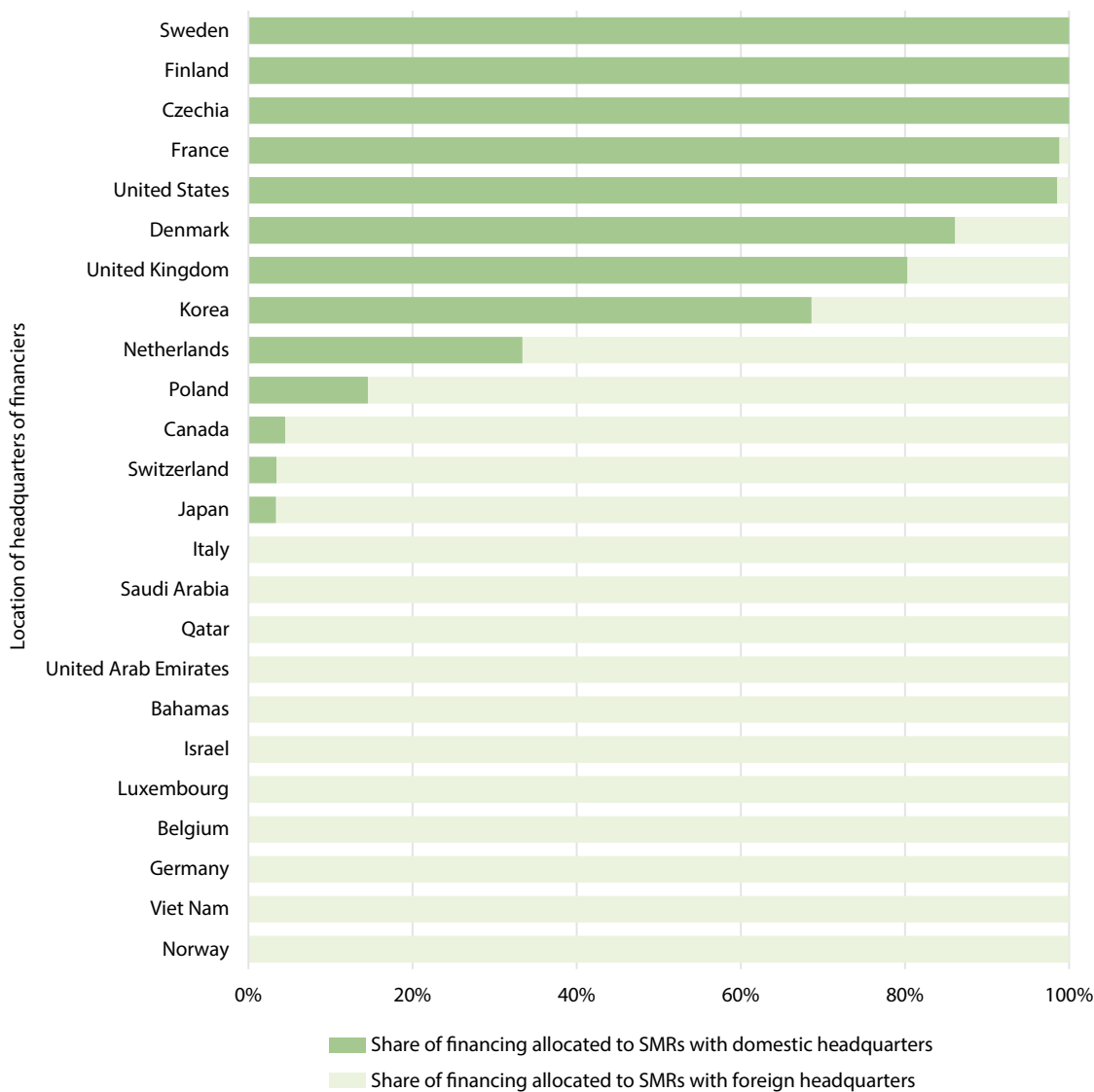


Figure 44A identifies countries where financiers are allocating a majority of financing to support SMRs with domestic headquarters, including for example financiers from the United States, Korea, France, and the United Kingdom. Figure 44B identifies countries where financiers are allocating a majority of financing to support SMRs with foreign headquarters, including for example financiers from Canada, Italy, Poland and Saudi Arabia.

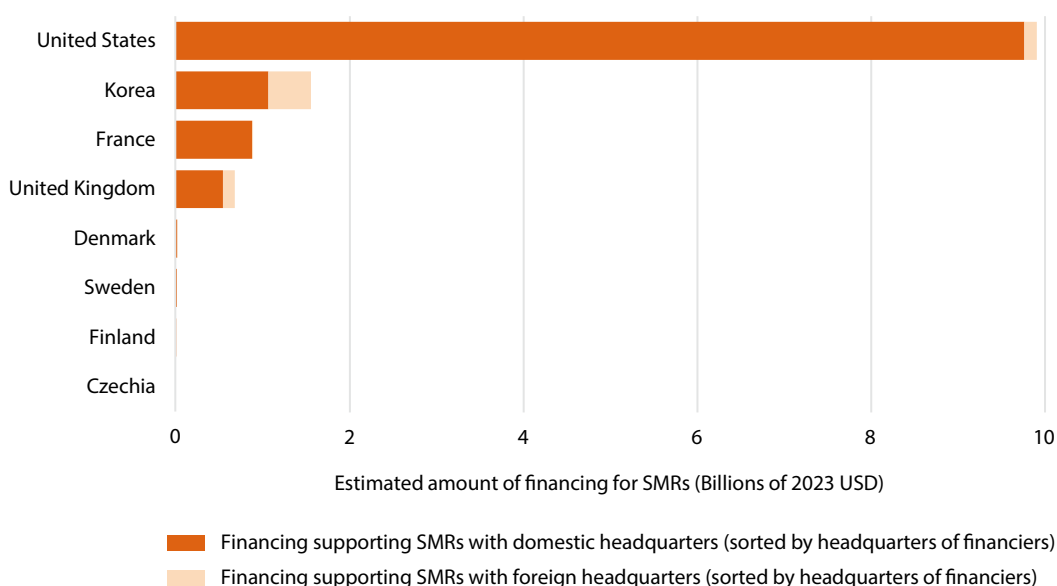
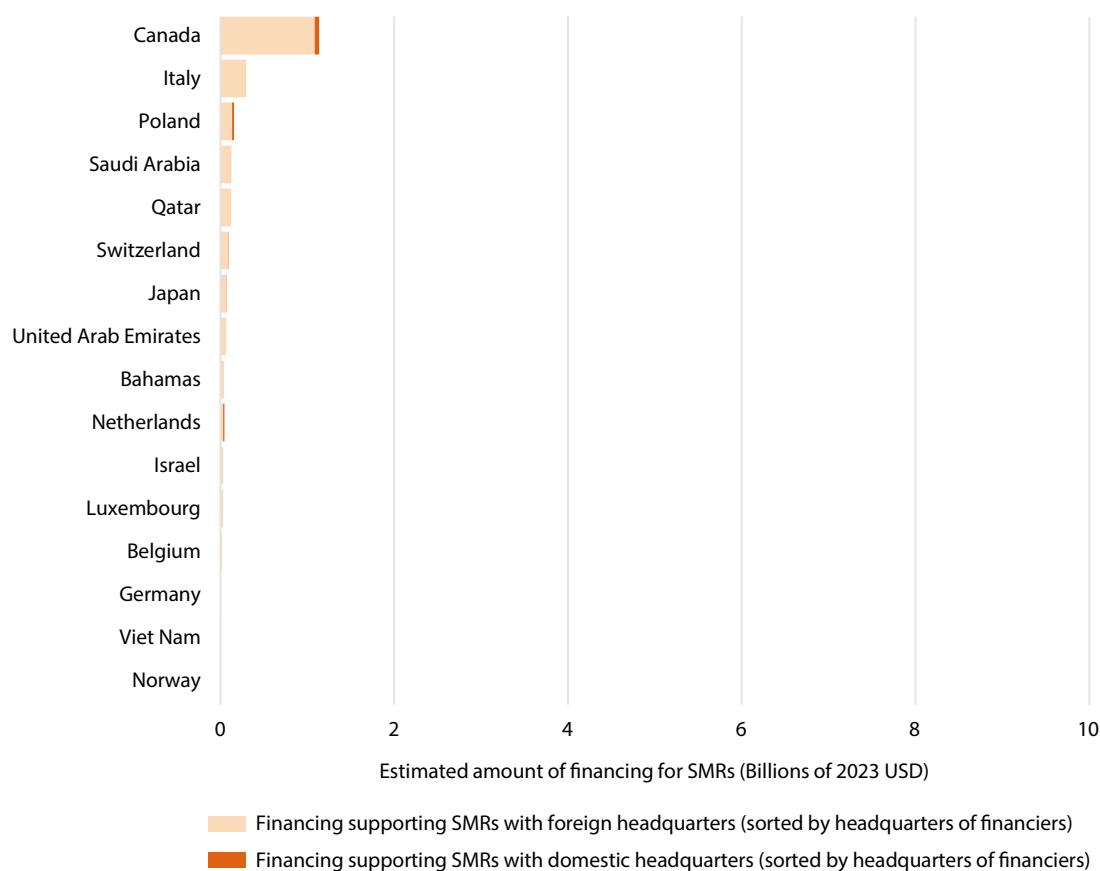
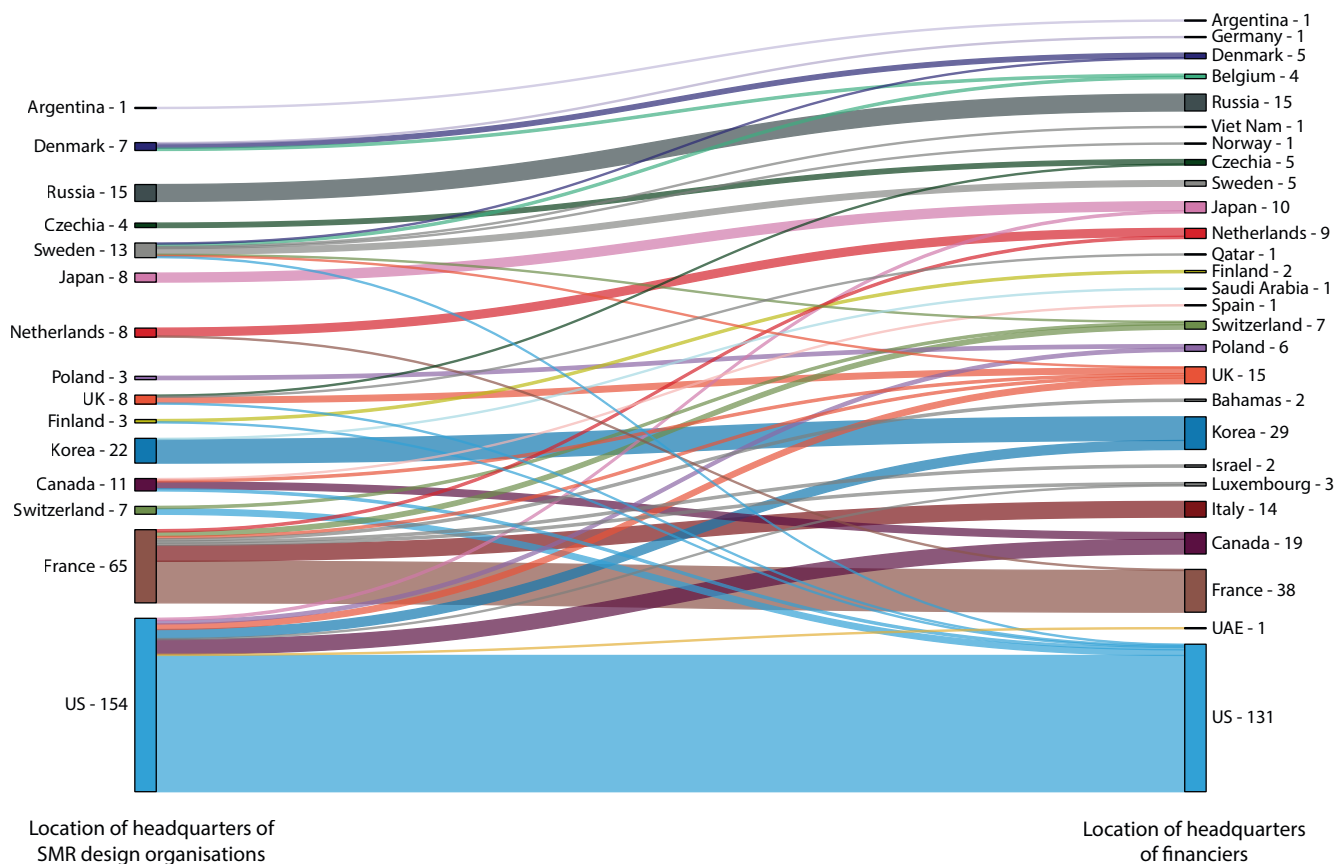
Figure 44A. **Financiers supporting SMRs with domestic headquarters**Figure 44B. **Financiers supporting SMRs with foreign headquarters**

Figure 45 presents another view with the combined data, in a Sankey diagram. The countries that are home to the headquarters of the SMR design organisations appear on the left. The countries that are home to the headquarters of the financiers appear on the right. The bars map the flows of financing to SMRs with domestic and foreign headquarters.

Figure 45. **SMR financing by domestic and foreign financiers**



Public and private financing from financiers in five countries – the United States, Korea, Canada, France, and the United Kingdom – account for up to 95% of global financial support for SMRs. In turn, approximately 90% of total SMR financing worldwide is redirected to contracts with supply chains in those five countries to support project development and deployment.

The United States is notable for both the scale of its public and private investments and the diversity of its leading companies represented by three SMRs that have secured the largest sums of financing: NuScale, TerraPower and X-energy, respectively water-cooled, metal-cooled and gas-cooled SMR concepts.

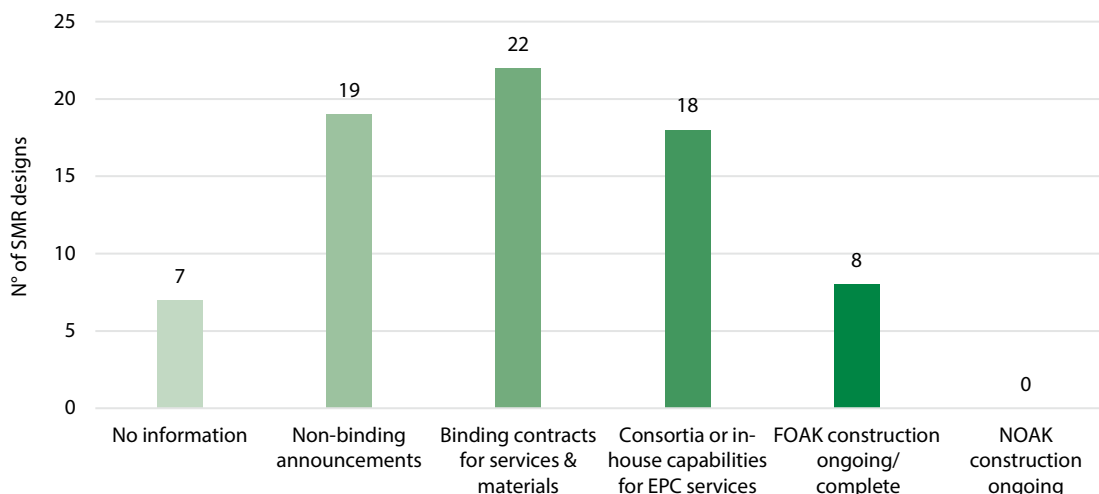
Similar trends are observed in the United Kingdom with Rolls-Royce SMR, France with NUWARD, and Korea with i-SMR and SMART100, where domestic SMRs receive domestic financing and in turn sign contracts with domestic supply chain companies.

Canada is unique in that most of its SMR supply chain is tied to its leadership in deployment, specifically, the BWRX-300 project at the Darlington site in Ontario, developed by the US-based GE Hitachi.

Italy also presents a notable case. Despite not operating any commercial nuclear power plants, significant private investment from financiers in Italy have been allocated to foreign SMRs, primarily to newcleo, a company headquartered in France. More than half of the capital raised from Italian sources has then been directed to contracts with supply chain companies in Italy via newcleo's activities in Italy.

Supply chain

Figure 46. **Supply chain progress**



SMR supply chains expand globally as developers secure more contracts and advance towards deployment

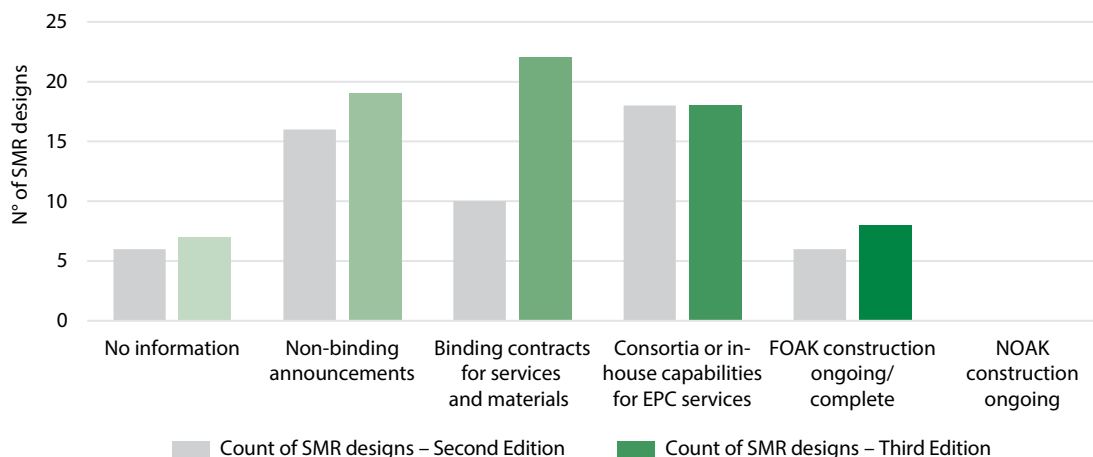
Figure 46 shows the pipeline of SMRs making progress with supply chain readiness around the world. Eight SMR designs have started construction on sites in Argentina, China and Russia. Eighteen SMR designs, not yet under construction, have partnered or started joint ventures or consortia specifically with engineering, procurement and construction (EPC) companies and 22 others have entered into binding contracts for services, intermediary goods and materials with supply chain companies. Nineteen SMR designs have started to make progress by entering into non-binding agreements to develop their supply chain. No information from verifiable public sources related to supply chain activities is available for the remaining seven SMR designs.

Table 8. **Supply chain highlights**

- Most SMR designs have made progress with respect to supply chain readiness, many starting with non-binding agreements and binding contracts for services, intermediary goods and materials.
- While water-cooled SMR designs remain the most represented in supply chain activities, there is a clear progress with supply chains for Generation IV and non-water-cooled technologies, as well as marine-based SMRs, mobile and micro designs.
- The United States has the highest number of SMR-related supply chain announcements. Russia and China have the most announcements tied to SMRs already under construction or in operation.
- Regional supply chains are beginning to take shape:
 - The United States, Canada, the United Kingdom, and Korea are developing closely connected SMR supply chains, particularly around NuScale, TerraPower and X-energy, all US-based designs.
 - Canada and the United States also share supply chain connections related to the deployment of the BWRX-300 in Canada.
- In Europe, supply chains are emerging among France, Italy, Belgium, and the United Kingdom, especially in relation to newcleo's LFR-AS-200 and LFR-AS-30 reactor designs.

Since the *NEA SMR Dashboard: Second Edition* (NEA, 2024), considerable progress on supply readiness has been made by SMR designers and project proponents around the world (Figure 47). This progress is enabled by growing access to early-stage capital (see *Financing* section). Additional details on global supply chain developments can be found in Table 9.

Figure 47. **Comparison of supply chain pipelines between the second (2024) and third (2025) editions of the NEA SMR Dashboard***



* The NEA SMR Dashboard: Second Edition is based on data as of 10 November 2023. The NEA SMR Dashboard: Third Edition is based on data as of 14 February 2025.

Table 9. **Supply chain progress worldwide, detailed**

Nth-of-a-kind construction ongoing	
There are no SMRs that have started construction of a fleet.	
First-of-a-kind construction ongoing or completed	
- Construction has been completed for three SMR designs: The floating commercial light-water reactor KLT-40S in Russia; the high-temperature, gas-cooled demonstrator HTR-PM in China; and the research high-temperature gas-cooled HTTR in Japan. - Five SMR designs are under construction. Three are water-cooled: CAREM in Argentina; the ACP100 in China; and the RITM-200S in Russia. There is also the metal-cooled BREST-OD-300 in Russia, and the molten-salt cooled demonstrator Hermes in the United States.	
Consortia or in-house capabilities for EPC services	
- Eighteen SMR designs, not yet under construction, have made progress in the area of EPC capabilities through partnerships, joint ventures or consortia, or with in-house capabilities. - Out of these 18 SMR designs, 13 are either incumbent civil nuclear companies or governmental research institutions with access to in-house EPC capabilities: - In China: China General Nuclear Power Group's (CGN) ACPR50S design, China's State Power Investment Corporation's (SPIC) HAPPY200 design, and Tsinghua University Institute of Nuclear and New Energy Technology's (INET) NHR-200-II design - In France: EDF and its subsidiary NUWARD for the NUWARD SMR design - In Korea: Korea Hydro and Nuclear Power's (KHNP) and Korea Atomic Energy Research Institute's (KAERI) and i-SMR design - In Russia: OKBM Afrikantov's RITM-200M and RITM-200N designs - In the United Kingdom: Rolls-Royce and its subsidiary Rolls-Royce SMR for the RR SMR design - In the United States: BWX Technologies' (BWXT) Project Pele design, GE Vernova Hitachi Nuclear Energy's (formerly GE Hitachi Nuclear Energy) BWRX-300 design, Holtec International's SMR-300 design, and Westinghouse Electric Company's AP300 and eVinci SMR designs - Three SMR designs at this stage are from the United States: - NuScale, which has developed a mature supply chain including through agreements with the EPC Fluor Corporation. In 2023, Doosan started forging upper reactor pressure vessels for NuScale and ordered steam generator tubes from Alleima in 2025.	

- TerraPower has been developing Natrium with GE Hitachi since 2020 and has partnerships with EPC companies, including Bechtel. In 2024, Bechtel and TerraPower began site preparation in Kemmerer, and placed orders for long-lead items with Teledyne Brown Engineering, ENSA, Doosan Enerbility, HD Hyundai, and Marmen. In 2025, TerraPower announced it has completed the procurement selection for all long lead items for its Natrium Reactor Plant. - X-energy has selected several EPC companies as constructors, including Zachry Group, Day & Zimmerman, and Burns & McDonnell. In 2023, X-energy and Energy Northwest signed a Joint Development Agreement to deploy Xe-100 small modular reactors at the Columbia site in Richland, Washington. In 2024, Energy Northwest selected AtkinsRéalis to serve as the Owner's Engineer for the Xe-100 project. • One SMR design at this stage is from Canada: - Moltex Energy's SSR-W. Moltex Energy announced a partnership with AtkinsRéalis to support the licensing and construction of the first SSR-W in New Brunswick in 2022. The engineering company IDOM, which has invested in Moltex's SSR-W technology, is also providing engineering support in various technical areas. • One SMR design at this stage is from Denmark: - Saltfoss' CMSR. In 2023, Saltfoss joined a consortium with Korea Hydro & Nuclear Power (KHNP) and Samsung Heavy Industries (SHI). SHI announced it completed the conceptual design for the CMSR Power Barge in January 2023.
Binding contracts for services and materials
• Twenty-two SMR designs from seven countries – Canada, Finland, France, Poland, Sweden, the United Kingdom and the United States – have entered into binding contracts for services, intermediary goods and materials with supply chain companies.
Supplier days/events/workshops/trade shows/non-binding agreements/MoUs/non-binding announcements
• Nineteen SMRs have entered into non-binding agreements to develop their supply chain.
No information available
• For seven SMRs, no information related to supply chains was readily available.

Real progress is being made in SMR supply chains across all reactor concepts

Water-cooled SMR designs have made the most progress on their supply chains, representing 66% of the 18 designs already accessing EPC capabilities. However, as set out in Figures 48 and 49, all types of reactor concepts and configurations are making progress on supply chain readiness, including: Two gas-cooled designs, Project Pele and Xe-100; two molten salt-cooled design, SSR-W and CSMR; one metal-cooled design, Natrium Reactor Plant; and one heat pipe-cooled design eVinci.

Figure 48. Supply chain progress, by SMR concept

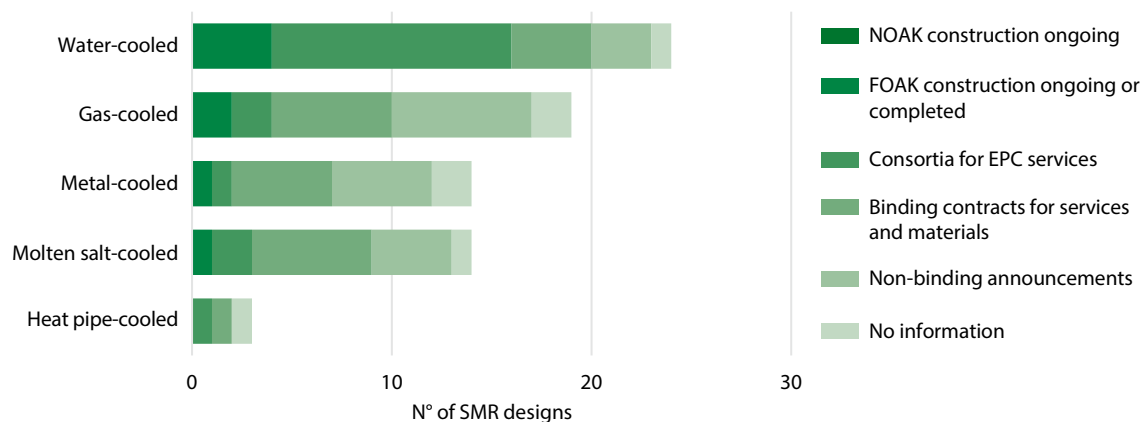
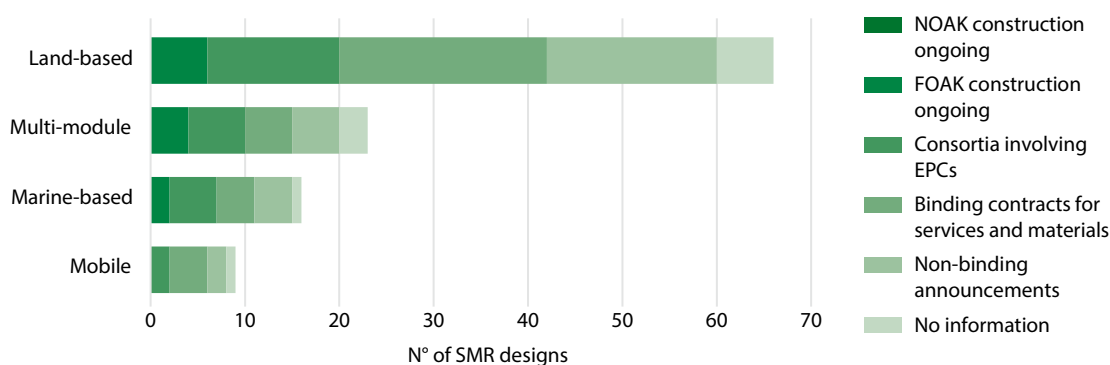


Figure 49. Supply chain progress, by SMR configuration

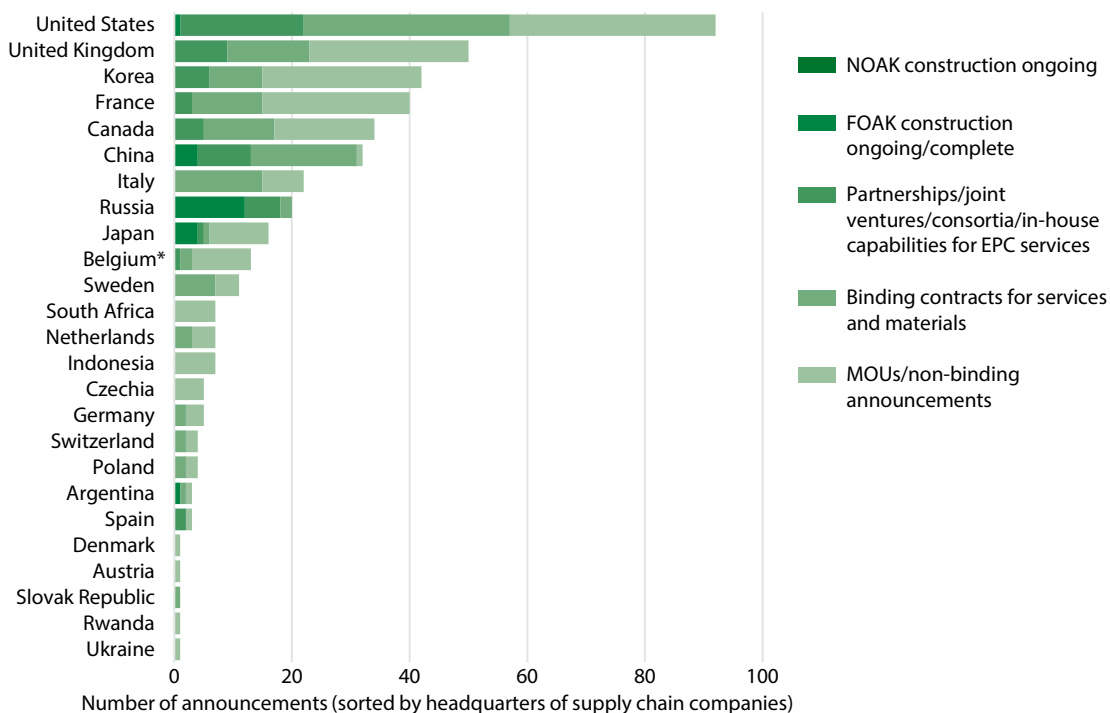


The United States leads in number of supply chain activities for SMRs, but Russia and China continue to lead in deployment readiness

The United States is home to the largest number of SMR-related supply chain announcements, with 92 tracked by the NEA. The United Kingdom, Korea and France follow, home to at least 40 announcements each (see Figure 50).

In the United States, the Hermes molten salt-cooled demonstration reactor is already under construction in Oak Ridge. Although fewer in number, announcements for supply chain activities in Russia and China reveal more advanced supply chain readiness all linked to construction of first-of-a-kind SMRs. In Russia, these include supply chain activities related to the operational KLT-40S; the lead-cooled BREST-OD-300 demonstration reactor; and the floating light-water RITM-200S. In China, these relate to the operational gas-cooled HTR-PM and the water-cooled ACP100.

Figure 50. Supply chain progress by country



* SMR designs selected to form a Project Working Group under the EU Industrial Alliance for SMRs are considered with one announcement (non-binding) located in Brussels, Belgium.

SMR Supply chain specificities vary widely by countries, indicating different deployment priorities

Russia and Sweden stand out for having a significant portion of their supply chain activity focused on marine-based SMR concepts. In Russia, this includes the KLT-40S, RITM-200M and RITM-200S designs. In Sweden, it includes the Sealer-55 and CMSR designs.

Canada and the United States are home to supply chain activities in support of mobile SMR configurations. In Canada, this includes the eVinci and LEUNR designs. In the United States, mobile SMR-related supply chain announcements involve Project Pele, Kaleidos, eVinci, ODIN, MN-1, R1 and the LOKI MMR.

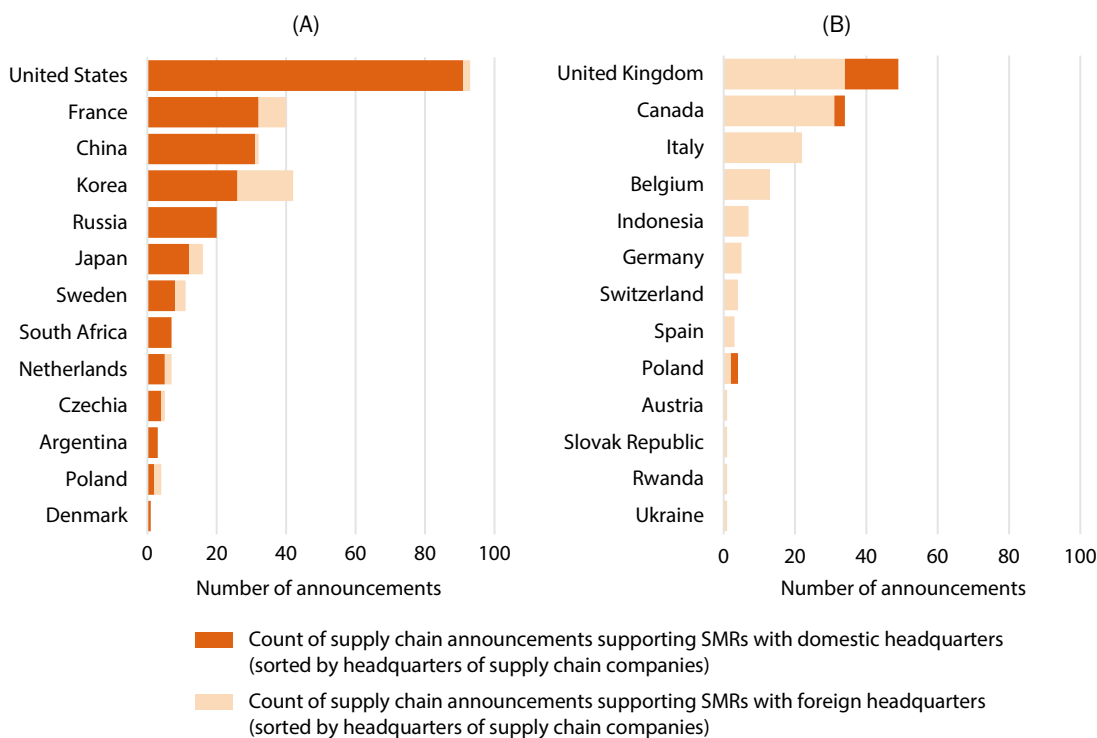
France and the United States are also home to supply chains for micro-SMRs. In the United States, this includes supply chains for BANR, PWR-20, Nuclearis N1 and the KRONOS MMR. In France, this includes supply chains for the CAL-30, Jimmy HTR and XAMR designs.

In France, Italy and Sweden, supply chain announcements are mainly in support of the development of fast spectrum SMRs, in particular: HEXANA's SMR, Naarea's XAMR, Transmutex's START, and Stellaria's Stellarium designs in France; newcleo's LFR-AS-200 and LFR-AS-30 designs in Italy; and Sealer's Sealer-One design in Sweden.

Regional SMR supply chains are beginning to emerge

Figure 51A identifies countries where supply chain activities support SMRs that have domestic headquarters. Figure 51B identifies countries where supply activities support SMRs with headquarters abroad. In the United States, France, China, Russia and Japan, the majority of SMR-related supply chain activities support SMR designs that have headquarters within their own borders. In contrast, in the United Kingdom, Canada, and Italy most SMR-related supply chain activities support SMR designs headquartered abroad. In some cases, domestic supply chains support SMR projects with domestic sites, although the SMR design organisation is headquartered abroad. For example, significant progress is seen from supply chain companies in Canada to support the GEVH BWRX-300 project at OPG's Darlington site in Canada, even though GEVH headquarters are located in Wilmington, United States. These data reveal that different countries are positioned to capture different types of benefits by participating differently in SMR global value chains.

Figure 51. Supply chains supporting SMRs with (A) domestic headquarters; (B) foreign headquarters



Figures 52 and 53 present additional views of the same data. They reveal whether SMR design organisations are relying on domestic supply chains or imports. SMR developers in China, Korea, Russia, and Japan primarily rely on domestic supply chain companies. In contrast, strong supply chain linkages are observed between the United States, Canada, the United Kingdom, and Korea, as well as between France, Italy, the United Kingdom and Belgium.

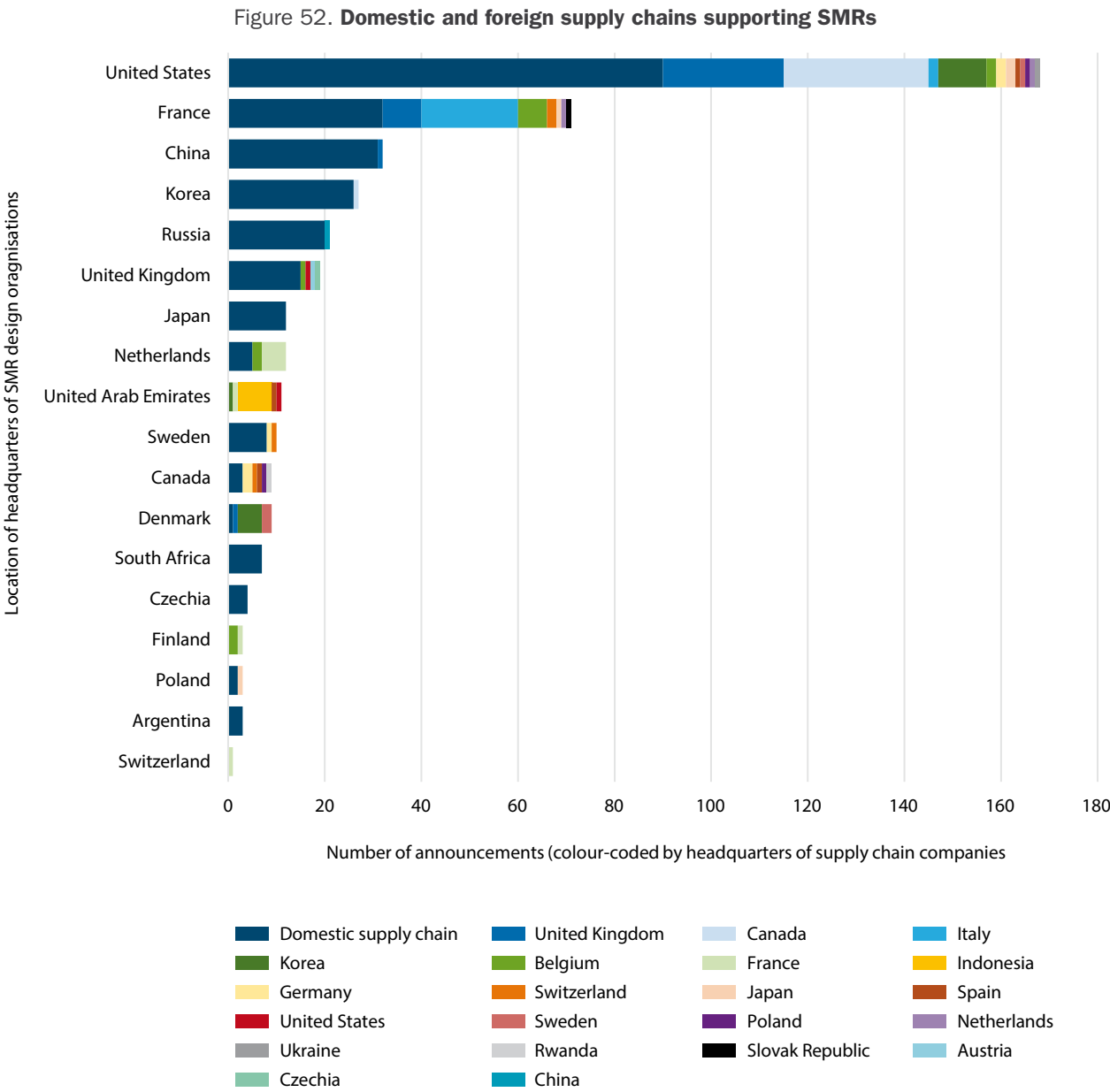
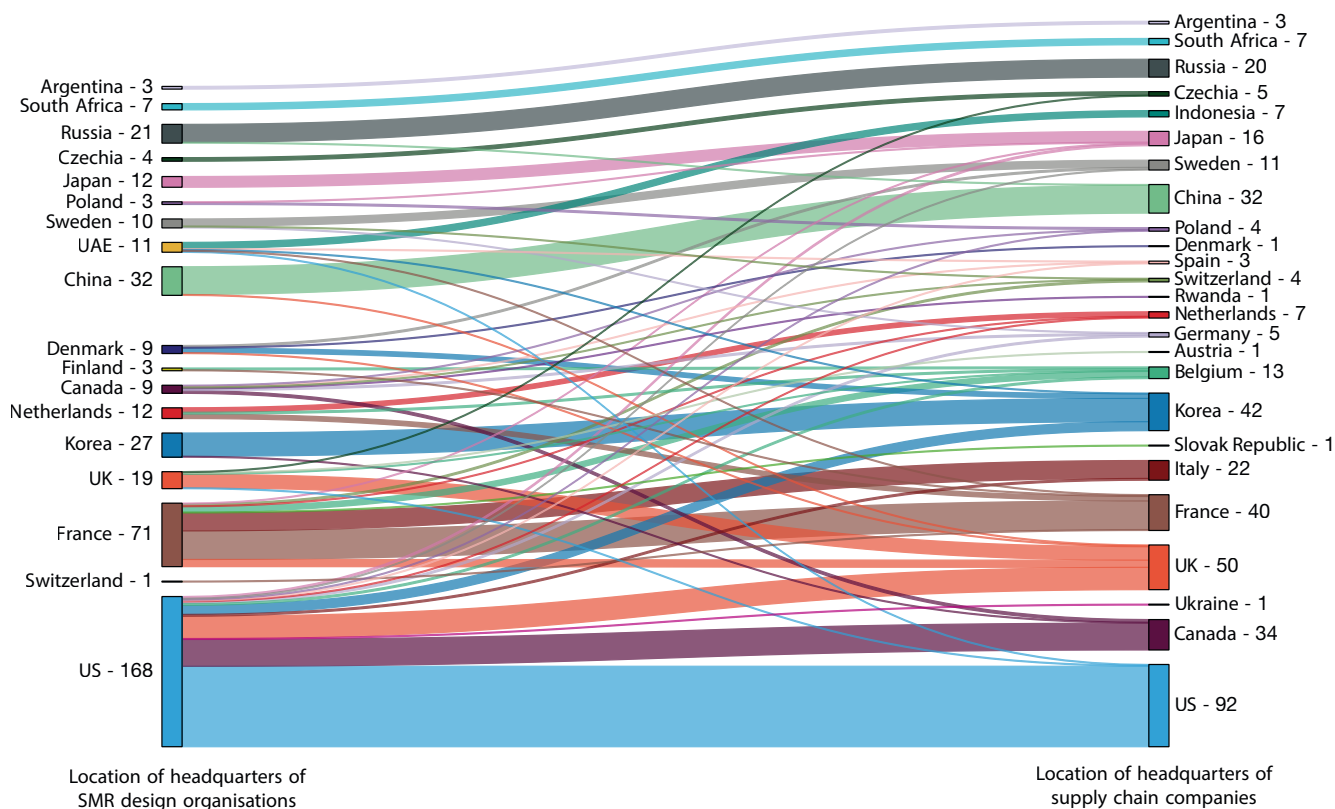
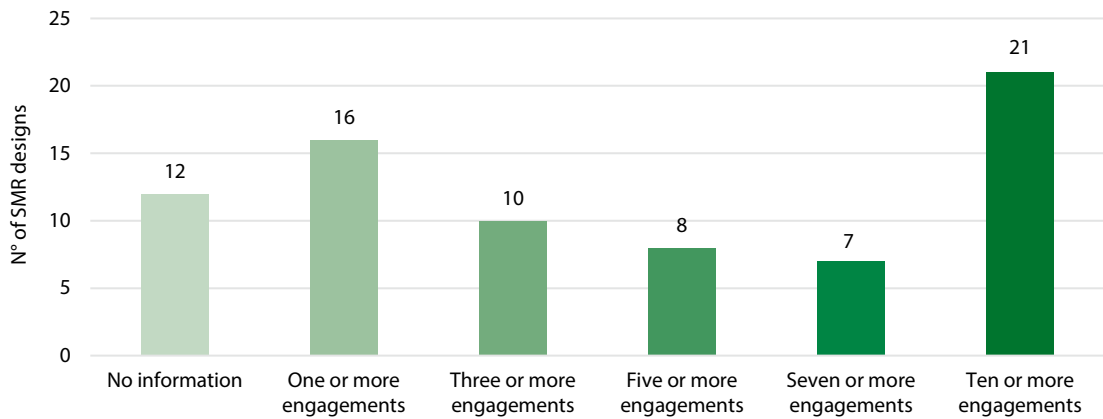


Figure 53. **SMRs working with domestic and foreign supply chain companies**

Engagement

Figure 54. Engagement progress worldwide



Engagement progress worldwide

Figure 54 shows the pipeline of SMRs making progress in the area of engagement with people and communities associated with their project. Of the SMR designs, 21 have advanced more than 10 engagement activities, 7 have advanced at least 7 engagement activities, 8 have advanced at least 5 engagement activities, 10 have advanced at least 3 engagement activities, and 16 have advanced at least 1 engagement activity. For 12 SMR designs, no recent information was readily available from verifiable public sources related to engagement activities.

Table 10. Engagement highlights

• Beyond traditional media outreach, SMR developers are adopting an increasingly diverse range of engagement strategies to advance their development and deployment efforts. • Communities and elected officials are expressing increasing support for nuclear energy in general, and SMRs in particular. • Globally, SMR developers are placing greater emphasis on building a talent pipeline. Raising awareness and interest in SMRs among students and young professionals is increasingly seen as a strategic priority.
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Figure 55 illustrates global progress in the area of engagement. Recent engagement from a growing number of SMR developers, spanning all major design concepts, configurations and countries are identified. Beyond public outreach through media, there has been an increasingly diverse range of engagement efforts. Notable examples include:

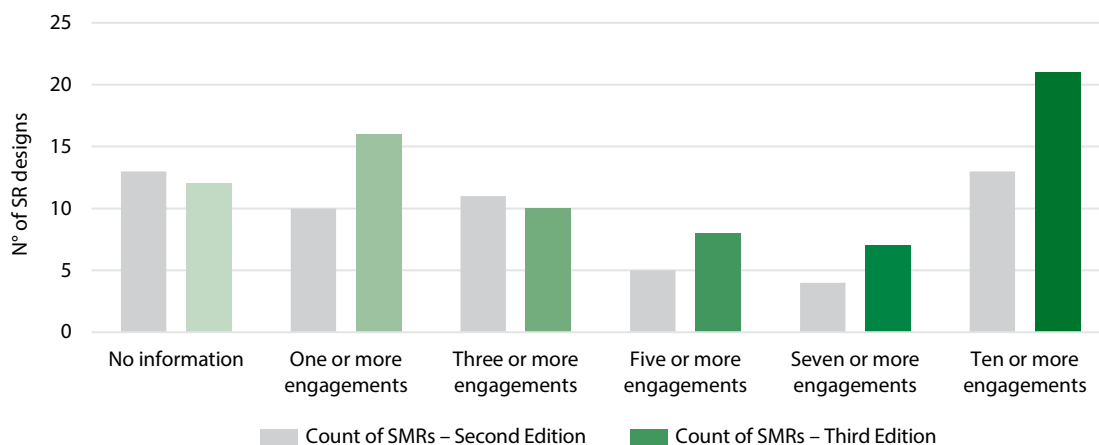
- **Promoting a positive narrative around nuclear energy:** SMR developers are contributing to broader public understanding and acceptance of nuclear energy technologies. This includes, for example, public screenings and special private screenings for elected officials of Oliver Stone’s *Nuclear Now* documentary, organised by newcleo’s CEO in Bratislava, Slovak Republic, and by Natura Resources’ CEO in Texas, United States.
- **Demonstration and educational tools:** Several SMR developers are investing in mock-ups and models as outreach tools, including traveling exhibits. Examples include a traveling exhibit by Last Energy, and the construction of full-scale, non-nuclear prototypes, like the one developed by newcleo in Italy. As of early 2025, NuScale had established eight Energy Exploration (E2) Centers across Africa, Asia, Europe and the United States, as educational tools in the use of modular reactors.

- **Educational collaborations:** SMR developers are engaging with future talent through educational partnerships. For example, Naarea in France has hosted delegations of students and young professionals at its facilities, while newcleo has provided direct support for students to participate in specialised SMR training programmes. In Indonesia, ThorCon International has active Memoranda of Understanding with three universities to support joint research and education initiatives in nuclear energy.
- **Global outreach and strategic partnerships:** SMR developers are actively engaging in international co-operation through strategic partnerships and multilateral forums. Argentina signed agreements with Bolivia and Indonesia to promote peaceful nuclear uses, including for the CAREM SMR; Thailand's state utility has shown interest in China's ACP100; GE Vernova Hitachi (GVH) signed Memoranda of Understanding with partners in Sweden and the United Arab Emirates; and Rolls-Royce SMR has Memoranda of Understanding with partners in Jordan, Finland, the Netherlands and Ukraine. Developers also leverage international platforms: newcleo contributed to the World Economic Forums' Annual Summit in Davos and co-hosted a high-level event at the G7 Energy Ministerial in Italy; NuScale has joined the United Nations' 24/7 Carbon-Free Energy Compact; Korea Hydropower and Nuclear Power (KHNP) showcased its i-SMR model at COP28.
- **Political and institutional endorsement:** As SMR projects advance toward deployment, they are receiving growing support from elected officials. For instance, Swedish Deputy Prime Minister Ebba Busch attended the groundbreaking ceremony for Blykalla's SEALER-E electrical prototype at the Oskarshamn site in February 2025. In the United States, Abilene Christian University and Natura Resources have received formal resolutions of support from both the Texas Senate and House.
- **Local community engagement and inclusion:** SMR developers are also advancing engagement with local and Indigenous communities. In Atlantic Canada, Moltex has supported outreach activities with Indigenous and civil society organisations. ARC Clean Technology, a long-standing sponsor of the Joint Economic Development Initiative's (JEDI) annual golf tournament in New Brunswick, reinforces its commitment to Indigenous economic development.

Other developers are similarly investing in community dialogue and partnerships. Blykalla has initiated engagement with residents in Sweden's Gävleborg region. Ontario Power Generation (OPG) regularly hosts public sessions, hearings and workshops. In the United States, Kairos Power created a virtual community platform and collaborates with the American Museum of Science and Energy. TerraPower is working with local Tribal representatives and Central Wyoming College to help build the future workforce for its Sodium project near Kemmerer, Wyoming.

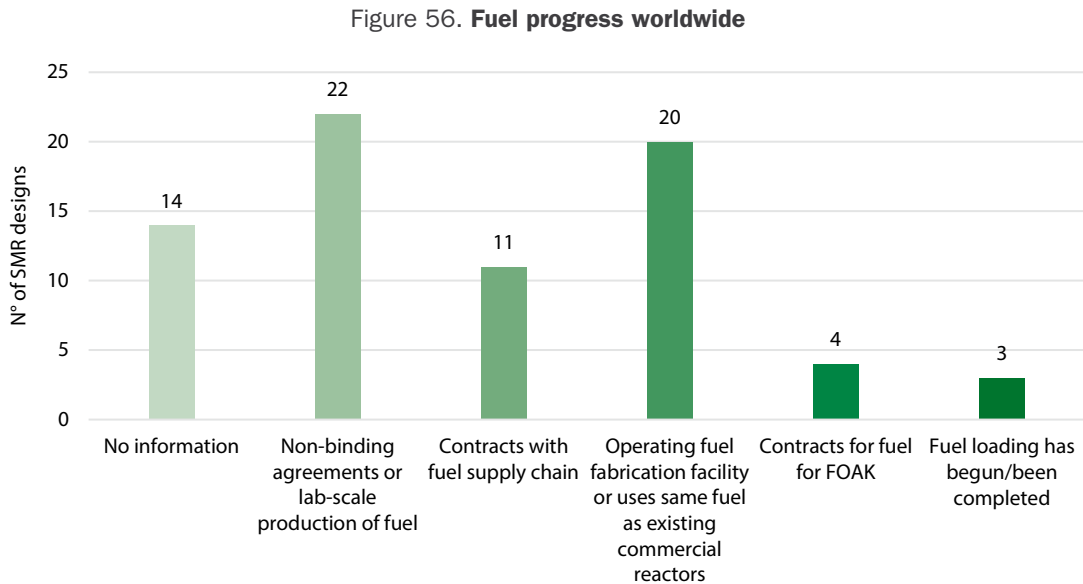
These examples reflect the expanding visibility and momentum of SMR initiatives across sectors, countries and institutions, signaling growing and diversified support for the technology's future.

Figure 55. **Comparison of engagement pipelines between the second (2024) and third (2025) editions of the NEA SMR Dashboard***



* The NEA SMR Dashboard: Second Edition is based on data as of 10 November 2023. The NEA SMR Dashboard: Third Edition is based on data as of 14 February 2025.

Fuel



Fuel readiness gaps persist for non-water-cooled SMRs

Figure 56 shows the pipeline of SMR designs making progress towards securing a supply of licensed and qualified fuel. Fuel loading has begun or has been completed for the first-of-a-kind of three SMR designs. Four SMR designs have secured contracts for fuel for their first-of-a-kind. Twenty SMR designs are associated with operating fabrication facilities producing fuel, or plan to use same fuel as existing commercial reactors. Eleven have entered contracts with fuel supply chain companies. Twenty-two SMR designs have non-binding agreements in place and/or studies with national laboratories related to their fuel. For 14 SMR designs, there is no information available from verifiable public sources related to fuel activities.

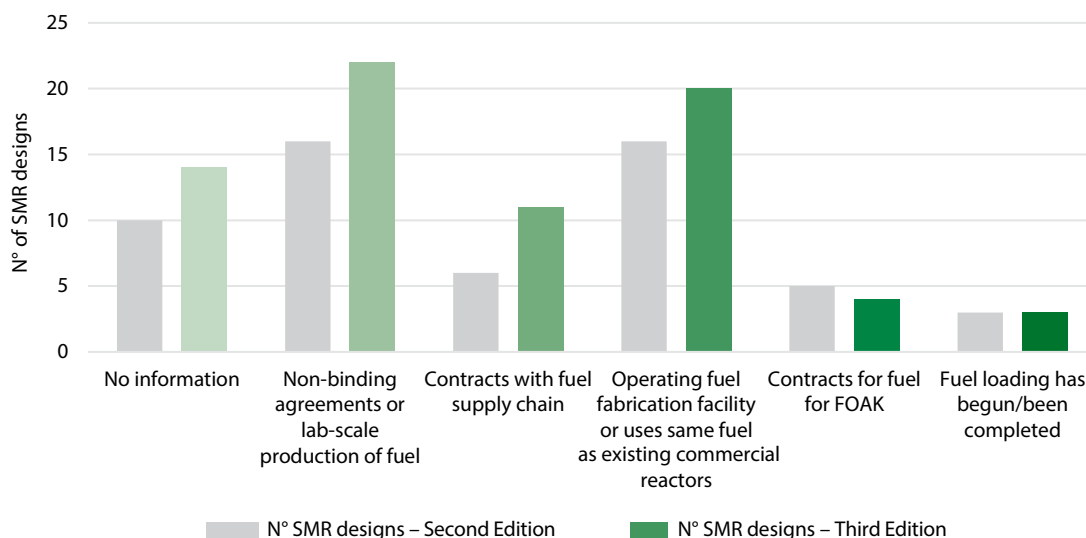
Table 11. **Fuel highlights**

• The <i>NEA SMR Dashboard</i> identifies more than 17 distinct fuel forms proposed across SMR designs. Only two, UO₂ and MOX, are currently qualified and available at commercial scale. • Many SMR developers plan to use high-assay low-enriched uranium (HALEU), enriched between 5% and 20%, but supply remains a major constraint. Only two US-based SMR designs have secured contracts for HALEU fuel for their first-of-a-kind deployments: – Project Pele (BWXT) will use HALEU derived from down-blended HEU stockpiles, supplied through the National Nuclear Security Administration (NNSA) and processed under contract with Idaho National Laboratory (INL). – Oklo’s Aurora Powerhouse will receive HALEU sourced from used fuel from the Experimental Breeder Reactor-II, also via INL. • More than 20 SMRs plan to use TRISO fuel. Among OECD countries, only Japan’s HTTR has completed TRISO fuel loading, using a prismatic design. • A growing number of SMR developers are investing in in-house TRISO fuel development and production capabilities, for example: – Kairos Power is advancing TRISO and pebble fuel manufacturing at its development labs in Albuquerque, New Mexico. – Jimmy Energy submitted a licence application in 2024 to build a TRISO fabrication facility near Le Creusot, France. – Westinghouse & Urenco are conducting a Pre-FEED study for TRISO fuel production at the Springfields facility in Preston, United Kingdom. – X-energy is constructing the TRISO-X Fuel Fabrication Facility in Oak Ridge, Tennessee (under construction since 2022).

There are a growing number of R&D initiatives focused on innovative fuel forms, along with increased engagement from SMR developers with fuel supply chain stakeholders. These developments indicate early, albeit modest, progress toward establishing reliable supplies of licensed and qualified fuel. More detailed information is provided in Table 12.

Figure 57 shows bimodal progress in the area of fuel readiness for SMR deployment, with one mode significantly more advanced than the other. The former represents SMRs for which the availability of qualified fuel is not expected to be a rate limiting factor for deployment. This includes water-cooled SMRs as well as few gas-cooled designs. The latter includes SMRs proposing the use of diverse and innovative fuels, many of which are yet to be qualified.

Figure 57. **Comparison of fuel pipelines between the second (2024) and third (2025) editions of the NEA SMR Dashboard**



* The *NEA SMR Dashboard: Second Edition* is based on data as of 10 November 2023. The *NEA SMR Dashboard: Third Edition* is based on data as of 14 February 2025.

Table 12. **Fuel progress worldwide, detailed**

Fuel loading has begun/been completed

- Three SMR designs have completed fuel loading: The HTR-PM in China; the floating KLT-40S in Russia; and the HTTR in Japan.
- The HTR-PM and the HTTR are both high-temperature gas cooled SMR designs, using TRISO fuel enriched between 5% and 10%. The HTR-PM is a commercial reactor. The HTTR is a research reactor.
- The KLT-40S is a floating commercial light-water reactor, using uranium-oxide ceramic (UO₂) enriched to 18.6%.

Contracts for fuel for first-of-a-kind

- Four SMR designs have contracts for fuel for their first-of-a-kind, including CNEA's CAREM in Argentina, GEVH's BWRX-300 in Canada, Oklo's Aurora Powerhouse and BWX Technologies' Project Pele in the United States.
- The BWRX-300 design progressed to this stage, as Ontario Power Generation selected suppliers for the first fuel contracts for its Darlington project in Canada in November 2023.
- Oklo's Aurora Powerhouse, a commercial metal-cooled SMR design, was selected by INL in 2020 to receive HALEU for its first-of-a-kind by processing used fuel from the Experimental Breeder Reactor-II.

Operating fuel fabrication facility, or uses same fuel as existing/Generation III commercial reactors
- All of the 20 SMR designs assessed at this stage of progress are water-cooled. Twelve plan to use the same fuel as existing commercial reactors, i.e. UO_2 fuel enriched at less than 5%. - Three SMR designs from the RITM series in Russia plan to use UO_2 fuel embedded in a ceramic-metallic (CERMET) composite matrix, enriched to 19.75%.
Contracts/agreements with fuel supply chain (uranium/conversion/enrichment/fabrication)
- Eleven SMR designs have contracts or agreements with fuel supply chain companies, up from six in the <i>NEA SMR Dashboard: Second Edition</i>. - SMR designs at this stage propose to use diverse fuel forms: BANR, Hermes, Jimmy, KRONOS MMR, LOKI MMR, and Xe-100 plan to use TRISO; Natrium Reactor Plant plans to use metallic alloy fuel; and IMSR plans to use a molten salt fuel. - The French SMR developer Jimmy, a high-temperature gas cooled design, contracted with Urenco for uranium supply enriched to 9.75% for its first-of-a-kind in October 2024.
Non-binding agreements and studies with national labs for RDD/lab-scale production of fuel
Twenty-two SMR designs have non-binding agreements in place and/or studies with national laboratories related to their fuel.
No information
For 14 SMR designs, no information was readily available from verifiable public sources related to fuel activities.

Figures 58 and 59 present progress of various SMR concepts and configurations toward securing qualified fuel supply. Water-cooled SMR concepts show more progress in securing licensed and qualified fuel supply. Within OECD countries, four non-water-cooled SMR designs have secured fuel for first-of-a-kind units, although through ad hoc agreements, rather than through established commercial supply chains.

Figure 60 illustrates the extent of reliance on domestic and foreign fuel supply chains. The countries that are home to the headquarters of the SMR design organisations appear on the left. The countries that are home to the headquarters of supply chain companies appear on the right.

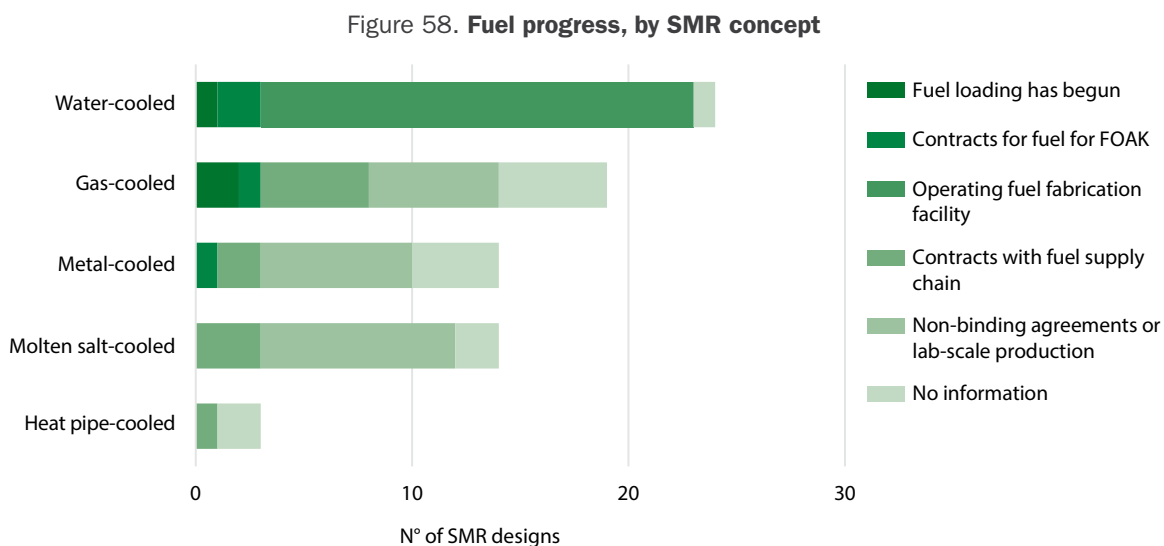
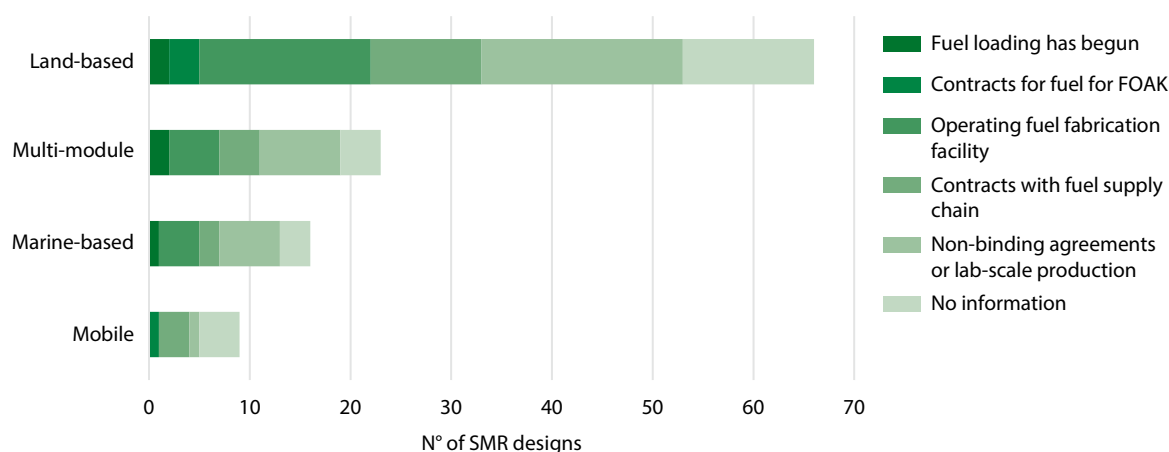
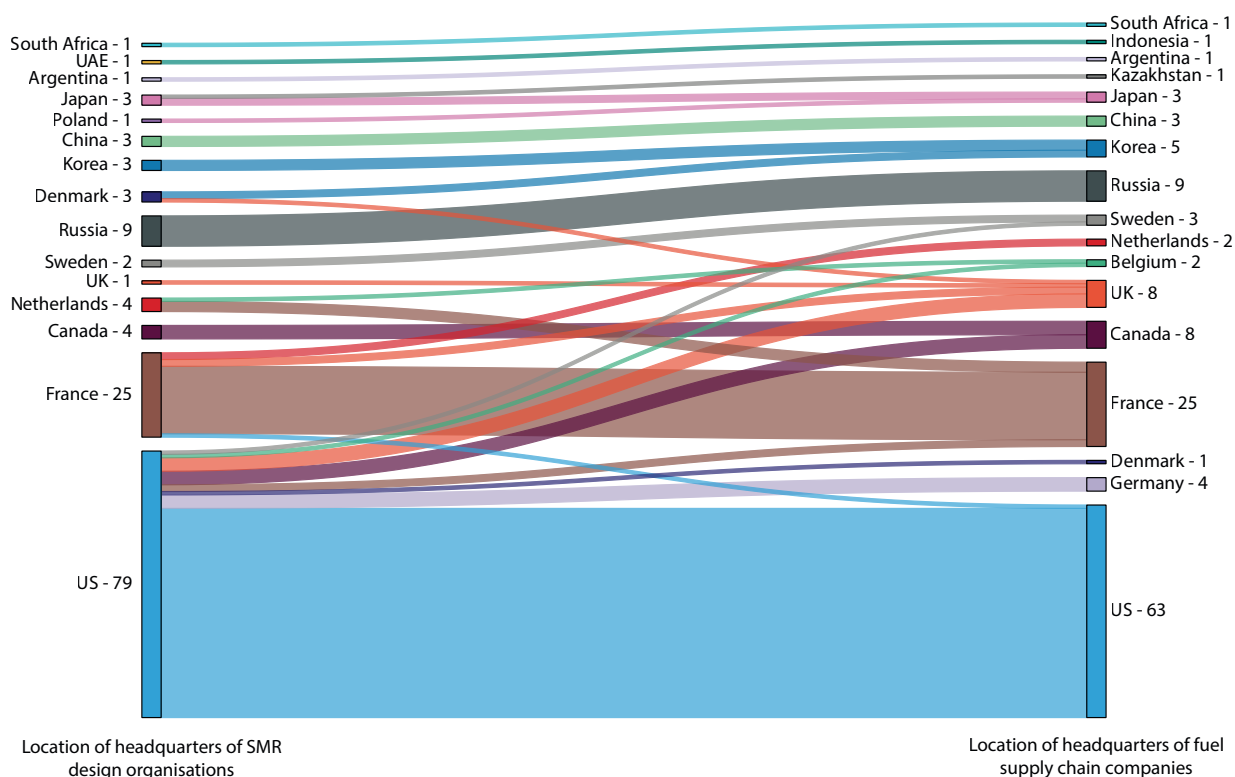


Figure 59. **Fuel progress, by SMR configuration**Figure 60. **SMRs supported by domestic and foreign fuel supply chain companies**

UK fuel supply chain companies are supporting SMRs from the United States, France, Denmark, and the United Kingdom. Canadian fuel supply chain companies are supporting SMRs from Canada and the United States. French fuel supply chain companies are supporting SMRs from France, the Netherlands and the United States. US fuel supply chain companies are supporting SMRs from the United States and France.

Conversely, SMRs with headquarters in the United States are working with fuel supply chain companies in the United States, Canada, Germany, the United Kingdom, France, Belgium, Denmark, and Sweden.

Diverse and innovative SMR fuels

The extensive data collected by the NEA in the preparation of the *Dashboard* creates a basis upon which to define a comprehensive taxonomy of SMR fuels.

The NEA SMR fuel taxonomy is presented in Figure 61. It accounts for every variation of fuel currently proposed by SMR designers around the world.

Figure 61. NEA SMR fuel taxonomy

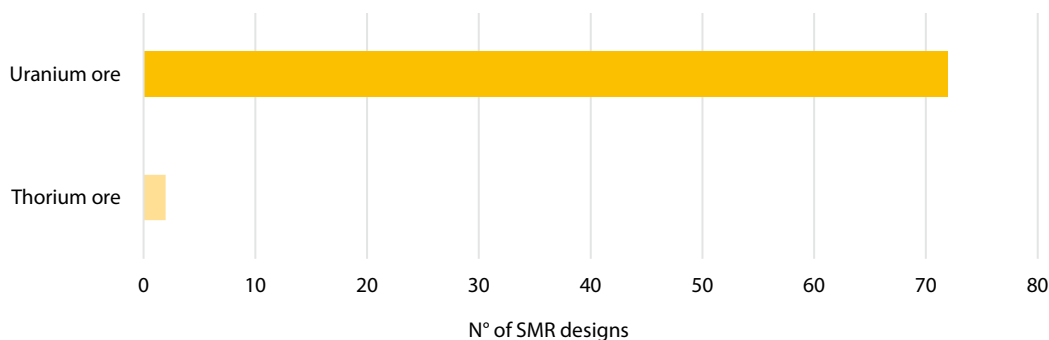
I. FISSILE MATERIAL		
Attribute	Possible values	Definition
A. Fuel cycle basis	▪ Uranium ore ▪ Thorium ore	Identifies the fertile base material of the fuel cycle – either uranium or thorium. This classification determines the long-term overarching cycle type and has implications for fuel supply, reactor design, potential reprocessing and waste composition.
B. Initial fissile material	▪ Uranium-235 ▪ Plutonium ▪ Uranium-233	Specifies the fissile isotopes present in the fresh fuel at the start of the cycle (e.g. U-235, Pu, U-233). This can define the startup strategy and licensing considerations. The following three subcategories allow to numerically specify the fissile percentage in the fuel. Specifies the fissile isotopes present in the fresh fuel at the start of the cycle (e.g. U-235, Pu, U-233). This can define the startup strategy and licensing considerations. The following three subcategories allow to numerically specify the fissile percentage in the fuel.
C. Percentage of initial fissile material	▪ Percentage of fissile plutonium (Pu-239 + Pu-241)	Indicates the percentage of fissile plutonium isotopes, which is particularly relevant for mixed oxide (MOX) fuel, fast reactors and recycled fuel cycles.
	▪ Percentage of uranium-233	Indicates the percentage of U-235 in the fuel, typically arising from thorium fuel cycles.
	▪ Percentage of uranium-235	Indicates the percentage of U-233 in the fuel. This directly impacts enrichment requirements.
D. Uranium-235 enrichment	▪ 5%: Low-enriched uranium ▪ $\geq 5\%$, $< 10\%$: Intermediate enrichment ▪ $\geq 10\%$, $< 20\%$: High-assay low-enriched uranium	Categorises U-235 enrichment into defined ranges which relate to different commercially available fuels and/or safeguard categories.

II. FUEL FORM		
Attribute	Possible values	Definition
A. Physical-chemical form	▪ Ceramics ▪ Metallic alloys ▪ Molten salts ▪ Other	Indicates the general material form of the fuel – oxide, metal, carbide, nitride, etc. – which influences fabrication, thermal conductivity and in-core performance.
B. Physical-chemical form specifics	▪ Uranium oxide ceramic ▪ Uranium-carbide ceramic ▪ Uranium-oxycarbide ceramic ▪ Uranium-silicide ceramic ▪ Uranium-nitride ceramic ▪ MOX ceramic ▪ Molten chlorides ▪ Molten fluorides ▪ Metal alloy: Specify	Provides specific families of physical or chemical forms of the fuel composition, differentiating uranium oxides (largely dominant today) from other forms.
C. Composite fuel architecture	▪ TRISO pebble ▪ TRISO prismatic ▪ TRISO-X pebble ▪ Ceramic-metal (CERMET) ▪ Ceramic-ceramic (CERCER) ▪ Other inert matrix fuel (IMF) ▪ (Not applicable)	Flags a more complex structural design of the fuel, useful to describe advanced composite fuels. Note that for conventional UO_2 fuels which do not have this composite structure this field is left blank (not applicable).

III. MINOR ACTINIDES		
Attribute	Possible values	Definition
A. Presence of minor actinides in fresh fuel	• Yes • No	Flags whether minor actinides (e.g., neptunium, americium, curium) are deliberately incorporated in the fresh fuel, for transmutation or waste reduction strategies.

The vast majority of SMRs are designed to use uranium-ore based fuels, though some aim to use thorium-ore based fuels (Figure 62). Only two SMR designs plan to use a thorium ore-based fuel cycle: Flibe Energy's LFTR in United States; and the Transmutex's START in Switzerland.

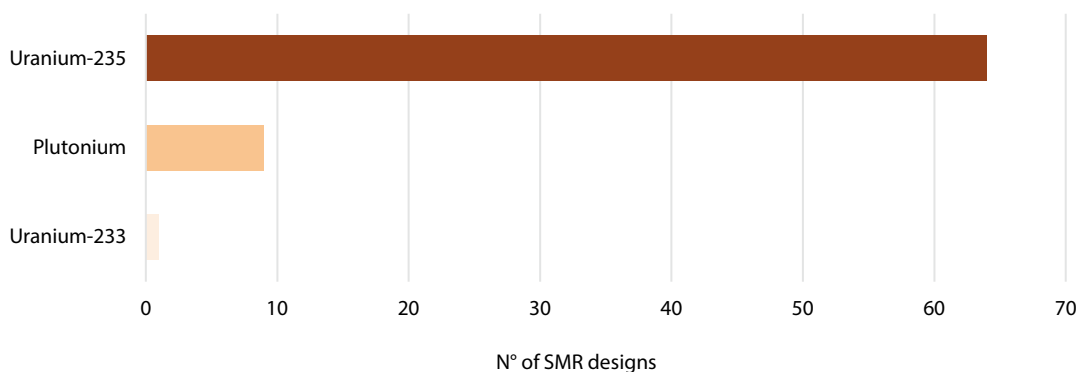
Figure 62. **Diverse SMR fuels: Fuel cycle basis**



Fissile materials for nuclear fuel include uranium-235 (U-235), plutonium-239 (Pu-239), and uranium-233 (U-233). U-235 is the only naturally occurring fissile material. Pu-239 and U-233 can be created from naturally occurring fertile materials, uranium-238 (U-238) and thorium-232 (Th-232) respectively.

Figure 63 shows that 65 SMR designs currently plan to use U-235 as the main fissile isotope present in the fresh fuel at the start of the cycle. Ten plan to use plutonium, and one, Flibe's LFTR, plans to use U-233.

Figure 63. **Diverse SMR fuels: Initial fissile material**



Plutonium-based fuels present an opportunity to harness untapped energy remaining in spent nuclear fuel stockpiles. However, their use remains limited, as only a few countries currently authorise plutonium for civil nuclear applications. Among SMR developers planning to use plutonium as initial fissile material, most aim to utilise mixed oxide (MOX) fuel, while a smaller number are pursuing molten-salt-based fuels. As interest in these fuel cycles grows, particularly for use in SMRs, their broader deployment would necessitate important policy discussions about nuclear security.

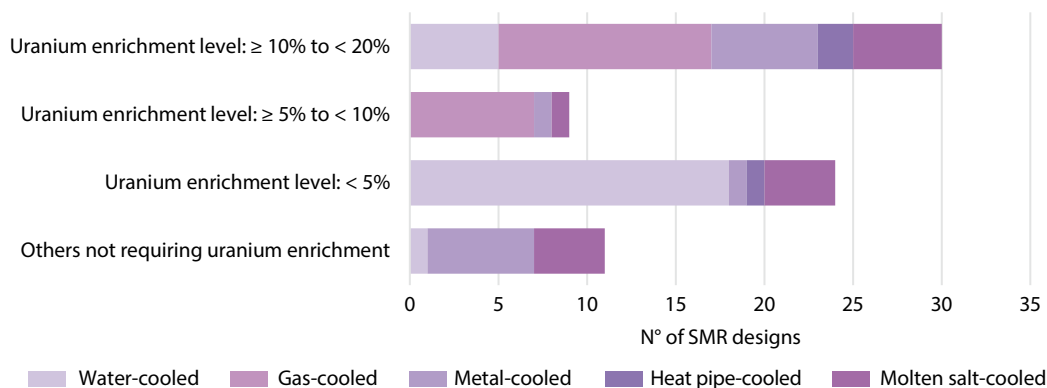
Out of nine SMR designs planning to use plutonium as their initial fissile material, eight show signs of real progress. Notably, five of these designs are based in France, a country with a well-established history in nuclear fuel reprocessing, and six have already initiated engagement with the French nuclear fuel company Orano. France also currently uses MOX fuel in part of its reactor fleet. The other two designs include Moltex Energy's SSR-W in Canada; and NIKIET's BREST-OD-300 in Russia.

Among European developers, newcleo plans to develop and operate its own MOX production facility, addressing current supply limitations for fast reactors. In 2024, newcleo submitted its Safety Option File for a fuel assembly testing facility to France's nuclear safety authority and is now advancing from Basic Design to Detailed Design of its MOX plant in 2025.

HALEU supply remains a key constraint for SMR deployment

More than half of SMR designs (39) require uranium enrichment levels higher than 5% (Figure 64). Of those, 30 plan to use uranium enriched at levels above 10% and below 20%. Nine plan to use uranium enriched at levels above 5% and below 10%. Nine plan to use uranium enriched at levels above 5% and below 10%.

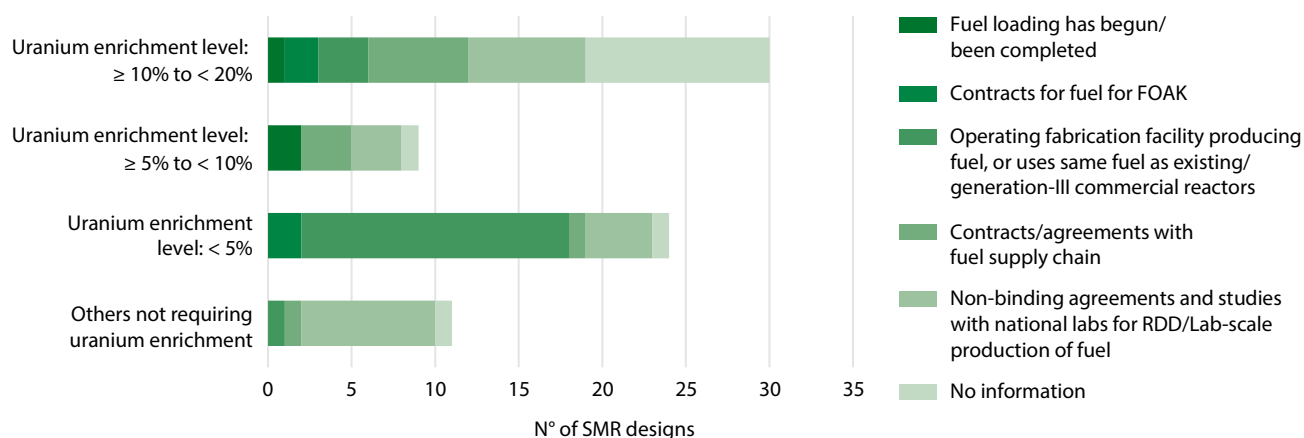
Figure 64. **Diverse SMR fuels: Uranium enrichment requirements**



While some SMR developers have begun engaging early to secure HALEU supplies for their first-of-a-kind reactors, the availability of HALEU, defined as uranium enriched above 5% and below 20%, remains a significant barrier to the deployment of many SMR designs.

Figure 65 illustrates the progress of SMR designs according to their planned uranium enrichment levels. Of the 39 SMR designs intending to use HALEU, more than half (22) have not yet progressed beyond non-binding agreements or collaborative studies with national laboratories related to fuel supply.

Figure 65. **Fuel progress, by uranium enrichment requirements**



Nine SMR designs that are planning to use HALEU fuel already have binding contracts with fuel supply chain companies within OECD countries. Among these, four designs have established binding agreements with entities specifically related to HALEU production, while the remaining five have binding agreements focused on fuel fabrication.

One of these four is the French SMR design Jimmy, which secured a contract with Urenco for uranium enriched to 9.75%, with first delivery expected in 2026. The other three, ODIN, KRONOS, and LOKI MMR, are being developed by NANO Nuclear Energy. To support HALEU production for its SMR designs, NANO Nuclear Energy invested in LIS Technologies, a developer of laser isotope separation (LIS) technology.

Some SMR developers are adjusting their first-of-a-kind fuel and reactor specifications in response to limited HALEU availability. For example, in 2024, Jimmy Energy contracted with Urenco to supply uranium enriched at 9.75%, lower than the originally intended 19.75% enrichment level planned for its later Nth-of-a-kind deployment.

Three SMR designs, Russia's RITM-200M, RITM-200N and RITM-200S, have access to operating fuel fabrication facilities that support uranium enrichment levels above 5%. Russia has longstanding commercial-scale HALEU production capabilities used for the KLT-40M reactors powering its nuclear icebreaker fleet and the KLT-40S SMR, which is currently docked in Pevek, Russia. Notably, the KLT-40S is the only SMR using uranium enriched above 10% for which fuel loading has already occurred, with first reloads completed in late 2023 and early 2024.

Two of the US-based SMR designs planning to use uranium enriched between 10% and 20% have already secured contracts for first-of-a-kind fuel:

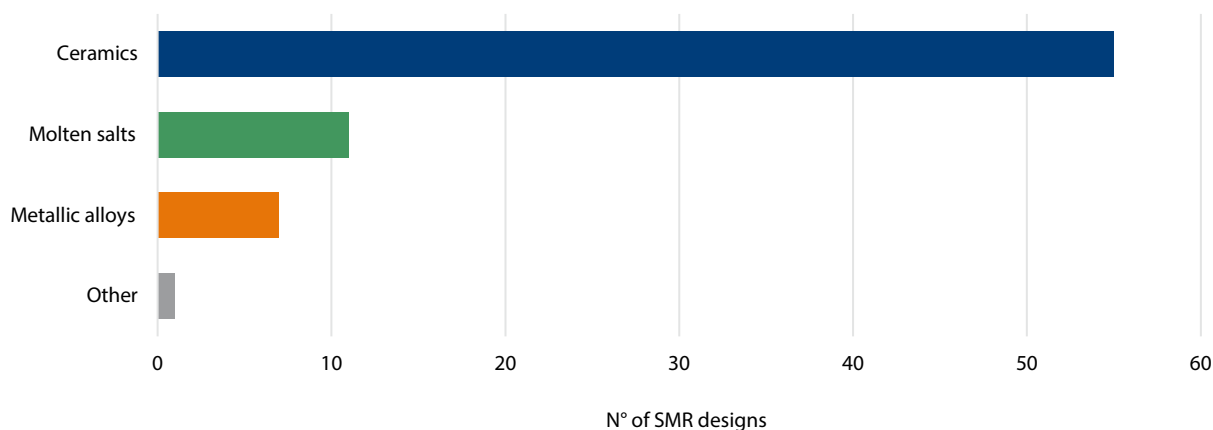
- Project Pele, developed by BWX Technologies (BWXT) for the US Department of Defense, will use HALEU produced by down-blending HEU stockpiles from the National Nuclear Security Administration. BWXT, which owns the only privately licensed US facilities to process HEU, will carry out this work under contract with Idaho National Laboratory (INL).
- Oklo's Aurora Powerhouse will receive HALEU produced from used fuel from the Experimental Breeder Reactor-II, also via INL. Oklo was selected by INL in 2020 for this first-of-a-kind fuel supply.

Additionally, two operating SMR designs using uranium enriched between 5% and 10% have completed fuel loading: the HTR-PM in China and the HTTR in Japan.

SMR designs are based on an increasingly diverse range of fuel forms, most of which have not yet been licensed or qualified for use

As illustrated in Figure 66, the majority of SMR developers plan to use fuels with ceramic physical-chemical forms, 11 SMR developers plan to use molten-salt fuel forms and 7 SMR designers plan to use fuels in metallic alloy forms.

Figure 66. **Diverse SMR fuels: Physical-chemical forms**

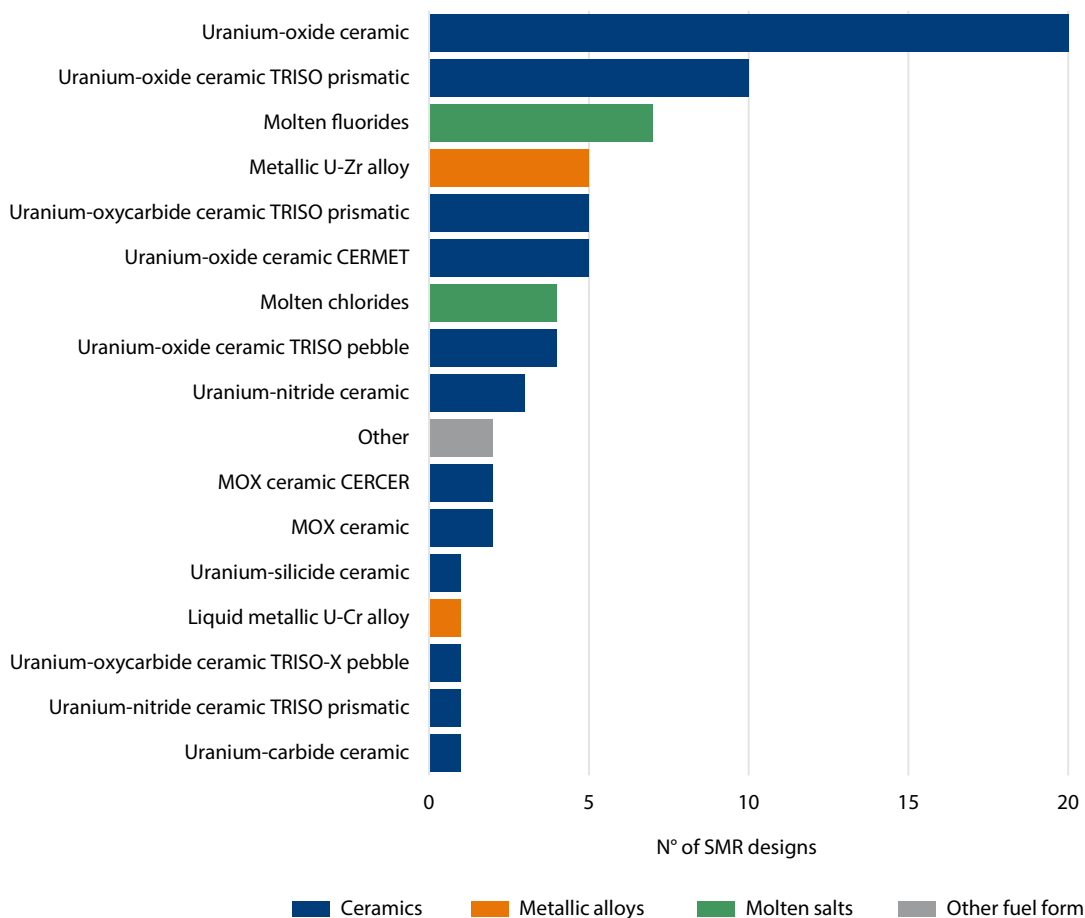


Nuclear fuel can be broadly classified into three main physical-chemical forms – ceramic, metallic and molten salt – each with distinct chemical and mechanical properties, described in Table 13.

Table 13. **Physical-chemical fuel forms**

Ceramics	Metallic alloys	Molten salts
Ceramic fuels, primarily uranium dioxide (UO ₂), are the most widely used in commercial reactors and are favoured by many water-cooled SMR designs. They offer stability under irradiation and high temperatures, though they have low uranium density and poor thermal conductivity.	Metallic fuels, such as uranium-zirconium (U-Zr) or uranium-chromium (U-Cr) alloys, provide higher fissile material density, superior thermal conductivity, and enhanced performance at high burnup, making them well-suited for fast-spectrum, liquid metal-cooled SMRs.	Molten salt fuels are unique as the only liquid fuel form, with fissile materials dissolved in a salt mixture. Their key advantage lies in the potential for online reprocessing, though they face challenges related to material corrosion. Each fuel type presents specific trade-offs in performance, manufacturability and compatibility with reactor designs.

Figure 67. **Diverse SMR fuels: Aggregate physical-chemical forms and composite fuel architectures**



The diversity of fuel forms used in SMR designs becomes most evident when examining both the physical-chemical forms and their composite architectures. While uranium oxide ceramics remain the dominant fuel in commercial reactors today, SMRs are introducing a wide range of novel fuels. Each unique combination of physical-chemical forms and composite fuel architectures requires different manufacturing processes and technologies.

Thirty-nine SMR designs plan to use standard uranium oxide ceramic fuel, making it the most common fuel among SMRs under active development. Within this group, 19 designs incorporate or plan to incorporate a composite fuel architecture, distinguishing them from the conventional fuel used in today's large-scale light-water reactors. These variations may significantly alter fuel fabrication requirements and performance characteristics. More broadly, 47 SMR designs, representing over 60% of the designs, rely on fuel forms that are not currently available at commercial scale.

While the supply chain for uranium oxide ceramic fuel, and to some extent MOX ceramic fuel, is well established, with multiple vendors and robust infrastructure for transport and regulation, the same cannot be said for novel fuels. Fuels such as TRISO particles, metallic alloys and molten salts require distinct fabrication technologies, and no commercial-scale production facilities currently exist for them.

Figure 68. **Fuel progress, by aggregate physical-chemical forms and composite architectures**

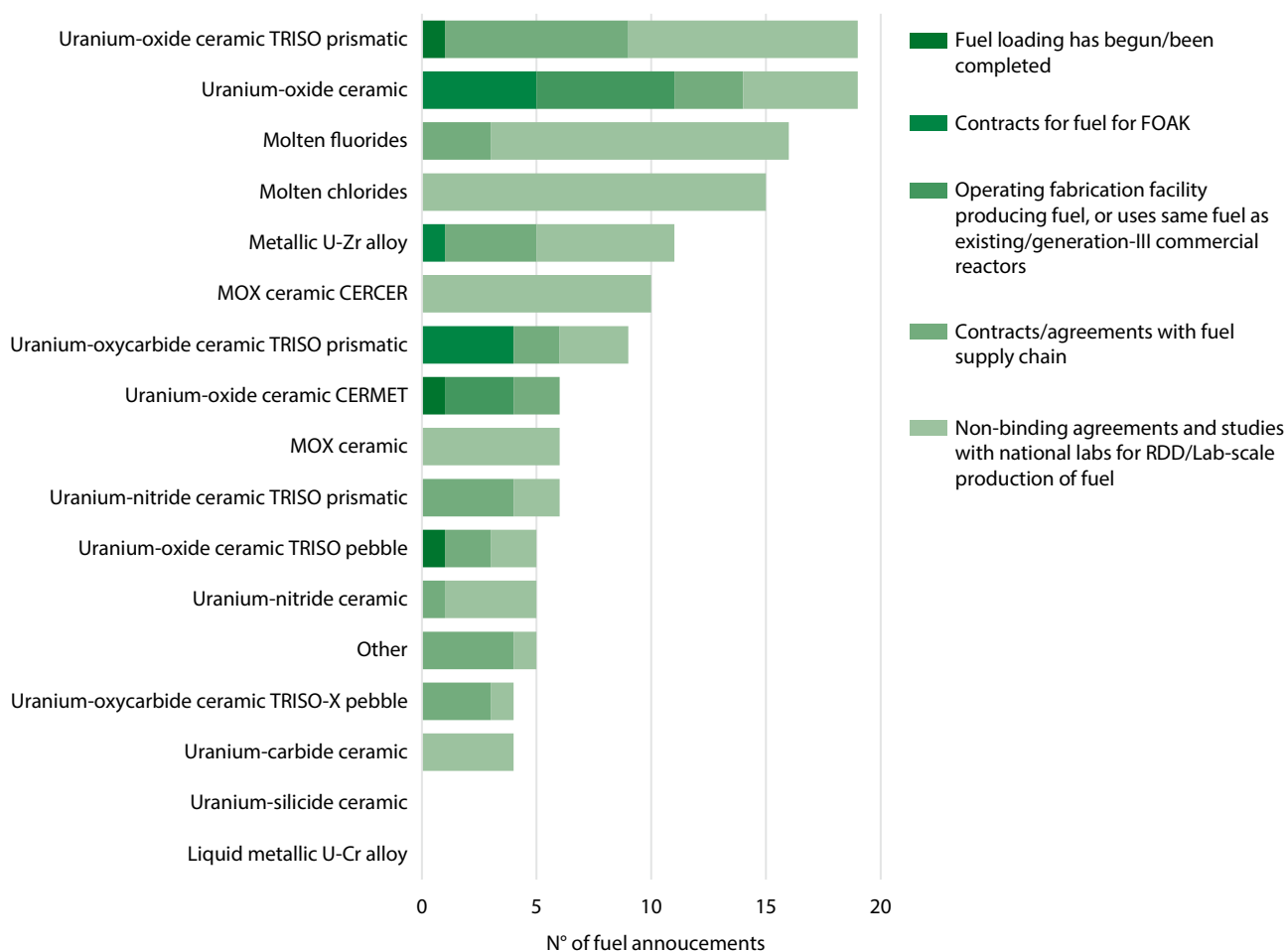


Table 14. **Fuel progress, by aggregate physical-chemical forms and composite architectures, detailed**

Ceramics
• The HTTR in Japan is the only SMR in an OECD country using TRISO fuel, in a prismatic form, that has completed fuel loading. • Contracts for first-of-a-kind TRISO fuel have been awarded under the US Department of Defense's Project Pele, developed by BWX Technologies (BWXT) in collaboration with the Idaho National Laboratory (INL) and BWXT itself. • BWXT operates commercial-scale TRISO fuel production facilities in Lynchburg, Virginia, United States. • Kairos Power is developing manufacturing processes at its TRISO and Pebble Development Labs in Albuquerque, New Mexico, United States. • Jimmy Energy submitted a licence application in 2024 to France's nuclear regulator (ASRN) to build an industrial facility near Le Creusot, which includes TRISO fuel manufacturing. • Westinghouse, in partnership with Urenco, is conducting a Pre-Front End Engineering Design (Pre-FEED) study to produce TRISO fuel at the Springfields facility in Preston, Lancashire, United Kingdom. • X-energy began construction of its TRISO-X Fuel Fabrication Facility in 2022, located in the Horizon Centre Industrial Park in Oak Ridge, Tennessee, United States. • In 2024, Ontario Power Generation (OPG) selected suppliers for the first fuel contracts for its Darlington project in Canada, where up to four BWRX-300 SMRs are planned. • In 2024, newcleo submitted its Safety Option File to France's nuclear safety authority (ASRN) for its fuel assembly testing facility. In 2022, newcleo also commissioned Orano to conduct feasibility studies on MOX fuel production.
Metallic alloys
• In 2022, TerraPower announced a binding agreement with Global Nuclear Fuel–Americas (GNF-A), a GE-led joint venture, to construct a fuel fabrication facility at GNF-A's existing site near Wilmington, North Carolina, United States.
Molten salts
• In 2023, Terrestrial Energy signed a manufacturing and supply agreement with Springfields Fuels Limited to design and build a pilot plant for Integral Molten Salt Reactor (IMSR) fuel. • In 2024, Saltfoss (formerly Seaborg) signed a non-binding agreement with Korean partners, including KEPCO Nuclear Fuel, to conduct a joint feasibility study for establishing a fuel salt production facility.

Different fuels for different attributes

Fuels are closely tied to the performance, safety, and operational characteristics of SMR designs. Figure 69 illustrates this relationship by highlighting the diversity of fuel used across the SMR concepts.

The majority of water-cooled SMR designs use uranium-oxide ceramic (UO₂) fuels, without specific composite fuel architecture. The only exceptions to this are the three Russian SMR designs from the RITM series which plan to use UO₂ fuel embedded in a ceramic-metallic (CERMET) composite matrix, enriched to 19.75%. UO₂ is a well-established nuclear fuel with a proven performance record in conventional LWRs that benefits from mature supply chains and historical technical expertise worldwide.

Gas-cooled SMR designs use diverse fuels, the majority in TRISO composite fuel architecture. TRISO fuel is designed for inherent safety characteristics, attributed to its multi-layer containment barrier. This multi-layered structure of TRISO fuel provides robust containment of fission products, maintaining integrity even within a high temperature environment.

Metal-cooled SMR designs use seven different fuel forms. Half (seven) plan to use metallic fuels, with improved thermal efficiency due to high thermal conductivity and dense fissile materials as well as enhanced safety features such as a higher thermal expansion coefficient. All SMR designs planning to use MOX ceramic fuels are also metal-cooled.

Molten salt-cooled designs mainly use molten salt fuels. This form offers unique advantages, including enhanced safety features, fuel flexibility, and the potential for waste minimisation. Two molten salt-cooled SMR designs, CVR's Energy Well and Kairos Power's Hermes, plan to use Uranium oxide TRISO prismatic and Uranium oxide TRISO pebble fuels respectively.

One heat-pipes cooled SMR design, Toshiba's MoveLuX, plans to use uranium-silicide, while the two others, Westinghouse's eVinci and Antares Nuclear's R1, plan to use uranium-oxycarbide (UCO) TRISO Prismatic fuel. UCO is a composite of uranium carbide and uranium oxide ceramic, designed to combine the beneficial properties of both fuel forms.

Figure 69. **Aggregate physical-chemical forms and composite fuel architectures, by SMR concept**

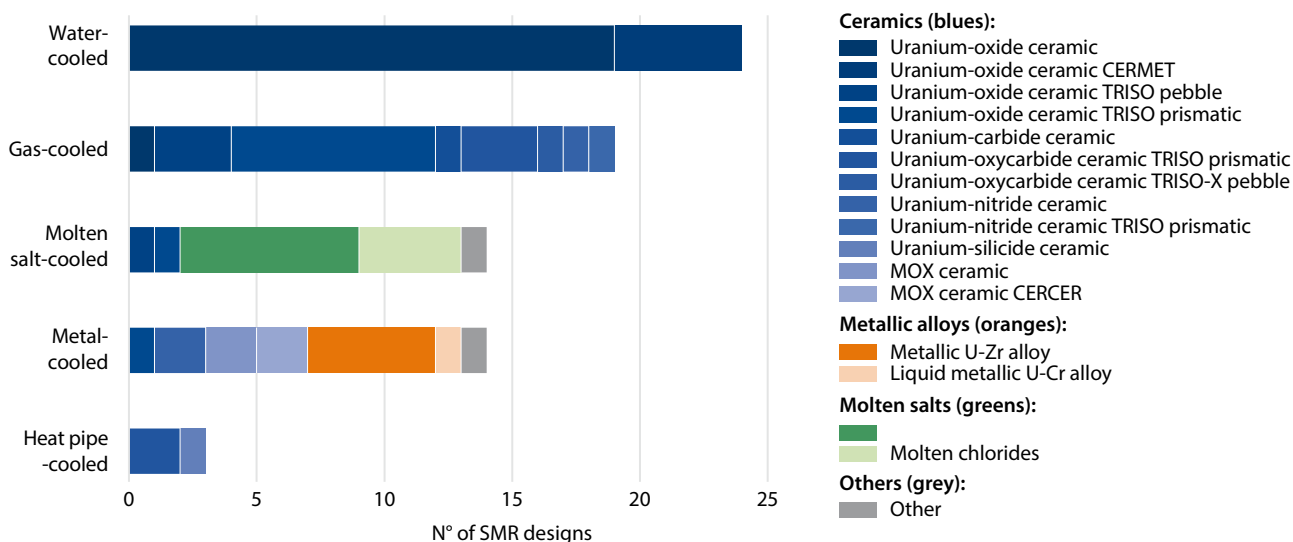
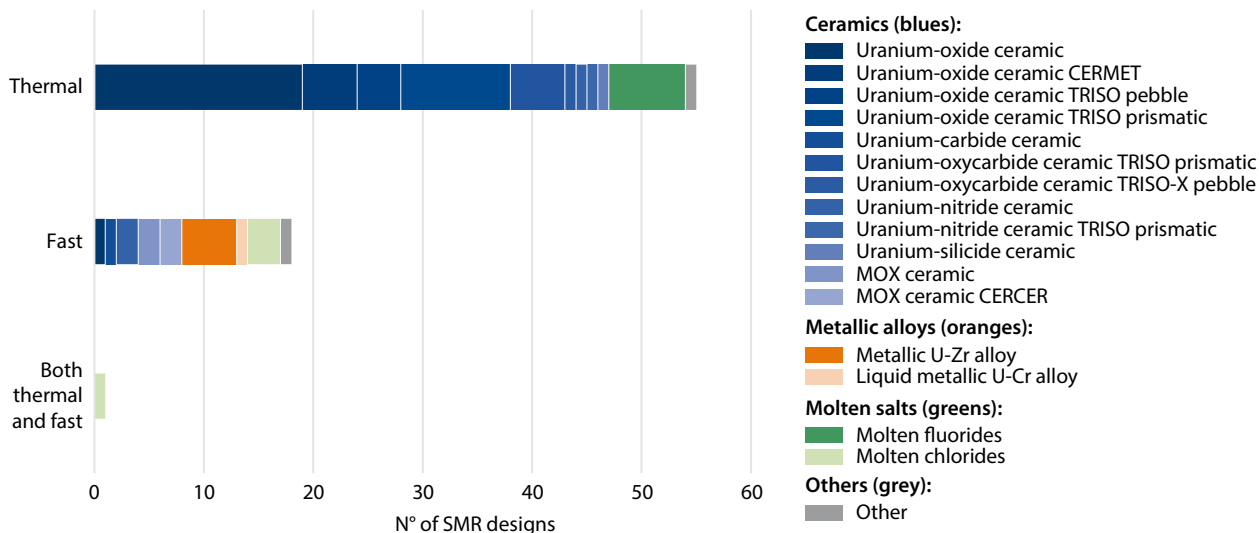


Figure 70 also illustrates the diversity of fuel forms across SMR designs based on their neutron spectrum, a key characteristic that directly impacts fuel cycle strategies, waste generation and disposal options. Fuels are particularly diverse among fast-spectrum SMR designs, which utilise nine different fuels. These include uranium oxide ceramics, plutonium-based fuels such as MOX ceramics, metallic alloys, and molten salt fuels.

Figure 70. **Aggregate physical-chemical forms and composite fuel architectures, by neutron spectrum**



Box 5. **Waste Integration for Small and Advanced Reactor Designs (WISARD)**

The innovative features of advanced reactors (ARs) and SMRs that enable their promised benefits will also have an inherent effect on the characteristics of their used fuel and waste streams. New fuel, moderator or coolant materials may lead to waste with different chemical, physical and radiological properties. Moreover, changes in reactor size may alter the volume and type of waste produced. There may also be implications for used fuel transportation due to factors such as higher burn-ups or increased end-of-cycle reactivity, elevated decay heat, or variations in fuel size that can affect cask compatibility. Whatever the design, waste from the next generation of nuclear power plants will differ in key ways from waste currently handled in national waste management programmes. These differences must be understood, evaluated and anticipated to provide sustainable, long-term waste management solutions.

Early consideration of the logistics and constraints of future waste streams allows waste management concerns to be actively factored into decisions on technology deployment and fuel cycle strategies. Some issues may be identified and “designed away” in this process, while early awareness and understanding of others will improve preparation for future disposal efforts. Another advantage of addressing waste management early on is the opportunity to co-ordinate with the multiple stakeholders involved in bringing a new reactor design to completion. While large light water reactors are a mature and well-understood technology, SMRs and ARs represent a diverse field with an active start-up culture and close collaboration with both academia and independent research organisations worldwide. With such a wide range of potential stakeholders, it is essential to involve all parties in the waste management process to minimise disconnects between the design, operation and decommissioning phases.

Recognising the importance of these challenges, the NEA has launched the Joint Project on Waste Integration for Small and Advanced Reactor Designs (WISARD). The core objective of this ambitious project is to explore and assess the back-end implications of innovative nuclear systems, with a focus on treatment and recycling, storage, transportation and disposal. Following a one-year preparatory phase throughout 2024, the project officially launched in April 2025. It brings together 12 signatory organisations across three continents, including national laboratories, research institutes, industry players and startups. Multiple third parties will also contribute to implementing the project's programme of work.

The NEA WISARD Joint Project will foster collaboration and shared understanding among stakeholders throughout the reactor life cycle – an essential foundation for building a successful, sustainable nuclear programme.

SMR assessments

Key information for each of the SMR designs assessed in this edition is listed in Table 15.

Table 15. **SMRs assessed in the NEA SMR Dashboard: Third Edition**

Name	Design organisation	Headquarters (city/region)	Country	Thermal power (MWth)	Core outlet temp. (°C)	Spectrum (thermal/fast)	Fuel form
R1	Antares Nuclear	Torrance, California	United States	1	800	Thermal	Uranium-oxycarbide ceramic TRISO prismatic
ARC-100	ARC Clean Technology	Washington DC	United States	286	510	Fast	Metallic U-Zr alloy
Blossom Energy SMR	Blossom Energy	Tokyo	Japan	180		Thermal	Uranium-oxide ceramic TRISO prismatic
Blue Capsule	Blue Capsule	Aix-en-Provence	France	150	750	Thermal	Uranium-oxide ceramic TRISO prismatic
SEALER-One	Blykalla	Stockholm	Sweden	70	550	Fast	Uranium-nitride ceramic
BANR	BWXT	Lynchburg, Virginia	United States	100	800	Thermal	Uranium-nitride ceramic TRISO prismatic
Project Pele	BWXT	Lynchburg, Virginia	United States	1 to 5		Thermal	Uranium-oxycarbide ceramic TRISO prismatic
CAL-30	Calogena	Paris	France	30	110	Thermal	Uranium-oxide ceramic
ACPR50S	CGN	Shenzhen	China	200	320	Thermal	
CAREM	CNEA	Buenos Aires	Argentina	100	326	Thermal	Uranium-oxide ceramic
ACP100	CNNC	Beijing	China	385	320	Thermal	Uranium-oxide ceramic
LEUNR	CSMC	Toronto, Ontario	Canada	0.1	130	Thermal	Uranium-oxide ceramic CERMET
Energy Well	CVŘ	Husinec	Czechia	20	700	Thermal	Uranium-oxide ceramic TRISO prismatic
DFBR-1	Deep Fission	Berkeley, California	United States	50	315	Thermal	Uranium-oxide ceramic

Table 15. **SMRs assessed in the NEA SMR Dashboard: Third Edition** (cont'd)

Name	Design organisation	Headquarters (city/region)	Country	Thermal power (MWth)	Core outlet temp. (°C)	Spectrum (thermal/fast)	Fuel form
DF300	Dual Fluid Energy	Vancouver, British Columbia	Canada	600	1 000	Fast	Liquid metallic U-Cr alloy
A-HTR-100	Eskom	Sandton	South Africa	100	1 200	Thermal	Uranium-oxide ceramic TRISO pebble
LFTR	Flibe Energy	Huntsville, Alabama	United States	600	650	Thermal	Molten fluorides
BWRX-300	GE Vernova Hitachi	Wilmington, North Carolina	United States	870	288	Thermal	Uranium-oxide ceramic
FMR	General Atomics	San Diego, California	United States	100	800	Fast	Uranium-oxide ceramic
EM ²	General Atomics	San Diego, California	United States	500	850	Fast	Uranium-carbide ceramic
HEXANA	HEXANA	Aix-en-Provence	France	800	530	Fast	MOX ceramic
SMR-300	Holtec	Camden, New Jersey	United States	1 050	321	Thermal	Uranium-oxide ceramic
HTR-PM	INET	Beijing	China	500	750	Thermal	Uranium-oxide ceramic TRISO pebble
NHR200-II	INET and CGN	Beijing	China	200	280	Thermal	Uranium-oxide ceramic
i-SMR	i-SMRDA, KHNP, and KAERI	Daejeon	Korea	520	321	Thermal	Uranium-oxide ceramic
GTHTR300	JAEA	Tokai	Japan	600	950	Thermal	Uranium-oxide ceramic TRISO prismatic
HTTR	JAEA	Tokai	Japan	30	850	Thermal	Uranium-oxide ceramic TRISO prismatic
HTR50S	JAEA	Tokai	Japan	50	750	Thermal	Uranium-oxide ceramic TRISO prismatic
Jimmy HTR	Jimmy	Paris	France	10	550	Thermal	Uranium-oxy-carbide ceramic TRISO prismatic
SMART100	KAERI	Daejeon	Korea	365	322	Thermal	Uranium-oxide ceramic
SALUS-100	KAERI	Daejeon	Korea	267	510	Fast	Metallic U-Zr alloy

Table 15. **SMRs assessed in the NEA SMR Dashboard: Third Edition** (cont'd)

Name	Design organisation	Headquarters (city/region)	Country	Thermal power (MWth)	Core outlet temp. (°C)	Spectrum (thermal/fast)	Fuel form
HECTAR	KAERI	Daejeon	Korea	90	750	Thermal	Uranium-carbide ceramic TRISO prismatic
Hermes	Kairos Power	Alameda, California	United States	35	650	Thermal	Uranium-oxide ceramic TRISO pebble
PWR-20	Last Energy	Washington DC	United States	80	300	Thermal	Uranium-oxide ceramic
MN-1	MobileNuclear Energy	Richmond, Virginia	United States	1	650	Thermal	Uranium-nitride ceramic
SSR-W	Moltex Energy	Saint John, New Brunswick	Canada	750	590	Fast	Molten chlorides
FLEX	MoltexFLEX	Stratford-upon-Avon	United Kingdom	60	700	Thermal	Molten fluorides
XAMR	NAAREA	Nanterre	France	80	625	Fast	Molten chlorides
ODIN	NANO Nuclear Energy	New York City, New York	United States	5	400	Thermal	Other
KRONOS MMR	NANO Nuclear Energy	New York City, New York	United States	10 to 45	660	Thermal	Uranium-oxide ceramic TRISO prismatic
LOKI MMR	NANO Nuclear Energy	New York City, New York	United States	5	727	Thermal	Uranium-oxide ceramic TRISO prismatic
MSR-1	Natura Resources and Abilene Christian University	Abilene, Texas	United States	1	650	Thermal	Molten fluorides
MSR-100	Natura Resources	Abilene, Texas	United States	250	650	Thermal	Molten fluorides
HTGR-POLA	NCBJ	Otwock	Poland	30	750	Thermal	Uranium-oxide ceramic TRISO prismatic
LFR-AS-200	newcleo	Paris	France	480	530	Fast	MOX ceramic CERCER
LFR-AS-30	newcleo	Paris	France	90	530	Fast	MOX ceramic CERCER
BREST-OD-300	NIKIET	Moscow	Russia	700	535	Fast	Uranium-nitride ceramic
Nuclearis N1	Nuclearis Energy	New York City, New York	United States	42	319.8	Thermal	Uranium-oxide ceramic

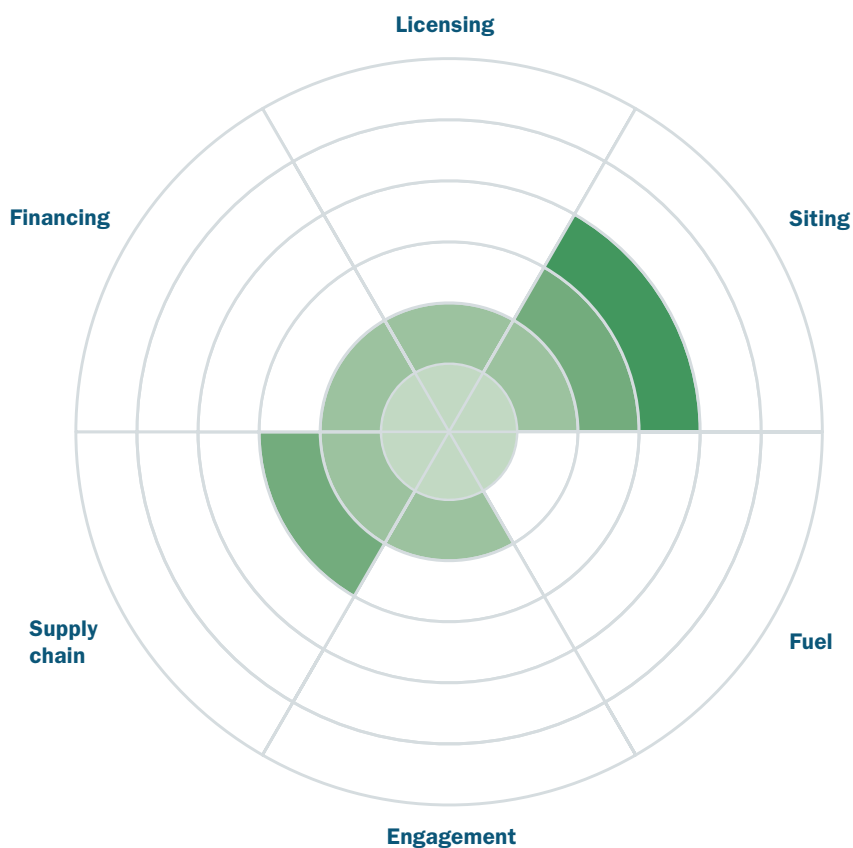
Table 15. **SMRs assessed in the NEA SMR Dashboard: Third Edition** (cont'd)

Name	Design organisation	Headquarters (city/region)	Country	Thermal power (MWth)	Core outlet temp. (°C)	Spectrum (thermal/fast)	Fuel form
ENTRA1 Energy Plants	NuScale Power	Portland, Oregon	United States	250	321	Thermal	Uranium-oxide ceramic
NUWARD SMR	NUWARD	Paris	France	1 000	307	Thermal	Uranium-oxide ceramic
KLT-40S	OKBM	Nizhny Novgorod	Russia	300	316	Thermal	Uranium-oxide ceramic CERMET
RITM-200M	OKBM	Nizhny Novgorod	Russia	396	321	Thermal	Uranium-oxide ceramic CERMET
RITM-200N	OKBM	Nizhny Novgorod	Russia	190	321	Thermal	Uranium-oxide ceramic CERMET
RITM-200S	OKBM	Nizhny Novgorod	Russia	396	318	Thermal	Uranium-oxide ceramic CERMET
Aurora Powerhouse	Oklo	Santa Clara, California	United States	205	500	Fast	Metallic U-Zr alloy
Otrera 2x300	Otrera	Aix-en-Provence	France	600	550	Fast	MOX ceramic
Kaleidos	Radiant	El Segundo, California	United States	3.5	700	Thermal	Uranium-oxide ceramic TRISO prismatic
RR SMR	Rolls-Royce SMR	Manchester	United Kingdom	1 358	325	Thermal	Uranium-oxide ceramic
CMSR	Saltfoss	Copenhagen	Denmark	250	650	Thermal	Molten fluorides
HAPPY200	SPIC	Beijing	China	200	120	Thermal	Uranium-oxide ceramic
LDR-50	Steady Energy	Espoo	Finland	50	150	Thermal	Uranium-oxide ceramic
Stellarium	Stellaria	Grenoble	France	1 080	750	Fast	Molten chlorides
HTMR-100	Stratek Global	Pretoria	South Africa	100	750	Thermal	Uranium-oxide ceramic TRISO pebble
Sodium Reactor Plant	TerraPower	Bellevue, Washington	United States	840	500	Fast	Metallic U-Zr alloy
IMSR	Terrestrial Energy	Charlotte, North Carolina	United States	884	700	Thermal	Molten fluorides
ThorCon 500	ThorCon International	Dubai	United Arab Emirates	1 114	704	Thermal	Molten fluorides

Table 15. **SMRs assessed in the NEA SMR Dashboard: Third Edition** (cont'd)

Name	Design organisation	Headquarters (city/region)	Country	Thermal power (MWth)	Core outlet temp. (°C)	Spectrum (thermal/fast)	Fuel form
Thorizon One	Thorizon	Amsterdam	Netherlands	250	550	Both thermal and fast	Molten chlorides
MoveLuX	Toshiba Energy Systems & Solutions Corporation	Kawasaki	Japan	10	685	Thermal	Uranium-silicide ceramic
4S	Toshiba Energy Systems & Solutions Corporation	Kawasaki	Japan	30	510	Fast	Metallic U-Zr alloy
START	Transmutex	Vernier	Switzerland	600	450	Fast	Other
TEPLATOR	UWB and CIIRC CTU	Prague	Czechia	50 to 170	180	Thermal	Uranium-oxide ceramic
AP300 SMR	Westinghouse Electric Company	Cranberry Township, Pennsylvania	United States	990	285	Thermal	Uranium-oxide ceramic
eVinci	Westinghouse Electric Company	Cranberry Township, Pennsylvania	United States	15	750	Thermal	Uranium-oxycarbide ceramic TRISO prismatic
Xe-100	X-energy	Rockville, Maryland	United States	200	750	Thermal	Uranium-oxycarbide ceramic TRISO-X pebble

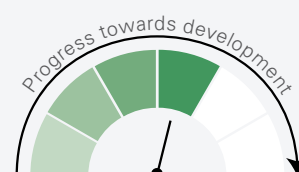
Antares Nuclear – R1



Licensing



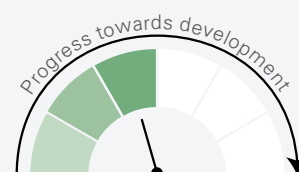
Siting



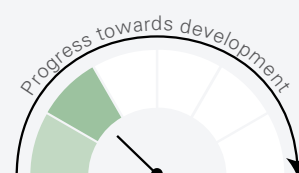
Financing



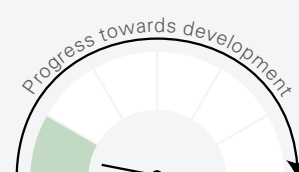
Supply chain



Engagement



Fuel

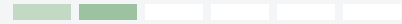


Reactor description: Kilowatt-scale heat-pipe reactor using TRISO fuel for earth, subsea, and outer space applications.

Thermal power (MWth)	1
Core outlet temperature (°C)	800
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxycarbide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level ≥ 10% to <20%

Assessment of R1's progress to deployment

Licensing



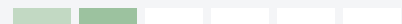
The National Reactor Innovation Center (NRIC), a US Department of Energy (DOE) programme led by Idaho National Laboratory (INL), has selected Antares Nuclear along with three other micro-reactors as a partner to begin a Front-End Engineering and Experiment Design (FEEED) study, which will include the preparation of safety documents and NEPA-compliance documentation for the R1 SMR. NRIC has been supporting regulatory engagement activities with the US Nuclear Regulatory Agency (NRC), the DOE, and reactor demonstrators to determine an appropriate regulatory strategy for the deployment and testing of microreactors, including the R1 by Antares, at the Demonstration Of Microreactor Experiments (DOME) facility at INL.

Siting



The Antares R1 kilowatt-scale SMR has been selected for demonstration at the DOME site under the NRIC programme by INL. Testing of microreactors in the DOME could start as early as mid-2026.

Financing



In 2024, Antares closed a Series A funding round of USD 30 million (USD₂₀₂₃ 29.1 million), led by a group of investors including Alt Capital and Caffeinated Capital. This follows a USD 8 million seed funding round in 2023, primarily led by Caffeinated Capital. In 2024, under the US Department of Defense Air Force Branch Small Business Innovation Research (SBIR) programme phases one and two, Antares has been awarded contracts totaling USD 2.5 million (USD₂₀₂₃ 2.4 million).

Supply chain



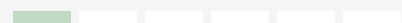
Antares joined the NRIC FEEED programme by INL to support the manufacturing, assembly, and evaluation of the R1 SMR. Antares Nuclear received a voucher from the US DOE GAIN programme to conduct an independent technical audit of the reactor's core neutronics and thermal hydraulics with Oak Ridge National Laboratory. Additionally, Savannah River National Laboratory and Antares signed a Cooperative R&D Agreement to advance studies on fuel cycle management, modular deployment, and security of SMRs. In 2024, Antares contracted Manufacturo, a manufacturing engineering software provider, for work on supply chain management. In February 2024, Antares further expanded its capabilities by opening a dedicated manufacturing and R&D facility in Torrance, California.

Engagement



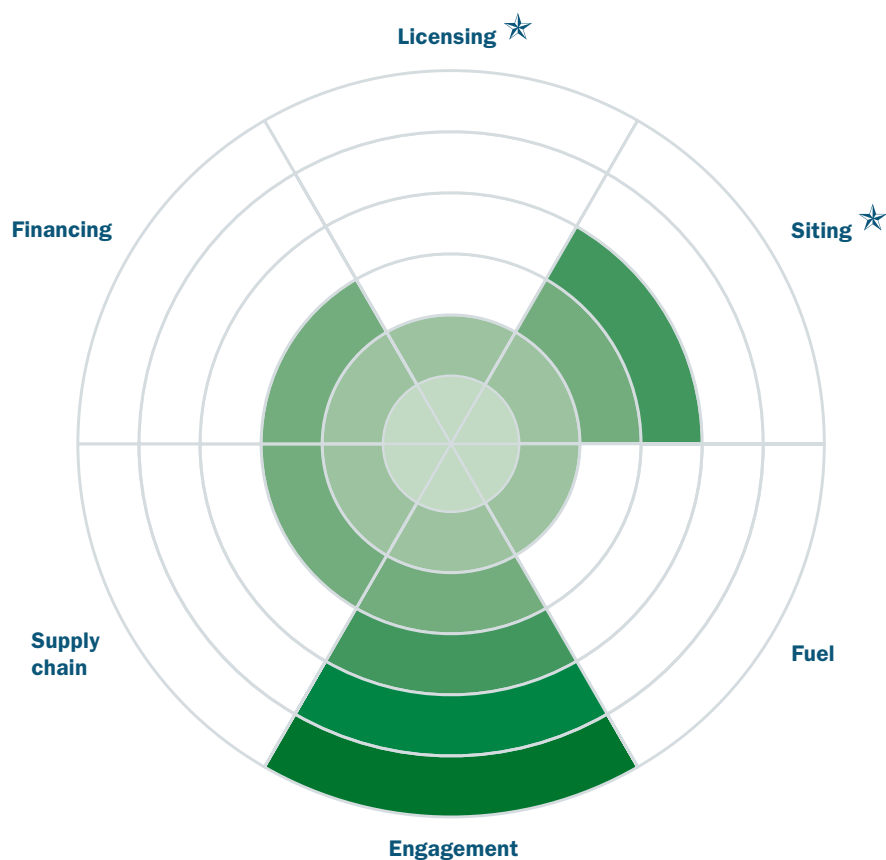
In 2024, Antares' co-founder and President, Julia DeWahl, participated in an episode of the *Age of Miracles* podcast with Packy McCormick.

Fuel



At the time of the assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

ARC Clean Technology – ARC-100



★ Active in multiple jurisdictions or countries.

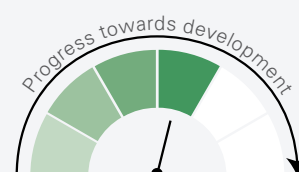
Reactor description: Sodium-cooled fast reactor using metallic uranium alloy fuel with a 20-year refuelling cycle.

Thermal power (MWth)	286
Core outlet temperature (°C)	510
Spectrum (thermal/fast)	Fast
Fuel form	Metallic U-Zr alloy
Base fissile material	U-235, enrichment level ≥ 10% to <20%

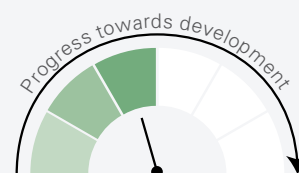
Licensing ★



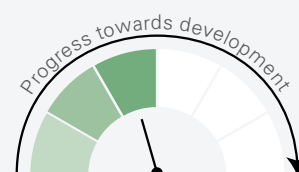
Siting ★



Financing



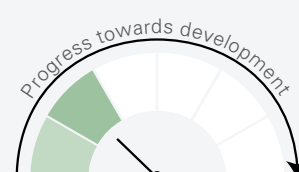
Supply chain



Engagement

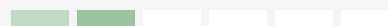


Fuel



Assessment of ARC-100's progress to deployment

Licensing ★



The ARC-100 reactor is in Phase 2 of the pre-licensing vendor design review (VDR) process with the Canadian Nuclear Security Commission (CNSC) and is engaged in pre-licensing activities with the US Nuclear Regulatory Commission (NRC). In June 2023, New Brunswick Power (NB Power) submitted a Licence to Prepare Site application to the CNSC as well as an Environmental Impact Assessment registration to the New Brunswick Department of Environment and Local Government for the ARC-100 project.

Siting ★



The ARC-100 reactor has been selected by New Brunswick Power (NB Power), the primary electric utility in the Canadian province of New Brunswick for deployment at their Point Lepreau Nuclear Generating Station. Canada's Belledune Port Authority also shortlisted ARC-100 to generate a minimum of 1 gigawatt of firm heat and electricity for industrial users at the port's announced Green Energy Hub. In November 2023, ARC Clean Technology signed a Memorandum of Understanding (MoU) with Korea Hydro and Nuclear Power Co (KHNP) and NB Power to explore the commercialisation of the ARC-100 in Canada, Korea, the United States and other jurisdictions.

Financing



Between 2018 and 2023, ARC Clean Technology was granted about USD 62 million in public funding from the United Kingdom's Advanced Modular Reactor competition (2018), Japan's Nuclear Energy X Innovative Promotion programme (2021), the US' Advanced Reactor Demonstration Program (2020), Canada's Clean Electricity Predevelopment Program (2023) as well as the Province of New Brunswick (2018 and 2021). ARC Clean Technology also announced the closing of its Series A funding round with CAD 30 million (USD₂₀₂₃ 24.7 million) from combined private and New Brunswick public funding (2022). In 2023, the North Shore Mi'kmaq Tribal Council (NSMTC) and its member communities made an equity investment of CAD 1 million (USD₂₀₂₃ 740 790) in ARC Clean Technology.

Supply chain



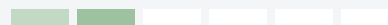
ARC Clean Technology, KHNP, and NB Power signed a MoU in 2023, and a trilateral cooperation agreement in 2024, to collaborate on the deployment of ARC-100 units. ARC Clean Technology has been collaborating with GE Hitachi Nuclear Energy (GEH) since 2017 under an agreement that includes in-kind contributions from GEH to provide engineering and design expertise. It also has commercial agreements with engineering companies including Hatch and AECOM. In the Canadian 2022 Provincial Strategic Plan for the Deployment of SMRs, the Government of New Brunswick set out its objective to advance supply chain readiness for ARC-100 in collaboration with Opportunities New Brunswick and Canadian Manufacturers and Exporters. ARC Clean Technology has also hosted multiple supply chain events.

Engagement



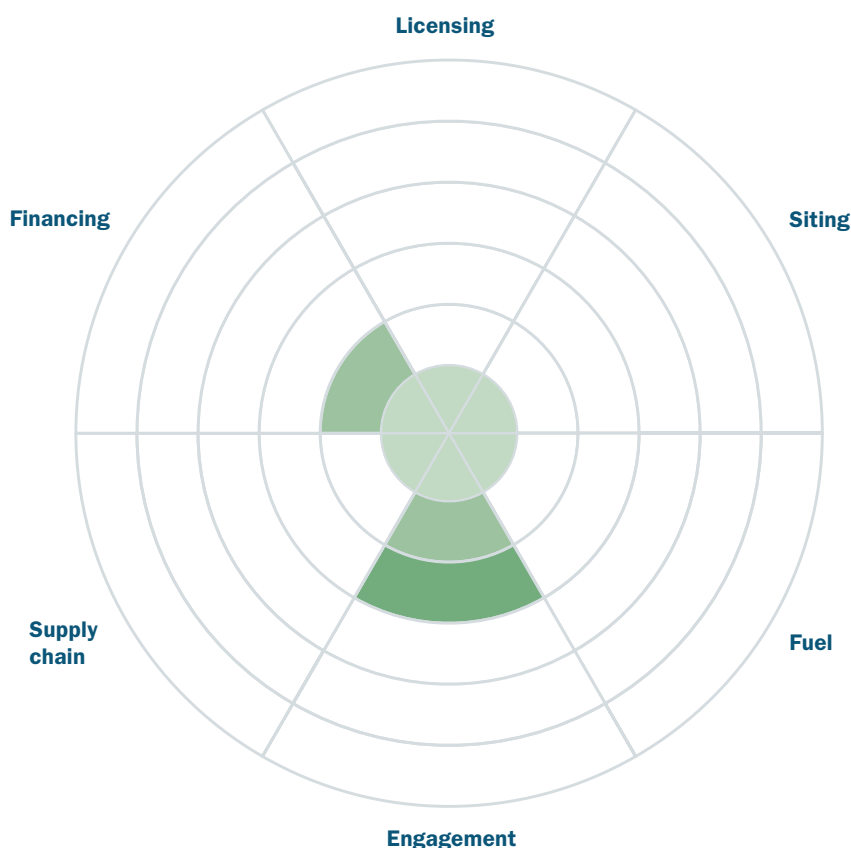
ARC Clean Technology has received endorsement from the province of New Brunswick for siting the ARC-100 reactor. ARC Clean Technology has sponsored engagement activities with various Indigenous communities, labour unions, and civil society organisations in the Atlantic Canada region. ARC Clean Technology has a partnership with NSMTC and entered into an equity agreement in September 2023 with NSMTC and its seven member communities for benefits sharing. It is also engaged with the University of New Brunswick's Centre for Nuclear Energy Research and Canada's Building Trades Unions to promote a talent pipeline for the ARC-100. ARC Clean Technology is a member of the Atlantic Clean Energy Alliance and the Atlantica Centre for Energy.

Fuel

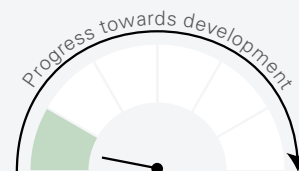


ARC Clean Technologies plans to use HALEU fuel for its ARC-100 SMR. ARC Clean Technology signed a letter of intent with Centrus for HALEU supply and is partnering with Argonne National Laboratory (ANL), Idaho National Laboratory (INL), and Canadian Nuclear Laboratories (CNL) to advance fuel qualification and develop its fuel manufacturing process. Fuel qualification efforts aim to leverage existing experimental data from the Experimental Breeder Reactor-II and Fast Flux Test Facility fast reactors. ARC Clean Technology has also been awarded a voucher under the Gateway for Accelerated Innovation in Nuclear (GAIN) initiative by the US DOE to support qualification of ARC's fuel in collaboration with the ANL and INL.

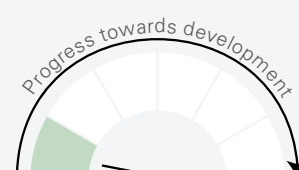
Blossom Energy – Cluster-type High-Temperature Gas-Cooled Reactor (Blossom Energy SMR)



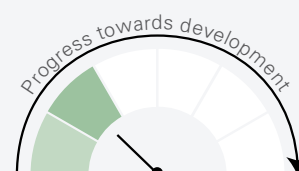
Licensing



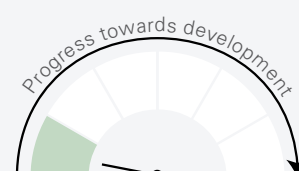
Siting



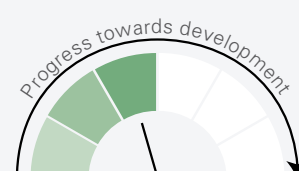
Financing



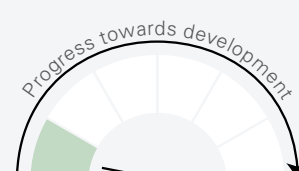
Supply chain



Engagement



Fuel



Reactor description: Cluster-type high-temperature gas-cooled reactor with two to eight modules coupling to one heat exchanger.

Thermal power (MWth)	180 (plant deployed with at least two modules of 90 MWth each)
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Core outlet temperature (°C)	No information
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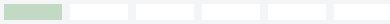
Spectrum (thermal/fast)	Thermal
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Fuel form	Uranium-oxide ceramic TRISO prismatic
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Base fissile material	U-235, enrichment level ≥ 10% to <20%
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Assessment of Blossom Energy SMR's progress to deployment

Licensing



At the time of assessment, there was no additional information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



At the time of assessment, no information related to siting was readily available from any site owners.

Financing



In 2024, Blossom Energy raised JPY 350 million (USD₂₀₂₃ 2.4 million) in a Pre-Series A funding round from private investors, including Incubate Fund, Animal Spirits No. 1 Fund, Tsuneishi Shoji, ReGACY Innovation Group, and Globis. This round brings the company's total funding to date to around JPY 450 million (USD₂₀₂₃ 3.1 million).

Supply chain



At the time of assessment, no information about supply chain readiness was readily available from verifiable public sources.

Engagement



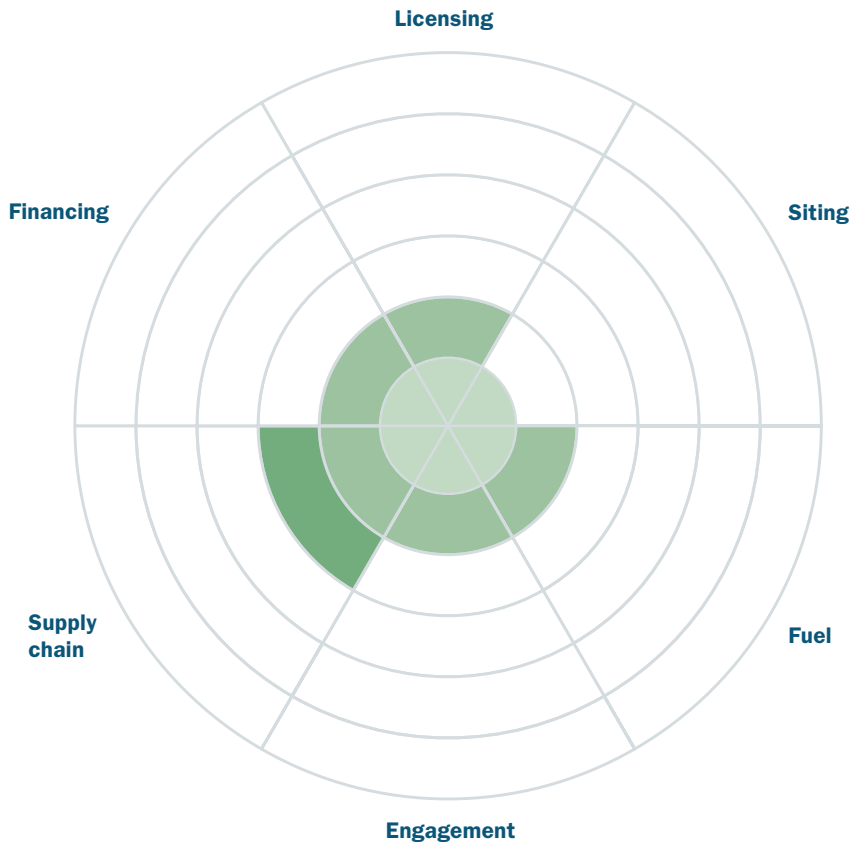
The Chief Executive Officer of Blossom Energy is engaging with the general public through journal contributions, media interviews, YouTube videos, and podcasts to raise awareness of nuclear technology, including high-temperature gas-cooled reactors (HTGR).

Fuel



The Cluster-type HTGR is designed to be fuelled with either LEU or HALEU. At the time of assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

Blue Capsule Technology – Blue Capsule



△ Changes since Second Edition (2024).

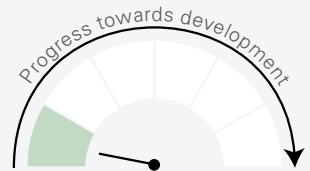
Reactor description: High-temperature, sodium-cooled, thermal spectrum SMR using TRISO fuel and air as its heat sink.

Thermal power (MWth)	150
Core outlet temperature (°C)	750
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level <5%

Licensing



Siting



Financing



Supply chain



Engagement

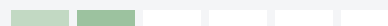


Fuel



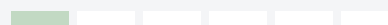
Assessment of Blue Capsule's progress to deployment

Licensing



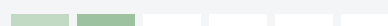
As of early 2025, Blue Capsule completed the screening phase of pre-licensing with France's Nuclear Safety and Radiation Protection Authority (ASNR) and is awaiting first meeting to initiate the preparatory review stage with the ASNR.

Siting



At the time of assessment, no information related to siting was readily available from any site owners.

Financing



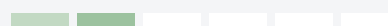
Blue Capsule received EUR 9.1 million (USD₂₀₂₃ 9.8 million) under the France 2030 programme in 2023 and raised EUR 2.0 million (USD₂₀₂₃ 2.1 million) in an equity round in 2024. This equity round involves public and private investors including: Exergon, EREN Groupe, Audacia Co-Investissement, and CEA Investissement.

Supply chain



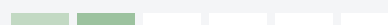
In 2024, Blue Capsule contracted the engineering company CSTI Group to design and build a test qualification loop. Blue Capsule has announced plans to run tests on this loop for several years starting in 2025, ahead of the launch of its non-nuclear prototype, scheduled by the end of 2028. Also in 2024, Blue Capsule broadened its collaboration with the French Atomic Energy and Alternative Energies Commission (CEA) to advance work on equipment, components, materials, calculation codes, and TRISO fuel. In 2022 and 2023, Blue Capsule signed non-binding collaboration agreements: with Elyse Technology to explore integrating the Blue Capsule SMR as a heat source for Elyse's biomass pyrolysis process; and with Egis to work on civil engineering and the assessment of external and seismic hazards.

Engagement



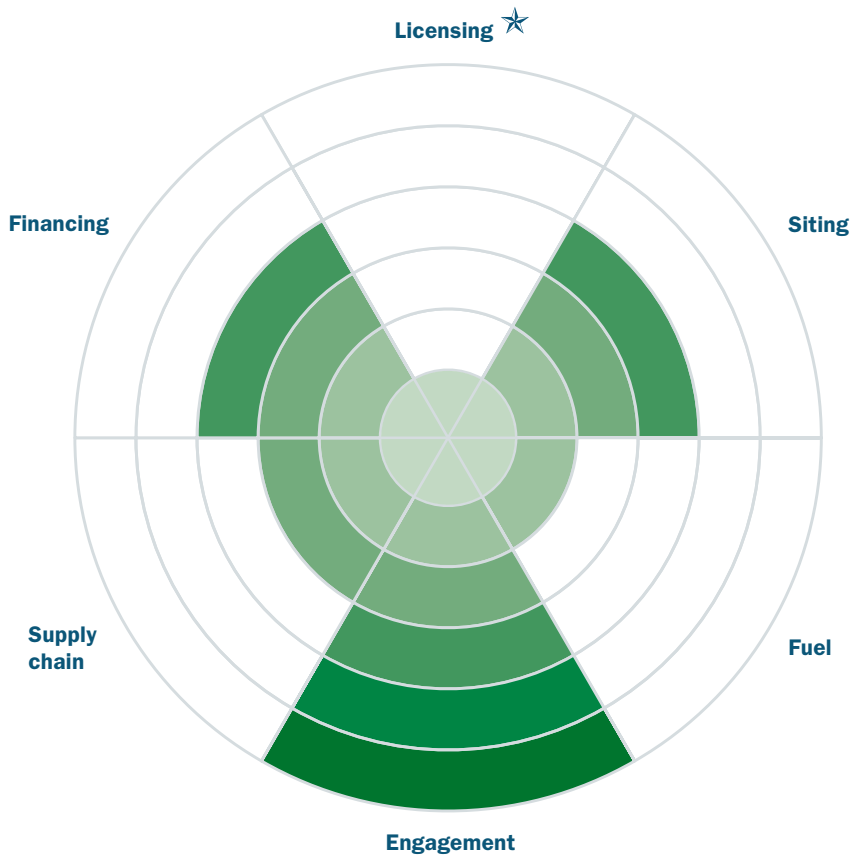
To raise public awareness about SMRs and their potential applications beyond electricity generation, Blue Capsule launched the *Blue Nuclear Pod* podcast series in November 2023. The company has also engaged with university and students in France, including at Sciences Po and the Institut National des Sciences et Techniques Nucléaires (INSTN).

Fuel



The Blue Capsule SMR is designed to use TRISO fuel enriched up to 4.95% with online refuelling. As part of their collaboration, Blue Capsule and CEA are conducting research and development on TRISO fuel for the Blue Capsule SMR.

Blykalla – SEALER-One



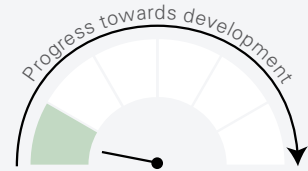
★ Active in multiple jurisdictions or countries.

△ Changes since Second Edition (2024).

Reactor description: Demonstration reactor for the SEALER-55 design, a lead-cooled fast reactor developed in Sweden.

Thermal power (MWth)	70
Core outlet temperature (°C)	550
Spectrum (thermal/fast)	Fast
Fuel form	Uranium-nitride ceramic
Base fissile material	U-235, enrichment level $\geq 5\%$ to $< 10\%$

Licensing ★



Siting △



Financing △



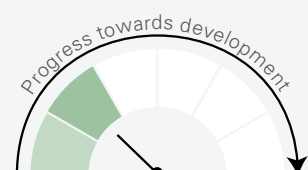
Supply chain



Engagement △

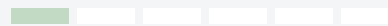


Fuel



Assessment of SEALER-one's progress to deployment

Licensing



In 2024, Janne Wallenius, Technical Director and Co-Founder of Blykalla (previously Leadcold) participated in a panel on nuclear development in Sweden during the National Radiation Safety Day event organised by the Swedish Radiation Safety Authority (SSM). The SSM reported meeting with Blykalla to discuss SMR pre-licensing engagement options in 2023. In support of the United Kingdom's Advanced Modular Reactor project, the UK Office for Nuclear Regulation (ONR) completed a three-stage engagement process with Blykalla to explore regulatory pathways for advanced nuclear technologies and build ONR capacities to license lead-cooled fast reactors. At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



Blykalla has formed a joint venture with Uniper to advance the SEALER reactor concept at the Oskarshamn Nuclear Power Plant, which is operated and owned by Oskarshamnskraftgrupp OKG in Sweden. Uniper owns the majority of OKG's shares with Fortum being the other shareholder. The development of the SEALER reactor is planned as an iterative process, starting with the construction of an electrically heated prototype (SEALER-E) at the Oskarshamn site, to be followed by the deployment of the nuclear demonstration reactor (SEALER-One) at the same site by 2030. Blykalla is aiming to deploy a commercial version of SEALER-55 by 2032. In February 2025, Blykalla announced it had started civil work for its SEALER-E electrical prototype at the Oskarshamn site.

Financing



In 2020-2024, Blykalla raised SEK 185 million (USD₂₀₂₃ 17.1 million) in seed and Series A funding led by Norrsken Launcher, SEK 99 million (USD₂₀₂₃ 9.9 million) from the Swedish Energy Agency (SEA), and grants from Euratom, the European Union and Vinnova. Blykalla is engaged in two national projects backed by over SEK 50 million (USD₂₀₂₃ 5.0 million) from SEA. The Swedish Strategic Research Foundation awarded SEK 50 million (USD₂₀₂₃ 5.0 million) over 2021-2026 to support lead-cooled reactor research at Oskarshamn in support of SEALER development. Blykalla secured a UK government contract for a SEALER-UK feasibility study as part of the Advanced Modular Reactor project. It also has an agreement with NewClearEnergy, allocating a share of electricity sales to support SEALER development.

Supply chain



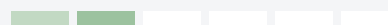
Blykalla is a spin-off from the Royal Institute of Technology (KTH) in Stockholm. The SUNRISE center, involving KTH, Uppsala University, Luleå University, and other partners, was established to advance lead-cooled SMR technology. SUNRISE aims to design a demonstration reactor for the SEALER-55 concept at Oskarshamn by 2030. In 2025, Blykalla signed contracts with Högåns for specialised materials and KSB for pump development. In 2024, a research project on SEALER SMR heat integration for bio-coal production was initiated with RISE, KTH, SSAB, and Envigas, along with a collaboration with Swerim on material research. Civil work for the SEALER-E prototype has started with the construction company NCC. Blykalla has signed a memorandum of understanding with ABB to explore automation and digital solutions.

Engagement



In 2025, Blykalla engaged with local communities in the Gävleborg region. Since 2023, Blykalla has actively engaged with government officials, receiving endorsements from Sweden Deputy Prime Minister Ebba Busch and participating in an expert group appointed by Minister Romina Pourmokhtari. Blykalla has presented to the Swedish parliament and advanced public engagement activities including interviews on SVT Nyheter and a YouTube video on Rejlers Play. In 2024, Blykalla's Head of Legal has been nominated for the "Power Women of the Year" award, and Blykalla was recognised as one of the top 100 impact companies by the Norrsken Foundation. Additionally, Blykalla is exploring partnerships with VARD and NGSI for potential deployments in Norway and Ukraine.

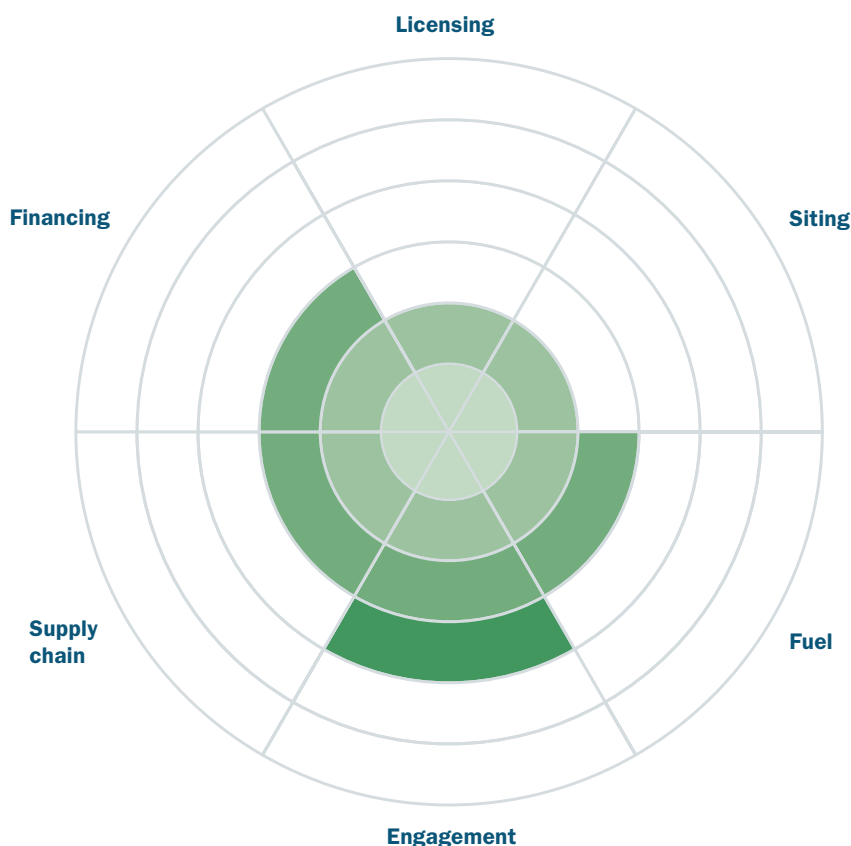
Fuel



Blykalla is engaged with KTH Royal Institute of Technology on studies related to the lab-scale production of fuel. This includes the fabrication of uranium nitride samples suitable for irradiation testing in a laboratory setting. Blykalla also participates in the EU funded FREDMANS Project with specific work packages on manufacturing, recyclability and management of nitride nuclear fuel.

BWX Technologies (BWXT)

– BWXT Advanced Nuclear Reactor (BANR)



△ Changes since Second Edition (2024).

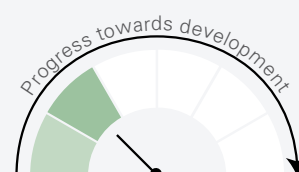
Reactor description: Transportable high-temperature gas-cooled microreactor using TRISO fuel.

Thermal power (MWth)	100 (plant deployed with minimum two modules of 50 MWth each)
Core outlet temperature (°C)	800
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-nitride ceramic and uranium-oxycarbide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level ≥10% to <20%

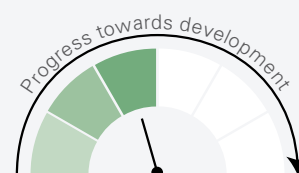
Licensing



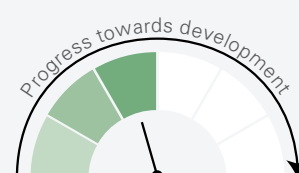
Siting



Financing



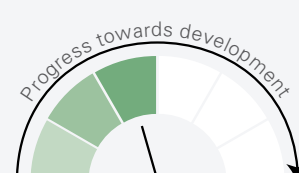
Supply chain



Engagement

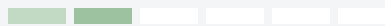


Fuel



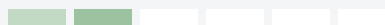
Assessment of BANR's progress to deployment

Licensing



The BWXT Advanced Nuclear Reactor (BANR) is one of two modular, transportable, high-temperature micro gas-cooled SMRs under development by BWXT, along with Project Pele at Idaho National Laboratory (INL) for the US Department of Defense. BANR can be configured with two to eight 50 MWth modules providing 100 to 400 MWth. In cogeneration mode, BANR can provide 15 to 60 MWe plus 55 to 220 MWth for process heat. Experience gained in Project Pele may help inform and accelerate work on BANR. BWXT submitted a pre-licensing regulatory engagement plan for BANR to the US Nuclear Regulatory Commission (NRC) in 2022. Additionally, BWXT submitted a Quality Assurance Topical Report in 2022 and a UN TRISO fuel qualification plan in 2023 to support fuel development activities for BANR.

Siting



In 2023, the Wyoming Energy Authority (WEA) contracted BWXT to evaluate the suitability of the BANR SMR for possible deployment in Wyoming, United States. Through a collaboration agreement in September 2023 and a Letter of Intent in December 2024, BWXT is working with Tata Chemicals Soda Ash Partners (TCSAP), a subsidiary of Tata Chemicals North America, to explore the feasibility and develop the commercial terms to deploy a series of BANR SMRs at the TCSAP's Green River site in Wyoming in the early 2030s. The BANR SMR would supply electricity and steam for trona ore mining and production of soda ash, used in the manufacture of glass and soap.

Financing



In 2020, it was announced that BWXT would receive a Risk Reduction award under the US Department of Energy (DOE) Advanced Reactor Demonstration Program (ARDP), with DOE contributing USD 85.3 million (USD₂₀₂₃ 100.4 million) and BWXT contributing USD 21.3 million (USD₂₀₂₃ 25.0 million). In 2023, BWXT received nearly USD 10 million from WEA through the Governor's Energy Matching Funds (GEMF) award for a USD 20.6 million project to evaluate the suitability and develop a business strategy for the deployment of the BANR SMR in the state of Wyoming. As of the end of 2024, BWXT had also received USD 227 000 (USD₂₀₂₃ 220 388) in tax incentives to establish an Innovation Campus in Lynchburg, Virginia, which includes BWXT's Advanced Technologies business unit developing BANR.

Supply chain



BWXT has contracted Burns & McDonnell to develop power cycle architecture and perform site integration design for the BANR SMR, including power distribution infrastructure and reactor building structures. In 2024, BWXT signed a Joint Development Agreement with L&H Industrial (L&H) to collaborate on the potential deployment of BANR SMRs as a cogeneration unit where energy would be leased to customers through L&H's Evercore Energy business unit. BWXT also has diverse nuclear supply chain capabilities, including engineering, procurement and construction capabilities. As part of the USD 20.6 million project supported by WEA, BWXT is developing supply chain capabilities for the BANR SMR in Wyoming, which includes fabrication of nuclear-quality components within the state.

Engagement



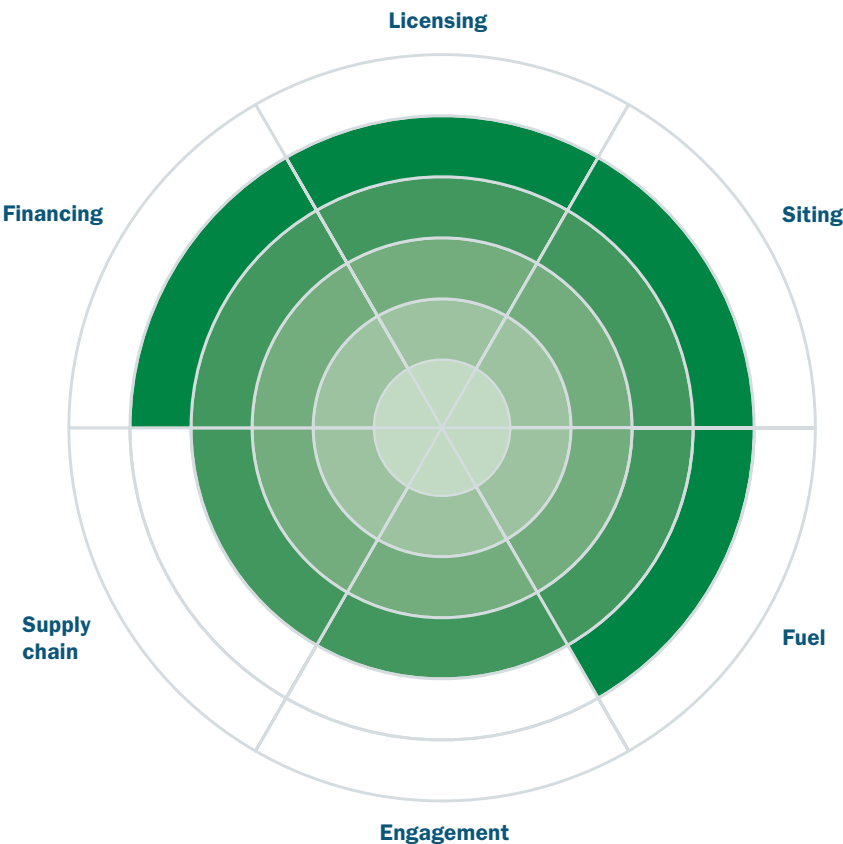
In 2023-2024, BWXT hosted stakeholder workshops in the state of Wyoming in the cities of Gillette, Cheyenne and Casper. In 2024, the Gillette City Council approved a co-operation agreement with BWXT, with endorsement from Gillette Mayor, Shay Lundvall. In 2024-2025, representatives from BWXT Advanced Technologies continued engagement about Wyoming's role in developing the BANR SMR, and the role of nuclear energy in space, defense, and industrial decarbonisation including through: the Alaska Sustainable Energy Conference in April 2024; the 2024 Wyoming's Energy Future Symposium hosted by the University of Wyoming's School of Energy Resources in September 2024; and a workshop hosted by the National Academies of Sciences, Engineering, and Medicine on nuclear energy in January 2025.

Fuel



BWXT intends to use HALEU for the BANR SMR. BWXT owns and operates the only private facilities in the United States licensed to down-blend high-enriched uranium into HALEU. In 2023, BWXT was awarded USD 47 million by the National Nuclear Security Administration to produce a limited quantity of HALEU. BWXT plans to use Uranium Oxycarbide (UCO) TRISO for the baseline core and Uranium Nitride (UN) TRISO for the upgraded core. BWXT plans to fabricate TRISO fuel at their facilities in Virginia and has also been working with the INL and Oak Ridge National Laboratory on the simulation, manufacturing and testing of their UN TRISO fuel. In July 2024, BWXT and WEA signed a co-operation agreement to evaluate potential sites for a TRISO fabrication facility in the state of Wyoming.

BWX Technologies (BWXT) – Project Pele



△ Changes since Second Edition (2024).

Reactor description: Micro, mobile gas-cooled demonstration reactor for government and possible later commercial applications.

Thermal power (MWth)	No information
Core outlet temperature (°C)	No information
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxycarbide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level ≥10% to <20%

Licensing



Siting



Financing



Supply chain



Engagement



Fuel



Assessment of Project Pele's progress to deployment

Licensing



Project Pele is a prototype commissioned by the US Department of Defense (DOD) Strategic Capabilities Office (SCO). Due to its classification, there is limited information available in public sources. It does not fall under the purview of the US Nuclear Regulatory Commission (NRC). The DOD and the US Department of Energy (DOE) have signed an Interagency Agreement for Project Pele to be tested and operated under DOE authorisation, with some technical support and advice from the NRC. The US Army Corps of Engineers is also engaged as the technical lead on the Environmental Impact Statement (EIS), which has been released by the DOD SCO. In September 2024, the DOD began site preparation for Project Pele at the Idaho National Laboratory's (INL) Critical Infrastructure Test Range Complex.

Siting



In 2022, the DOD SCO released the final EIS validating its plans for the Project Pele SMR to be constructed off-site and delivered to INL's Critical Infrastructure Test Range Complex, where the fuel will be loaded, and DOE will test and operate it. In September 2024, the DOD began site preparation for Project Pele at the INL's site and announced that the fully assembled reactor module will be transported to the complex in 2026 for safety reviews and initial planning and testing.

Financing



The DOD has announced that the Project Pele prototype reactor will be completed under a cost-type contract valued at USD 300 million (USD₂₀₂₃ 312.3 million). Additionally, BWXT has received USD 42.5 million (USD₂₀₂₃ 47.8 million) from the DOD to develop the reactor design and fuel, as well as USD 37 million (USD₂₀₂₃ 38.5 million) from INL to manufacture the Project Pele core.

Supply chain



DOD has announced the SMR will be manufactured by BWXT then transported to INL for fuelling and testing. BWXT operates a diverse supply chain specialised in the nuclear sector – including engineering, procurement and construction. BWXT has been selected by the DOD to be the prime contractor and integration lead for Project Pele. The project team also includes Northrop Grumman, Rolls-Royce LibertyWorks, and Torch Technologies. In 2023, BWXT announced that it and its sub-suppliers had begun fabricating parts for Project Pele, including forging the containment vessel and making moderator blocks.

Engagement



Dr Jeff Waksman, Project Pele Program Manager, was invited to participate in the Energy Communities Alliance (ECA) 2022 forum. The ECA is a non-profit organisation that aims to facilitate dialogue between the US DOE and local government representatives. In 2022, he also appeared on Boise State Public Radio News to discuss public engagement about the project. The DOD also undertook public engagement through a public comment period during 2021-2022. In 2023 and 2024, Dr Waksman provided an update on Project Pele to the Frontiers Project Meeting in Anchorage, Alaska, hosted by the Atlantic Council in partnership with INL and the University of Alaska, and to the 2024 Alaska Sustainable Energy Conference, hosted by the Office of Governor Mike Dunleavy.

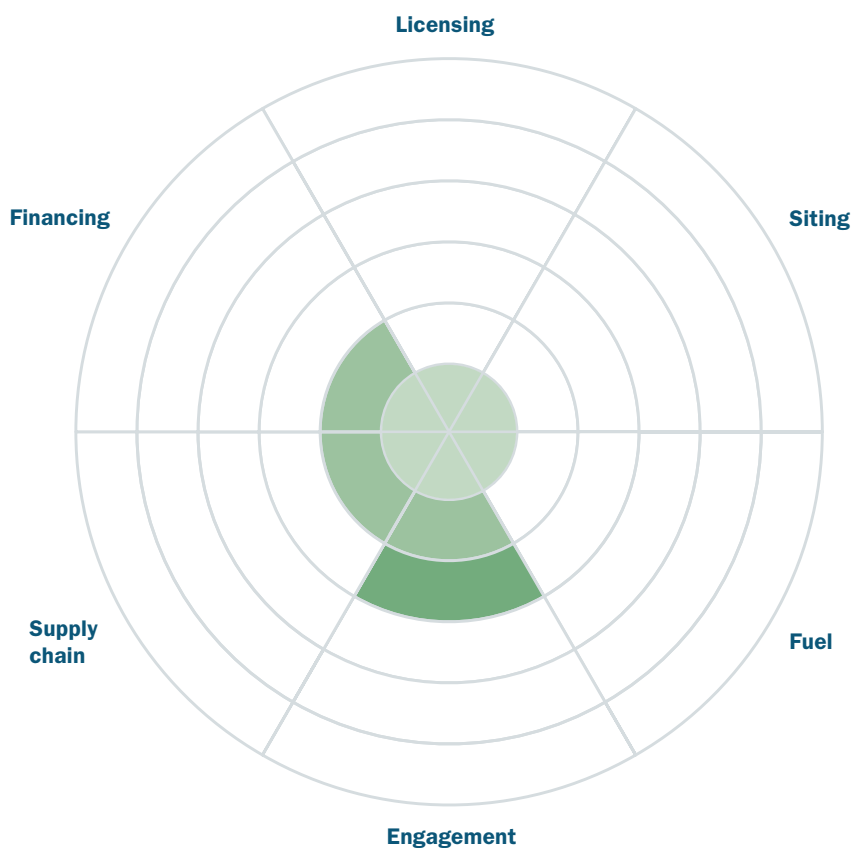
Fuel



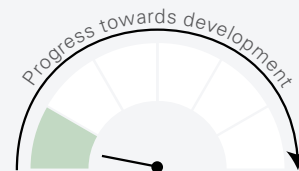
INL has contracted BWXT to produce the necessary TRISO fuel for the Project Pele first-of-a-kind prototype demonstration unit. As of 2025, BWXT owns and operates the only privately-owned facilities in the United States licensed to possess and process HEU for down-blending into HALEU. The BWXT facilities are located in Lynchburg, Virginia. Under the contract with INL, BWXT will down-blend HEU stockpiles from National Nuclear Security Administration to produce HALEU for Project Pele. In July 2024, BWXT and Wyoming Energy Authority signed a cooperation agreement to evaluate potential sites for a TRISO fabrication facility in the state of Wyoming.

Canadian Space Mining Corporation (CSMC)

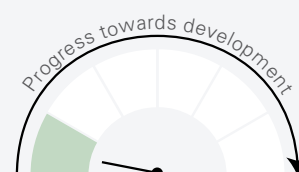
– Low Enriched Uranium Nuclear Reactor (LEUNR)



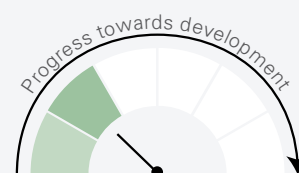
Licensing



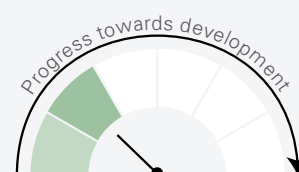
Siting



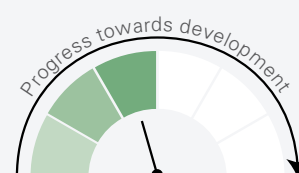
Financing



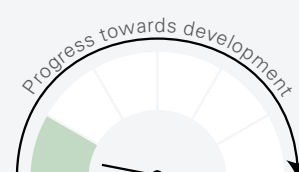
Supply chain



Engagement



Fuel



Reactor description: Pressurised light water microreactor for earth and space applications.

Thermal power (MWth) 0.1

Core outlet temperature (°C) 130

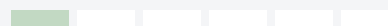
Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic CERMET

Base fissile material U-235, enrichment level ≥ 10% to < 20%

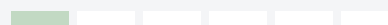
Assessment of LEUNR's progress to deployment

Licensing



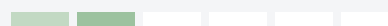
Canadian Space Mining Corporation (CSMC) signed a licencing term sheet with Canadian Nuclear Laboratories (CNL) in November 2024 to commercially develop the SLOWPOKE-2 research reactor as the technical basis for the LEUNR SMR. At the time of assessment, there was no additional information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



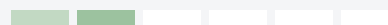
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



In 2022, CSMC received nearly CAD 860 000 (USD₂₀₂₃ 662 000) from the Canadian Space Agency to conduct technology concept studies and prototyping of their fission energy system in support of the Lunar Surface Exploration Initiative. In 2023, CSMC received CAD 1.0 million (USD₂₀₂₃ 741 000) funding from the Canadian Space Agency through the Space Research and Development programme to advance its solution for fission power for both space and terrestrial purposes.

Supply chain



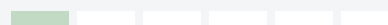
In November 2024, CSMC announced that they had signed a Memoranda of Understanding and licensing term sheet with CNL to develop the SLOWPOKE-2 research reactor in the commercial LEUNR SMR, planning demonstration in the near future. The LEUNR SMR is expected to benefit from operating experience from the three SLOWPOKE-2 reactors currently operating at the Ecole Polytechnique in Montreal and the Royal Military College of Canada in Kingston, Canada, as well as the International Centre for Environmental and Nuclear Sciences, in Jamaica.

Engagement



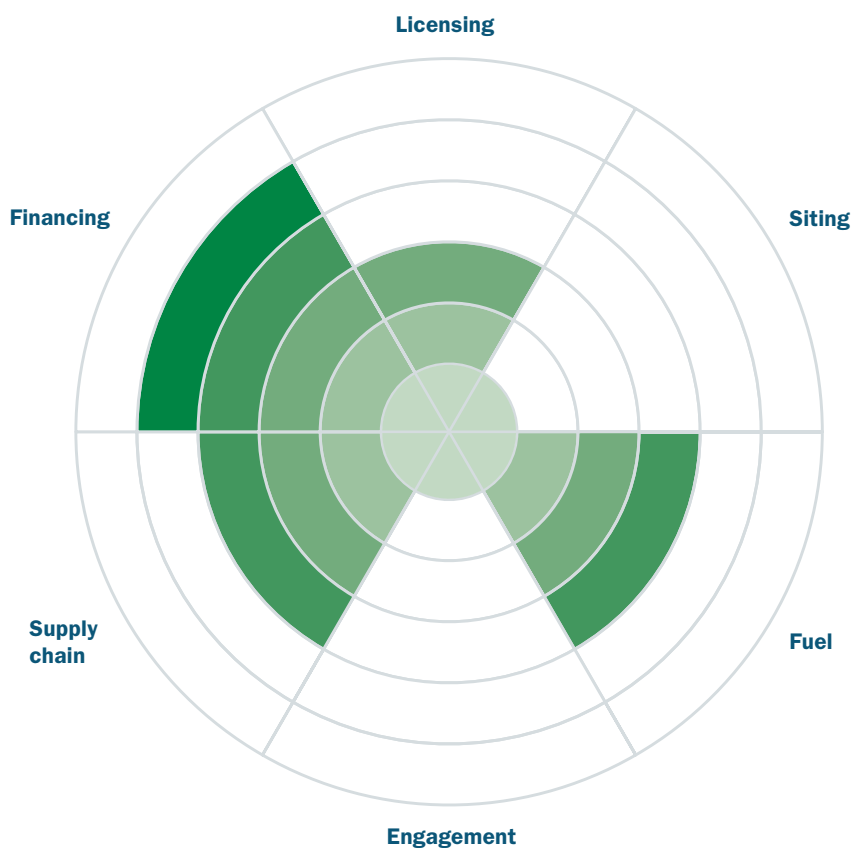
CSMC has engaged potential end-users for the LEUNR SMR outside the nuclear sector, including: at the annual Prospectors & Developers Association of Canada Convention in Toronto, to discuss applications of SMRS for mining; and at the Spacebound 2024, ministerial-level conference, in Ottawa, Canada, to discuss applications of SMRs for outer space. In 2024, an interview with Daniel Sax, CMSC's CEO, was published in BetaKit, an online Canadian start-up and tech innovation news source.

Fuel



CMSC plans to use HALEU for its LEUNR SMR. At the time of the assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

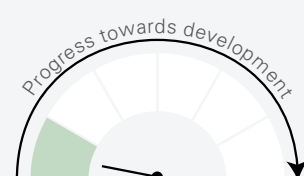
China General Nuclear Power Group (CGN) – ACPR50S



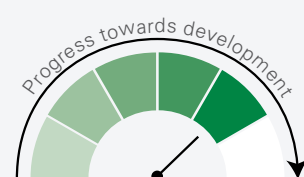
Licensing



Siting



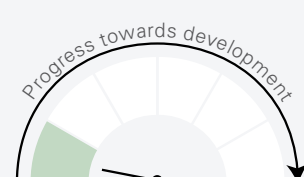
Financing



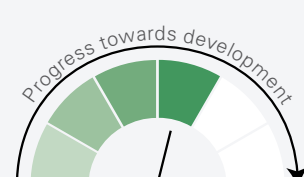
Supply chain



Engagement



Fuel



Reactor description: Floating pressurised light water reactor.

Thermal power (MWth) 200

Core outlet temperature (°C) 320

Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic

Base fissile material U-235, enrichment level <5%

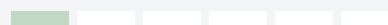
Assessment of ACPR50S's progress to deployment

Licensing



In 2017, China General Nuclear Power Group (CGN) submitted the first version of the preliminary safety analysis report for the ACPR50S to China's National Nuclear Safety Administration (NNSA). In 2022, the NNSA approved the assembly and manufacturing activities of the pressure vessel and reactor module for the ACPR50S. CGN is also engaged with the China Classification Society and Lloyd's Register to define safety requirements for floating nuclear power plants.

Siting



At the time of assessment, no information related to siting was readily available from any site owners.

Financing



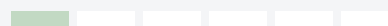
The ACPR50S demonstration project is wholly owned by CGN, a state-owned enterprise under direct management of the Chinese central government. China's National Development and Reform Commission approved the development of the ACPR50S reactor design in January 2016 as part of the 13th Five-Year Plan for innovative energy technologies.

Supply chain



The supply chain for the ACPR50S demonstration project involves: CGN as owner, main designer, and operator; China Nuclear Power Technology Research Institute as technical support organisation for design requirements; Dongfang Electric as pressure vessel manufacturer and reactor module assembler; China Shipbuilding Industrial Corporation as shipbuilder; China National Offshore Oil Corporation to advance design and address maritime safety aspects; and Lloyd's Register to investigate handling severe internal and external events. In 2016, a purchase agreement for the pressure vessel was signed between CGN and Dongfang Electric. In 2023, CGN completed the experimental research for the design of a part of the ACPR50S reactor safety injection system in collaboration with Chongqing University.

Engagement



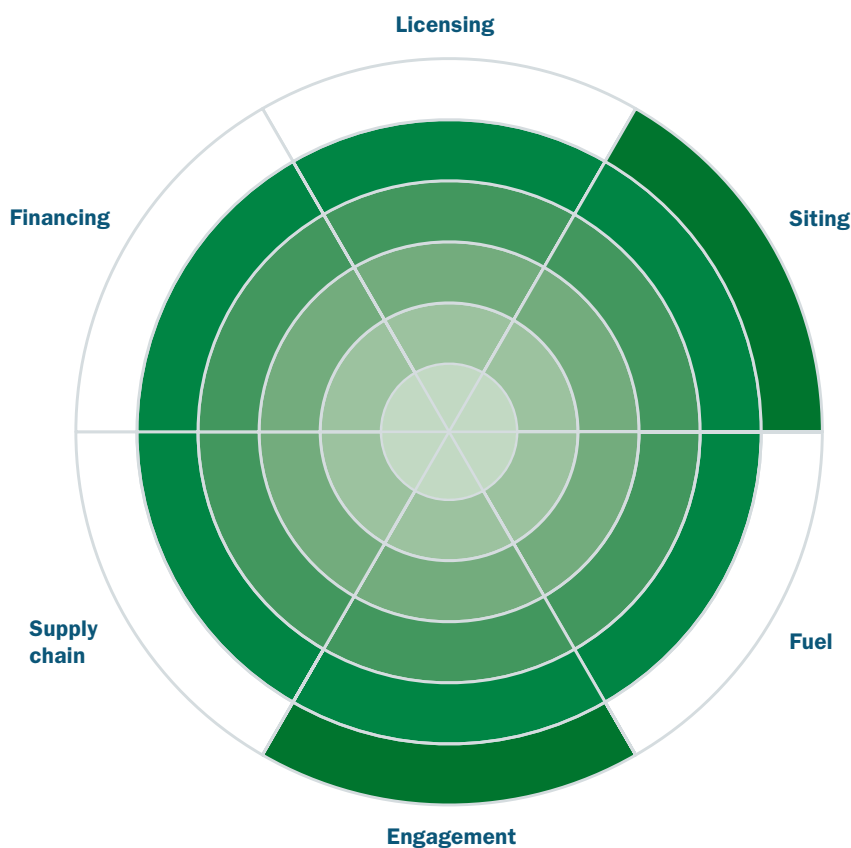
At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

Fuel

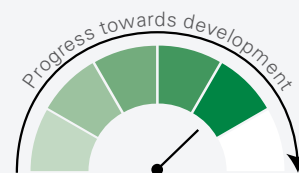


The ACPR50S is designed to use low-enriched fuel, either in the form of UO_2 pellets, or UO_2 pellets containing gadolinium oxide ($\text{UO}_2\text{-Gd}_2\text{O}_3$). The low-enriched UO_2 pellets are the current industry standard for water-cooled power reactor technologies and therefore rely on well-established fuel supply chains. CNNC Jianzhong Nuclear Fuel, a subsidiary of CNNC, operates a powder metallurgical fabrication line for Gd_2O_3 and UO_2 sintered pellets in Yibin, Sichuan.

Argentina's National Atomic Energy Commission (CNEA) – CAREM



Licensing



Siting



Financing



Supply chain



Engagement



Fuel



Reactor description: Integral pressurised water reactor.

Thermal power (MWth) 100

Core outlet temperature (°C) 326

Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic

Base fissile material U-235, enrichment level <5%

Assessment of CAREM's progress to deployment

Licensing



The CAREM prototype reactor licence for construction was approved in 2010. In 2013, the CAREM prototype reactor was authorised to start construction, which began in 2014.

Siting



The CAREM reactor is under construction on the nationally owned site of Atucha, where Atucha I and II are already located. The Argentina's Nuclear Regulatory Authority (ARN) granted the authorisation for site use and construction to the Argentina's National Atomic Energy Commission (CNEA) in September 2013. The first concrete was poured in February 2014.

Financing



The Argentinian government has written into national law that the government will fully finance the construction of the CAREM first-of-a-kind reactor. The national law does not specify the timeline to finance and to complete the first-of-a-kind reactor.

Supply chain



The supply chain for the CAREM project involves the following Argentine companies: CNEA as the responsible authority ensuring overall co-ordination along the supply chain; Nucleoelectrica Argentina as the provider of construction services and technical assistance; and Industrias Metalúrgicas Pescarmona Sociedad Anónima as the manufacturer of auxiliary components. Under current planning, up to 70% of the components are to be manufactured in Argentina. In March 2024, CNEA signed a memorandum of understanding (MoU) with INVAP to explore commercial opportunities for the CAREM reactor. In September 2024, Buenos Aires Herald, an Argentine news outlet, reported that CNEA had decided to put the construction of the CAREM reactor on hold and focus on the final stages of engineering.

Engagement



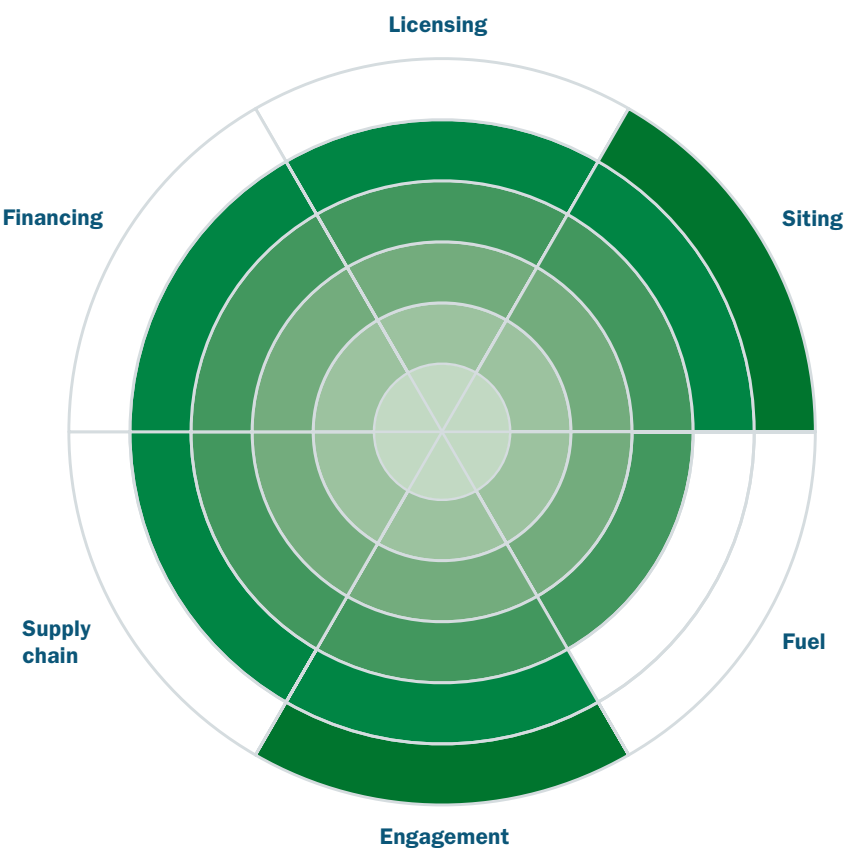
Argentina has co-operation agreements with Bolivia for activities in the field of peaceful uses of nuclear energy, including but not limited to engagement about CAREM. Argentina also has a MoU with Indonesia for co-operation on nuclear energy, including CAREM. CNEA regularly hosts national and subnational government officials, including the President of Argentina and the governor of Río Negro Province, to discuss the CAREM project. In 2022, the Deliberative Council of Zárate, the city where CAREM is being constructed, met in the facilities of the Atucha Nuclear Complex and unanimously agreed to support of the CAREM project.

Fuel



Combustibles Nucleares Argentinos was contracted to manufacture fuel for CAREM in 2013. This contract for fuel supply is ongoing.

China National Nuclear Corporation (CNNC) – ACP100

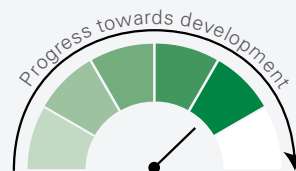


△ Changes since Second Edition (2024).

Reactor description: Integrated pressurised light water reactor, also known as the Linglong One.

Thermal power (MWth)	385
Core outlet temperature (°C)	320
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level <5%

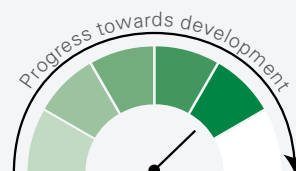
Licensing



Siting



Financing



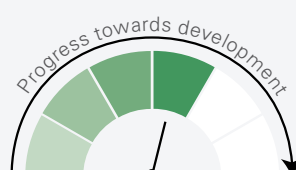
Supply chain



Engagement



Fuel



Assessment of ACP100's progress to deployment

Licensing



China's National Nuclear Safety Administration (NNSA) approved the ACP100 preliminary safety analysis report in 2020. In June 2021, the NNSA approved the Quality Assurance Outline (Construction Phase) of the ACP100 demonstration reactor project, followed by the issuance of the construction licence in July 2021. In 2021, China's Ministry of Ecology and Environment approved the Environmental Impact Assessment (Construction Phase) of the demonstration project. In 2025, the China National Nuclear Corporation (CNNC) submitted the Environmental Impact Assessment (Operation Phase) to the Ministry of Ecology and Environment for approval.

Siting



In 2017, the Hainan provincial government agreed to site one ACP100 unit on the existing Changjiang Nuclear Power Plant site. Construction started in 2021. The unit is planned to be commercially operational in 2026.

Financing



The People's Republic of China's National Development and Reform Commission approved the construction of a ACP100 demonstration reactor in June 2021. This ACP100 demonstration reactor is wholly owned by CNNC's China National Nuclear Power, a state-owned enterprise under direct management by the Chinese central government.

Supply chain



The supply chain for the ACP100 demonstration reactor includes China National Nuclear Power (a subsidiary of CNNC) as owner-operator, Nuclear Power Institute of China (NPIC) as reactor designer, China Nuclear Power Engineering Co. as plant construction company, and China First Heavy Industry as supplier of the reactor vessel. Other suppliers for reactor parts and materials include Harbin Electric Corporation, Dongfang Electric Corporation, Sinomach Heavy Equipment Group, Advanced Technology & Materials, Jiangsu Shentong Valve, and Nanjing Iron & Steel, among others. In 2023-2024, CNNC announced progress in the construction of the ACP100 demonstration project, including the installation of the reactor core, steel containment dome, digital control system, and reactor core internals.

Engagement



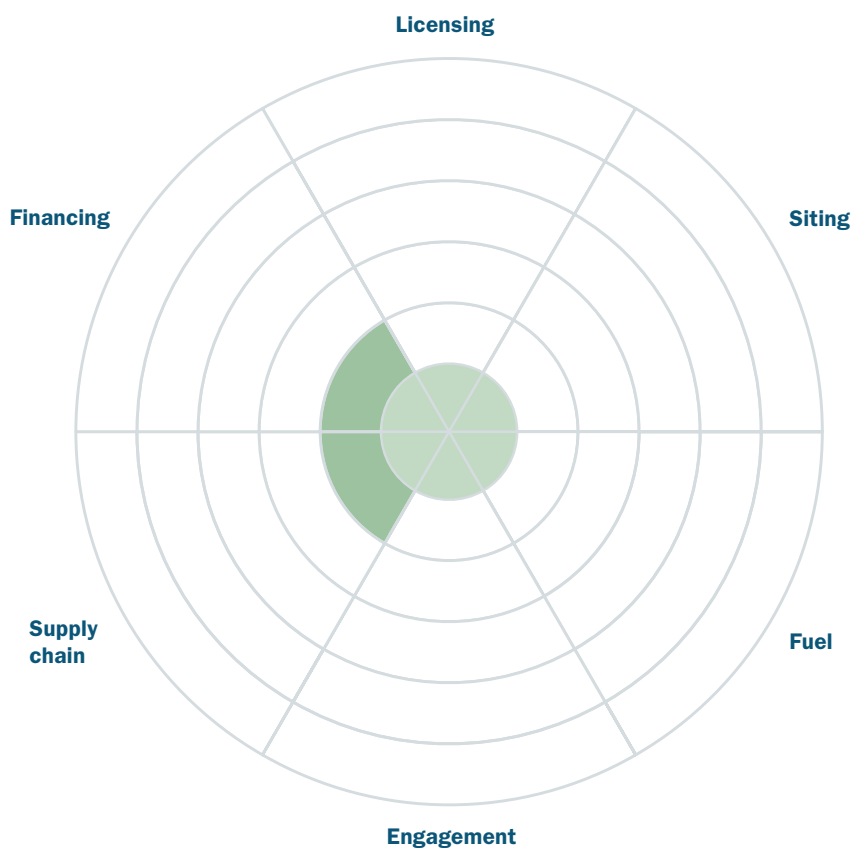
National and local Chinese governments have been endorsing CNNC's ACP100 demonstration project since 2014, through policy statements, media interviews and site visits. In 2024, the China Atomic Energy Authority, China's Ministry of Foreign Affairs, and China's Permanent Mission in Vienna organised a tour of the ACP100 construction site, which was attended by ambassadors and diplomats from eleven countries. CNNC has also advanced public outreach through media interviews to enhance public understanding of the ACP100 demonstration project. In 2024, the Electricity Generating Authority of Thailand, the country's largest state-owned utility, expressed interest in the potential deployment of the ACP100 in Thailand.

Fuel

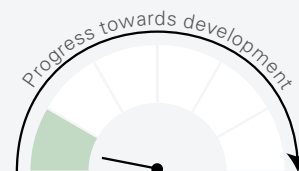


The ACP100 design is based on existing Generation II/III light water reactor technology and therefore relies on well-established fuel supply chains. In 2024, Xi'an Nuclear Equipment announced that it has manufactured a prototype of the fuel transport container for ACP100 fuel and passed a joint acceptance inspection by CNNC Jianzhong Nuclear Fuel and China Nuclear Power Engineering Co.

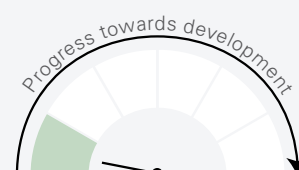
Research Centre Řež (CVŘ) – Energy Well



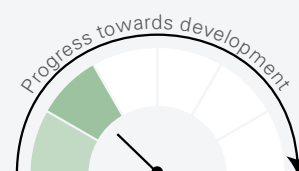
Licensing



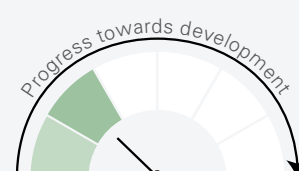
Siting



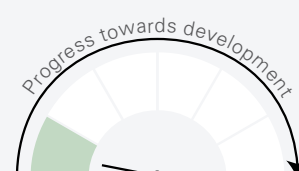
Financing



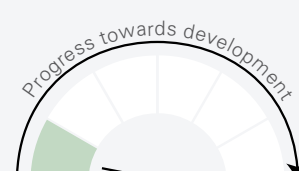
Supply chain



Engagement



Fuel



Reactor description: Micro molten salt-cooled reactor fuelled by HALEU TRISO fuel in a solid prismatic fuel assembly.

Thermal power (MWth) 20

Core outlet temperature (°C) 700

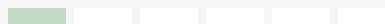
Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic TRISO prismatic

Base fissile material U-235, enrichment level ≥ 10% to < 20%

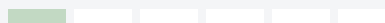
Assessment of Energy Well's progress to deployment

Licensing



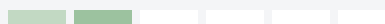
At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



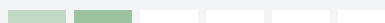
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



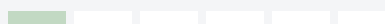
In its 2020 Annual Report, Research Centre Řež (CVŘ) highlighted that the Energy Well SMR received funding from the Technology Agency of the Czech Republic to support work towards the pre-conceptual design.

Supply chain



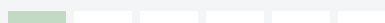
CVŘ has reported that ŠKODA JS (since acquired by ČEZ) is a key partner in the development of the transport container. CVŘ has stated its intention to work with the supply chain within Czechia and foresees potentially working with MICO and ATEKO for the main reactor components, with ZAT and DEL for instrumentation and control, with the Division of ENERGOPROJEKT within ÚJV Řež for primary engineering, and with the Sigma Group for pumps and valves.

Engagement



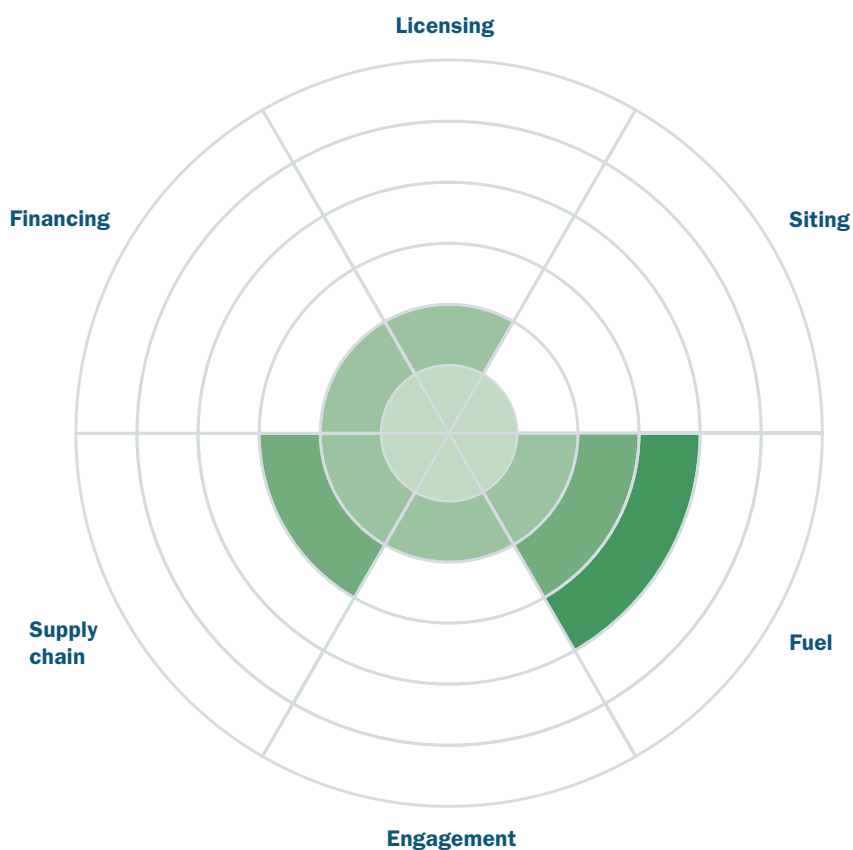
At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

Fuel

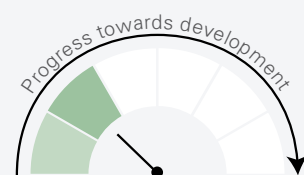


Energy Well plans to use TRISO fuel in a solid prismatic fuel assembly. At the time of assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

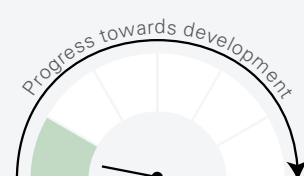
Deep Fission – Deep Fission Borehole Reactor (DFBR-1)



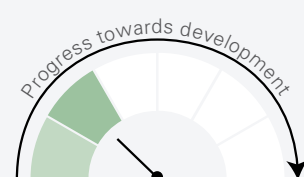
Licensing



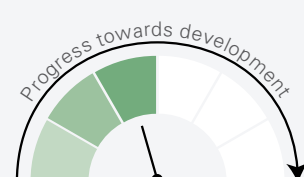
Siting



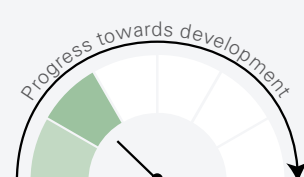
Financing



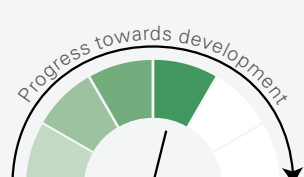
Supply chain



Engagement



Fuel



Reactor description: Underground, borehole-deployed, pressurised water SMR.

Thermal power (MWth)	50
Core outlet temperature (°C)	315
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level < 5%

Assessment of DFBR-1's progress to deployment

Licensing



Deep Fission is engaged with the US Nuclear Regulatory Commission (NRC) in pre-application activities for a standard design approval application for the Deep Borehole Pressurized Water Reactor (DB-PWR). In March 2024, Deep Fission submitted a regulatory engagement plan to the NRC.

Siting



At the time of the assessment, no information related to siting was readily available from any site owners.

Financing



8VC, an investment firm specialising in technology innovation, led a pre-seed funding round for Deep Fission that raised USD 4.0 million (USD₂₀₂₃ 3.9 million) in 2024.

Supply chain



Deep Fission has been awarded a GAIN Voucher from the US Department of Energy to support advanced thermal hydraulic modeling and validation. In January 2025, Deep Fission entered a binding partnership with Endeavour Energy to co-develop up to 2 gigawatts electric of nuclear power using Deep Fission's Borehole Reactor technology, aiming to supply electricity to Endeavour Energy's data centres.

Engagement



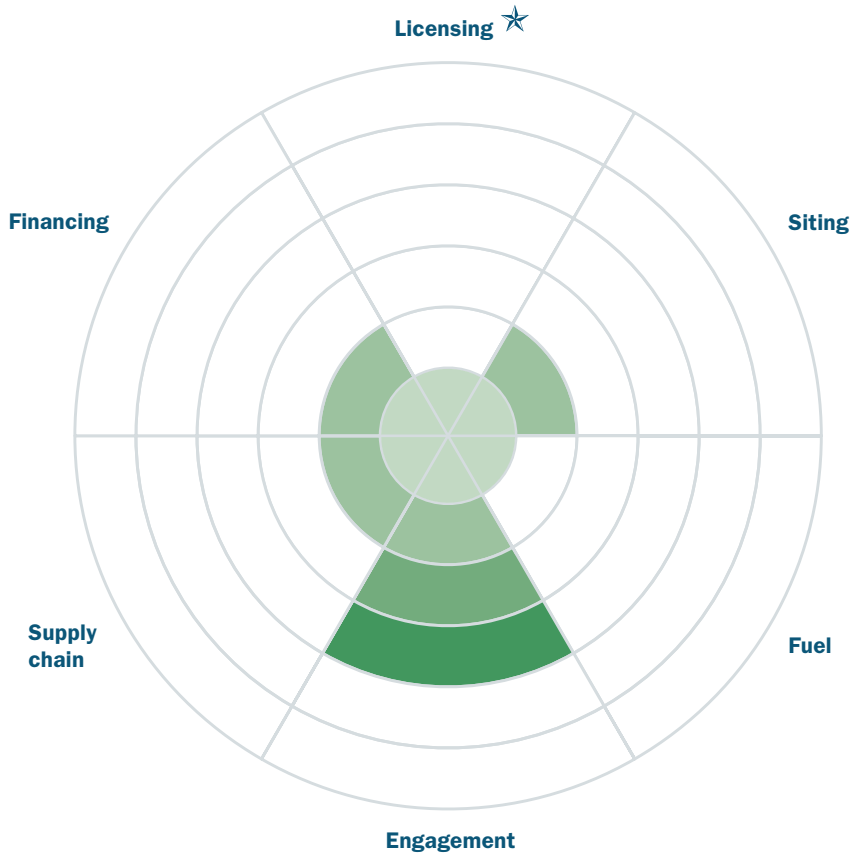
Deep Fission has begun some engagement and outreach activities including an interview with Fox Business on Fox News, a video interview with "MOXY: The Voice of Women in Infrastructure" online, as well as a podcast with Fire2Fission.

Fuel



The Deep Fission Borehole Reactor plans to utilise UO₂ pellets for fuel, which is currently the industry standard for water-cooled reactor technologies and therefore relies on well-established supply chains.

Dual Fluid Energy – Dual Fluid 300 (DF300)

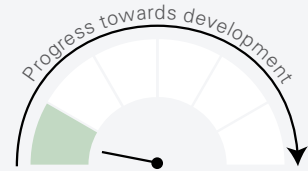


- ★ Active in multiple jurisdictions or countries.
- Δ Changes since Second Edition (2024).

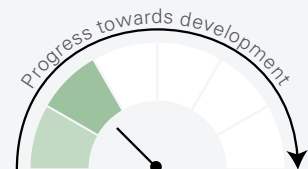
Reactor description: Liquid lead-cooled SMR using liquid metallic fuel with a high actinide density.

Thermal power (MWth)	600
Core outlet temperature (°C)	1 000
Spectrum (thermal/fast)	Fast
Fuel form	Liquid metallic U-Cr alloy
Base fissile material	U-235, enrichment level ≥ 10% to < 20%

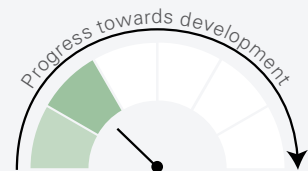
Licensing ★



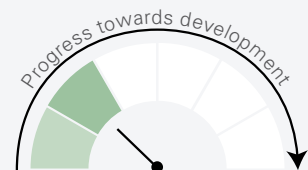
Siting



Financing



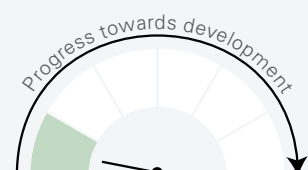
Supply chain



Engagement Δ



Fuel



Assessment of DF300's progress to deployment

Licensing

The development of the Dual Fluid 300 (DF300) reactor proposes a three-step approach. It begins with the construction of the Critical Demonstration Experiment (CDE) in Rwanda, a 0.1 MWth non-electrical reactor to demonstrate the Dual Fluid concept, followed by a full-scale prototype DF300 reactor, before moving to serial production. The formal review process of the Initial Safety Assessment Report for the CDE project started in February 2025. Dual Fluid Energy (Dual Fluid) also joined Canada's SMR Action Plan in 2021 with a commitment to engage the Canadian Nuclear Safety Commission through its pre-licensing process. At the time of assessment, there was no additional information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting

In 2023, Dual Fluid signed a partnership agreement with the Rwanda Atomic Energy Board (RAEB) to build a 0.1 MWth non-electrical reactor, called the CDE, to demonstrate the Dual Fluid concept with a criticality experiment at a site located on a new nuclear research campus in Nyamata, south of Rwanda's capital, Kigali. The CDE is expected to be operational by 2026, with subsequent technology testing to be completed by 2028. Following a successful criticality experiment with the CDE, RAEB may support the deployment of a prototype DF300 reactor on its site. In 2024, a preliminary site survey report for the CDE project, commissioned by the RAEB and conducted by expert companies, concluded that the site is suitable for nuclear installations.

Financing

In June 2021, Dual Fluid announced that it had raised almost CAD 7 million (USD₂₀₂₃ 5.7 million) in its first round of private financing for collaborations with research institutions on safety analysis in accordance with international regulatory standards, company build-up and laboratory setup. In October 2023, Dual Fluid announced that the company had started a financing round to raise an additional EUR 70 million (USD₂₀₂₃ 75.6 million) for the test reactor project in Rwanda.

Supply chain

The Dual Fluid Reactor concept was developed by the Institute for Solid State Nuclear Physics in Germany. In 2021, Dual Fluid established its headquarter in Vancouver, Canada. In 2023, Dual Fluid signed a Memorandum of Understanding with the TRIUMF particle accelerator centre in British Columbia, Canada, for materials research. Dual Fluid is engaged with research institutions such as the Technical University of Munich, the Technical University of Dresden, the Polish National Centre for Nuclear Research and the Swiss Paul Scherrer Institute on safety aspects. In 2023, the Government of Rwanda agreed to provide the necessary infrastructure for the CDE project in Nyamata, Rwanda. Dual Fluid has agreed to provide practical training with Rwandan scientists in the field of nuclear technology.

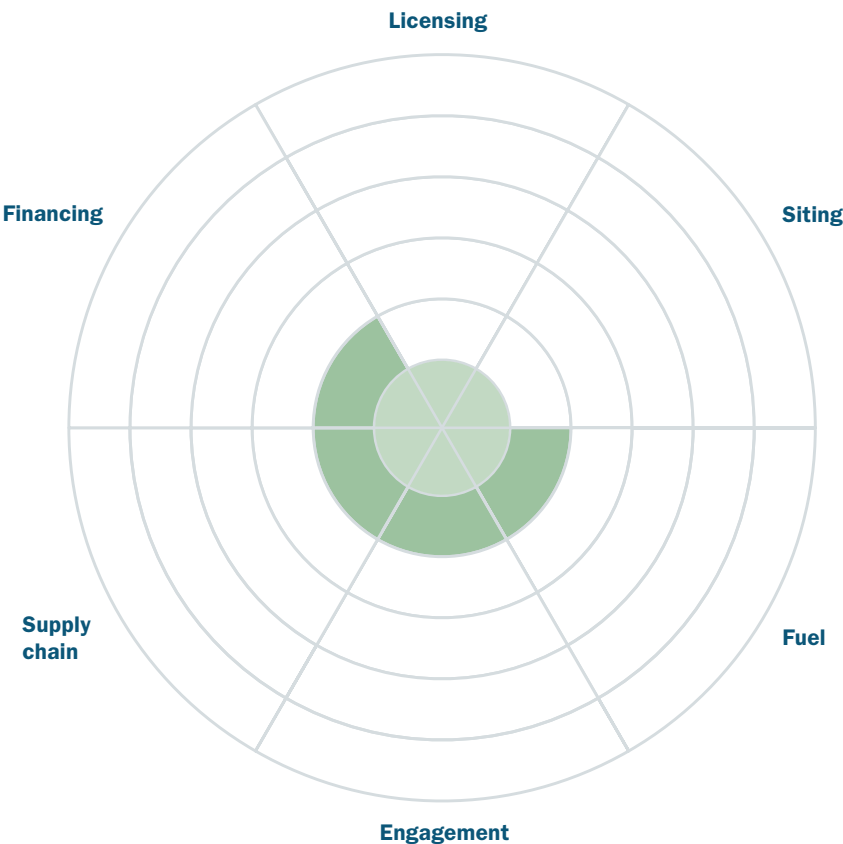
Engagement

In Rwanda, the CDE project has been endorsed by Minister of Infrastructure Jimmy Gasore and RAEB's CEO Fidele Ndahayo. In 2024, Dual Fluid met with Lassina Zerbo, the Chairperson of the RAEB and an energy advisor to Rwandan President Paul Kagame, to discuss the CDE project. Dual Fluid has also received multiple nominations for its DF300 design, including recognition as one of the top three technologies in Germany's GreenTec Awards, a national environmental competition, and as one of the Deep Tech Pioneers for the 2023 Global Summit, organised by Hello Tomorrow, a startup accelerator. In 2024 and 2025, Dual Fluid engaged with the general public through multiple interviews on German television and a media article to raise awareness of small modular reactor technology, including the DF300.

Fuel

The Dual Fluid design principle allows for a high density of fissile material in the liquid fuel. For the DF300, a mixture of 80% actinide and 20% chromium is intended for an initial fuel inventory utilising either enriched uranium or reactor-grade plutonium. At the time of assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

Eskom – A-HTR-100

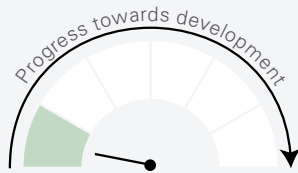


△ Changes since Second Edition (2024).

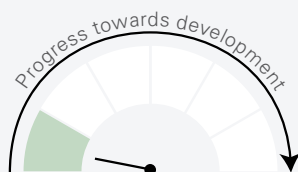
Reactor description: Very high-temperature pebble-bed gas reactor using intermediary heat storage for plant flexibility.

Thermal power (MWth)	100
Core outlet temperature (°C)	1 200
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic TRISO pebble
Base fissile material	U-235, enrichment level ≥ 10% to < 20%

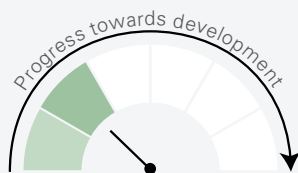
Licensing



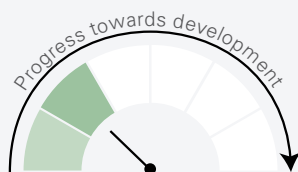
Siting



Financing



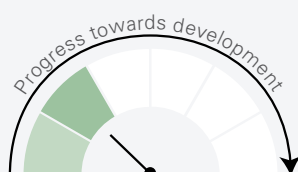
Supply chain



Engagement

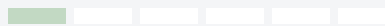


Fuel



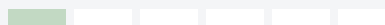
Assessment of A-HTR-100's progress to deployment

Licensing



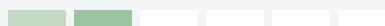
Eskom is currently exploring the feasibility of reactivating the A-HTR-100 SMR, which is based on the South African Pebble Bed Modular Reactor (PBMR) project, which was led by a consortium with Eskom during the 1990s and 2000s. The PBMR project has been in a state of “care and maintenance” since 2010, when the government decided to cease investments. In November 2024, the South African Department for Mineral Resources and Energy announced its intention to recommend that the Cabinet approve lifting the care and maintenance status on the PBMR programme. At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



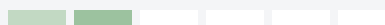
In 2008, the Department of Minerals and Energy (DMRE) published its Nuclear Energy Policy requiring nuclear power plants in South Africa to be owned and operated by Eskom or a consortium where Eskom has controlling share. In its 2019-2024 Research, Testing and Development Research Direction Report, Eskom anticipated a demonstration of the A-HTR-100 at the Pelindaba Nuclear Facility. In 2023, DMRE released its draft updated Integrated Resource Plan (IRP), which is awaiting cabinet approval. The IRP proposes 2500 MW of new nuclear energy capacity by 2035, including demonstration of a 100 MW SMR at the Komati power station, a decommissioned coal site in Mpumalanga, South Africa. At the time of assessment, no additional information related to siting was readily available from any site owners.

Financing



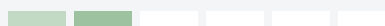
Research activities within the Eskom Research, Testing and Development (RT&D) business unit are publicly funded, as Eskom is wholly owned by the Government of South Africa. In 2017, Eskom stated that the ongoing research to develop the A-HTR-100 SMR was being funded through Eskom's internal research and development budget. The DMRE's 2023 IRP is currently pending cabinet approval which, if approved, would enable funding from the National Treasury in South Africa for a 2500 MW nuclear new-build programme that includes a 100 MW SMR demonstration.

Supply chain



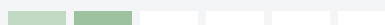
Eskom is exploring the feasibility of deploying the A-HTR-100 as a research initiative through its RT&D group and in collaboration with academia and research institutions, primarily through the South African Nuclear Engineering, Science and Technology collaborative forum. Eskom has conducted joint research on heat exchangers and materials for the A-HTR-100 design with the North-West University, and has also advanced research with Nelson Mandela Metropolitan University, and the University of Cape Town. In 2016, Eskom RT&D signed a Strategic Partnership Agreement with the Council for Scientific and Industrial Research to enhance skills and knowledge related to the A-HTR-100 in both organisations.

Engagement



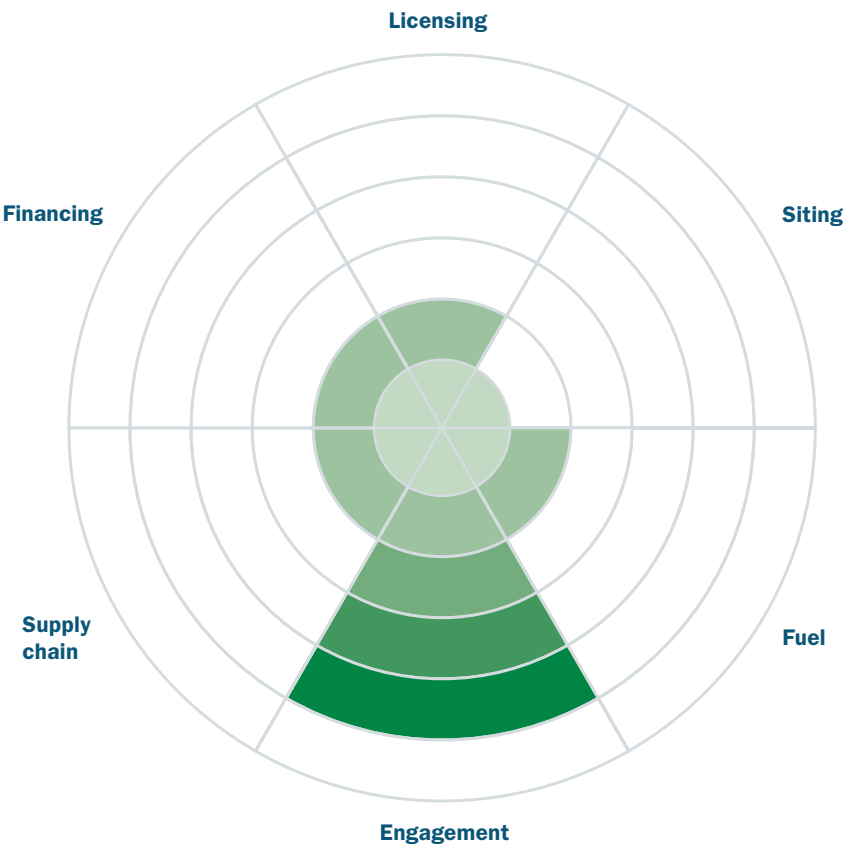
In November 2024, South African's Minister for Electricity and Energy Kgosientsho Ramokgopa highlighted the role of SMRs, including Eskom's A-HTR-100, as central to South Africa's energy future. During the 2023 United Nations Climate Change Conference (COP28) Eskom and ROSATOM, a state-owned nuclear energy corporation in Russia, signed an agreement to collaborate on joint education programmes including joint degree programmes among ROSATOM and Eskom partner universities, and business education.

Fuel



Eskom plans to use HALEU fuel for its A-HTR-100 SMR. The South African Nuclear Energy Corporation (NECSA) has a TRISO pebble fuel manufacturing laboratory near Pretoria, South Africa, which has been in care and maintenance since 2010. In 2017, NECSA announced that it is working to restart this facility to serve new potential customers for TRISO. As part of the Nuclear Grand Challenge, Eskom RT&D is advancing nuclear fuel components for the A-HTR-100. Eskom will leverage previous validation experiments on TRISO manufactured by NECSA for the PBMR programme, which includes irradiation performance testing at the Idaho National Laboratory in the United States.

Flibe Energy – Lithium Fluoride Thorium Reactor (LFTR)



△ Changes since Second Edition (2024).

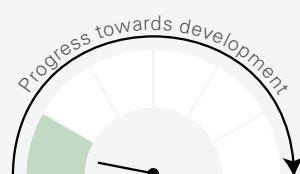
Reactor description: Lithium fluoride molten salt reactor to produce power and isotopes using uranium fuel and breeding thorium.

Thermal power (MWth)	600
Core outlet temperature (°C)	650
Spectrum (thermal/fast)	Thermal
Fuel form	Molten fluorides
Base fissile material	U-233, not requiring uranium enrichment

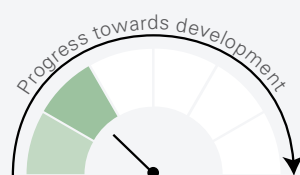
Licensing



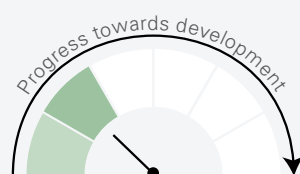
Siting



Financing



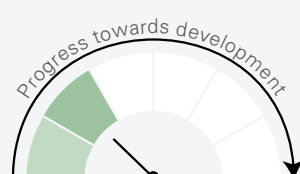
Supply chain



Engagement

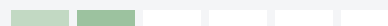


Fuel



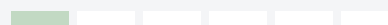
Assessment of LFTR's progress to deployment

Licensing



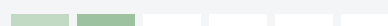
Flibe Energy has been engaged in pre-licensing activities with the US Nuclear Regulatory Commission (NRC) for the LFTR since 2013, when Flibe Energy first submitted its notice of intent to submit a licence application to the NRC for the LFTR and radioisotope production facility.

Siting



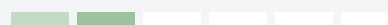
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



In July 2018, Flibe Energy raised over USD 2.6 million (USD₂₀₂₃ 3.2 million) in funding for a molten salt fuel study in collaboration with Pacific Northwest National Laboratory, which included over USD 2.1 million (USD₂₀₂₃ 2.5 million) from the US Department of Energy (DOE). In October 2018, Flibe Energy raised more than USD 4.7 million (USD₂₀₂₃ 5.7 million) in Series A private funding. In December 2023, the company raised an additional USD 4.5 million in convertible notes against future shares of Flibe Energy.

Supply chain



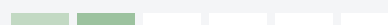
In 2019, Flibe Energy was awarded a voucher under the US DOE Gateway for Accelerated Innovation in Nuclear (GAIN) initiative to assess the LFTR design from a safeguard's perspective in partnership with Oak Ridge National Laboratory.

Engagement



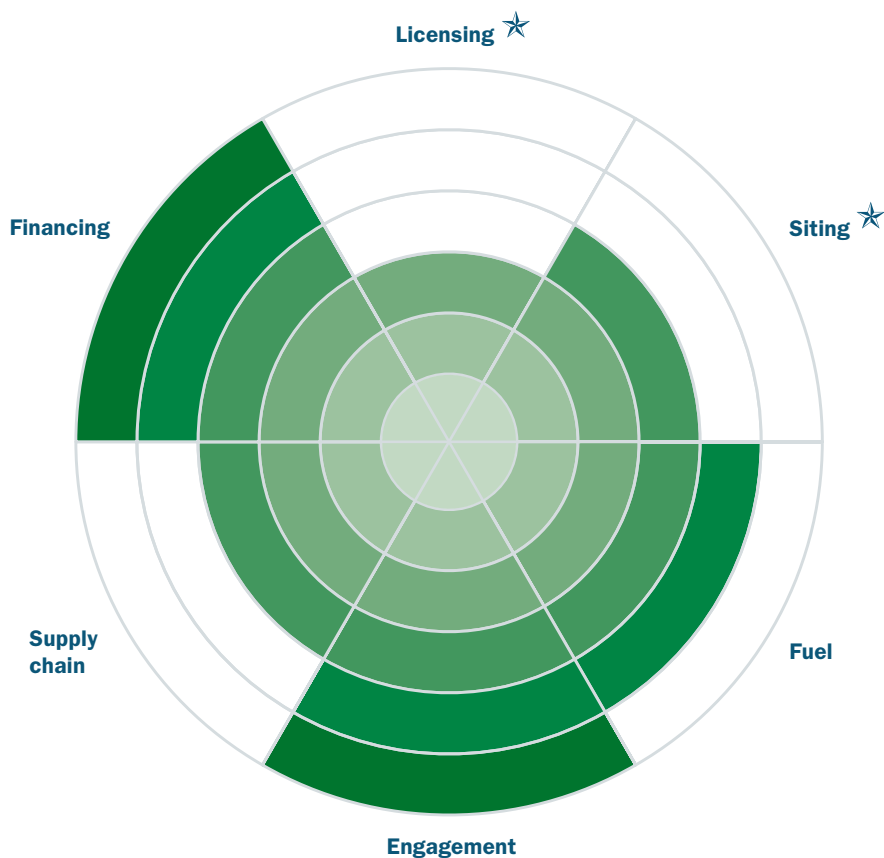
The founder of Flibe Energy, Kirk Sorensen, has engaged public audiences extensively through TED Talks, Google TechTalks and YouTube. Kirk Sorensen also engages university students and the general public through seminars and lectures around the world. In 2019, Flibe Energy discussed LFTR with the Wyoming Legislative Joint Committee on Minerals, Business & Economic Development. In 2023 and 2024, Flibe Energy participated in the 2023 Next Frontier Energy Summit and Posidonia 2024 (The International Shipping Exhibition) to discuss the LFTR technology and the maritime use of nuclear energy, respectively. Flibe Energy has also developed online tools and apps to raise public awareness about nuclear energy.

Fuel



LFTR is designed to use thorium in a blanket salt to breed U-233, a fissile material that can be used as new fuel. The reactor's startup fuel could include U-233, plutonium, transuranics, or enriched uranium. Flibe Energy has been awarded two vouchers under the US DOE GAIN initiative to work on research and development of the LFTR fuel in partnership with the Argonne National Laboratory and Pacific Northwest National Laboratory. In 2024, Flibe Energy signed a Memorandum of Understanding with Savannah River National Laboratory to advance research and development activities, including on the LFTR design and thorium-based fuel development.

GE Vernova Hitachi Nuclear Energy – BWRX-300



★ Active in multiple jurisdictions or countries.

△ Changes since Second Edition (2024).

Reactor description: Land-based boiling water reactor.

Thermal power (MWth) 870

Core outlet temperature (°C) 288

Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic

Base fissile material U-235, enrichment level: < 5%

Licensing ★



Siting ★



Financing △



Supply chain



Engagement



Fuel △



Assessment of BWRX-300's progress to deployment

Licensing ★



In 2024, the Canadian Nuclear Safety Commission (CNSC) confirmed the environmental assessment for BWRX-300 at Ontario Power Generation's (OPG) Darlington site. OPG's 2022 application for a licence to construct is under review by CNSC. In 2023, GE Vernova Hitachi (GVH) completed CNSC's vendor design review Phases 1-2. The US Nuclear Regulatory Commission (NRC) has approved five topical reports for BWRX-300; two more are under review. GVH has completed United Kingdom Generic Design Assessment Step 1 and started Step 2. In Poland, the National Atomic Energy Agency has issued a General Opinion that BWRX-300 is compliant with national nuclear safety requirements and the Ministry of Climate and Environment has issued decisions-in-principle for the construction of 24 BWRX-300s at six sites.

Siting ★



In 2024, OPG completed early-phase site preparation for the first BWRX-300 at its Darlington site; site preparation for three additional units is underway. BWRX-300 has also been selected for potential deployment by: SaskPower in Canada; ORLEN Synthos Green Energy in Poland; and American Electric Power at the Rockport coal-fired plant in Indiana, United States. Tennessee Valley Authority (TVA) has partnered with GVH and announced it will work with Bechtel and Sargent & Lundy to evaluate potential deployment at the Clinch River site in the United States. BWRX-300 has been shortlisted by: Great British Nuclear in the United Kingdom; and Vattenfall for its Ringhals nuclear site in Sweden.

Financing



GVH has secured CAD 970 million (USD₂₀₂₃ 746.6 million) from the Canadian Infrastructure Bank, and USD 7.2 million (USD₂₀₂₃ 8.5 million) from the US Department of Energy (DOE) Advanced Research Projects Agency-Energy. In 2021, the Province of Ontario enabled OPG to recover project costs from the electricity ratepayer through a regulatory amendment. In 2023, GVH, TVA, OPG and Synthos Green Energy (SGE) committed to invest USD 400 million. In 2024, GVH received a GBP 33.6 million (USD₂₀₂₃ 40.7 million) grant from the UK Future Nuclear Enabling Fund for development in the United Kingdom. In 2025, TVA announced that it will lead, along with a coalition of partners, an application for up to USD 800 million from the US DOE's Generation III+ Small Modular Reactor Program to support BWRX-300 in the United States.

Supply chain



In Canada, GVH is partnering with OPG, Aecon and AtkinsRéalis, and has selected: E.S. Fox for site preparation; BWXT for reactor pressure vessel; Velan for industrial valves; and Worley Chemetics for the isolation condenser system. In the United Kingdom, GVH has signed Memorandums of Understanding (MoUs) with suppliers, including Aecon UK, AtkinsRéalis UK, Amentum Clean Energy (formerly Jacobs Clean Energy), Laing O'Rourke, Bocard UK, Cavendish Nuclear, and Sheffield Forgemasters, and has contracted Instance BV to provide input on the design. In 2024, GVH formed a global supplier's group with BWXT Canada being the first to join, followed by Worley Chemetics. BWRX-300 was also selected for a Project-Working Group under the European Industrial Alliance on SMRs.

Engagement



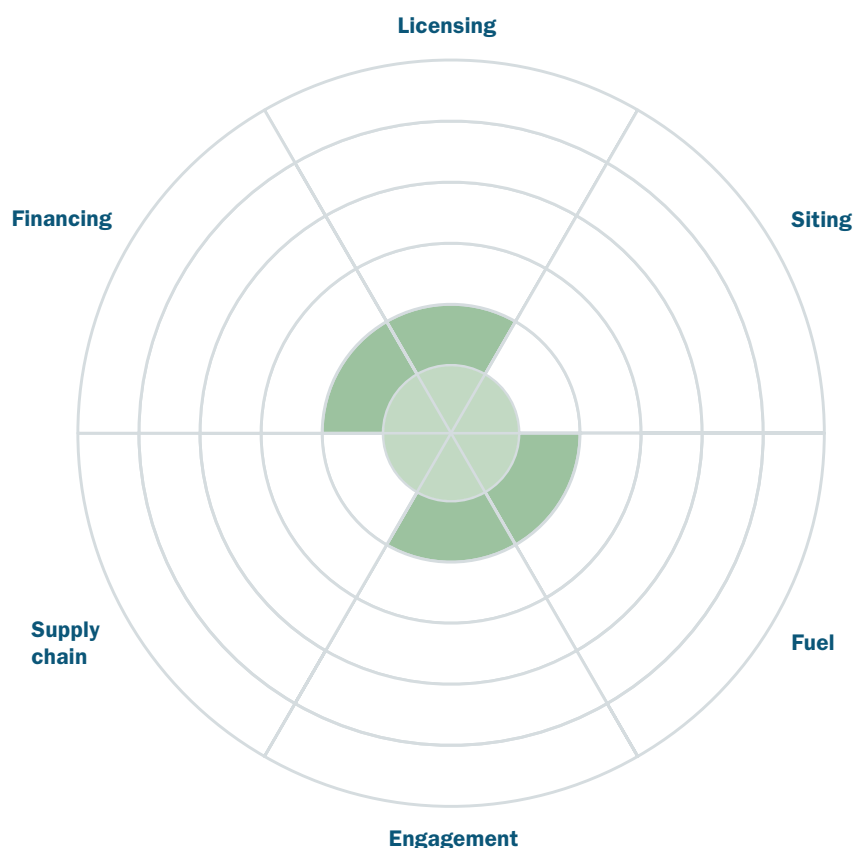
In Canada, the Darlington project has been endorsed by both provincial and federal governments. OPG engages the public through information sessions, workshops, hearings, and community events. GVH also has an agreement in place with the Saskatchewan Industrial and Mining Suppliers Association to engage with local suppliers to advance the potential deployment in Saskatchewan. Internationally, GVH has established a series of MoUs including with Kärnfull Next in Sweden and with Emirates Nuclear Energy Corporation in the United Arab Emirates for the potential deployment of BWRX-300. In 2025, Fermi Energia submitted an application to Estonia's Ministry of Economic Affairs and Communications to initiate a state planning process for a nuclear power plant based on BWRX-300 technology.

Fuel



The BWRX-300 reactor uses fuel that is proven, commercially available and currently being used in operating boiling water reactors. OPG has selected suppliers for the first fuel contracts for its Darlington project in Canada, where up to four BWRX-300 SMRs are planned. These suppliers include: Cameco for natural uranium hexafluoride; Urenco US for uranium enrichment services; Orano for additional enriched uranium product; and Global Nuclear Fuel-Americas, a joint venture led by GVH, for fuel fabrication and fuel assemblies. In 2025, GE Vernova, the parent company of GVH, announced plans to invest more than USD 50 million (USD₂₀₂₃ 47.1 million) in GVH's Wilmington site in North Carolina to expand fuel manufacturing capacity for boiling water reactors, including the BWRX-300.

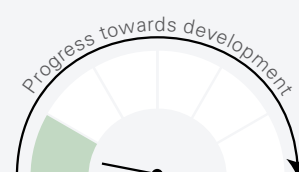
General Atomics – Energy Multiplier Module (EM²)



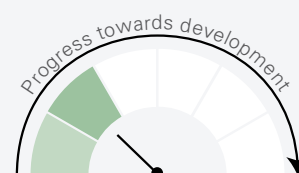
Licensing



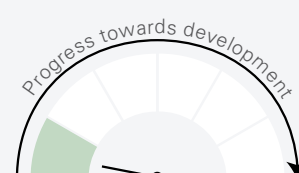
Siting



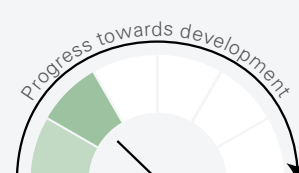
Financing



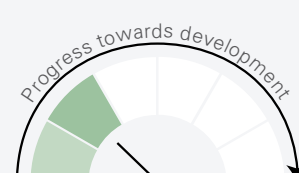
Supply chain



Engagement



Fuel

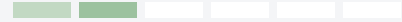


Reactor description: Gas-cooled fast reactor with one 500-MWth module coupling to one gas turbine, designed as a breeder.

Thermal power (MWth)	500
Core outlet temperature (°C)	850
Spectrum (thermal/fast)	Fast
Fuel form	Uranium-carbide ceramic
Base fissile material	U-235, enrichment level ≥ 10% to < 20%

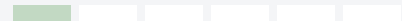
Assessment of EM²'s progress to deployment

Licensing



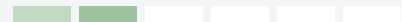
General Atomics (GA) has started pre-licensing activities for the Energy Multiplier Module (EM²) with the US Nuclear Regulatory Commission (NRC) in 2021. Since then, GA has docketed multiple white papers with the NRC on the fuel qualification strategy of the EM².

Siting



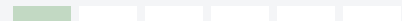
At the time of the assessment, no information related to siting was readily available from any site owners.

Financing



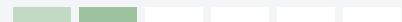
In 2018, General Atomics received over USD 3.2 million (USD₂₀₂₃ 3.9 million) from the US Department of Energy (DOE) through the Advanced Reactor Development Projects pathway and a Regulatory Assistance Grant. This funding supported the development of fuel performance modelling and simulation tools, as well as pre-licensing engagement with the regulator for a silicon carbide composite. This composite serves as cladding for uranium carbide fuel in the EM².

Supply chain



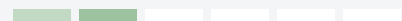
At the time of the assessment, no information about supply chain readiness was readily available from verifiable public sources.

Engagement



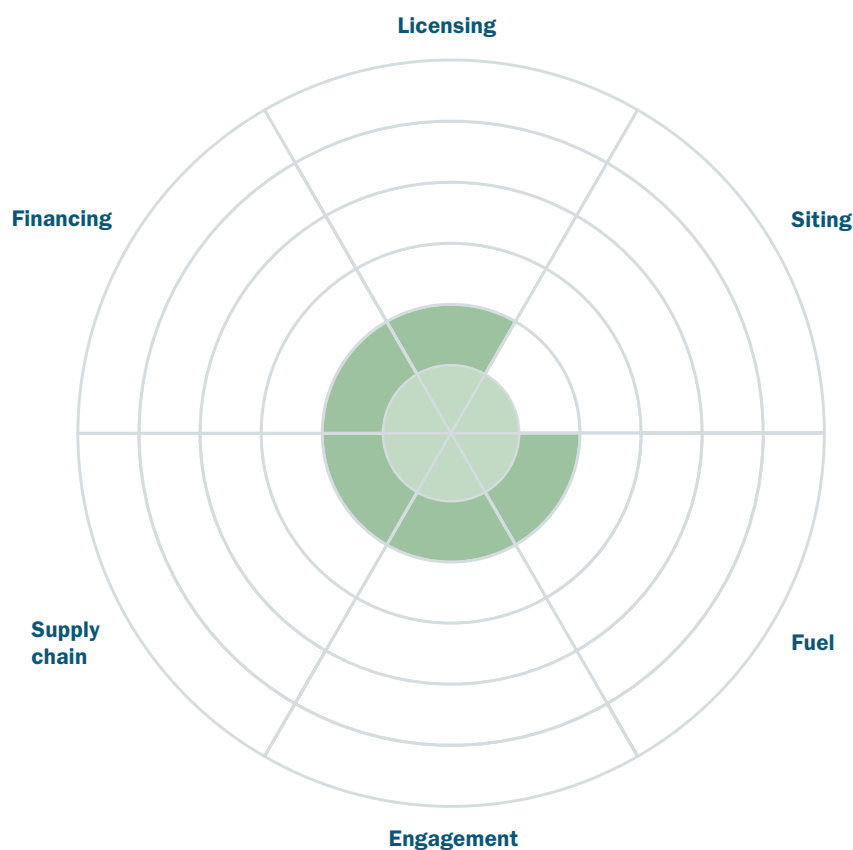
In 2024, General Atomics and the Emirates Nuclear Energy Corporation (ENEC) signed a Memorandum of Understanding to collaborate on the development and application of advanced nuclear technologies.

Fuel



The EM² is designed to use uranium carbide pellets enclosed in a silicon carbide cladding material called SiGA, which is under development by General Atomics. In 2024, GA produced its first batch of full-length SiGA tubes to be used in pressurised water reactors and developed preliminary performance models for SiGA. To advance research and development of EM² fuel, General Atomics has collaborated with both private and public partners, including Westinghouse Electric Company, Idaho National Laboratory (INL), Oak Ridge National Laboratory (ORNL), and the University of Tennessee.

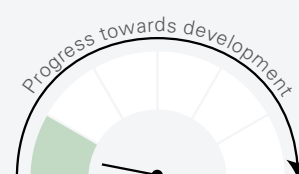
General Atomics – Fast Modular Reactor (FMR)



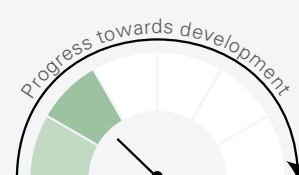
Licensing



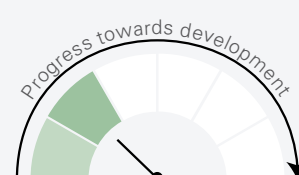
Siting



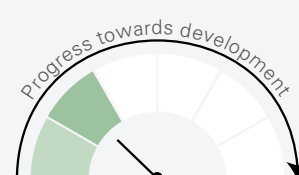
Financing



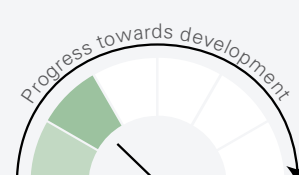
Supply chain



Engagement



Fuel

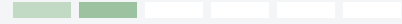


Reactor description: Gas-cooled fast reactor with one 100-MWth module coupling to one gas turbine.

Thermal power (MWth)	100
Core outlet temperature (°C)	800
Spectrum (thermal/fast)	Fast
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level ≥ 10% to < 20%

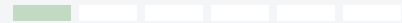
Assessment of FMR's progress to deployment

Licensing



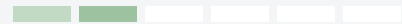
General Atomics (GA) has started pre-licensing activities for the Fast Modular Reactor (FMR) with the US Nuclear Regulatory Commission (NRC) in 2022. Since then, GA has met with the NRC on a variety of topics, and has docketed multiple reports, licensing project plans and white papers with the NRC on topics related to the design and safety of the FMR.

Siting



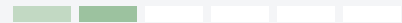
At the time of the assessment, no information related to siting was readily available from any site owners.

Financing



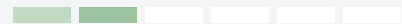
Since 2020, General Atomics received approximately USD 33 million (USD₂₀₂₃ 32.8 million) in grants from the US Department of Energy (DOE) to support advanced nuclear research and development. This includes USD 31.1 million (USD₂₀₂₃ 36.6 million) in 2020 through the Advanced Reactor Demonstration Program (ARDP) and USD 1.1 million (USD₂₀₂₃ 1.1 million) in 2024 under the Nuclear Energy University Program, in partnership with the University of Michigan, to advance the FMR design.

Supply chain



In 2020, General Atomics and Framatome announced a collaboration to develop the Fast Modular Reactor (FMR). That same year, General Atomics joined a collaboration through the Center of Excellence for Thermal Fluids Applications in Nuclear Energy that aims to accelerate the deployment of advanced nuclear reactors. This initiative, based at Idaho National Laboratory (INL) and Argonne National Laboratory (ANL), involves collaborations with several US universities, including Pennsylvania State University, as well as key stakeholders from the nuclear industry. In 2024, DOE funded research between General Atomics and the University of Michigan on silicon carbide composite rods.

Engagement



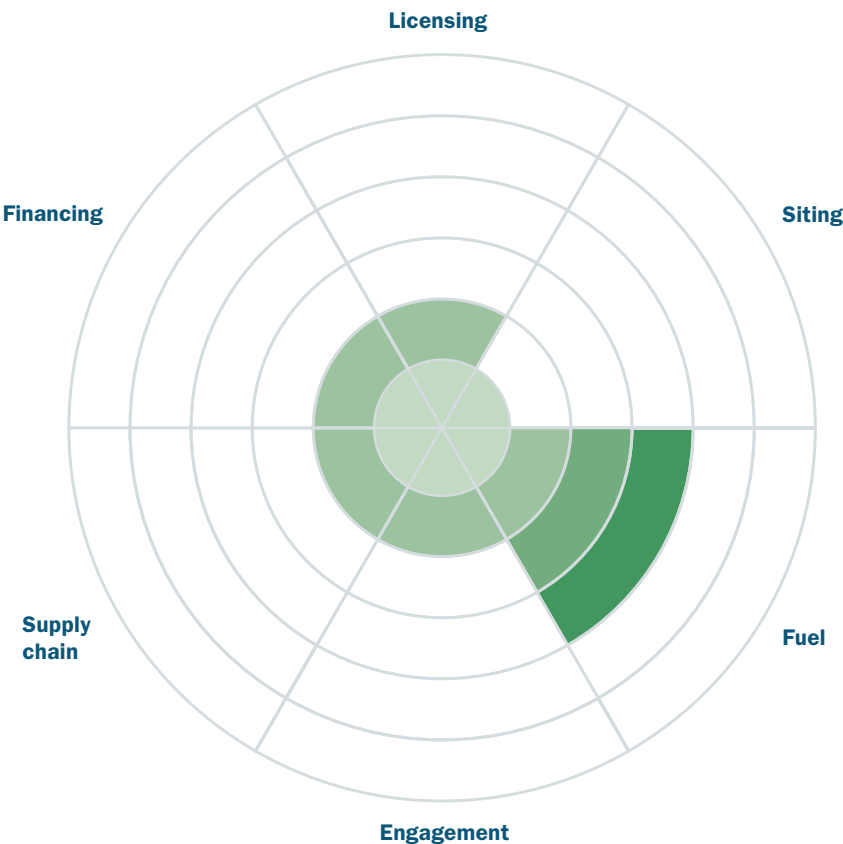
In 2024, General Atomics and the Emirates Nuclear Energy Corporation (ENEC) signed a Memorandum of Understanding to collaborate on the development and application of advanced nuclear technologies.

Fuel



The FMR is designed to use uranium dioxide (UO₂) pellets enclosed in a silicon carbide cladding material called SiGA, which is under development by General Atomics. In 2023, DOE awarded funding to GA for an irradiation test in the Advanced Test Reactor at INL. This test aims to validate the integrity of FMR's fuel and cladding under high-temperature and high-burnup conditions. In 2024, GA produced its first batch of full-length SiGA tubes to be used in pressurised water reactors and developed preliminary performance models for SiGA. To advance research and development of FMR fuel, General Atomics has collaborated with both private and public partners, including Westinghouse Electric Company, Idaho National Laboratory (INL) and Oak Ridge National Laboratory (ORNL).

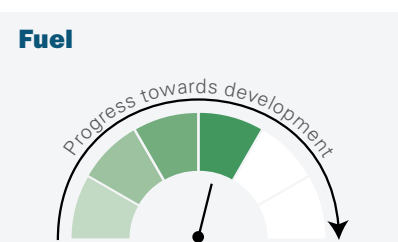
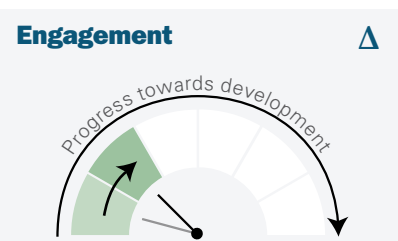
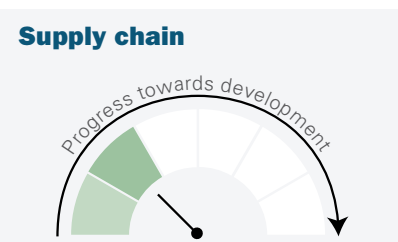
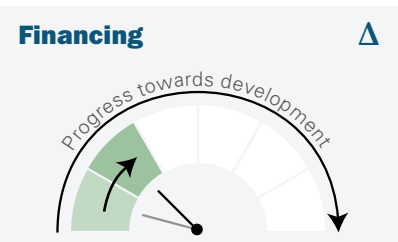
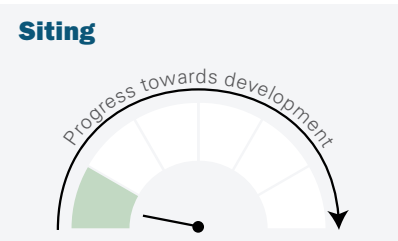
Gorgé-Calogena (Calogena) – CAL-30



△ Changes since Second Edition (2024).

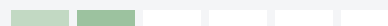
Reactor description: Closed-core, pool-type water-cooled microreactor intended for district heating.

Thermal power (MWth)	30
Core outlet temperature (°C)	110
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level: < 5%



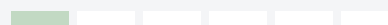
Assessment of CAL-30's progress to deployment

Licensing



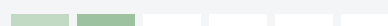
In October 2024, Gorgé-Calogena (Calogena) began the pre-instruction phase of pre-licensing, which provides support for licence application preparations, with France's Nuclear Safety and Radioprotection Authority (ASNR). In 2024, Calogena submitted the first round of materials for its Safety Option File (DOS) to the ASNR and, in 2025, submitted additional documents for its DOS. In January 2025, Calogena met with the Finnish Radiation and Nuclear Safety Authority (STUK) about licensing in Finland.

Siting



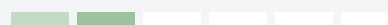
Calogena has expressed interest in deploying the CAL-30 SMR in both France and Finland. To support its development in Finland, Calogena established a local subsidiary in Helsinki in November 2024. At the time of assessment, no information related to siting was readily available from any site owners.

Financing



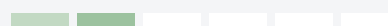
In 2023, Calogena received EUR 5.2 million (USD₂₀₂₃ 5.6 million) under the France 2030 programme. Groupe Gorgé, the parent company of Calogena, is also developing its fund-raising strategy for the Calogena project. Groupe Gorgé has stated its intention to invest several tens of millions of euros of its own capital for the initial phases of the project.

Supply chain



Groupe Gorgé has experience engineering and manufacturing equipment for nuclear power plants and has 200 engineers working in the nuclear power and innovation sector. In 2024, Calogena was selected to form a joint Project-Working Group, alongside another SMR developer, under the European Industrial Alliance on Small Modular Reactors. Additionally, Calogena is a member of the Nuclear Valley network, a group of over 400 companies involved in the French nuclear supply chain.

Engagement



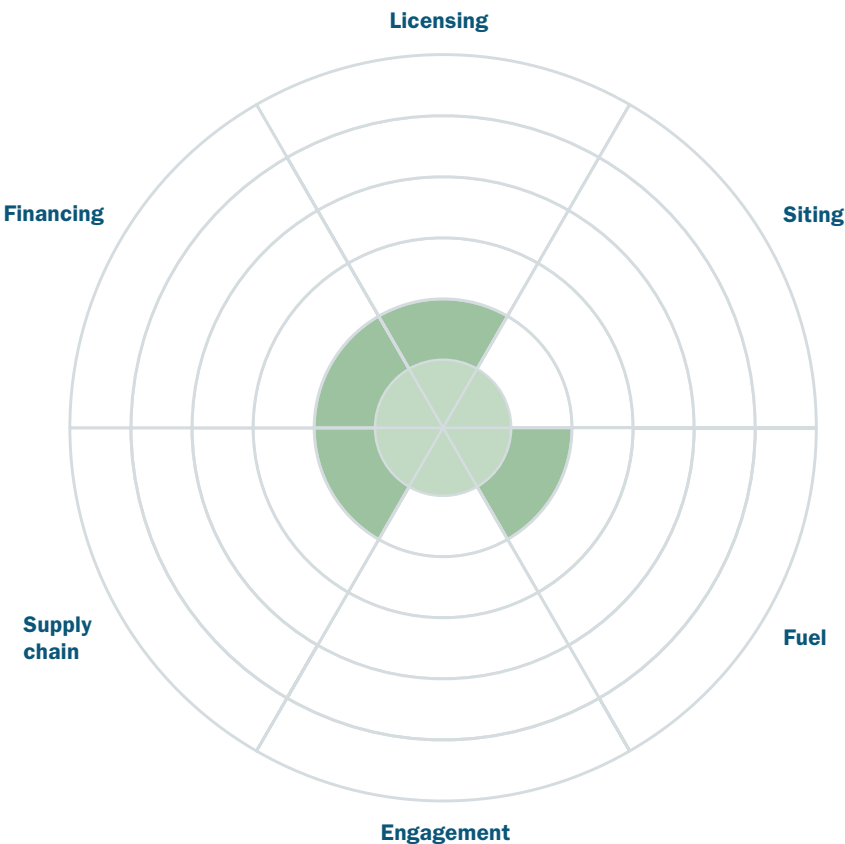
The French National Association of Local Information Committees and Commissions (ANCCLI), which aims to raise public awareness about nuclear energy, featured both Calogena and Jimmy SMR in its 2024 "Grand Dossier" on SMRs. In 2024, Calogena also participated in a public workshop by ANCCLI dedicated to SMRs.

Fuel



The Calogena reactor plans to use low-enriched uranium dioxide (UO₂) fuel, enriched to 3.4%, the current industry standard for light water-cooled reactors, packaged in a shortened version of the fuel rods used in pressurised water reactors (PWRs). Calogena has announced that it has chosen to use the same fuel as the existing power plants in France after studying the matter with Orano and Framatome. Calogena plans to use the existing French nuclear fuel.

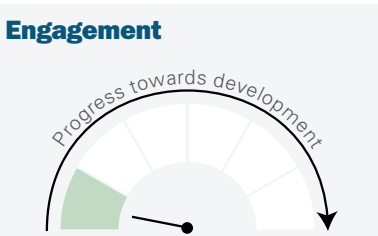
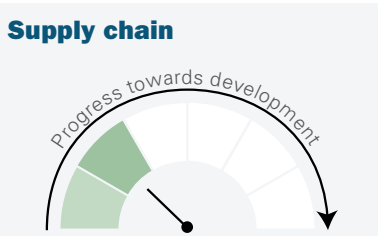
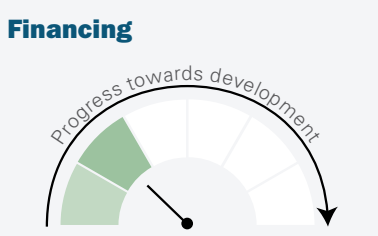
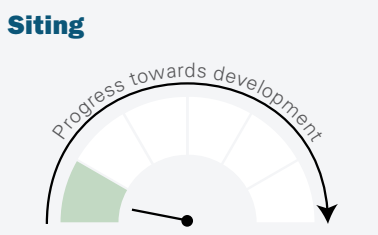
Hexana – HEXANA



△ Changes since Second Edition (2024).

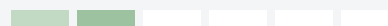
Reactor description: Sodium-cooled fast reactor with two 400 MWth reactor modules and a high-temperature heat storage system.

Thermal power (MWth)	800 (plant deployed with two modules of 400 MWth each)
Core outlet temperature (°C)	530
Spectrum (thermal/fast)	Fast
Fuel form	MOX ceramic
Base fissile material	Plutonium, not requiring uranium enrichment



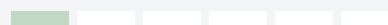
Assessment of HEXANA's progress to deployment

Licensing



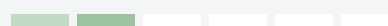
In early 2025, HEXANA started the screening phase of pre-licensing with France's Nuclear Safety and Radiation Protection Authority (ASNR). HEXANA plans to enter the preparatory review phase of pre-licensing with the ASNR in 2025.

Siting



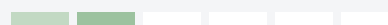
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



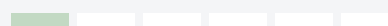
In 2023, HEXANA received EUR 10.0 million (USD₂₀₂₃ 10.8 million) under the France 2030 programme and an additional EUR 15.0 million (USD₂₀₂₃ 16.2 million) in an equity round. This round included EUR 2.6 million (USD₂₀₂₃ 2.8 million) from Blast.Club and EUR 12.4 million (USD₂₀₂₃ 13.4 million) from Eren Industries, CEA Investissement, and Région Sud Investissement.

Supply chain



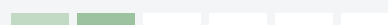
In 2023, HEXANA signed non-binding collaboration agreements with several partners: Bureau Veritas to conduct regulatory studies for deploying its SMR near industrial sites; Egis and Bouygues to incorporate construction constraints early in the reactor's design; and Laborelec to explore integration of the HEXANA SMR with a high-temperature thermal storage system. Also in 2023, HEXANA held its first supply chain workshop with Framatome and announced plans to leverage Framatome's experience in developing primary circuit components. In 2023, the French Atomic Energy and Alternative Energies Commission (CEA) announced it would contribute its expertise to support HEXANA's development. HEXANA was one of two CEA spin-offs launched in March 2023.

Engagement



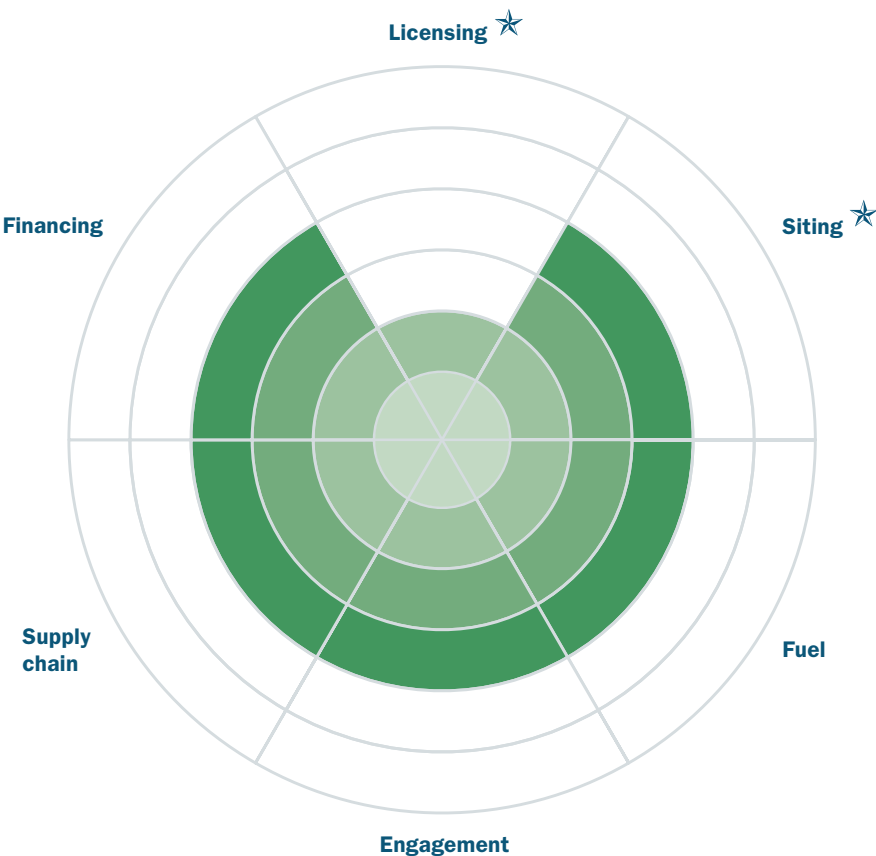
At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

Fuel



HEXANA plans to use mixed plutonium-uranium oxide fuel (MOX). MOX fuel is a technically proven fuel type; however, there are presently limited suppliers, in particular for fast reactors. In 2024, Orano launched a working group with HEXANA and two other French fast-spectrum SMRs to develop a joint strategy for MOX fuel supply and reprocessing in France.

Holtec International – SMR-300



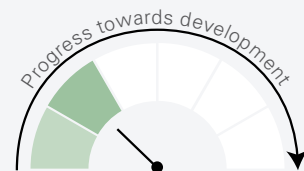
★ Active in multiple jurisdictions or countries.

Δ Changes since Second Edition (2024).

Reactor description: Land-based pressurised light water reactor.

Thermal power (MWth)	1 050
Core outlet temperature (°C)	321
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level: < 5%

Licensing ★



Siting ★ Δ



Financing



Supply chain



Engagement Δ

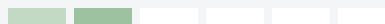


Fuel



Assessment of SMR-300's progress to deployment

Licensing ★



In 2023, Holtec International (Holtec) renamed its “SMR-160” to “SMR-300” with a power increase from 525 to 1 050 MWth. The SMR-300 is expected to benefit from pre-licensing activities for the SMR-160. Prior to this change, Holtec had completed Phase 1 of the pre-licensing vendor design review with the Canadian Nuclear Safety Commission (CNSC) in 2020. It continues pre-licensing activities with the US Nuclear Regulatory Commission (NRC) with its new design. This includes meetings with the NRC regarding deployment at its Palisade site for which it plans to submit a construction permit application in 2026. In 2024, Holtec completed step 1 of UK Office for Nuclear Regulation's (ONR) Generic Design Assessment (GDA). In 2024, Holtec also started GDA step 2, which it expects to complete in 2025.

Siting ★



In the United States, Holtec plans two SMR-300 units at the Palisades site it acquired in Michigan in 2022. Holtec has signed an agreement with Enercon to support the preparation of their application to the NRC for deployment of the two SMR-300 at the Palisades site. Holtec is also considering deploying an SMR at its Oyster Creek nuclear site in New Jersey. Holtec has a Memorandum of Agreement to assess SMR deployment feasibility at Entergy Corporation sites. Holtec also has two co-operation agreements with NAEK Energoatom to plan for the deployment of SMRs in Ukraine. SMR-300 was shortlisted with three other SMRs by Great British Nuclear (GBN) for the next phase of the innovative nuclear technology competition for potential deployment in the United Kingdom.

Financing ★



In 2020, Holtec secured USD 116 million (USD₂₀₂₃ 136.5 million) over seven years from the US Department of Energy (DOE) Advanced Reactor Demonstration Program for early-stage design, engineering and licensing. In 2021, Holtec stated it had invested USD 400 million (USD₂₀₂₃ 449.7 million) in its SMR programme. Holtec is a member of the Mid-Atlantic Clean Hydrogen Hub selected by DOE to receive up to USD 750 million. In 2023, the Korea Trade Insurance Corporation and the Export-Import Bank of Korea signed agreements with Holtec and Hyundai Engineering & Construction to support financing of Holtec SMR projects. In 2024, the UK Department for Energy Security and Net Zero (DESNZ) awarded GBP 30 million (USD₂₀₂₃ 36.4 million) to Holtec to support the GDA pre-licensing process for the SMR-300.

Supply chain



Holtec has in-house nuclear engineering and construction capabilities. Holtec has contracted Mitsubishi Electric, and Hyundai Engineering and Construction (Hyundai E&C). In 2022, DOE approved part 1 of a loan application for Holtec to expand manufacturing in the United States. Starting in 2018 and renewed in 2024, Holtec has been working with NAEK Energoatom on a manufacturing hub in Ukraine. In 2024: Holtec selected South Yorkshire, United Kingdom, for its SMR factory, a GBP 1.5 billion (USD₂₀₂₃ 1.9 billion) investment pending final decision; Holtec and Hyundai E&C signed partnership agreements with Balfour Beatty and Mott MacDonald for the UK GBN competition; and Holtec signed Memoranda of Understanding with Sheffield Forgemaster and Sheffield Advanced Manufacturing Research Centre.

Engagement



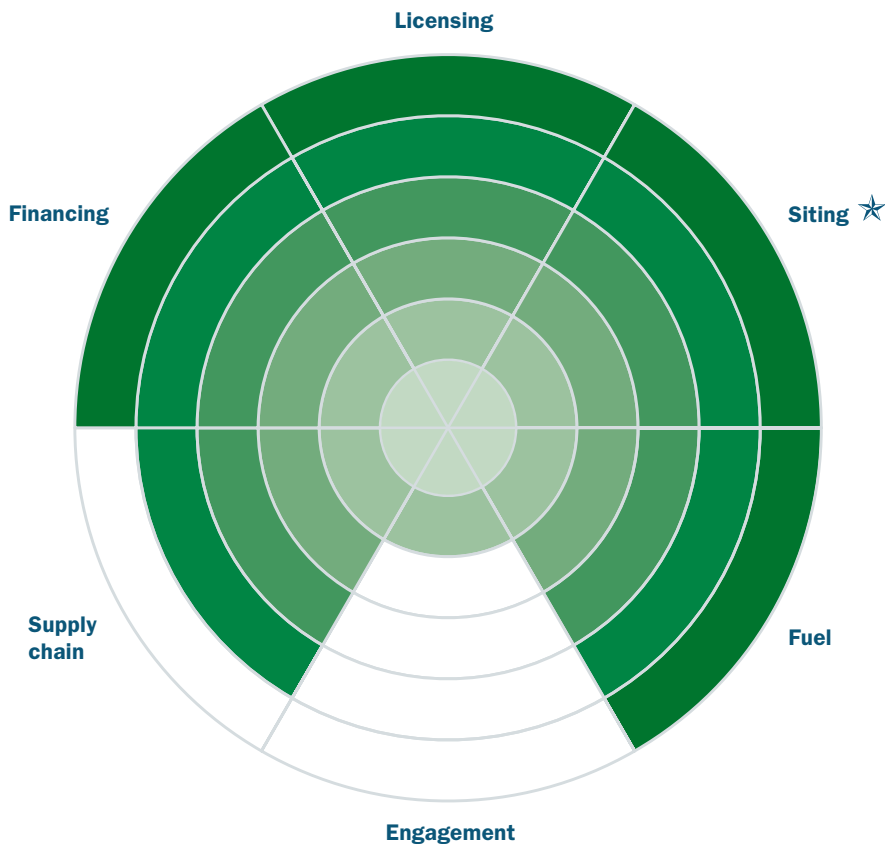
In 2022, Holtec presented its SMR activities to the Van Buren County Council, Michigan, United States. In 2023, Holtec participated in the Southwest Public Policy Institute's podcast dedicated to SMRs. Holtec senior staff have also participated in various other podcasts and TV interviews to discuss SMR plans. In 2024, Holtec CEO met with India's Prime Minister in New York and discussed possible cooperation on nuclear energy. In 2024, South Yorkshire Mayor Oliver Coppard publicly welcomed plans to manufacture SMRs in South Yorkshire, United Kingdom.

Fuel



The SMR-300 uses LEU fuel that is the current industry standard for water-cooled reactor technologies. In 2018, Holtec International signed an MoU with Global Nuclear Fuel-Americas, a joint venture led by GE Hitachi Nuclear Energy, to develop nuclear fuel and control rod drive mechanisms for Holtec's SMR. In 2020, Holtec signed a contract with Framatome, a French nuclear fuel manufacturer, to supply nuclear fuel the SMR-300 in the form of the Framatome's GAIA 17x17 fuel assembly that is already used in operating pressurised light-water reactors.

Tsinghua University Institute of Nuclear and New Energy Technology (INET) – HTR-PM



★ Active in multiple jurisdictions or countries.

Δ Changes since Second Edition (2024).

Reactor description: 500 MWth plant with two 250 MWth pebble bed high temperature gas reactor modules coupling to one steam turbine.

Thermal power (MWth)	500 (plant deployed with two modules of 250 MWth each)
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Core outlet temperature (°C)	750
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Spectrum (thermal/fast)	Thermal
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Fuel form	Uranium-oxide ceramic TRISO pebble
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Base fissile material	U-235, enrichment level: ≥ 5% to < 10%
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Licensing



Siting ★



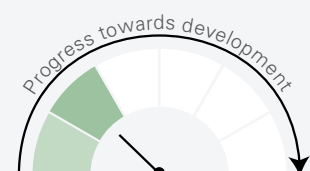
Financing Δ



Supply chain



Engagement



Fuel



Assessment of HTR-PM's progress to deployment

Licensing



The HTR-PM is a commercial demonstration of INET's pebble-bed High-Temperature Gas-cooled Reactor (HTGR) module. HTR-PM contains two-HTGR modules coupled to a nuclear steam supply system (NSSS). In August 2021, China's National Nuclear Safety Administration (NNSA) approved the licence to operate for the HTR-PM reactor, which operated at low power initially, and was connected to the grid in December 2021, before entering commercial operations in December 2023. In February 2025, the Ministry of Ecology and Environment in China announced that it has accepted the environmental impact assessment documents for the HTR-PM600S, composed of six HTGR modules to be constructed on the Lianyungang Petrochemical Park site as part of the Jiangsu Xuwei Nuclear Energy Heating Power Plant project.

Siting ★



In 2021, the HTR-PM at the Shidaowan Nuclear Power Plant site in Shandong province was connected to the electrical grid and entered commercial operation in December 2023. In March 2024, it was connected to a local district heating system. In August 2024, the Chinese Government approved the deployment of the HTR-PM600S as part of Phase 1 of the Jiangsu Xuwei Nuclear Energy Heating Power Plant project. The HTR-PM600S consists of six HTGR modules using three NSSS. The plan for Jiangsu Xuwei Nuclear Energy Heating Power Plant project includes the HTR-PM600S alongside two gigawatt-scale HPR1000 light-water-reactors to provide a large volume of steam to the Lianyungang Petrochemical Park for a range of processes by 2030.

Financing



The first-of-a-kind (FOAK) HTR-PM is operating and has been fully financed. In August 2024, the Chinese Government approved the phase one of the Jiangsu Xuwei Nuclear Heating Power Plant project, which will include the HTR-PM600S. Pre-construction began at the Lianyungang Petrochemical Park site in February 2025. China National Nuclear Corporation Suneng Nuclear Power is responsible for the investment, construction and operation.

Supply chain



The HTR-PM was developed by Tsinghua University's Institute of Nuclear and New Energy Technology (INET) and constructed by China National Nuclear Corporation (CNNC). China Huaneng Group operates the FOAK HTR-PM, which is owned by a consortium composed of China Huaneng, China Nuclear Engineering Corporation, and INET. Up to 93.4% of the equipment was manufactured domestically. First concrete is expected to be poured for the HTR-PM600S at the Jiangsu Xuwei Nuclear Energy Heating Power Plant in 2026.

Engagement



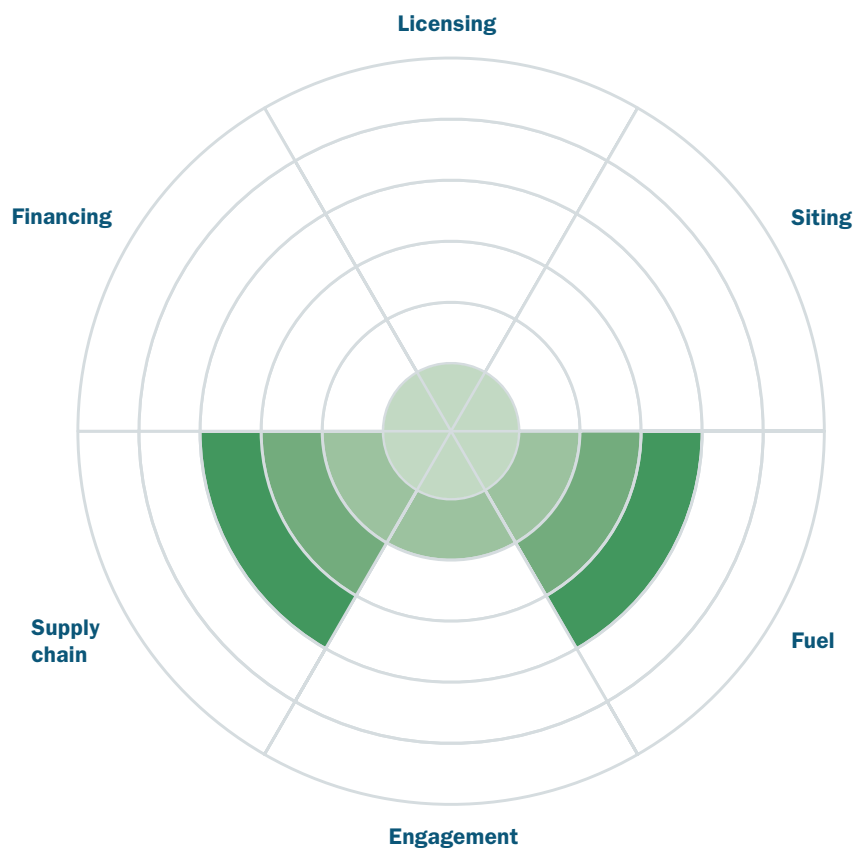
Tsinghua University's Institute of Nuclear and New Energy Technology, which is part of the consortium that owns the HTR-PM project, creates opportunities for engagement with students. In December 2021, an event to celebrate the first grid connection of the HTR-PM was held at the Shidaowan site, which featured participation from heads of ministries and commissions, provincial and municipal leaders, and representatives from industry and research institutions.

Fuel

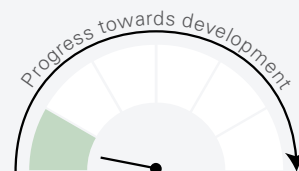


The HTR-PM uses TRISO fuel, which is manufactured and supplied by CNNC in co-operation with INET and the China Huaneng Group. HTR-PM fuel is licensed for operation.

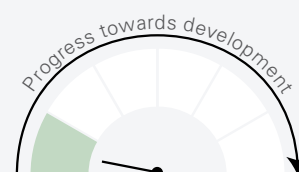
INET and China General Nuclear Power Group (CGN) – NHR200-II



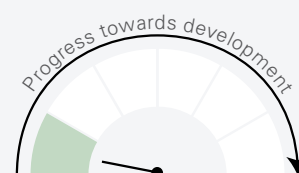
Licensing



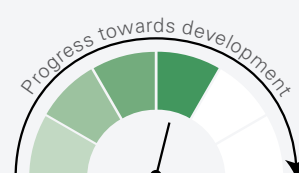
Siting



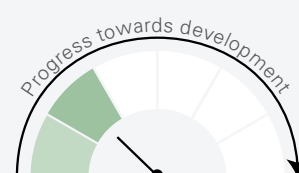
Financing



Supply chain



Engagement



Fuel

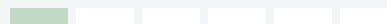


Reactor description: Integrated pressurised water-cooled reactor.

Thermal power (MWth)	200
Core outlet temperature (°C)	280
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level < 5%

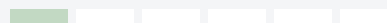
Assessment of NHR200-II's progress to deployment

Licensing



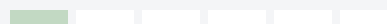
The NHR200-II design is based on the NHR-5, a 5 MWth experimental reactor developed and constructed by the Institute of Nuclear and New Energy Technology (INET) at Tsinghua University in the 1980s. In 2016, INET completed the basic design and engineering verification tests for the NHR200-II. At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



In 2018, China's National Energy Administration approved a feasibility study for a nuclear heating demonstration project using the NHR200-II in northern China. It also selected China General Nuclear Power Group (CGN) and INET to conduct the preliminary work on the construction of a demonstration project, including planning and site selection. At the time of assessment, no additional information related to siting was readily available from any site owners.

Financing



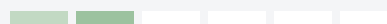
The NHR200-II demonstration project is led by CGN and INET. CGN is a state-owned enterprise under direct management of the Chinese central government, which may create opportunities to access support from national funding. At the time of the assessment, no information about financing was readily available from verifiable public sources.

Supply chain



CGN has extensive experience in designing, constructing, and operating pressurised water-cooled reactors in China. INET has experience in developing and operating the NHR-5, the prototype reactor of the NHR200-II. Given the experience of CGN and INET, there is an opportunity to leverage China's existing supply chains for pressurised water-cooled reactors for NHR200-II development and deployment.

Engagement



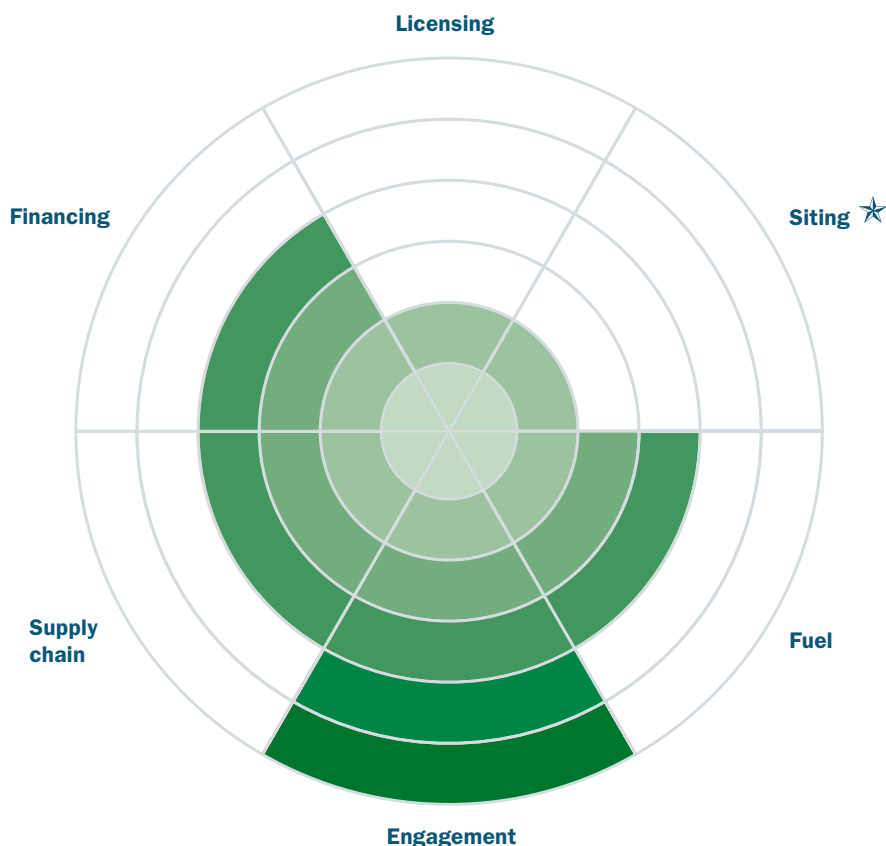
Development of the NHR200-II design was carried out by INET, which created opportunities for engagement with students.

Fuel



The NHR200-II is designed to use low-enriched fuel in the form of UO_2 pellets, which are the current industry standard for water-cooled power reactor technologies.

i-SMR Development Agency (i-SMRDA), KHNP, and KAERI – i-SMR



★ Active in multiple jurisdictions or countries.

Reactor description: Integrated pressurised water reactor.

Thermal power (MWth) 520

Core outlet temperature (°C) 321

Spectrum (thermal/fast) Thermal

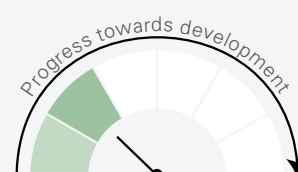
Fuel form Uranium-oxide ceramic

Base fissile material U-235, enrichment level < 5%

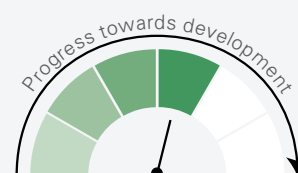
Licensing



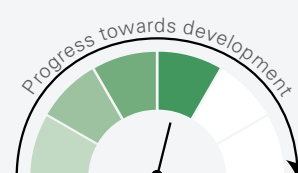
Siting ★



Financing



Supply chain



Engagement

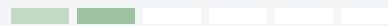


Fuel



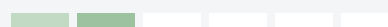
Assessment of i-SMR's progress to deployment

Licensing



The i-SMR integrated pressurised water reactor is jointly developed by the i-SMR Development Agency (i-SMRDA), Korea Hydro & Nuclear Power (KHNP) and the Korea Atomic Energy Research Institute (KAERI). The i-SMR can be deployed in configurations ranging from a single module to up to eight modules of 170 MWe each. Since 2022, Korea's Nuclear Safety and Security Commission (NSSC) has been working towards the establishment of regulatory pathways for SMRs, including approaches for full life cycle analysis and verification of SMR technologies. In 2023, i-SMRDA and KHNP submitted preliminary design reports for the i-SMR to NSSC as part of the pre-licensing process. The NSSC implementation plan for the i-SMR anticipates the Standard Design Approval (SDA) process to begin in 2026.

Siting ★



Korea's 11th Basic Plan for Electricity Supply and Demand allocates up to 700 MWe of power for the commercial demonstration of the i-SMR by 2036. KHNP is collaborating with local governments, including in the cities of Daegu and Gyeongju, on the deployment of the i-SMR and the development of the SMR Smart Net-Zero City (SSNC). The SSNC model proposes a carbon-neutral city powered by i-SMRs and renewable energy, and supporting hydrogen production, desalination, and district heating.

Financing



In 2021-2023, KHNP invested KRW 50 billion (USD₂₀₂₃ 38.3 million) to complete the i-SMR basic design. In 2023, the government of Korea announced plans to invest KRW 399.2 billion (USD₂₀₂₃ 305.7 million) from 2023 to 2028 to develop the standard design. In 2024, the government of Korea also announced plans to invest KRW 400 billion (USD₂₀₂₃ 306.3 million) from 2026 to 2030. In 2024, the Korean Ministry of Trade, Industry and Energy (MOTIE) announced that it will invest up to KRW 80 billion (USD₂₀₂₃ 61.3 million) to support the SMR industry in Korea, including the i-SMR. NSSC is investing KRW 36 billion (USD₂₀₂₃ 27.6 million) over 2022 to 2028 to establish regulatory pathways for SMRs, and KRW 47.4 billion (USD₂₀₂₃ 36.3 million) over 2025 to 2032 towards verification of SMR technologies.

Supply chain



KHNP has partnerships and in-house nuclear engineering, manufacturing and construction capabilities. To support i-SMR deployment, the government of Korea launched the SMR Alliance in 2023, bringing together 42 public and private entities, including MOTIE, KHNP, major Engineering, Procurement, and Construction (EPC) companies, suppliers, and industrial end-users. In 2024, KHNP launched an i-SMR simulator for design validation and operator training. Also in 2024, KHNP and MOTIE opened the SMR Smart Net-Zero City Control and Simulator Centre, at KHNP's Central Research Institute in Daejeon. KHNP is also working with Korean Advanced Institute of Science & Technology (KAIST) on the i-SMR development and Busan Research Institute on the development of the SSNC model.

Engagement



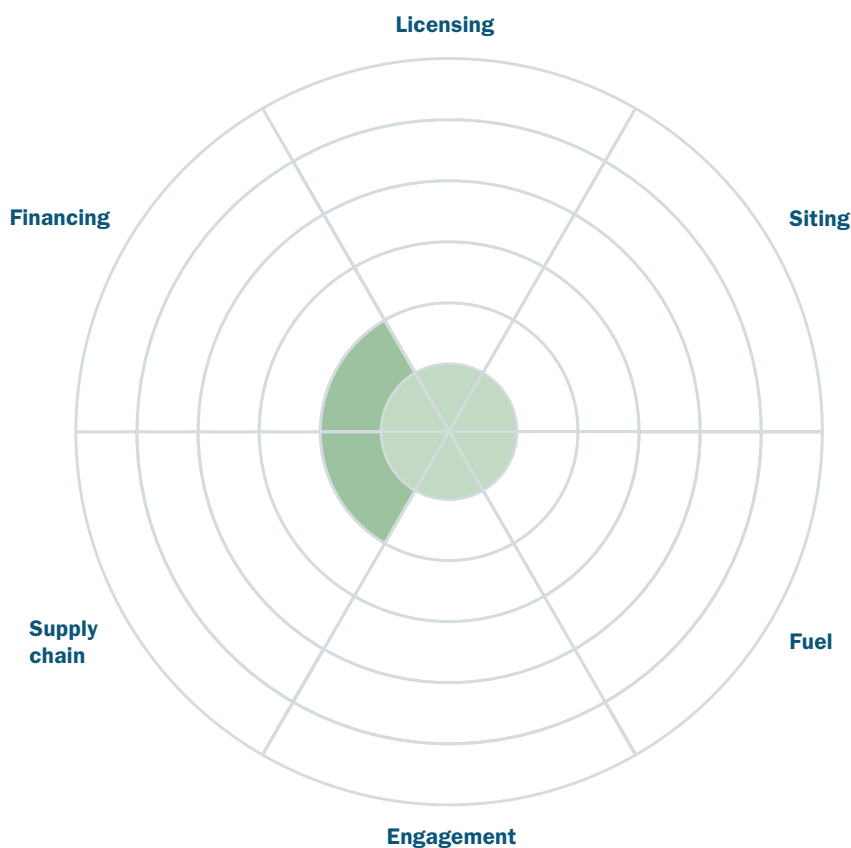
KHNP has signed Memoranda of Understandings (MoUs) with global partners in Indonesia, Jordan, Viet Nam, Norway, Sweden, and Finland to collaborate on i-SMR deployment. Domestically, KHNP is engaged with LS Electric on the potential development of data centres powered by i-SMR with the SSNC model. The i-SMR Development Agency is also engaged public outreach through media interviews and YouTube contents to raise awareness on the i-SMR. KHNP engaged international audiences at the 28th United Nations Climate Change Conference (COP28) with the public launch of its i-SMR powered SSNC model.

Fuel

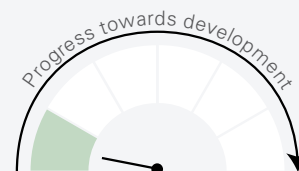


The i-SMR uses the same low-enriched uranium fuel as existing Generation III commercial reactors, manufactured by KEPCO Nuclear Fuel.

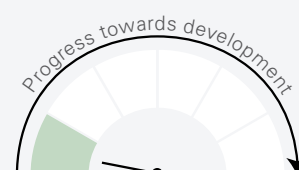
Japan Atomic Energy Agency (JAEA) – GTHTR300



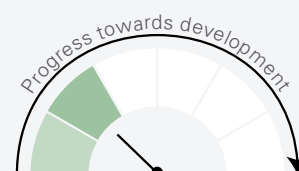
Licensing



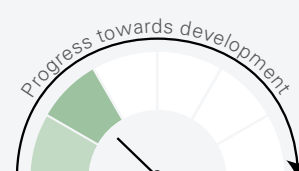
Siting



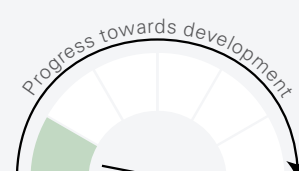
Financing



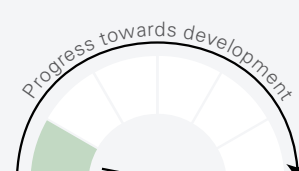
Supply chain



Engagement



Fuel

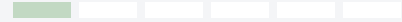


Reactor description: High temperature gas-cooled reactor designed as a commercial plant concept with a helium gas turbine.

Thermal power (MWth)	600
Core outlet temperature (°C)	950
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%

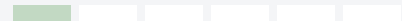
Assessment of GTHTR300's progress to deployment

Licensing



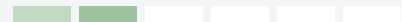
At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



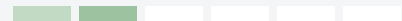
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



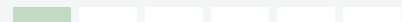
The basic design for the GTHTR300 was completed in 2003 by the Japan Atomic Energy Agency (JAEA) with funding from Japan's Ministry of Education, Culture, Sports, Science and Technology. At the time of assessment, no additional information about financing was readily available from verifiable public sources.

Supply chain



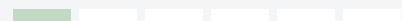
Mitsubishi Heavy Industries, Toshiba, IHI Corp, Fuji Electric and domestic nuclear fuel industries in Japan have existing supply chain capabilities developed under the HTTR project.

Engagement



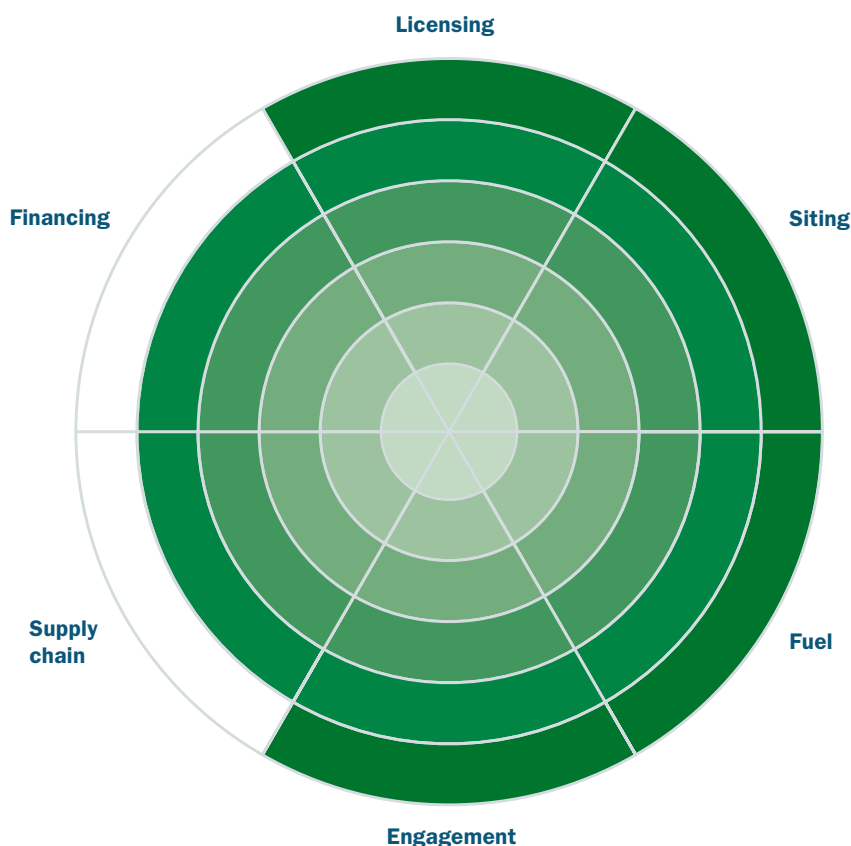
At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

Fuel



The GTHTR300 proposes to use HALEU fuel. In 2024, JAEA and the UK National Nuclear Laboratory signed a collaboration agreement to develop fuel manufacturing technology for an HTGR demonstration reactor. The work under this agreement may help inform and accelerate the development of fuel for the GTHTR300. At the time of assessment, no additional information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

Japan Atomic Energy Agency (JAEA) – HTTR



Licensing



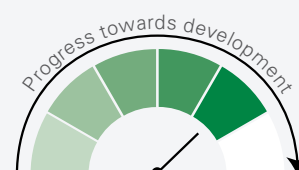
Siting



Financing



Supply chain



Engagement



Fuel



Reactor description: High temperature gas-cooled reactor designed as a commercial plant concept with a helium gas turbine.

Thermal power (MWth)	30
Core outlet temperature (°C)	850 or 950 in high-temperature test operation mode
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level: ≥ 5% to < 10%

Assessment of HTTR's progress to deployment

Licensing



The High Temperature Test Reactor (HTTR) is fully licensed by the Nuclear Regulatory Authority (NRA) of Japan. The first criticality of the HTTR was achieved on 10 November 1998. The reactor outlet coolant temperature of 850°C under full thermal power of 30 MWth was achieved on 6 December 2001. The reactor outlet coolant temperature of 950°C under full thermal power of 30 MWth was achieved on 19 April 2004. The Japan Atomic Energy Agency (JAEA) has been engaged in discussions with the NRA since 2023 about a hydrogen production facility utilising heat generated by the HTTR. JAEA aims to seek regulatory approval in 2025 and start construction of the facility in 2026 for operations in 2028.

Siting



The HTTR initially started operating at the site of the Oarai Research and Development Institute of the Japan Atomic Energy Agency (JAEA) in Oarai, Ibaraki Prefecture. The HTTR stopped operating in 2011 and was restarted in 2021. In June 2024, the HTTR was taken offline for maintenance.

Financing



The HTTR is a one-of-a-kind high temperature gas-cooled reactor (HTGR). The construction of the HTTR was fully financed by the Japanese government. The Ministry of Economy, Trade and Industry (METI) plans to invest a total of JPY 130.4 billion (USD₂₀₂₃ 958.8 million) over five years (2022-2026) in the HTGR Demonstration Reactor Development project, including a technology demonstration for large-scale hydrogen production utilising the HTTR.

Supply chain



The HTTR is a one-of-a-kind HTGR. The construction finished in 1997, led by Toshiba, Hitachi, Fuji Electric and Mitsubishi Heavy Industries (MHI), with MHI as the managing company. The JAEA is working with MHI on a project for mass production of hydrogen utilising the HTTR commissioned by the Ministry of Economy, Trade and Industry's Agency for Natural Resources and Energy.

Engagement



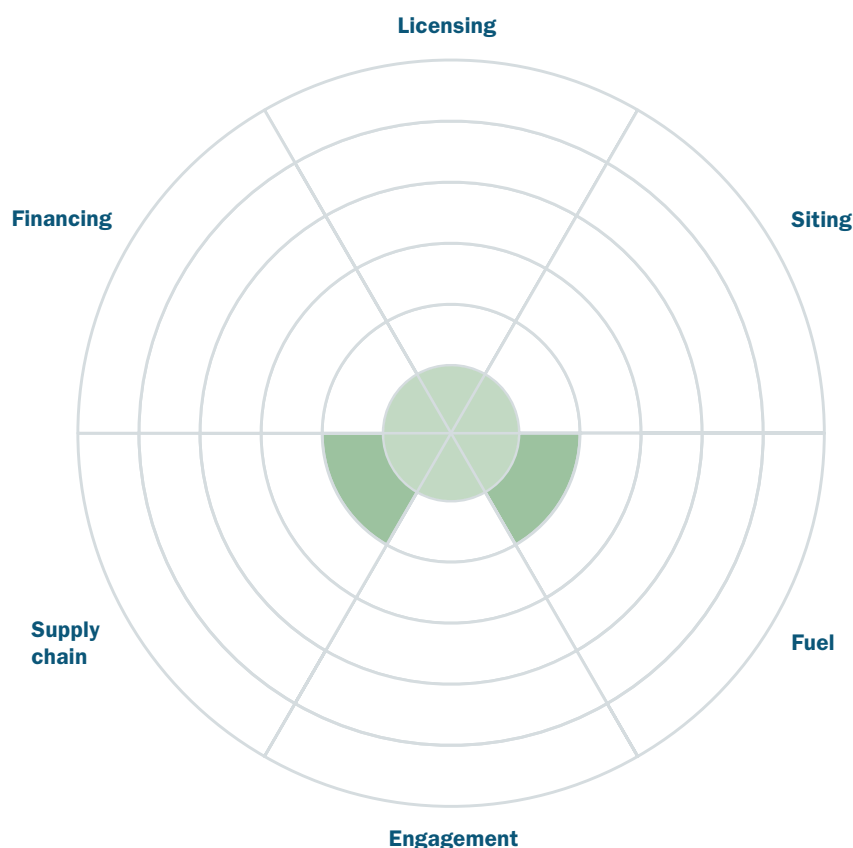
The JAEA has regularly hosted "Open Lab" events at its sites, including the Oarai site where the HTTR is located. Through these events, the JAEA has engaged with thousands of members of the public on its nuclear science and technology programme, including the HTTR. In addition, JAEA Oarai staff regularly deliver science lectures for the public, including discussions about the capabilities of the HTTR and the JAEA's research facilities. The JAEA also makes presentations and participates in discussions during meetings of the local government through the Ibaraki Prefecture Nuclear Safety Committee. In 2023, JAEA held multiple administrative consultation sessions with the local government to discuss plans for a hydrogen production facility connected to the HTTR.

Fuel

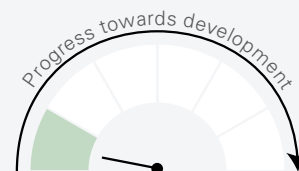


The first and second core fuels of the HTTR were manufactured by Nuclear Fuel Industries. The first core fuel was loaded in 1998, and the second core fuel is stored by the JAEA on site and is ready for the HTTR.

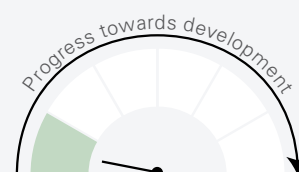
Japan Atomic Energy Agency (JAEA) – HTR50S



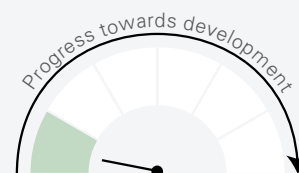
Licensing



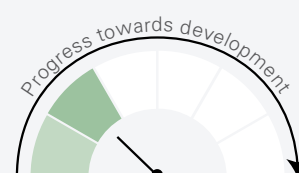
Siting



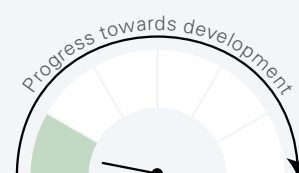
Financing



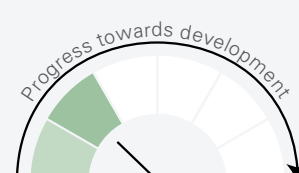
Supply chain



Engagement



Fuel



Reactor description: High temperature gas-cooled reactor for electricity production or cogeneration.

Thermal power (MWth) 50

Core outlet temperature (°C) 750

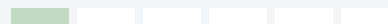
Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic
TRISO prismatic

Base fissile material U-235, enrichment level:
≥ 5% to < 10%

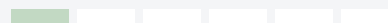
Assessment of HTR50S's progress to deployment

Licensing



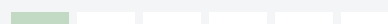
The HTR50S, designed by the Japan Atomic Energy Agency (JAEA), is a small-sized high temperature gas-cooled reactor (HTGR) intended primarily for deployment in developing countries for electricity production or cogeneration. JAEA plans to use conventional and proven technologies demonstrated through the construction and operation of the High Temperature Test Reactor (HTTR) in Oarai, Japan. In 2013, JAEA completed the conceptual design and preliminary safety analysis of the HTR50S. Licensing for the HTR50S will benefit from JAEA's prior experience in licensing and operating the HTTR. At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities for the HTR50S.

Siting



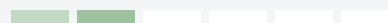
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



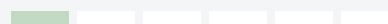
At the time of assessment, no information about financing was readily available from verifiable public sources.

Supply chain



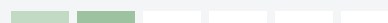
The conceptual design of the HTR50S was completed with support from Toshiba, Fuji Electric, Kawasaki Heavy Industries, and Nuclear Fuel Industries. Mitsubishi Heavy Industries, Toshiba, IHI Corp, Fuji Electric and domestic nuclear fuel Industries in Japan have existing supply chain capabilities developed under the HTTR project. The HTR50S is expected to benefit from Japan's existing HTGR supply chain capabilities developed under the HTTR project.

Engagement



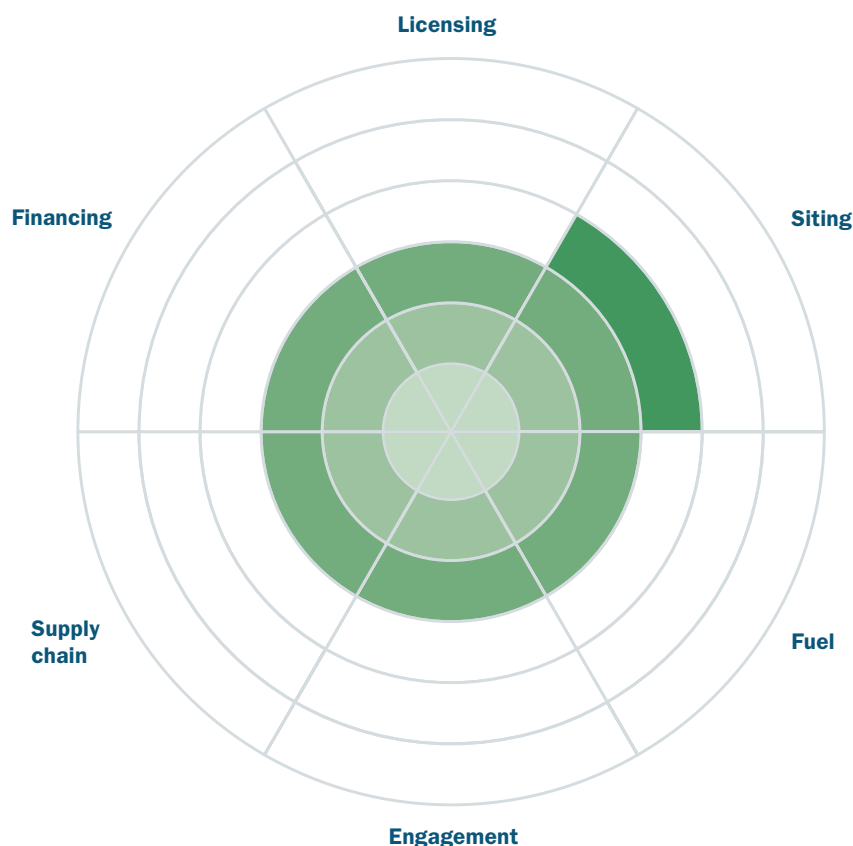
At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

Fuel



JAEA plans to use newly designed TRISO prismatic HALEU fuel with a higher burnup rate than the TRISO fuel used in the HTTR. A test sample of the new fuel particles has been manufactured by Nuclear Fuel Industries and irradiation testing was performed in the WWR-K reactor at the Institute of Nuclear Physics in Kazakhstan. In 2024, JAEA and the UK National Nuclear Laboratory signed a collaboration agreement to develop fuel manufacturing technology for an HTGR demonstration reactor. The work under this agreement may help inform and accelerate the development of fuel for the HTR50S.

Jimmy – Jimmy High Temperature Reactor (Jimmy HTR)



△ Changes since Second Edition (2024).

Reactor description: Gas-cooled prismatic block microreactor for the production of industrial process heat.

Thermal power (MWth)	- 10 for First-of-a-kind (FOAK) - 20 for Nth-of-a-kind (NOAK)
Core outlet temperature (°C)	550
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxycarbide ceramic TRISO prismatic
Base fissile material	- U-235, enrichment level: ≥ 5% to < 10% for FOAK - U-235, enrichment level: ≥ 10% to < 20% for NOAK

Licensing △



Siting △



Financing △



Supply chain △



Engagement △



Fuel △



Assessment of Jimmy HTR's progress to deployment

Licensing



In May 2024, Jimmy submitted two licensing applications to the French Nuclear Safety and Radiation Protection Authority (ASNR): a licence application to deploy its SMR at the Bazancourt site in France to supply heat to Cristal Union's food processing plant; and a licence application to build an assembly plant, including for nuclear fuel elements, near Le Creusot in France. Jimmy started pre-licensing activities with the French nuclear regulatory agencies in 2022.

Siting



Jimmy has submitted a licence application (Demande d'Autorisation de Création) to the French regulator to deploy its SMR at Cristal Union's food processing plant in Bazancourt, France.

Financing



In 2023, Jimmy was awarded EUR 32.0 million (USD₂₀₂₃ 34.6 million) from the French government under the France 2030 programme. In 2021-2022, Jimmy raised more than EUR 17.2 million (USD₂₀₂₃ 20 million) in private sector financing from investors that included Eren Group, Noria, Omnes, Otium Capital, Polytechnique Ventures, and Serena. Jimmy was one of the 127 awardees of the BPI France 2021-2022 competition for innovation under the i-Nov programme, which co-finances startups with up to EUR 5.0 million (USD₂₀₂₃ 5.7 million). Jimmy was also one of the awardees for two government-sponsored startup incubator programmes: French Tech Green20 and French Tech 2030.

Supply chain



In 2024, Jimmy signed a contract with Toyo Tanso, a Japanese company experienced in supplying graphite for nuclear applications, to provide graphite for Jimmy's HTR prismatic blocks. Earlier in 2024, Jimmy submitted a licence application to the ASNR to build its own industrial facility near Le Creusot, France. Jimmy plans for the facility to feature three plants for manufacturing of TRISO fuel, assembly of graphite blocks, and storage, testing and assembly of reactors components. Additionally, Jimmy is a member of the Nuclear Valley network, a group of over 400 companies involved in the French nuclear supply chain.

Engagement



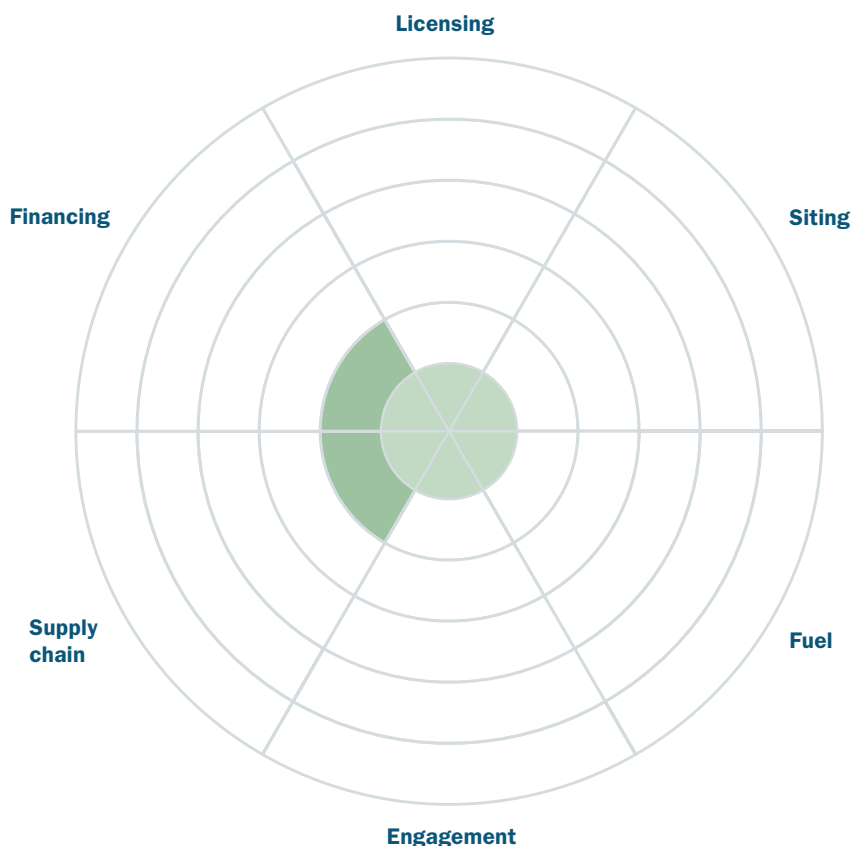
Jimmy has been engaging with the public through podcasts and interviews, including with Frenchweb, L'Union, Le Parisien, France 3 BFC, JSL, Les Échos, Business O Féminin, and Forbes. In 2024, the company also connected with local communities and government representatives in Le Creusot, where it plans to operate its own SMR manufacturing site. The French National Association of Local Information Committees and Commissions (ANCCLI), which aims to raise public awareness about nuclear energy, featured both Jimmy SMR and Calogena in its 2024 "Grand Dossier" on SMRs. In 2024, Jimmy also participated in a public workshop by ANCCLI dedicated to SMRs.

Fuel

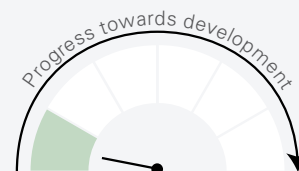


In 2024, Jimmy contracted with Urenco for uranium supply enriched to 9.75%. Urenco will produce the enriched uranium in its operating Eunice plant in New Mexico, United States, and plans its first delivery in 2026. Jimmy plans to use HALEU once it is commercially available.

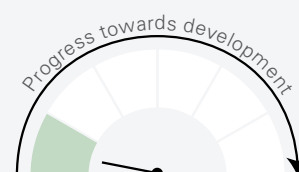
Korea Atomic Energy Research Institute (KAERI) – HElium Cooled Thermal Application Reactor (HECTAR)



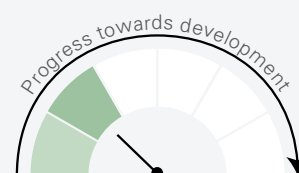
Licensing



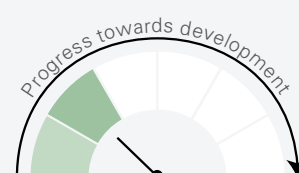
Siting



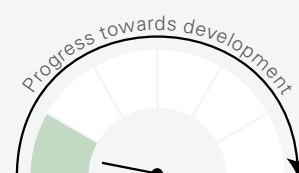
Financing



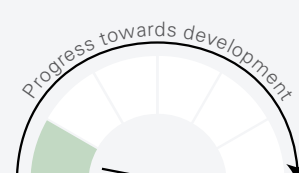
Supply chain



Engagement



Fuel

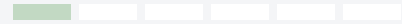


Reactor description: Land-based high temperature gas-cooled reactor for high temperature steam production.

Thermal power (MWth)	90
Core outlet temperature (°C)	750
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxycarbide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%

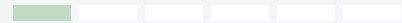
Assessment of HECTAR's progress to deployment

Licensing



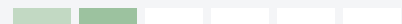
At the time of the assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



At the time of the assessment, no information related to siting was readily available from any site owners.

Financing



In 2024, the Korean government announced a project to support the development of the HECTAR SMR design by KAERI and five private companies: POSCO E&C, Daewoo E&C, Lotte Chemical, SK Ecoplant, and Smart Power. The total project budget is 44.5 billion KRW (USD₂₀₂₃ 34.1 million), with the government investing 22.5 billion KRW (USD₂₀₂₃ 17.2 million), and the five companies collectively contributing 22 billion KRW (USD₂₀₂₃ 16.9 million).

Supply chain



KAERI will lead on reactor design, while POSCO E&C, Daewoo E&C, Lotte Chemical, SK Ecoplant, and Smart Power will be responsible for designing and building the power plant. In 2024, KAERI and POSCO E&C signed a Memorandum of Understanding to establish a cooperative framework for the development of the HECTAR design.

Engagement



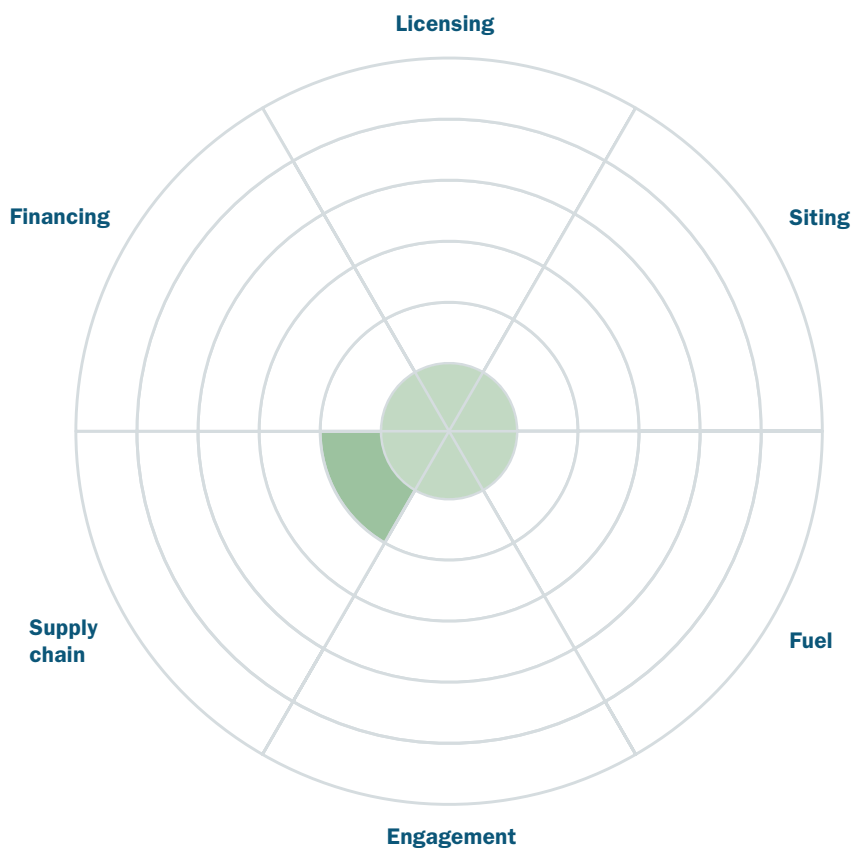
At the time of the assessment, no information was readily available from verifiable public sources related to engagement activities.

Fuel

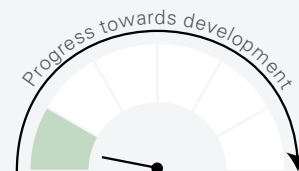


KAERI plans to use HALEU TRISO fuel for its HECTAR SMR. KAERI performed the irradiation testing of TRISO-coated particle fuel in 2018 using the research reactor HANARO. At the time of the assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

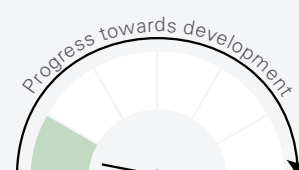
Korea Atomic Energy Research Institute (KAERI) – Small Advanced Long-cycled and Ultimate Safe SFR (SALUS-100)



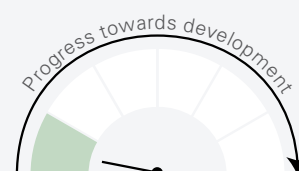
Licensing



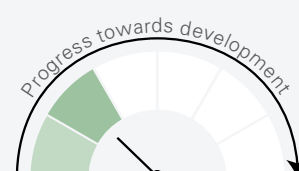
Siting



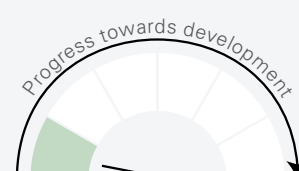
Financing



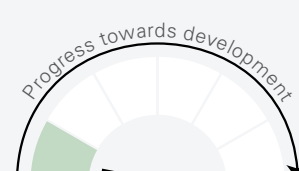
Supply chain



Engagement



Fuel

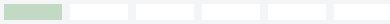


Reactor description: Sodium-cooled fast reactor using metallic fuel with a long refuelling cycle.

Thermal power (MWth)	267
Core outlet temperature (°C)	510
Spectrum (thermal/fast)	Fast
Fuel form	Metallic U-Zr alloy
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%

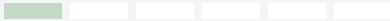
Assessment of SALUS-100's progress to deployment

Licensing



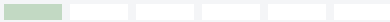
The SALUS-100 is a sodium-cooled fast reactor based on research and development for the Prototype Gen IV Sodium Cooled Fast Reactor (PGSFR) design by KAERI that started in 2012. In 2024, the SALUS-100 design completed the International Atomic Energy Agency's (IAEA) Technical Safety Review. At the time of the assessment, there was no additional information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



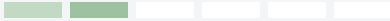
At the time of the assessment, no information related to siting was readily available from any site owners.

Financing



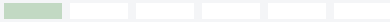
At the time of the assessment, no information about financing was readily available from verifiable public sources.

Supply chain



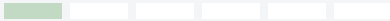
In 2024, KAERI and Hyundai Engineering & Construction signed a Memorandum of Understanding to establish a co-operative framework for the development of the SALUS-100 design.

Engagement



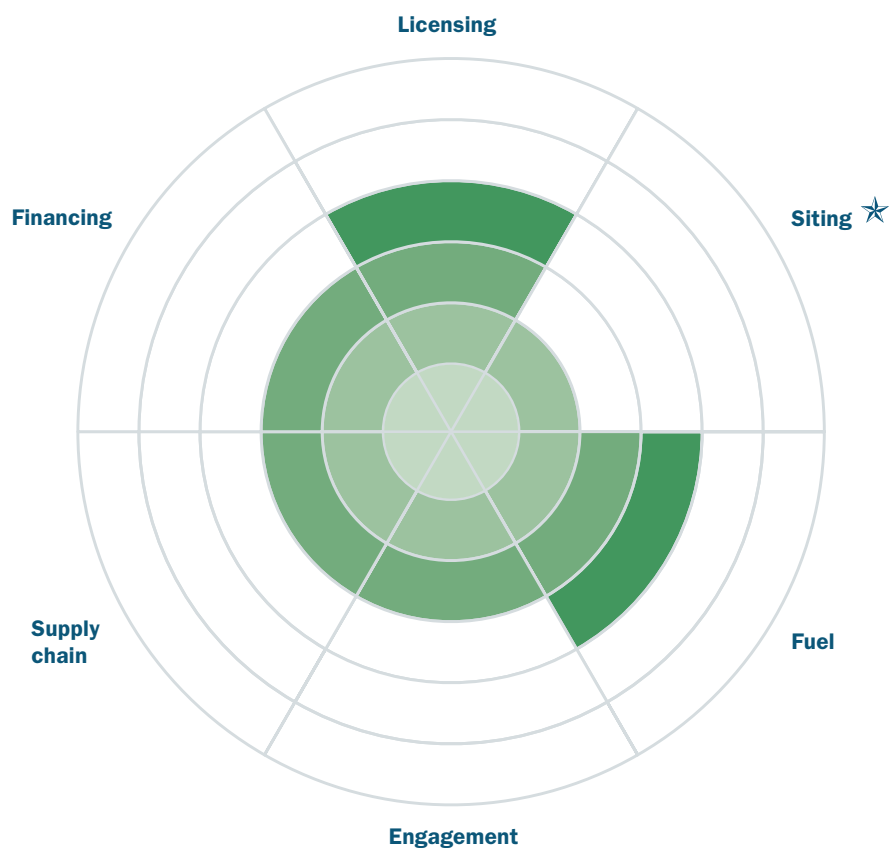
At the time of the assessment, no information was readily available from verifiable public sources related to engagement activities.

Fuel



The SALUS-100 proposes to use metallic uranium-zirconium alloy fuel enriched up to 19.0% with an eighteen-year refuelling cycle. At the time of the assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

Korea Atomic Energy Research Institute (KAERI) – SMART100



★ Active in multiple jurisdictions or countries.

Reactor description: Pressurised light water reactor.

Thermal power (MWth) 365

Core outlet temperature (°C) 322

Spectrum (thermal/fast) Thermal

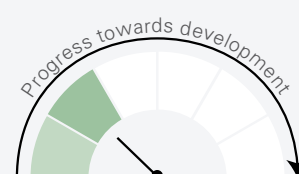
Fuel form Uranium-oxide ceramic

Base fissile material U-235, enrichment level: < 5%

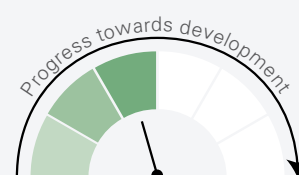
Licensing



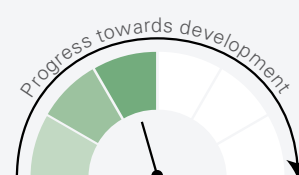
Siting ★



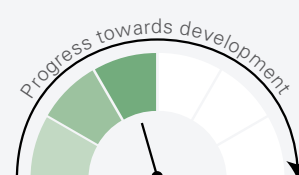
Financing



Supply chain



Engagement



Fuel



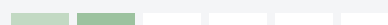
Assessment of SMART100's progress to deployment

Licensing



In 2024, Korea Atomic Energy Research Institute (KAERI), Saudi Arabia's King Abdullah City for Atomic and Renewable Energy (K.A.CARE), and Korea Hydro & Nuclear Power (KHNP) received Standard Design Approval (SDA) for the 110 MWe System-integrated Modular Advanced Reactor (SMART100) from the Korean Nuclear Safety and Security Commission (NSSC). SMART100 is an updated version of the SMART SMR that was originally designed by KAERI and received an SDA from NSSC in 2012. In 2019, K.A.CARE signed a Memorandum of Understanding (MoU) with Korea's Ministry of Science and ICT (MSIT) to create the SMART Partnership for Joint Commercialization, which facilitates training for Saudi engineers at the Korea Institute of Nuclear Safety and supports regulatory readiness in Saudi Arabia.

Siting ★



In 2015, KAERI signed a MoU with K.A.CARE in Saudi Arabia to assess the potential of siting multiple SMART SMRs in Saudi Arabia. In 2021, KAERI announced Yanbu as the candidate site for the first SMR100 in Saudi Arabia. In 2023, KAERI signed an MoU with the Atomic Energy of Canada Limited (AECL) to accelerate potential demonstration of the SMART100 reactor at Canadian Nuclear Laboratories in Chalk River, Ontario.

Financing



Development of the SMART100 benefits from financing related to the original SMART design. The Government of Korea, Korea Electric Power Corporation (KEPCO), and various others, including POSCO, Daewoo and STX Heavy Industries have contributed KRW 310 billion (USD₂₀₂₃ 315.7 million) for development of the SMART and the original SDA process. In 2015, MSIT announced that KAERI was partnering with K.A.CARE on pre-project engineering to construct SMART units in Saudi Arabia, supported by investments by the two partners totaling USD 130 million (USD₂₀₂₃ 167 million). The Government of Korea has also supported the development of the SMART through the National Research Foundation of Korea.

Supply chain



In 2020, KAERI and K.A.CARE agreed to establish the joint venture "SMART EPC" to develop SMART100 for Saudi Arabia. Hyosung Goodsprings developed the reactor coolant pumps; BHI designed the fuel handling system; Soosan ENS verified the reactor protection system and engineered the safety feature component control system; and Doosan Enerbility is providing design and engineering services for major components. KEPCO E&C and POSCO conducted the balance of plant design for the SMART SMR, with KEPCO Nuclear Fuel responsible for fuel design. KAERI also has an MoU with AECL for R&D and with Hyundai Engineering to advance deployment opportunities internationally.

Engagement



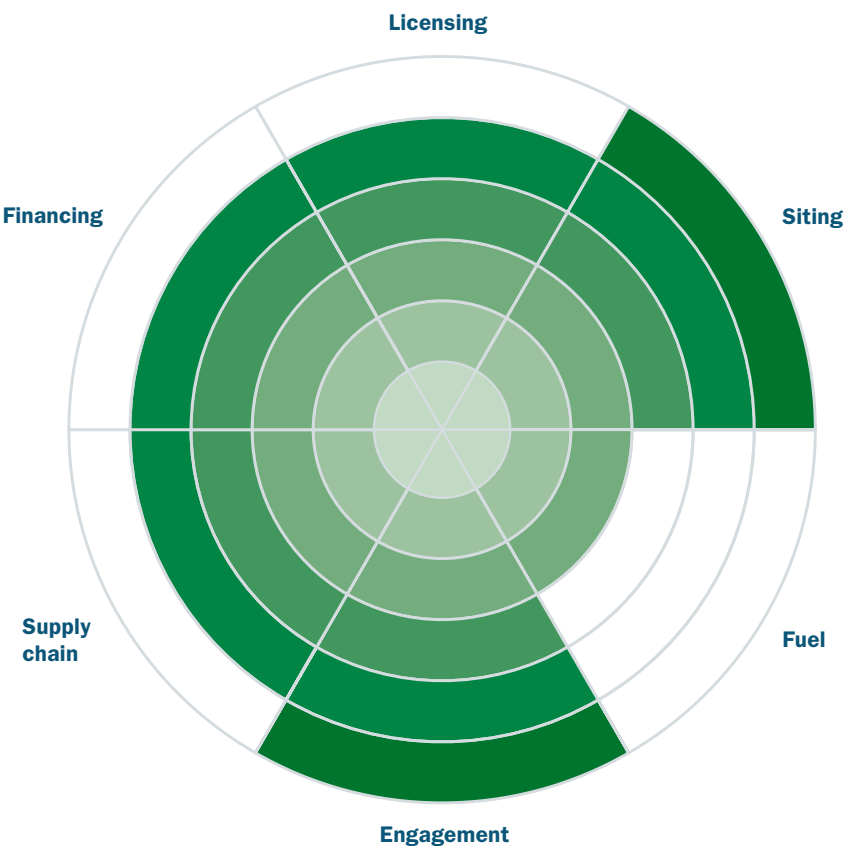
In 2023, KAERI and the Government of Alberta, Canada, signed an MoU to support the development of SMRs, including SMART100, for emissions reduction in Alberta. KAERI signed an MoU with K.A.CARE in 2019 that included collaborative efforts to deploy SMART100 in other countries. Through their ongoing collaboration, KAERI and K.A.CARE established a joint R&D centre which includes a Learning Pool for students in Saudi Arabia to learn about ongoing research projects at KAERI. In 2009, the Korea Atomic Energy Cultural Foundation and KAERI signed an MoU to collaborate on enhancing public understanding and awareness of nuclear technology, including the SMART SMR.

Fuel



KEPCO Nuclear Fuel designed the fuel for the SMART100, which is based on the Korea fuel assembly (KOFA) that is currently used in commercial Pressurised Water Reactors in Korea.

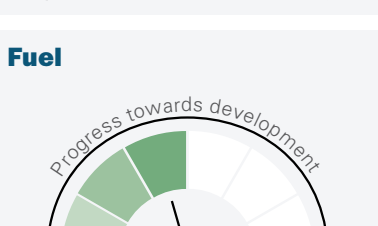
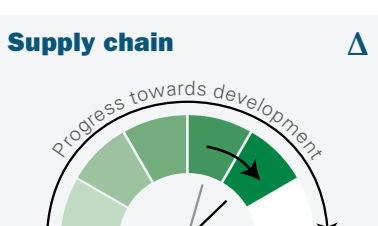
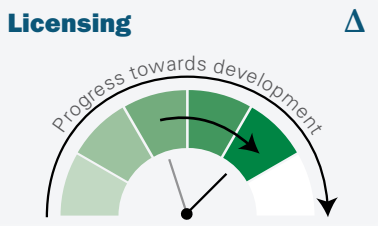
Kairos Power – Hermes



Δ Changes since Second Edition (2024).

Reactor description: High temperature fluoride salt-cooled demonstration reactor using TRISO pebble fuel.

Thermal power (MWth)	35
Core outlet temperature (°C)	650
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic TRISO pebble
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%



Assessment of Hermes' progress to deployment

Licensing



Hermes is a 35 MWth, non-power demonstration reactor planned on the East Tennessee Technology Park campus in Oak Ridge, Tennessee (United States), supporting the development of Kairos Power's 320 MWth (75 MWe) commercial KP-FHR technology. Kairos Power submitted a Construction Permit Application (CPA) to the US Nuclear Regulatory Commission (NRC) under a regulatory pathway distinct from commercial reactors' licensing process. In December 2023, the NRC issued a Construction Permit to Kairos Power for the Hermes reactor.

Siting



Kairos Power is working to deploy the Hermes test reactor on 30 acres of a larger plot purchased on the East Tennessee Technology Park campus in Oak Ridge, which was previously reclaimed from the US Department of Energy (DOE). In December 2023, the NRC issued a Construction Permit to Kairos Power to build the one-of-a-kind Hermes reactor on this site. Kairos Power has engaged Barnard Construction Company, a heavy-civil construction firm, to carry out site work and excavation at the Hermes site in Oak Ridge, with work commencing in July 2024.

Financing



In 2024, Kairos Power signed a power purchase agreement with Google for up to 500 MWe. Also in 2024, Kairos Power and the US Department of Energy (DOE) signed a Technology Investment Agreement to support the design, construction, and operation of the Hermes demonstration reactor. This agreement includes milestones and performance-based criteria to facilitate the payment of a USD 303 million (USD₂₀₂₃ 356.7 million) grant that Kairos Power was awarded in 2020 under the US Advanced Reactor Demonstration Program. Kairos Power is also providing private investments supporting the project and its associated infrastructure. Kairos Power was selected to receive over USD 10 million (USD₂₀₂₃ 12 million) in matching grants from various DOE initiatives in 2018-2019 to support the design and licensing of the KP-FHR.

Supply chain



Barnard Construction started work on the Hermes Oak Ridge site in 2024. Kairos Power is enhancing its in-house capabilities on its Albuquerque campus, focusing on system design, manufacturing, testing, salt production, and TRISO fuel and pebble development. Kairos Power is manufacturing non-nuclear Engineering Test Units (ETUs). The ETU 2.0 reactor vessel was completed in 2025. Kairos Power received ASME certification for U-stamped vessels in 2022 and began constructing a Salt Production Facility in 2024. A collaboration between Materion and Kairos Power produced 14 tons of salts by late 2023. The company has received multiple DOE GAIN vouchers. In 2022, Kairos Power launched its Operations, Manufacturing, and Development Alliance (KP-OMADA), an advisory group of North American utilities.

Engagement



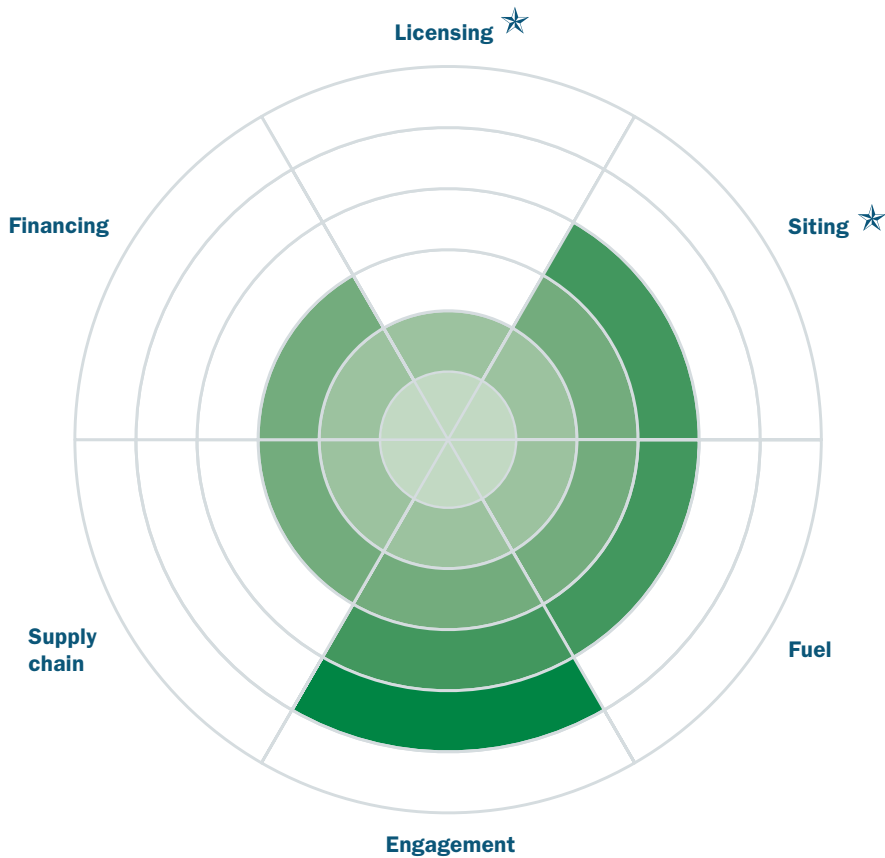
Kairos Power launched a virtual meeting space and partnered with the American Museum of Science and Energy to engage local communities. Kairos Power projects have been endorsed by government officials, including New Mexico Governor Michelle Grisham, Senator Martin Heinrich, Albuquerque Mayor Tim Keller, Tennessee Lieutenant Governor Randy McNally, and Oak Ridge Mayor Warren Gooch. Kairos Power has sponsored initiatives like Nuclear Science Week and Discover STEAM Week and participated in the UC Berkeley Clean Energy Speaker Series. In 2024, Texas A&M University System Chancellor John Sharp announced the university's potential interest in hosting SMRs from Kairos Power and other companies. Kairos Power received the BloombergNEF Pioneer Award in 2022 for advancements in nuclear technology.

Fuel



In 2023, Kairos Power and Urenco US signed a Memorandum of Understanding to explore future collaboration areas towards securing HALEU supply for Kairos KP-FHR commercial reactors. The Hermes test reactor is designed to use TRISO fuel pebbles, which will be produced at the New Mexico lab's Low Enriched Fuel Fabrication Facility (LEFFF) under an agreement between Los Alamos National Laboratory and Kairos Power. The LEFFF will host several fuel manufacturing processes being developed by Kairos Power at its TRISO and Pebble Development Labs in Albuquerque, New Mexico.

Last Energy – PWR-20



★ Active in multiple jurisdictions or countries.

△ Changes since Second Edition (2024).

Reactor description: Micro pressurised light water reactor.

Thermal power (MWth)	80
Core outlet temperature (°C)	300
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level: < 5%

Licensing ★



Siting ★



Financing



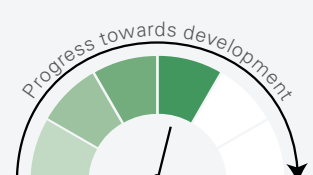
Supply chain



Engagement

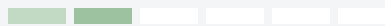


Fuel



Assessment of PWR-20's progress to deployment

Licensing ★



Last Energy has completed Tier 1 and Tier 2 of the early regulatory engagement framework for new nuclear projects with the UK Office of Nuclear Regulation (ONR). Last Energy has started Tier 3 with a Preliminary Design Review ongoing. As of January 2025, Last Energy entered the ONR nuclear site licensing process pre application phase. Last Energy is also engaged in pre-application activities with the US Nuclear Regulatory Commission (NRC) regarding a future application at Last Energy's Haskell, Texas site, and the review of white papers.

Siting ★



In 2024, Last Energy obtained site control in Bridgend County, Wales, where it plans to develop four PWR-20 micro modular nuclear reactors as part of the Llynfi Clean Energy Project. In 2022, Romania's Prime Minister announced plans to deploy Last Energy's SMR at RATEN's Mioveni site. In 2022 and 2023, Last Energy signed agreements in Poland, including virtual PPAs with the Katowice and Legnica Special Economic Zones and Letters of Intent with Enea for PWR-20 deployment. In 2023, it secured three PPAs in the United Kingdom – one virtual and two physical.

Financing



Last Energy secured a USD 100 million (USD₂₀₂₃ 97.1 million) Letter of Interest from the US Export-Import Bank for project financing to support its Llynfi Clean Energy Project in the United Kingdom. The company raised USD 40 million (USD₂₀₂₃ 38.8 million) in its Series B round in 2024, bringing total corporate financing to USD 66 million, in 2023 values, from investors including Gigafund and Autodesk Foundation. Last Energy has also announced power purchase agreements with industrial customers in Poland, Romania, and the United Kingdom.

Supply chain



In 2024, Last Energy engaged with 170 interested suppliers through an event co-organised with Industry Wales and the Welsh Nuclear Forum. Since 2022, Last Energy contracted Hydrock in the United Kingdom to review key components, started working with AtkinsRéalis UK (formerly Atkins UK) and Wood on the development of the PWR-20, joined Britain's Energy Coast Business Cluster, and was selected to form a Project-Working Group under the European Industrial Alliance on SMRs. In 2023, Last Energy built a full-scale mechanical prototype of the PWR-20 module in Texas, United States. Last Energy signed a letter of intent with the Polish manufacturer Energoinstal S.A. for the construction of a prototype module and presented it at the Polish Investment and Trade Agency Business Forum 2023 in Warsaw.

Engagement



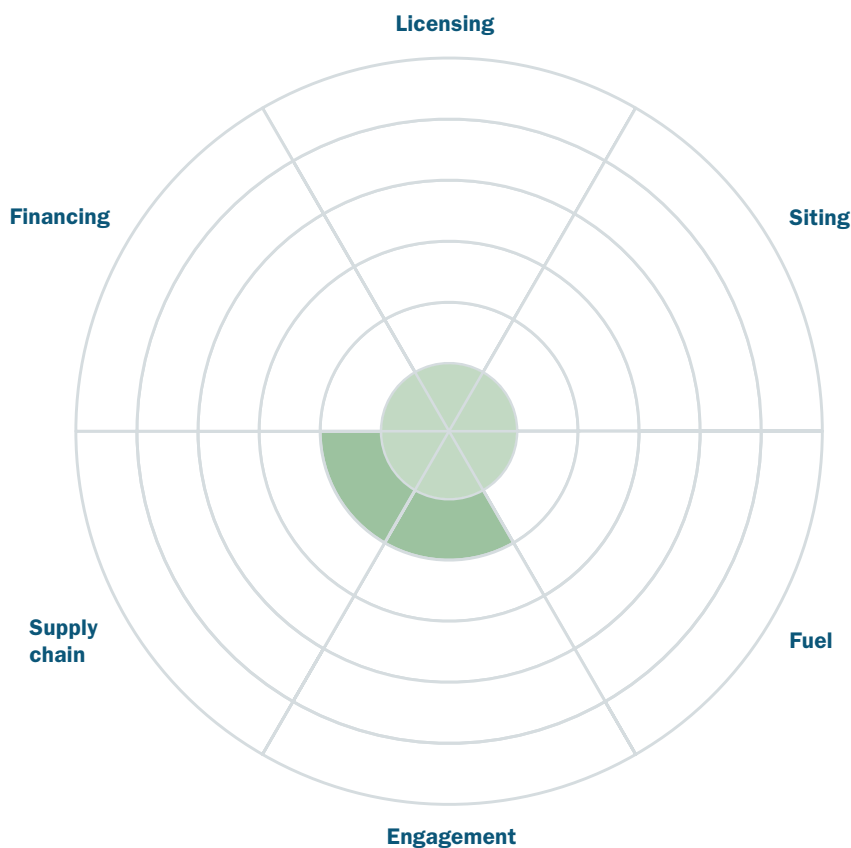
In December 2024, Last Energy began holding public consultations in Bridgend, Wales, United Kingdom, for the Llynfi Clean Energy Project. Lord Hunt, UK Minister for Energy Security and Net Zero, and Great British Nuclear expressed support for the launch of the Last Energy public engagement website for the Llynfi Clean Energy Project. In 2022, Last Energy testified in front of the UK Parliamentary Science and Technology Committee on modular manufacturing techniques. Last Energy has received endorsements from local and federal governments in Romania and Poland, including Prime Minister Nicolae Ciuca of Romania and Enea's CEO, Pawel Majewski. Last Energy has travelling exhibits with full scale non-nuclear, non-electrical mock ups that it has presented in the United States and Poland.

Fuel

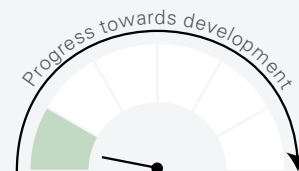


The PWR-20 SMR plans to use fuel that is the current industry standard for water-cooled reactor technologies of similar design. There are no barriers expected in the fuel supply chain for this SMR.

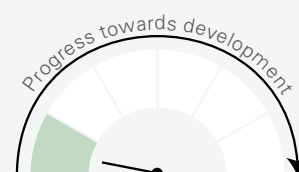
MobileNuclear Energy – MN-1



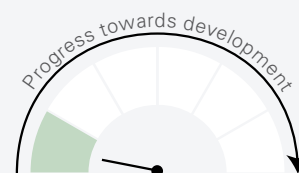
Licensing



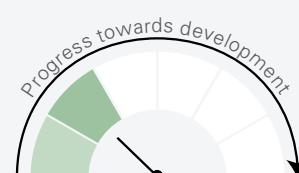
Siting



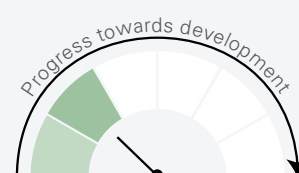
Financing



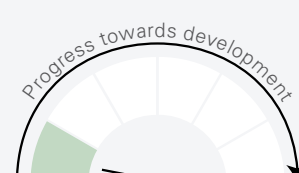
Supply chain



Engagement



Fuel



Reactor description: Mobile 1-MWth high-temperature gas-cooled reactor.

Thermal power (MWth) 1.0

Core outlet temperature (°C) 650

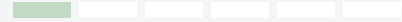
Spectrum (thermal/fast) Thermal

Fuel form Uranium-nitride ceramic and uranium-carbide ceramic

Base fissile material U-235, enrichment level $\geq 10\%$ to $< 20\%$

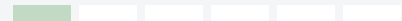
Assessment of MN-1's progress to deployment

Licensing



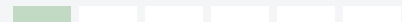
The MN-1 from MobileNuclear Energy is designed to provide energy for military installations, emergency response applications, off-grid sites, industrial sites, and marine propulsion. The MN-1 design couples a gas-cooled reactor with a super-critical carbon-dioxide (CO₂) turbine-generator. At the time of the assessment, no information about licensing readiness was readily available from verifiable public sources.

Siting



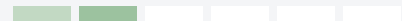
The MN-1 microreactor is designed to be a mobile system that can be transported and deployed at various types of sites. At the time of the assessment, no information related to specific deployment sites was readily available from any site owners.

Financing



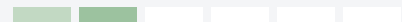
At the time of the assessment, no information about financing readiness was readily available from verifiable public sources.

Supply chain



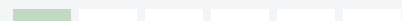
In 2024, MobileNuclear Energy partnered with Peregrine Turbine Technologies' Nuclear Energy Systems (PTT NES) to integrate supercritical carbon-dioxide energy conversion systems with the MN-1. In January 2025, MobileNuclear Energy signed a non-binding strategic partnership with Fenice Group, a company that proposes to use nuclear energy for applications such as luxury real estate and superyachts.

Engagement



The CEO of MobileNuclear Energy, Chris Pehrson, has advanced engagement with various audiences including through participation in a webinar organised in collaboration with the Society of American Military Engineers (SAME), focusing on advancements in mobile nuclear energy solutions and their applications across various sectors.

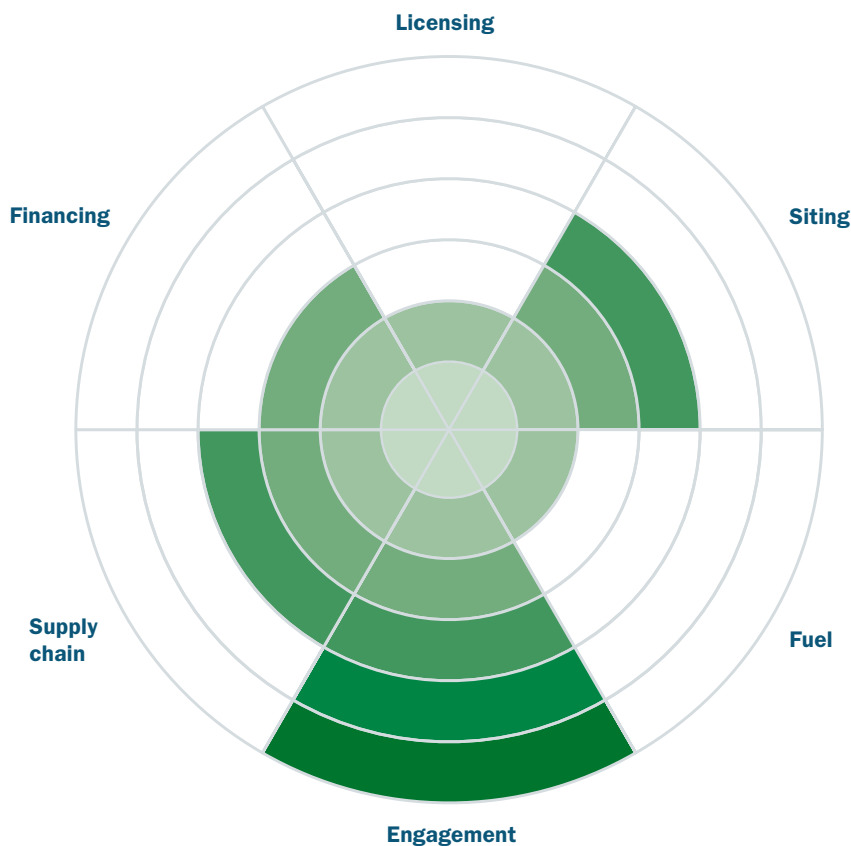
Fuel



At the time of the assessment, no information about fuel was readily available from verifiable public sources.

Moltex Energy

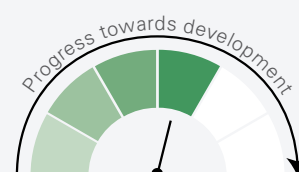
– Stable Salt Reactor-Wasteburner (SSR-W)



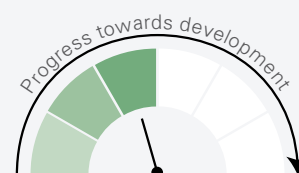
Licensing



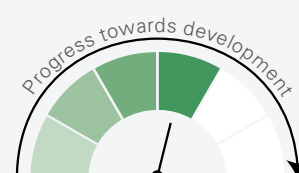
Siting



Financing



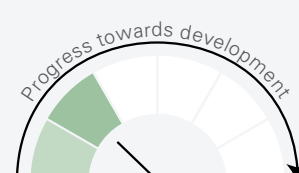
Supply chain



Engagement



Fuel



Reactor description: Molten salt reactor that plans to use recycled spent nuclear fuel.

Thermal power (MWth) 750

Core outlet temperature (°C) 590

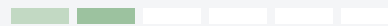
Spectrum (thermal/fast) Fast

Fuel form Molten chlorides

Base fissile material Plutonium, not requiring uranium enrichment

Assessment of SSR-W's progress to deployment

Licensing



The Stable Salt Reactor – Wasteburner (SSR-W) design from Moltex Energy (Moltex) has been engaged in pre-licensing activities with the Canadian Nuclear Safety Commission (CNSC) since 2017. Phase 1 of CNSC's pre-licensing vendor design review (VDR) was completed with Moltex submitting 50 documents in eight work packages to cover 19 technical review focus areas. Moltex has a service agreement in place with the CNSC for Phase 2 of the VDR.

Siting



The SSR-W has been selected by New Brunswick Power (NB Power) for deployment in the Canadian province of New Brunswick. Moltex has a Memorandum of Understanding (MoU) with NB Power to evaluate the construction of its first SSR-W at the Point Lepreau Nuclear Generating Station site. Moltex expects to have both its spent fuel recycling facility and reactor in operation by early- to mid-2030s. The company is also supporting NB Power with site evaluation and preparation activities.

Financing



Moltex has obtained funding from multiple public and private sources. Public grants have been received from the US Department of Energy's Advanced Research Projects Agency-Energy, the Canadian government, the Ontario Power Generation's Centre for Canadian Nuclear Sustainability and the UK government's Advanced Reactor Initiative. Multimillion investments have been secured from engineering company IDOM and a crowdfunding process via the Shadow Foundry platform. In March 2023, SNC-Lavalin (now AtkinsRéalis) made a minority equity investment. In September 2023, the North Shore Mi'kmaq Tribal Council and its 7 member communities made an equity investment of CAD 2.0 million (USD₂₀₂₃ 1.5 million).

Supply chain



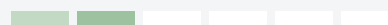
In the Canadian 2022 Provincial Strategic Plan for the Deployment of SMRs, the Government of New Brunswick set out its objective to advance supply chain readiness for SMRs, including Moltex, in collaboration with Canadian Manufacturers and Exporters and Opportunities New Brunswick. In April 2022, Moltex announced a partnership with AtkinsRéalis to support the licensing and construction of the first SSR-W in New Brunswick. IDOM, which has invested in Moltex's SSR-W technology, is also providing engineering support in various technical areas. Moltex has also been awarded a voucher under the Gateway for Accelerated Innovation in Nuclear (GAIN) initiative by the US DOE to study conjugate heat transfer in the SSR-W in partnership with Argonne National Laboratory.

Engagement



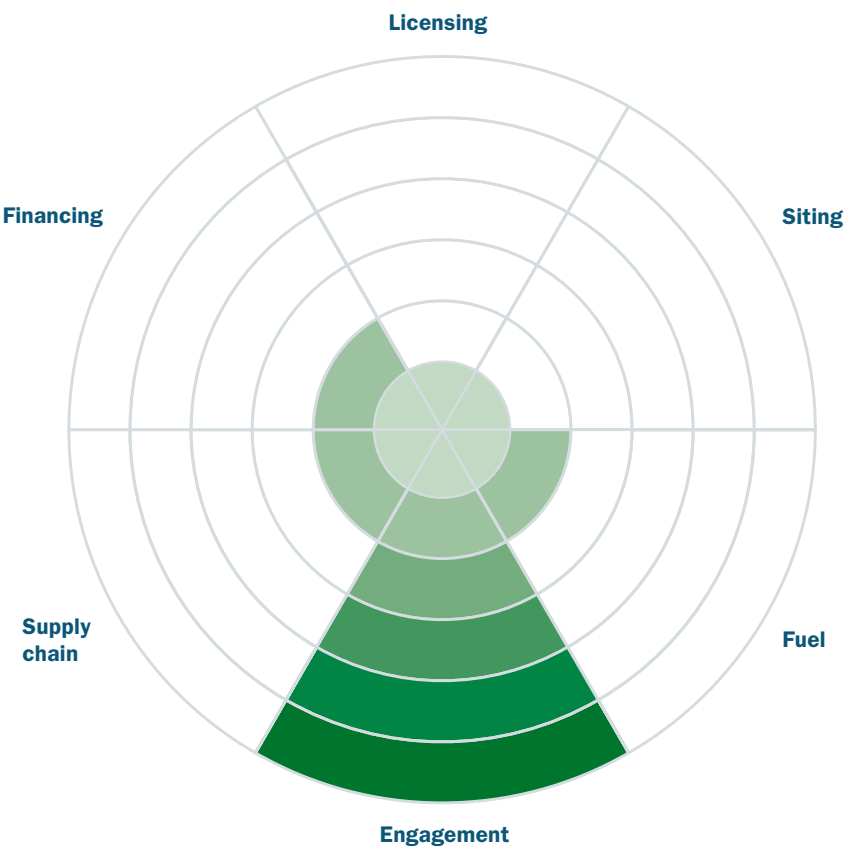
Moltex has received endorsements from provincial and federal governments in Canada. Moltex has sponsored engagement activities with various Indigenous and civil society organisations in the Atlantic Canada region. Moltex has a partnership with the North Shore Mi'kmaq Tribal Council and entered into an equity agreement in September 2023 with the North Shore Mi'kmaq Tribal Council and its member communities for benefits sharing. Moltex has advanced collaborations with the First Nations Power Authority to deepen Indigenous engagement on nuclear energy broadly and SMR technology specifically. The Centre for Nuclear Energy Research at the University of New Brunswick (UNB) is also engaged with Moltex on SMR research and talent development.

Fuel



Moltex Energy plans to fuel the SSR-W with spent Canada Deuterium Uranium (CANDU) reactor fuel recycled through the company's patented Waste to Stable Salt (WATSS) process, which has been experimentally validated in partnership with Canadian Nuclear Laboratories (CNL). The successful demonstration of the WATSS process was supported by a CAD 1.0 million (USD₂₀₂₃ 821 000) investment from Ontario Power Generation. UNB and CNL are supporting SSR-W fuel development through CNL's Canadian Nuclear Research Initiative. Moltex plans to produce the fuel for the SSR-W through the WATSS process using spent fuel from NB Power's Point Lepreau Nuclear Generating Station in Canada.

MoltexFLEX – FLEX

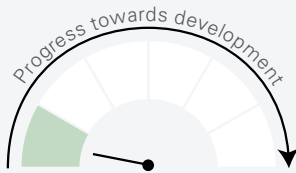


△ Changes since Second Edition (2024).

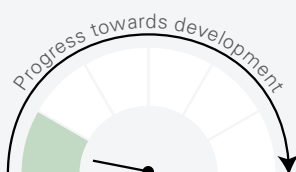
Reactor description: Molten High-temperature graphite moderated molten salt reactor using separate molten salts as the fuel and coolant.

Thermal power (MWth)	60
Core outlet temperature (°C)	700
Spectrum (thermal/fast)	Thermal
Fuel form	Molten fluorides
Base fissile material	U-235, enrichment level enrichment: < 5%

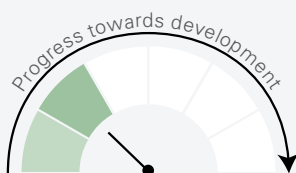
Licensing



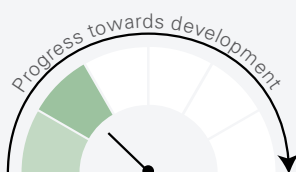
Siting



Financing



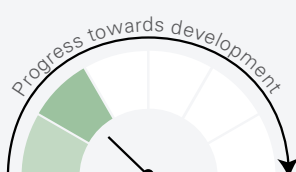
Supply chain



Engagement

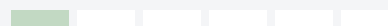


Fuel



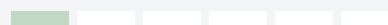
Assessment of FLEX's progress to deployment

Licensing



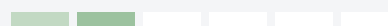
In 2024, MoltexFLEX met with the UK Office for Nuclear Regulation (ONR) and the UK Environment Agency (EA) to initiate early engagement on the regulatory process. At the time of assessment, there was no additional information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



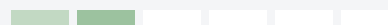
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



In February 2023, MoltexFLEX received a grant from the Henry Royce Institute to work with the University of Manchester's Nuclear Graphite Research Group to study the interactions between molten coolant salt and graphite.

Supply chain



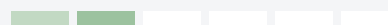
In 2020, MoltexFLEX selected Jacobs, an American engineering, procurement and construction company, to develop experimental capabilities at Jacob's Birchwood Park facility in Warrington, Cheshire (United Kingdom), to test thermal transfer behaviours of fuel and coolant molten salts. In 2023, the test rig for the FLEX in Warrington became operational with instruments supplied by manufacturer Anton Paar. The Hydrock engineering consultancy is developing the safety case and business strategy for the FLEX reactor. MoltexFLEX has also been collaborating with the University of Manchester's Nuclear Graphite Research Group to advance research on the FLEX reactor's graphite moderator, both at MoltexFLEX's laboratory in Warrington and at the University of Manchester in the United Kingdom.

Engagement



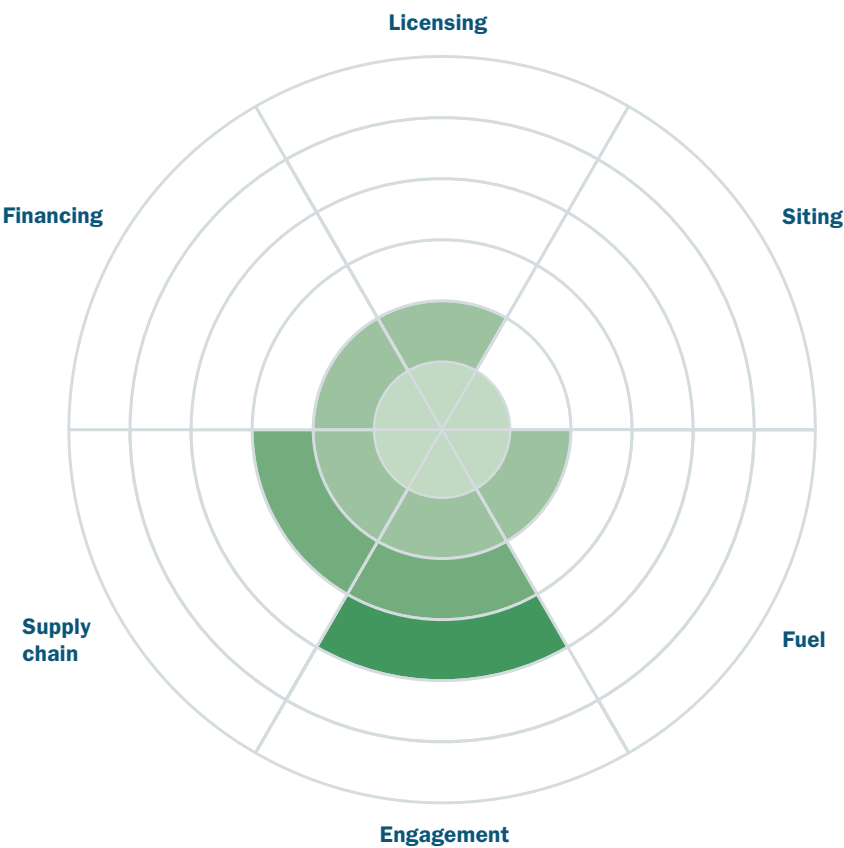
Since 2021, MoltexFLEX has hosted tours of its facility in Birchwood for Grant Shapps, the former UK Secretary of State for Energy Security and Net Zero, and UK Members of Parliament. In 2023, MoltexFLEX engaged the UK Parliament by providing oral and written evidence to the Energy Security and Net Zero Committee, and through the All-Party Parliamentary Group on Small Modular Reactors. MoltexFLEX won the UK government's Green Builder of Tomorrow competition in 2023. Also in 2023, MoltexFLEX signed a Memorandum of Understanding with Emirates Nuclear Energy Corporation for the protentional deployment of the FLEX reactor. MoltexFLEX has advanced public outreach through webinars, radio, and podcasts, and was recognised by the BBC in October 2022 as one of 39 ways to save the planet.

Fuel



The FLEX reactor proposes to use a mixture of low-enriched uranium fluoride and sodium fluoride salts as a fuel, and a separate fluoride salt as a coolant. In 2023, MoltexFLEX was offered a GBP 1.3 million (USD₂₀₂₃ 1.6 million) matching-grant as a part of the UK government's Nuclear Fuel Fund to develop capabilities to produce molten salt fuel for the FLEX SMR and secured equipment to support research on molten salt fuel properties.

NAAREA – XAMR



△ Changes since Second Edition (2024).

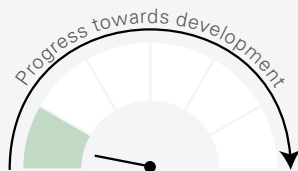
Reactor description: Transportable fast spectrum chloride molten salt reactor.

Thermal power (MWth)	80
Core outlet temperature (°C)	625
Spectrum (thermal/fast)	Fast
Fuel form	Molten chlorides
Base fissile material	Plutonium, not requiring uranium enrichment

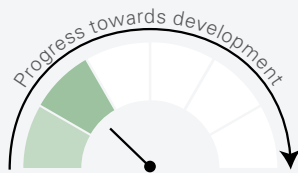
Licensing



Siting



Financing



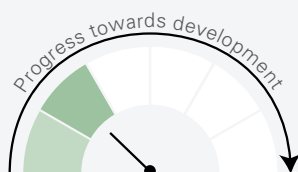
Supply chain



Engagement

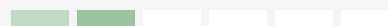


Fuel



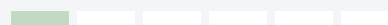
Assessment of XAMR's progress to deployment

Licensing



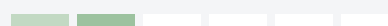
In 2023, NAAREA began pre-licensing engagement with the France's Nuclear Safety and Radiation Protection Authority (ASNR) for its XAMR design. In 2024, NAAREA completed the preparatory review stage of pre-licensing with ASNR.

Siting



At the time of assessment, no information related to siting was readily available from any site owners.

Financing



NAAREA raised USD 57.1 million (USD₂₀₂₃ 64.2 million) in 2021 and USD 5.2 million (USD₂₀₂₃ 5.5 million) in 2024 through early-stage venture capital rounds, including from EREN Group and Total Eren. In 2023, NAAREA was awarded EUR 10 million (USD₂₀₂₃ 10.8 million) as part of the France 2030 initiative to advance research and development activities for its XAMR design. NAAREA was also one of the awardees for the first edition of the French Tech 2030 programme, a government-sponsored startup incubator.

Supply chain



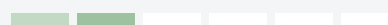
NAAREA signed contracts: in 2021 with Assystem for a digital twin; in 2024 with Phoenix Manufacture for additive manufacturing; and in 2024 with QGEMS for plant operating optimisation. In 2023, NAAREA built and operated the world's first silicon carbide molten salt loop in collaboration with Mersen Group and ICAR-CM2T laboratory. In 2024, NAAREA launched: the Innovation Molten Salt Lab with CNRS-IJCLAB laboratory and Université Paris-Saclay; and a collaboration with CNRS-IJCLAB lab and iUMTEK on tools for monitoring molten salts. NAAREA is also developing in-house capabilities with the opening in 2025 of its I-Lab, a facility for non-nuclear testing of XAMR materials and components. NAAREA has Memoranda of Understanding with fast-spectrum SMR developers newcleo and Thorizon.

Engagement



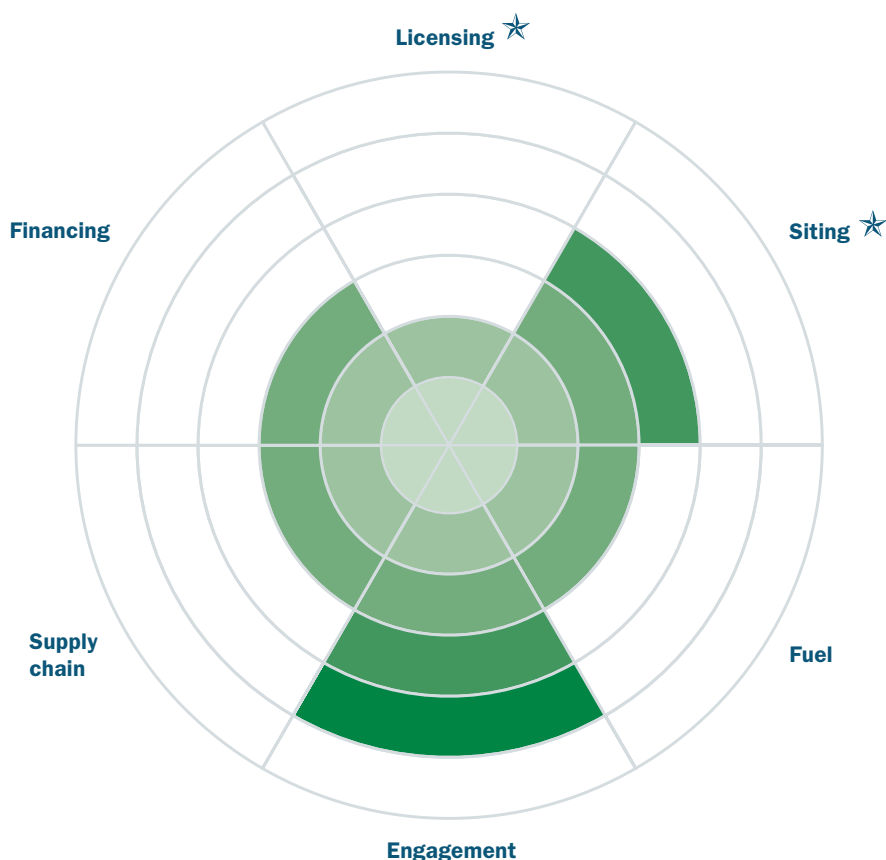
Throughout 2023, NAAREA invited several French Members of Parliament, including Maud Bregeon, Paul Midy, Antoine Armand and Raphael Schellenberger, to visit their office and present NAAREA's XAMR. Jean-Luc Alexandre, NAAREA's Chief Executive Officer, also gave interviews on SMRs and the XAMR to mainstream media outlets, including L'Opinion, Sciences et Avenir and Europe 1. NAAREA has also advanced engagement with end-users, including with EO Concept for hydrogen and synthetic fuel production, and with Automotive Cells Company for its gigafactories power needs. In 2024, NAAREA hosted delegations of students and young professionals to learn about the XAMR in collaboration with the OECD Nuclear Energy Agency.

Fuel



In 2023, NAAREA and ROBATEL, a nuclear equipment manufacturer, announced a collaboration to conduct feasibility studies for XAMR fuel management. In 2024, NAAREA joined a working group with Orano and two other fast spectrum molten-salt fuelled SMRs to develop a joint strategy for liquid molten-salt fuel supply, logistic and processing in France.

NANO Nuclear Energy – KRONOS MMR

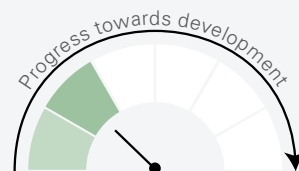


★ Active in multiple jurisdictions or countries.

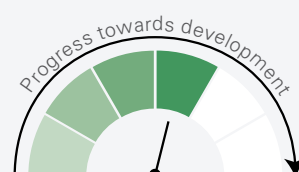
Reactor description: High-temperature gas-cooled reactor using TRISO fabricated into fully ceramic microencapsulated fuel pellets.

Thermal power (MWth)	10 to 45
Core outlet temperature (°C)	660
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic TRISO prismatic
Base fissile material	- U-235, enrichment level ≥ 5% to < 10% for the FOAK - U-235, enrichment level ≥ 10% to < 20% for NOAK

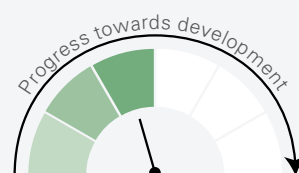
Licensing ★



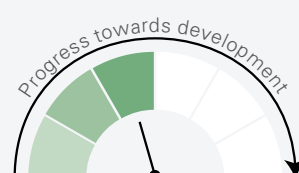
Siting ★



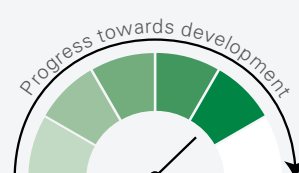
Financing



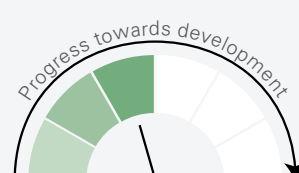
Supply chain



Engagement

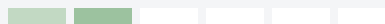


Fuel



Assessment of KRONOS MMR's progress to deployment

Licensing ★



NANO Nuclear Energy (Nano) acquired the patented Micro Modular Reactor (MMR) design from Ultra Safe Nuclear Corporation (USNC) in December 2024 and is continuing to progress the design as the KRONOS MMR. In Canada, the MMR design completed Phase 1 of the Canadian Safety Nuclear Commission's (CNSC) pre-licensing vendor design review process. In the United States, Nano is continuing to collaborate with the University of Illinois at Urbana-Champaign (UIUC) to deploy the KRONOS MMR as a research reactor on the campus and is engaged in pre-licensing activities with the US Nuclear Regulatory Commission (NRC). In November 2024, Nano and Everstar AI completed a successful pilot of Everstar's "Gordian" model aiming to simplify and streamline Nano's regulatory and licensing workflows.

Siting ★



In the United States, a Letter of Intent was submitted to the NRC in May 2021 to submit an application to construct the MMR design developed by USNC at the University of Illinois Urbana-Champaign (UIUC). In January 2025, Nano re-affirmed their commitment to collaborate with UIUC to construct the first KRONOS MMR on the university's campus for research applications. In Canada, Canadian Nuclear Laboratories (CNL) established a project host agreement in 2020 to deploy the MMR design at Chalk River Laboratories (CRL). Nano also reaffirmed its intent to conduct demonstrations of the KRONOS MMR at CRL.

Financing



In December 2024, Nano acquired USNC for USD 8.5 million (USD₂₀₂₃ 8.2 million), which included the patented MMR design. After an initial public offering with NASDAQ Capital Markets, Nano became publicly traded in May 2024. Since April 2022, Nano has raised more than USD 120 million in both public markets and through private financing to support four microreactor designs: KRONOS MMR, LOKI MMR, ODIN, and ZEUS.

Supply chain



Nano is establishing a facility in Westchester County, New York, United States to demonstrate non-nuclear components of the company's microreactor technologies, including the KRONOS MMR. In February 2025, Nano announced that it had engaged aRobotics Company to oversee construction of the facility.

Engagement



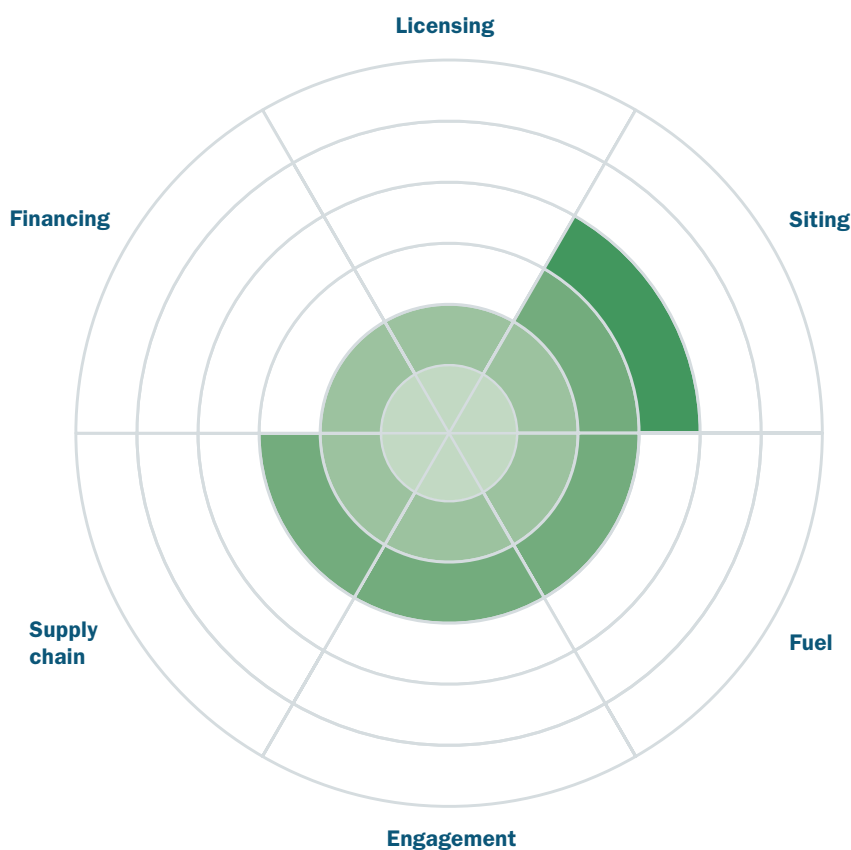
In 2024, Nano signed a Memorandum of Understanding (MoU) with the Rwanda Atomic Energy Board to strengthen Rwanda's educational programmes in nuclear energy in collaboration with the United Kingdom's Cambridge Nuclear Energy Centre. Also in 2024, Nano responded to a New York State Energy Research and Development Authority Request for Information on advanced nuclear energy and announced an MoU with the Government of the Togolese Republic to evaluate opportunities and licensing in Togo. UIUC professors and students have discussed the KRONOS MMR project on local news outlets, radio stations, podcasts, and through monthly public meetings, and the Illinois Institute for Sustainability, Energy, & Environment held a community event in 2020 about the KRONOS MMR project at UIUC.

Fuel

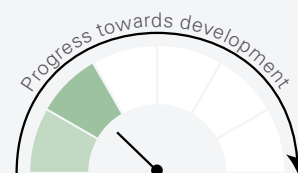


KRONOS MMR plans to use HALEU. UIUC announced it will source KRONOS MMR fuel from the US Research Reactor Infrastructure Program. In 2024, Nano invested USD 2 million (USD₂₀₂₃ 1.9 million) into LIS Technologies (LIST) laser isotope separation enrichment technology. In 2024, the US DOE LEU Acquisition Program awarded up to USD 2 million (USD₂₀₂₃ 1.9 million) for LIST to develop enrichment capabilities, and for Nano to develop deconversion, fabrication, and transportation capabilities. Nano is working through its subsidiaries: HALEU Energy Fuel, to develop HALEU fuel supply in the US; American Uranium, to advance exploration and mine development; and Fuel Transportation, to work on HALEU transportation systems. In 2024, Nano signed an agreement with GNS Gesellschaft für Nuklear-Service to develop HALEU transportation technologies.

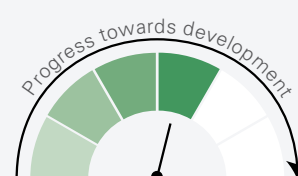
NANO Nuclear Energy – LOKI MMR



Licensing



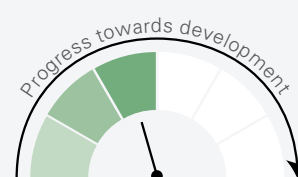
Siting



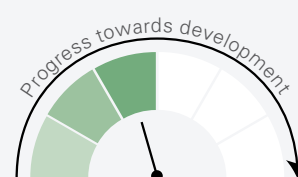
Financing



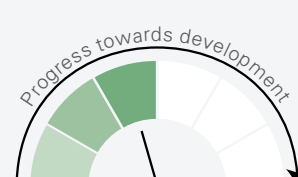
Supply chain



Engagement



Fuel



Reactor description: Transportable high-temperature gas-cooled reactor for off-grid applications and space.

Thermal power (MWth) 5

Core outlet temperature (°C) 727

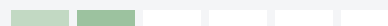
Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic
TRISO prismatic

Base fissile material U-235, enrichment level
≥ 5% to < 10%

Assessment of LOKI MMR's progress to deployment

Licensing



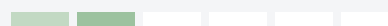
NANO Nuclear Energy (Nano) acquired the Pylon D1 space reactor design from Ultra Safe Nuclear Corporation (USNC) in December 2024 and is continuing to progress the design as the LOKI MMR. Nano has announced its intent to continue preparations for the planned Pylon D1 reactor demonstration, which was selected by the National Reactor Innovation Center (NRIC), a US Department of Energy (DOE) programme led by Idaho National Laboratory (INL), for deployment at the Demonstration Of Microreactor Experiments (DOME) facility at INL. NRIC has been supporting regulatory engagement activities with the US Nuclear Regulatory Commission (US NRC), the DOE, and reactor demonstrators to determine an appropriate regulatory strategy for the deployment and testing of microreactors at the DOME facility.

Siting



NRIC has selected the LOKI MMR (formerly called Pylon D1) to be demonstrated at INL's DOME facility in Idaho, United States. Nano intends to continue preparations for the reactor demonstration to deploy the LOKI MMR at the DOME facility by 2027.

Financing



In December 2024, Nano acquired USNC for USD 8.5 million (USD₂₀₂₃ 8.2 million), which included the Pylon D1 design. After an initial public offering with NASDAQ Capital Markets, Nano became publicly traded in May 2024. Since April 2022, Nano has raised more than USD 120 million in both public markets and through private financing to support four microreactor designs: KRONOS MMR, LOKI MMR, ODIN, and ZEUS.

Supply chain



Nano is establishing a facility in Westchester County, New York to demonstrate non-nuclear components of the company's microreactor technologies, including the LOKI MMR. In February 2025, Nano announced that they had engaged aRobotics Company to oversee construction of the facility. Through the NRIC FEEED process, Nano will develop an experimental programme plan for the LOKI MMR, Class 4 cost estimates, and the contract for the work to be performed at INL.

Engagement



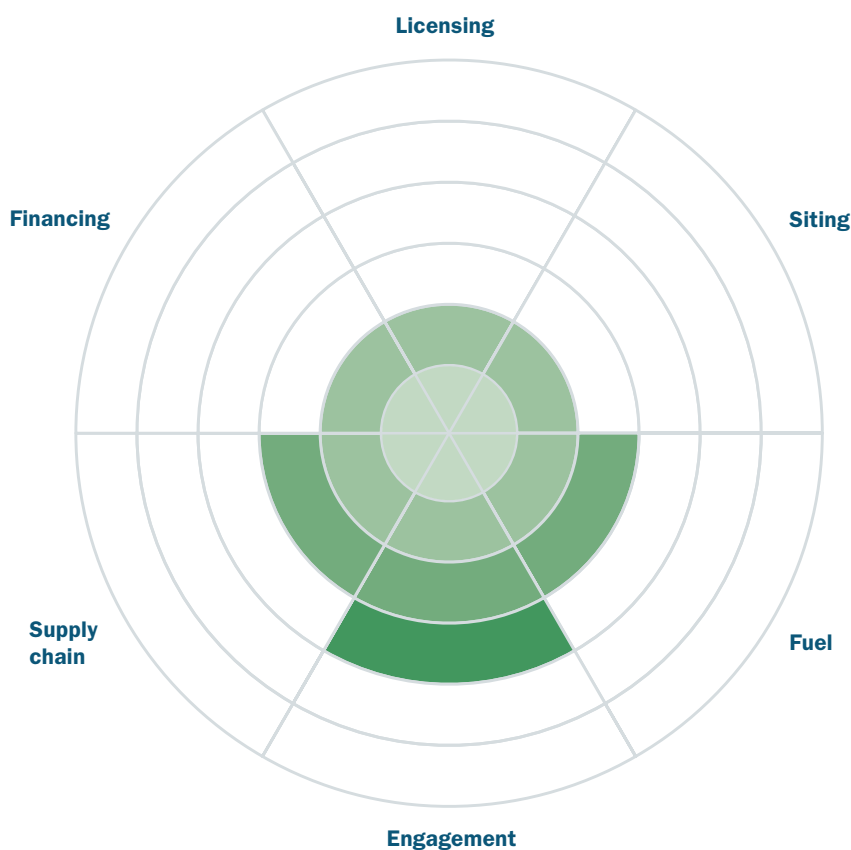
In 2024, Nano signed a Memorandum of Understanding (MoU) with the Rwanda Atomic Energy Board to strengthen Rwanda's educational programmes in nuclear energy in collaboration with the United Kingdom's Cambridge Nuclear Energy Centre. In 2024, Nano announced an MoU with the Government of the Togolese Republic to evaluate opportunities in Togo. The LOKI MMR concept was presented and discussed as part of the Space Technology Industry-Government-University Roundtable in October 2023, which convened senior representatives to explore research issues and options for public-private partnerships for space technologies.

Fuel

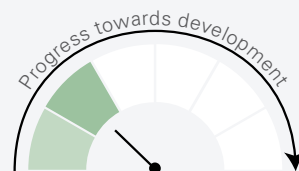


Nano plans to use HALEU for its LOKI MMR. In 2024, Nano invested USD 2.0 million (USD₂₀₂₃ 1.9 million) into LIS Technologies (LIST) laser isotope separation enrichment technology. In 2024, the US DOE LEU Acquisition Program awarded up to USD 2.0 million (USD₂₀₂₃ 1.9 million) for LIST to develop enrichment capabilities, and for Nano to develop deconversion, fabrication, and transportation capabilities. Nano is working through its subsidiaries: HALEU Energy Fuel, to develop HALEU fuel supply in the US; American Uranium, to advance exploration and mine development; and Fuel Transportation, to work on HALEU transportation systems. In 2024, Nano signed an agreement with GNS Gesellschaft für Nuklear-Service to develop HALEU transportation technologies.

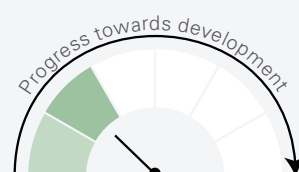
NANO Nuclear Energy – ODIN



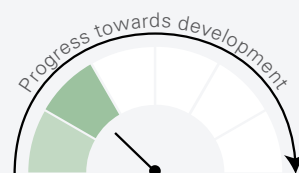
Licensing



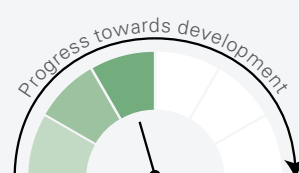
Siting



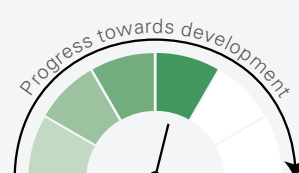
Financing



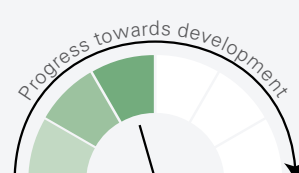
Supply chain



Engagement



Fuel

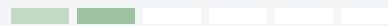


Reactor description: Low-pressure, salt-cooled microreactor using thermal convection to circulate heat.

Thermal power (MWth)	5
Core outlet temperature (°C)	400
Spectrum (thermal/fast)	Thermal
Fuel form	Other (uranium hydride fuel)
Base fissile material	U-235, enrichment level ≥ 10% to < 20%

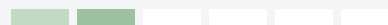
Assessment of ODIN's progress to deployment

Licensing



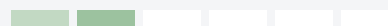
In October 2024, NANO Nuclear Energy Inc (Nano) formally requested to initiate the pre-application review and engagement process for the ODIN SMR with the US Nuclear Regulatory Commission (NRC), with the intent to submit a Regulatory Engagement Plan and begin Technical Information Exchange meetings. In November 2024, Nano and Everstar AI completed a successful pilot of Everstar's "Gordian" model aiming to simplify and streamline Nano's regulatory and licensing workflows.

Siting



In December 2024, Nano announced that it had signed a Memorandum of Understanding (MoU) with the US Department of Energy DOE to evaluate the feasibility of siting an ODIN SMR at Idaho National Laboratory (INL) in Idaho, United States.

Financing



After an initial public offering with NASDAQ Capital Markets, Nano became publicly traded in May 2024. Since April 2022, Nano has raised more than USD 120 million in both public markets and through private financing to support four microreactor designs: KRONOS MMR, LOKI MMR, ODIN, and ZEUS.

Supply chain



Nano is establishing a facility in Westchester County, New York to demonstrate non-nuclear components of the company's microreactor technologies, including the ODIN SMR. In February 2025, Nano announced that they had engaged aRobotics Company to oversee construction. In June 2024, Nano acquired intellectual property for an annular linear induction pump necessary to circulate the salt coolant and, in January 2025, announced that it had contracted Thermal Engineering International to support design and fabrication of heat exchangers for the ODIN SMR. In February 2024, INL completed an external pre-conceptual review of the ODIN SMR concept.

Engagement



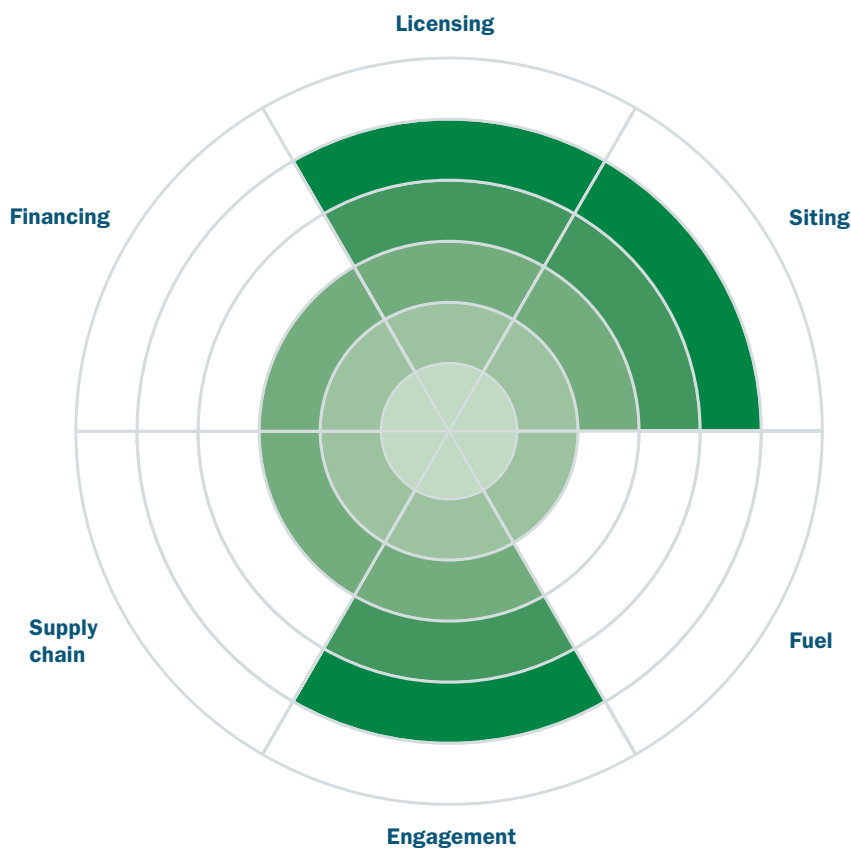
In August 2024, Nano signed an MoU with the Rwanda Atomic Energy Board to strengthen Rwanda's educational programmes in nuclear energy in collaboration with the United Kingdom's Cambridge Nuclear Energy Centre. In December 2024, Nano responded to a New York State Energy Research and Development Authority Request for Information on advanced nuclear energy, announced an MoU with the Government of the Togolese Republic to evaluate opportunities and licensing in Togo, and signed an MoU with Vert2Grow Energy Solutions to explore the feasibility of integrating microreactors with vertical farming with a pilot project targeting regions across sub-Saharan Africa, South America, Southeast Asia, and Northern Canada. Nano has an Executive Advisory Board composed of military, scientific and governmental experts.

Fuel

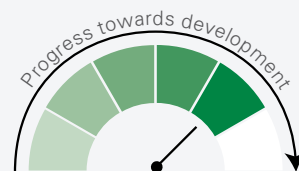


Nano plans to use HALEU for ODIN. In 2024, Nano invested USD 2 million (USD₂₀₂₃ 1.9 million) into LIS Technologies (LIST), a proprietary developer of laser isotope separation enrichment technology. In 2024, the US DOE LEU Acquisition Program awarded up to USD 2 million (USD₂₀₂₃ 1.9 million) for: LIST to develop enrichment capabilities; and Nano to develop deconversion, fabrication, and transportation capabilities. Nano is also working through its subsidiaries: HALEU Energy Fuel, to develop HALEU fuel supply in the United States; American Uranium, to advance exploration and mine development; and Fuel Transportation, to work on HALEU transportation systems. In 2024, Nano signed an agreement with GNS Gesellschaft für Nuklear-Service in Germany on a range of HALEU transportation technologies.

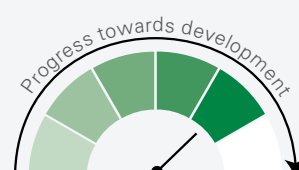
Natura Resources and Abilene Christian University (ACU) – MSR-1



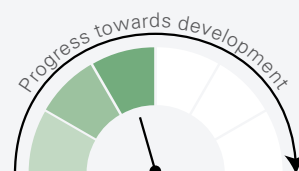
Licensing



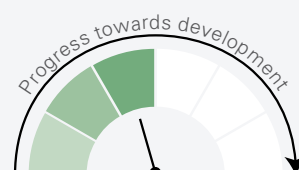
Siting



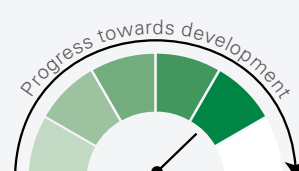
Financing



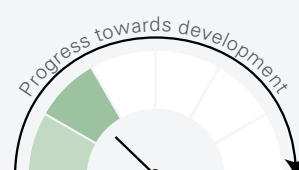
Supply chain



Engagement



Fuel



Reactor description: Liquid-fuelled molten salt research reactor to support development of the MSR-100 design.

Thermal power (MWth)	1
Core outlet temperature (°C)	Up to 650
Spectrum (thermal/fast)	Thermal
Fuel form	Molten fluorides (LiF-BeF ₂ salt)
Base fissile material	U-235, Uranium enrichment level: ≥ 10% to < 20%

Assessment of MSR-1's progress to deployment

Licensing



In 2020, Abilene Christian University (ACU) began the licensing process for the MSR-1 non-power reactor with the US Nuclear Regulatory Commission (NRC). In September 2024, the NRC granted a construction permit for the MSR-1, approximately two years after the initial application was submitted. The following month, in October 2024, ACU submitted a revised Regulatory Engagement Plan to the NRC in preparation for an operating licence application, which it plans to submit in 2025.

Siting



In September 2024, the NRC issued a construction permit for the MSR-1, which will be housed at ACU's Nuclear Energy eXperimental Testing Laboratory (NEXT Lab). In September 2023, ACU opened the Gayle and Max Dillard Science and Engineering Research Center (SERC), which includes the NEXT Lab.

Financing



In 2018, NEXT Lab received over USD₂₀₂₃ 6.0 million in funding to support its development, including USD 3.2 million (USD₂₀₂₃ 3.9 million) in donations from the Robison Excelsior Foundation and USD 300 000 (USD₂₀₂₃ 364 000) from the Development Corporation of Abilene (DCOA). In 2020, Doug Robison, founder of both the Robison Excelsior Foundation and Natura Resources, committed USD 30.5 million (USD₂₀₂₃ 37.0 million) through a Sponsored Research Agreements to the Natura Resources Research Alliance, including USD₂₀₂₃ 25.3 million for ACU. In 2021, DCOA partnered with NEXT Lab and approved a USD 2.9 million (USD₂₀₂₃ 3.5 million) incentive package under "Project Future Hope" to support the expansion of NEXT Lab's research and development facility at the former Taylor Elementary School site.

Supply chain



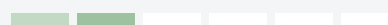
The Natura Resources Research Alliance (NRRA) was established in 2019 between ACU, Georgia Institute of Technology, Texas A&M University, and the University of Texas at Austin to advance molten salt reactor research. In 2025, Natura Resources entered a Memorandum of Understanding with ACU, Texas Tech University, and the Texas Produced Water Consortium to demonstrate integration of molten salt reactor technology with water desalination systems. In 2023, Zachry Nuclear Engineering was awarded an Engineering Services Agreement to complete the MSR-1 detailed design. Earlier in 2023, ACU opened the SERC, which houses the NEXT Lab. In 2022, Teledyne Brown Engineering was awarded a contract to perform front-end engineering and design of the MSR-1 system.

Engagement



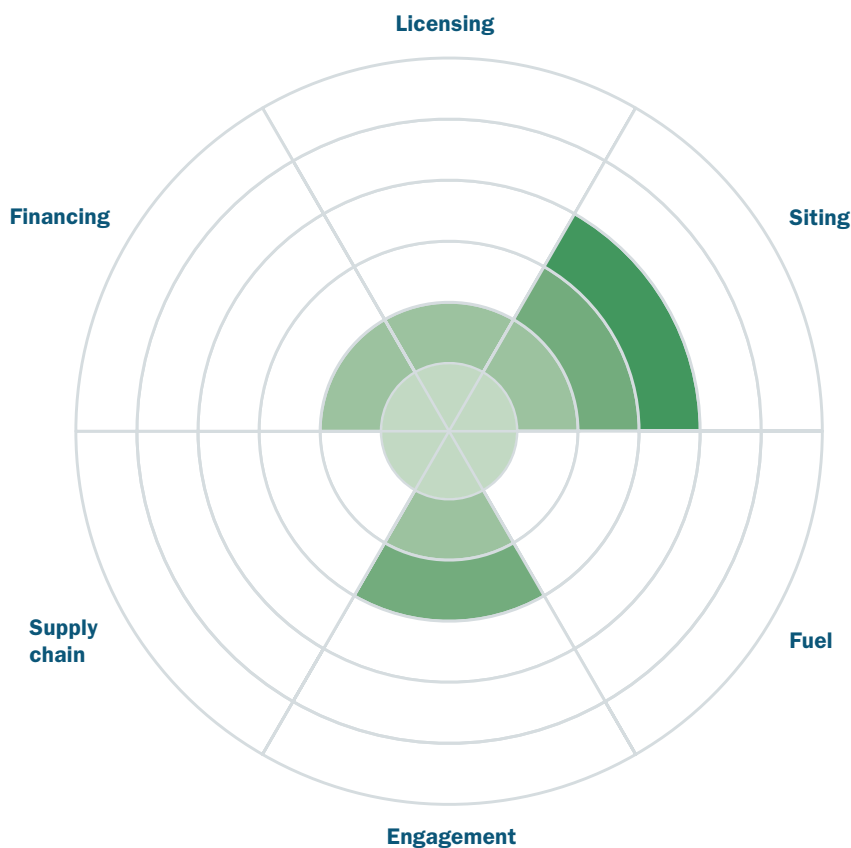
ACU and Natura Resources have received endorsements from several elected officials including formal resolutions from both the Texas Senate and House recognising their achievements in advancing the establishment of a molten salt research reactor in Texas. In 2023, Abilene Mayor Weldon Hurt attended the ribbon-cutting ceremony for ACU's SERC grand opening. Additionally, US Senators Ted Cruz and John Cornyn, US Congressman Jodey Arrington, and Texas Representative Cody Harris toured the NEXT Lab alongside Doug Robison, founder of Natura Resources. In 2024, Natura Resources expanded its public engagement efforts, including Doug Robison's participation in Elemental Energy and POWER Magazine podcasts.

Fuel

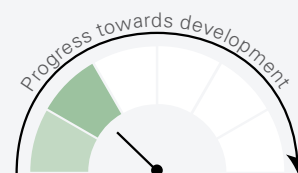


The liquid-fuelled molten salt reactor MSR-1 will use a LiF-BeF₂-UF₄ salt mixture. In 2024, ACU commissioned the Molten Salt Test System (MSTS) to support research on materials, components, and operations for MSR-1 development. That same year, Copenhagen Atomics and Materion delivered 250 kilograms of highly purified salt to ACU for use in the MSTS. In 2024, Idaho National Laboratory (INL) hosted a three-day workshop with the Natura Resources team to assess the fuel requirements for the Natura MSR-1, evaluate the feasibility of producing that fuel at INL, and outline needs for INL to support fuelled operation of the reactor at ACU. Previously, in 2019, the US Department of Energy (DOE) issued a programmatic letter of support for the MSR-1, including ACU's request for fuel-related services.

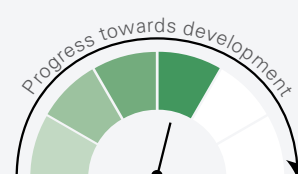
Natura Resources – MSR-100



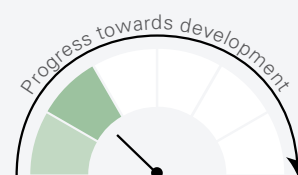
Licensing



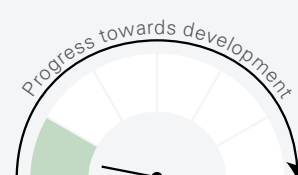
Siting



Financing



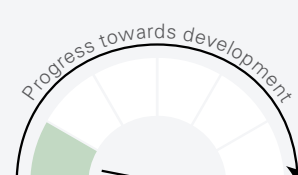
Supply chain



Engagement



Fuel

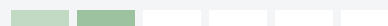


Reactor description: Land-based liquid-fuelled molten salt reactor.

Thermal power (MWth)	250
Core outlet temperature (°C)	Up to 650
Spectrum (thermal/fast)	Thermal
Fuel form	Molten fluorides (LiF-BeF ₂ salt)
Base fissile material	- U-235, enrichment level: ≥ 5% to < 10% (initial loading) - U-235, enrichment level: ≥ 10% to < 20% (refuelling)

Assessment of MSR-100's progress to deployment

Licensing



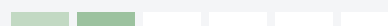
In January 2024, Natura Resources submitted a Regulatory Engagement Plan for the MSR-100 to the US Nuclear Regulatory Commission (NRC), officially initiating the pre-licensing process with the NRC.

Siting



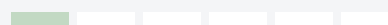
In February 2025, John Sharp, Chancellor of the Texas A&M University System – a network of universities and state agencies in Texas – announced that Texas A&M University System had selected Natura Resources' MSR-100 for deployment at its RELLIS campus. This decision is part of "The Energy Proving Ground" project, which also includes the deployment of reactors from Kairos Power, Terrestrial Energy, and Aalo Atomics at the RELLIS campus.

Financing



Natura Resources, founded in 2020, has secured USD₂₀₂₃ 78 million through early-stage venture capital funding.

Supply chain



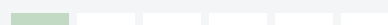
Natura Resources' MSR-100 will benefit from the supply chain established for the Natura MSR-1 system, which is being deployed at ACU. At the time of the assessment, no information about supply chain readiness was readily available from verifiable public sources for the MSR-100 design.

Engagement



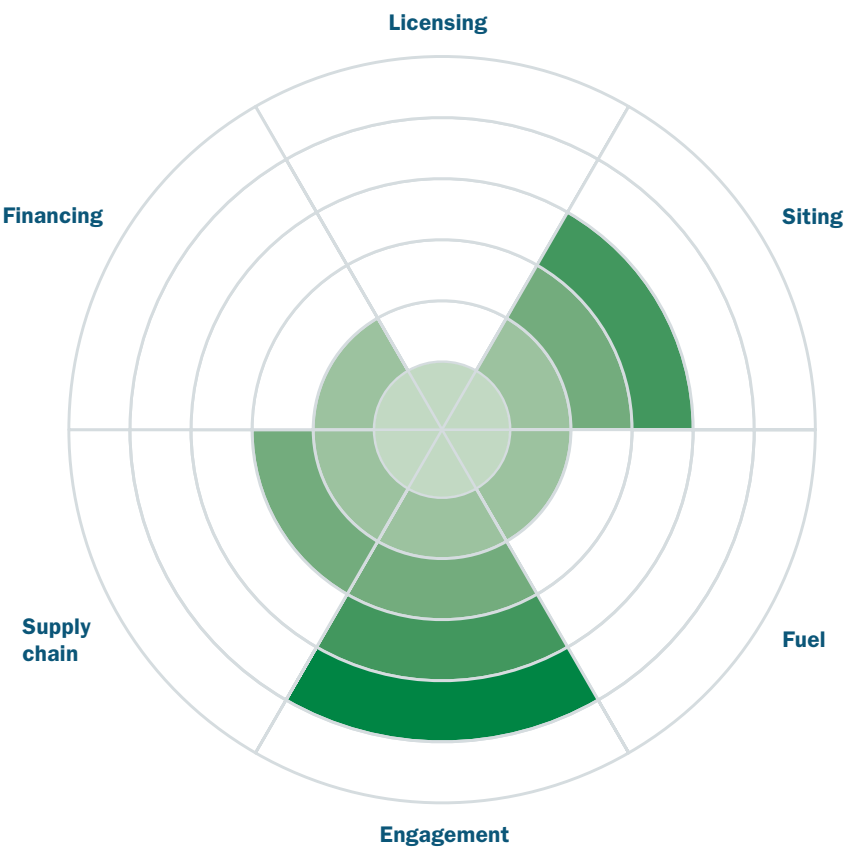
In 2024, Doug Robison, founder of Natura Resources, participated in a screening of Oliver Stone's documentary *Nuclear Now* in Texas and joined a post-screening panel discussion alongside Texas Public Utility Commissioner Jimmy Glotfelty and State Representative Cody Harris. In 2025, Natura Resources engaged with potential end-users of the MSR-100, signing a Memorandum of Understanding with ACU, Texas Tech University, and the Texas Produced Water Consortium to explore the integration of molten salt reactor (MSR) technology with water desalination systems. In 2024, Natura Resources expanded its public engagement efforts, including Doug Robison's participation in Elemental Energy and POWER Magazine podcasts.

Fuel



The liquid-fuelled molten salt reactor MSR-100 will operate using a LiF-BeF₂-UF₄ salt mixture. Its fuel development and qualification will leverage advancements made for the MSR-1. At the time of the assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

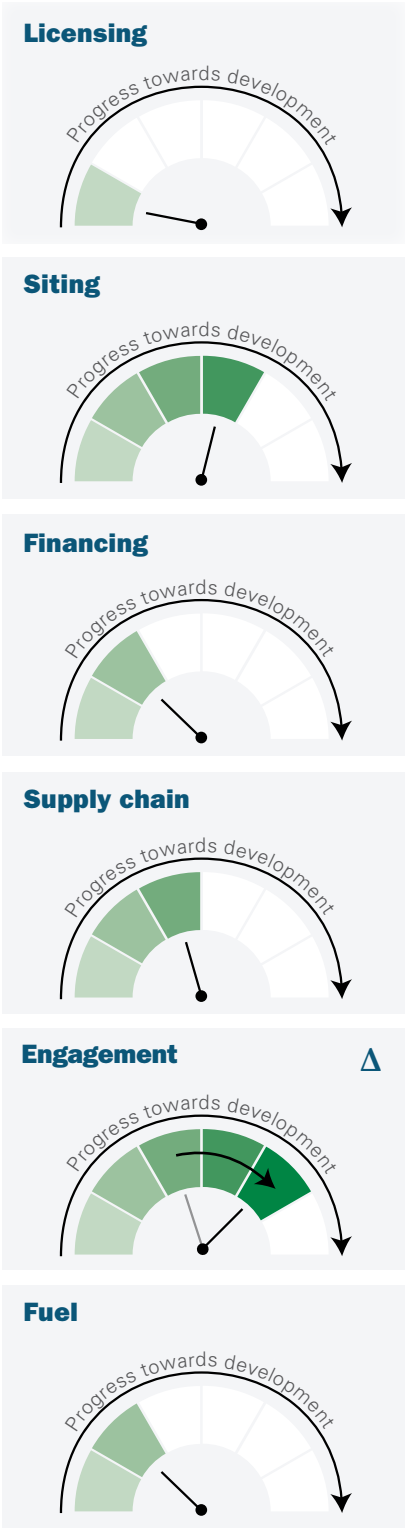
National Centre for Nuclear Research (NCBJ) – HTGR-POLA



 Changes since Second Edition (2024).

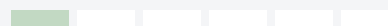
Reactor description: Gas-cooled high temperature demonstration reactor.

Thermal power (MWth)	30
Core outlet temperature (°C)	750
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level: ≥ 5% to < 10%



Assessment of HTGR-POLA's progress to deployment

Licensing



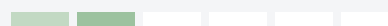
In 2019, the Polish Ministry of Energy and the National Centre for Research and Development (NCBiR) launched the GoHTR project to support HTGR development in Poland. In 2021, the National Centre for Nuclear Research (NCBJ) signed a contract with the Polish Ministry of Education and Science (MEiN) for the conceptual and basic designs and a Preliminary Safety Report for the HTGR-POLA. The conceptual design was completed in 2022, and the basic design was completed in 2024. The NCBJ expects to begin the licensing process in 2025 following the completion of the Preliminary Safety Report. At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



NCBJ plans to build an HTGR-POLA reactor at the site it owns in Świerk, a district of Otwock, Poland, which also hosts NCBJ's research pool type reactor MARIA. NCBJ intends to scale up the HTGR-POLA design to 180 MWth units by the early- to mid-2030s to support cogeneration applications, including to power the chemical industry.

Financing



In 2019, the NCBiR committed PLN 18 million (USD₂₀₂₃ 6.1 million) in public funding to the GoHTR project to support the HTGR-POLA pre-conceptual design development. In 2021, the MEiN committed PLN 60 million (USD₂₀₂₃ 16.6 million) over 3.5 years for the HTGR-POLA conceptual and basic design development, which was completed in December 2024.

Supply chain



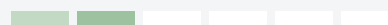
In 2022, NCBJ updated its 2019 Implementation Arrangement with Japan Atomic Energy Agency (JAEA) for co-operation on research and development (R&D) in the field of HTGR technology, which included preparation of a basic design of HTGR-POLA leveraging JAEA's experience with the High Temperature Test Reactor (HTTR) in Oarai, Japan. In August 2023, the HTGR-POLA project team hosted supply chain companies from Japan, including Mitsubishi Heavy Industries and Toshiba Energy Systems and Solutions, to discuss technical details of the SMR. The development of the basic design has been supported by the AGH University of Science and Technology in Kraków, Poland, and Energoprojekt-Katowice, a Polish engineering, procurement and construction company.

Engagement



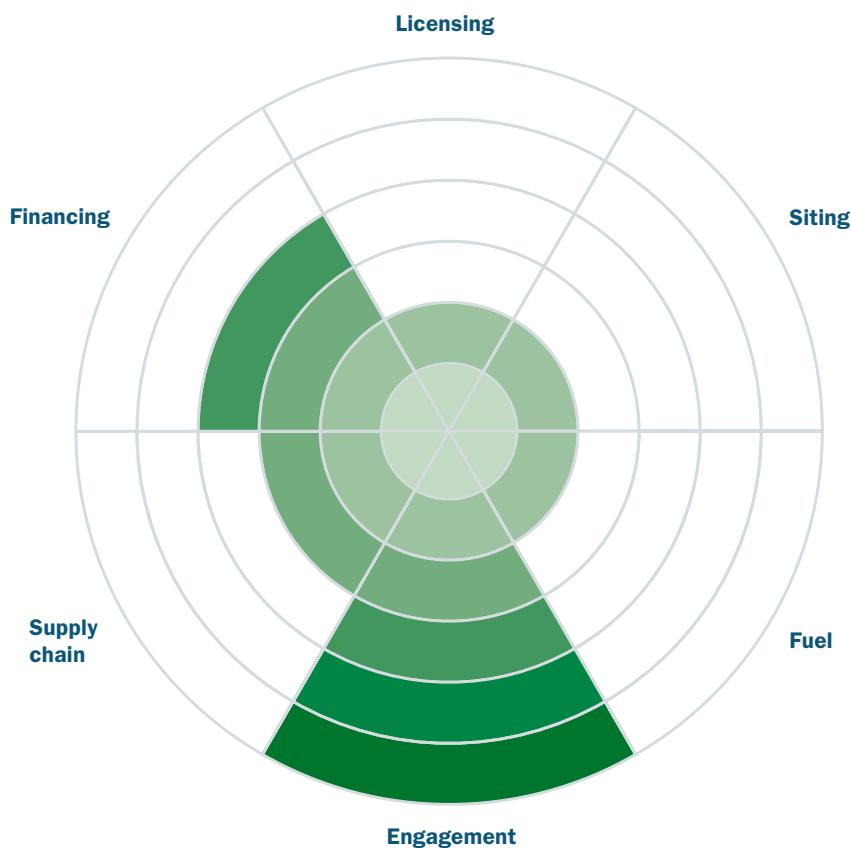
NCBJ has engaged the public on the HTGR-POLA and on nuclear energy in general, including: a webinar with the Institute of Economic Diplomacy in 2022; publications through online media outlets in 2023 such as Energetyka24 and Energy Market Agency; lectures in schools; YouTube videos; and a board game. Between September and December 2024, NCBJ engaged the Polish Senate Group for Nuclear Energy and Renewable Energy Sources, as well as the Poland's Parliamentary Teams for Nuclear Energy and for Energy and Hydrogen.

Fuel

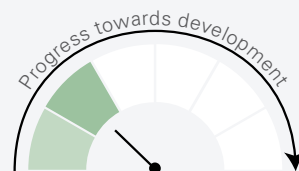


NCBJ plans to use the same HALEU fuel that has already been qualified and demonstrated by JAEA in the HTTR reactor JAEA operates in Oarai, Japan. The HTGR-POLA project will benefit from JAEA's knowledge and expertise under NCBJ's research collaboration contract with JAEA.

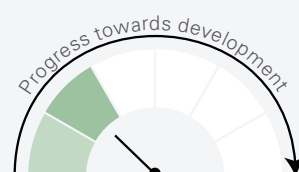
newcleo – LFR-AS-30



Licensing



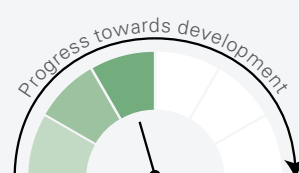
Siting



Financing



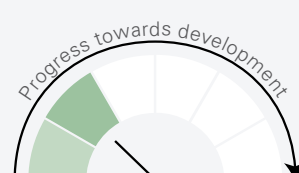
Supply chain



Engagement



Fuel

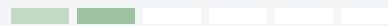


Reactor description: Fast spectrum, lead-cooled reactor using mixed plutonium-uranium oxide (MOX) fuel.

Thermal power (MWth)	90
Core outlet temperature (°C)	530
Spectrum (thermal/fast)	Fast
Fuel form	MOX ceramic CERCER
Base fissile material	Plutonium, not requiring uranium enrichment

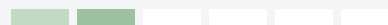
Assessment of LFR-AS-30's progress to deployment

Licensing



Newcleo plans a staged approach: first, the establishment of test facilities and an electrical prototype in Italy; the development of a 30 MWe Lead-cooled Fast Reactor (LFR-AS-30) in France; then commercial deployment of LFR-AS-200. LFR-AS-30 will enable materials testing for the LFR-AS-200. In 2024, newcleo completed the preparatory review stage of pre-licensing with France's Nuclear Safety and Radiation Protection Authority (ASNR) and plans to apply for a licence for its LFR-AS-30 design in 2026. In 2024, newcleo submitted its Safety Option File (DOS) to the ASNR for its MOX fuel assembly testing facility. In 2024, ASNR held a technical seminar on the LFR-AS-30, with the Slovak Republic's Nuclear Regulatory Authority and Sweden's Radiation Safety Authority.

Siting



Newcleo has begun discussions with national and local French authorities to acquire land for its LFR-AS-30 reactor within the Chinon Vienne et Loire community of municipalities.

Financing



By the end of 2024, newcleo had raised EUR 537 million (USD₂₀₂₃ 580 million) in private funding, building on its initial EUR 300 million (USD₂₀₂₃ 345 million) raised by mid-2022, and EUR 100 million (USD₂₀₂₃ 126 million) secured in 2021. This funding will support newcleo's global development strategy, including the development of the LFR-AS-30. In June 2023, the French government selected newcleo's LFR-AS-30 as one of two SMRs to be awarded EUR 24.9 million (USD₂₀₂₃ 26.9 million) for advancing research and development activities as part of the "Réacteurs nucléaires innovants" programme. As of September 2024, the company had hired more than 850 employees at 19 locations across France, Italy, the United Kingdom, Switzerland and the Slovak Republic.

Supply chain



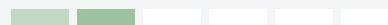
In 2021, newcleo acquired Hydromine Nuclear Energy. In 2023-2024, newcleo acquired Servizi Ricerche e Sviluppo, Fucina Italia, and Rüttschi Group, and signed a work-for-equity agreement with Tosto Group. In partnership with ENEA, newcleo commissioned its CAPSULE and CORE facilities for materials corrosion testing at ENEA Brasimone Centre. Newcleo has agreements with Enel, Nuclear AMRC, Ingerop, Onet Technologies, Assystem, Maire, Saipem, Nuclear Transport Solutions, and Naarea. In 2024, newcleo formed a European SMR Project-Working Group. Newcleo is also funding research positions at Politecnico di Torino and the London School of Economics. Additionally, newcleo is a member of the Nuclear Valley network, a group of over 400 companies involved in the French nuclear supply chain.

Engagement



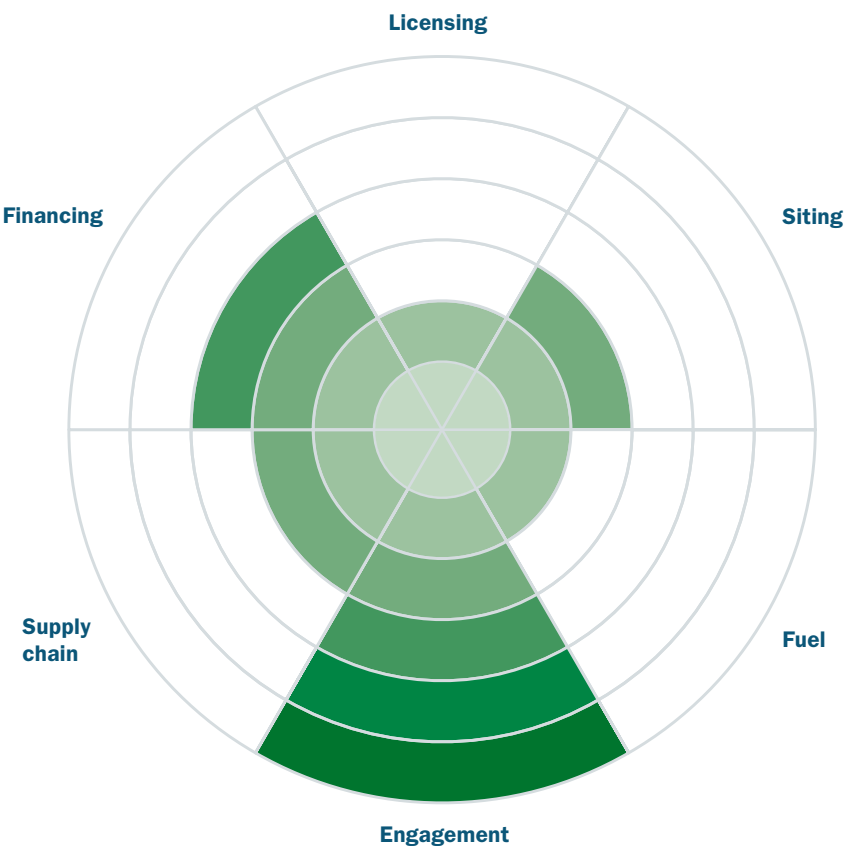
Newcleo engages students in universities in Italy and France. Newcleo has engaged the public through podcasts, media interviews, and public meetings in Italy to discuss investment in the Bologna region and partnership with ENEA. In 2024, newcleo engaged elected officials through the World Economic Forum, the Global Energy Transition Congress and Exhibition, Italy's G7 Energy Ministerial, Abu Dhabi's World Future Energy Summit, and the EU-Italy Energy Days. In 2024, newcleo hosted a screening of Oliver Stone's Nuclear Now documentary in Bratislava with Slovak government officials. In 2023, newcleo signed an agreement with Fincantieri and RINA for a feasibility study on LFR-AS-30 for naval propulsion.

Fuel



Newcleo plans to design, build, and operate a MOX fuel production facility. In 2022, newcleo commissioned Orano for feasibility studies on MOX production. Newcleo collaborates with the UK National Nuclear Laboratory and the French Atomic Energy and Alternative Energies Commission (CEA) to advance research on LFRs, including MOX fuel development, that builds on CEA's experience with Superphénix. Newcleo is part of a working group with Orano and two other French fast-spectrum SMRs to develop a joint strategy for MOX fuel supply and reprocessing in France. In 2024, the company submitted its DOS to ASNR for its MOX fuel assembly testing facility. In 2025, newcleo's Fuel Cycle and LFR Materials teams also visited Japan to engage with the Japan Atomic Energy Agency (JAEA) fuel experts.

newcleo – LFR-AS-200



△ Changes since Second Edition (2024).

Reactor description: Fast spectrum, lead-cooled reactor using mixed plutonium-uranium oxide (MOX) fuel.

Thermal power (MWth)	480
Core outlet temperature (°C)	530
Spectrum (thermal/fast)	Fast
Fuel form	MOX ceramic CERCER
Base fissile material	Plutonium, not requiring uranium enrichment

Licensing



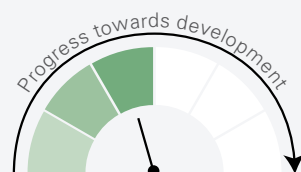
Siting



Financing



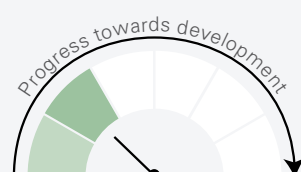
Supply chain



Engagement

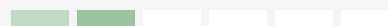


Fuel



Assessment of LFR-AS-200's progress to deployment

Licensing



Newcleo is developing LFR-AS-30 in France as a demonstration and to test materials to support commercial deployment of its 200 MWe Lead-cooled Fast Reactor (LFR-AS-200). In 2024, newcleo completed the preparatory review stage of pre-licensing with France's Nuclear Safety and Radiation Protection Authority (ASNR). In 2024, newcleo applied to the UK Department for Environment, Food and Rural Affairs (DEFRA) for a justification decision on lead-fast reactors, as part of the regulatory process for any new class of nuclear technologies in the United Kingdom. Additionally, the company completed Tier 1 of the early engagement process with the UK Office for Nuclear Regulation (ONR) and applied to enter the Generic Design Assessment (GDA) pre-licensing process for LFR-AS-200.

Siting



In 2025, newcleo entered into an agreement with JAVYS, a state-owned company responsible for managing spent nuclear fuel in the Slovak Republic. The agreement facilitates the creation of a joint venture to develop up to four reactors at the Bohunice nuclear site, which is owned by the government and located in the west of the Slovak Republic.

Financing



By the end of 2024, newcleo had raised EUR 537 million (USD₂₀₂₃ 580 million) in private funding, building on its initial EUR 300 million (USD₂₀₂₃ 345 million) raised by mid-2022, and EUR 100 million (USD₂₀₂₃ 126 million) secured in 2021. This funding will support newcleo's global development strategy, including the development of the LFR-AS-200. As of September 2024, the company had hired more than 850 employees at 19 locations across France, Italy, the Slovak Republic, Switzerland and the United Kingdom.

Supply chain



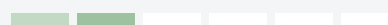
In 2021, newcleo acquired Hydromine Nuclear Energy. In 2023-2024, newcleo acquired Servizi Ricerche e Sviluppo, Fucina Italia, and Rüttschi Group, and signed a work-for-equity agreement with Tosto Group. In partnership with ENEA, newcleo commissioned its CAPSULE and CORE facilities for materials corrosion testing at ENEA Brasimone Centre. Newcleo has agreements with Enel, Nuclear AMRC, Ingerop, Onet Technologies, Assystem, Maire, Saipem, Nuclear Transport Solutions, and Naarea. In 2024, newcleo formed a European SMR Project-Working Group and is funding research positions at Politecnico di Torino and the London School of Economics. In 2025, newcleo signed an agreement with VUJE for engineering services for up to four LFR-AS-200 reactors at the Bohunice plant.

Engagement



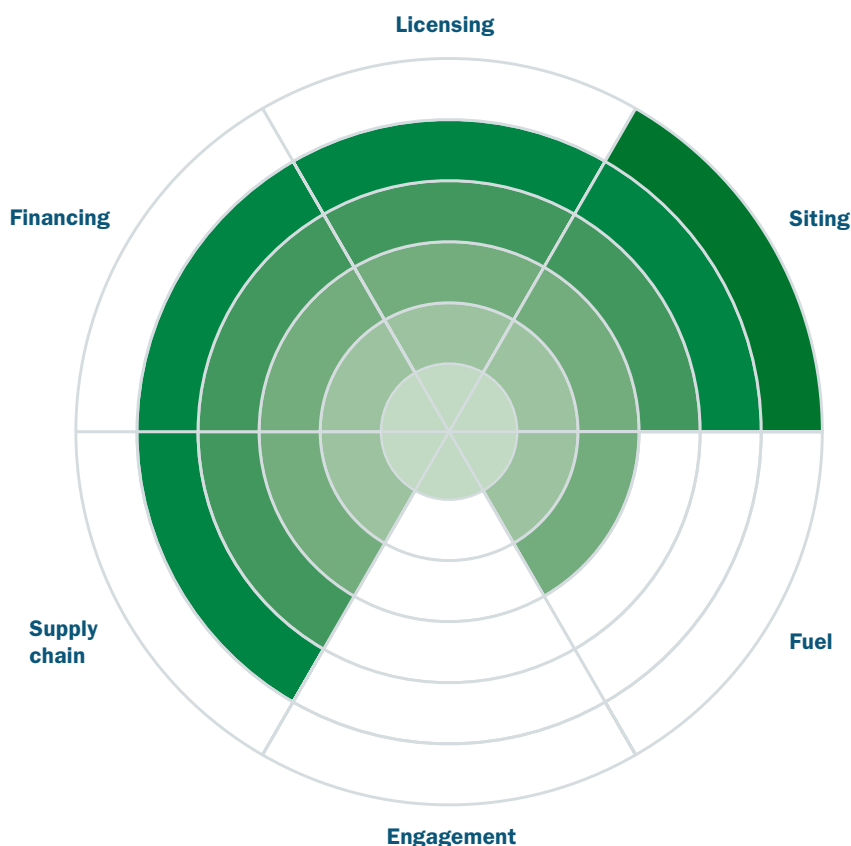
Newcleo engages students in universities in Italy and France. Newcleo has engaged the public through podcasts, media interviews, and public meetings in Italy to discuss investment in the Bologna region and partnership with ENEA. In 2024, newcleo engaged elected officials through the World Economic Forum, the Global Energy Transition Congress and Exhibition, Italy's G7 Energy Ministerial, Abu Dhabi's World Future Energy Summit, and the EU-Italy Energy Days. In 2024, newcleo hosted a screening of Oliver Stone's Nuclear Now documentary in Bratislava with Slovak government officials.

Fuel

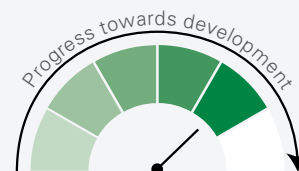


Newcleo plans to design, build, and operate a MOX fuel production facility. In 2022, newcleo commissioned Orano for feasibility studies on MOX production. Newcleo collaborates with the UK National Nuclear Laboratory and the French Atomic Energy and Alternative Energies Commission (CEA) to advance research on LFRs, including MOX fuel development, that builds on CEA's experience with Superphénix. Newcleo is part of a working group with Orano and two other French fast-spectrum SMRs to develop a joint strategy for MOX fuel supply and reprocessing in France. In 2024, the company submitted its DOS to ASNR for its MOX fuel assembly testing facility. In 2025, newcleo's Fuel Cycle and LFR Materials teams also visited Japan to engage with the Japan Atomic Energy Agency (JAEA) fuel experts.

N.A. Dollezhal Research and Design Institute of Power Engineering (NIKIET) – BREST-OD-300



Licensing



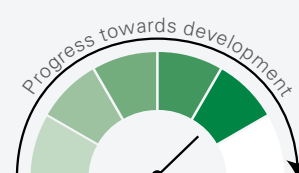
Siting



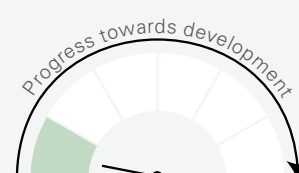
Financing



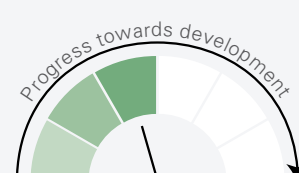
Supply chain



Engagement



Fuel



Reactor description: Lead-cooled fast spectrum demonstration reactor using mixed uranium-plutonium nitride fuel (MNUP fuel).

Thermal power (MWth)	700
Core outlet temperature (°C)	535
Spectrum (thermal/fast)	Fast
Fuel form	Uranium-nitride ceramic
Base fissile material	Plutonium, not requiring uranium enrichment

Assessment of BREST-OD-300's progress to deployment

Licensing



The Brest-OD-300 lead-cooled fast reactor is a demonstration unit of one of the technologies that ROSATOM has selected for its “Waste-free Atom” strategy, which seeks to deploy a mix of thermal and fast spectrum reactors to close the nuclear fuel cycle. The Brest-OD-300 demonstration will support future efforts to develop the 1200 MWe BR-1200 technology, a larger design with the same fuel cycle. A licence to construct the BREST-OD-300 was issued by the Russian nuclear regulator Rostekhnadzor in February 2021. In expectation of the reactor's startup, the Russian nuclear regulator and its Technical Support Organisation (TSO) are working on developing specific operational regulations for this type of lead-cooled fast reactor.

Siting



Construction of the BREST-OD-300 started in 2021 at the Siberian Chemical Combine site in Seversk, Russia. First criticality of the reactor is expected in 2026. The BREST-OD-300 demonstration reactor is part of a larger project called the Proryv, which includes a fuel fabrication plant, the BREST Reactor, and a fuel reprocessing plant. These will form the pilot-demonstration energy complex (PDEC), focused on demonstrating new technologies for the production and recycling of nuclear fuel owned by the Siberian Chemical Combine. The full PDEC is expected to be commissioned in 2030. The goal of Proryv is to close the fuel cycle by collocating thermal and fast spectrum reactors to expand the nuclear sector's raw material base while reducing the volumes of waste generated by the industry.

Financing



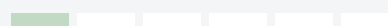
In 2010, the government of Russia announced RUB 110 billion (USD₂₀₂₃ 3.1 billion) over 10 years for advanced nuclear technology development, which included RUB 60 billion (USD₂₀₂₃ 1.7 billion) for fast spectrum reactors such as BREST-OD-300. In 2021, the Russian government approved a project cost of RUB 506 billion (USD₂₀₂₃ 7.1 billion) for new nuclear technologies by 2030, including the construction of BREST-OD-300 in Seversk. In 2022, the Russian government announced that it would invest RUB 100 billion (USD₂₀₂₃ 1.2 billion) in new nuclear technologies, including BREST-OD-300. The government has approved RUB 64.2 billion (USD₂₀₂₃ 798 million) for the Proryv project, which includes BREST-OD-300.

Supply chain



Construction of the BREST-OD-300 reactor began in June 2021. The Siberian Chemical Combine (SCC), a subsidiary of ROSATOM's fuel company TVEL, is responsible for building the pilot demonstration power complex. The SCC contracted the Concern Titan-2 engineering company for RUB 26.3 billion (USD₂₀₂₃ 340 million) to undertake construction and installation works. ZiO-Podolsk, part of ROSATOM's machine-building unit, will provide and install the steam generators for the reactor facility. CKBM, part of ROSATOM's mechanical engineering division Atomenergomash, is responsible for prototyping and qualifying the main circulation pumps.

Engagement



At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

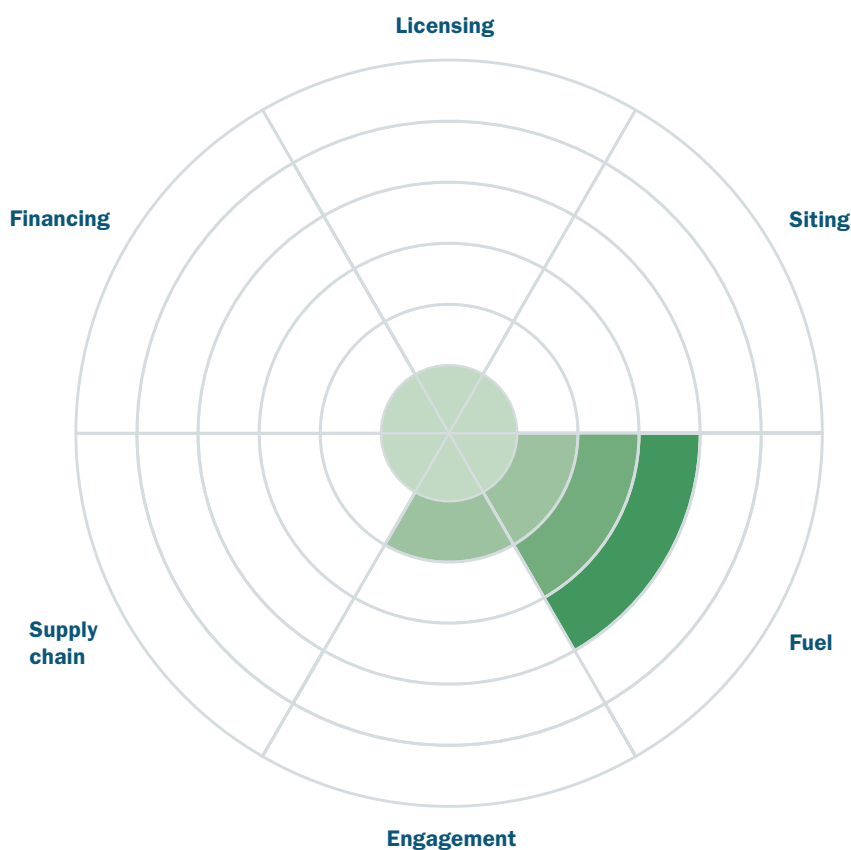
Fuel



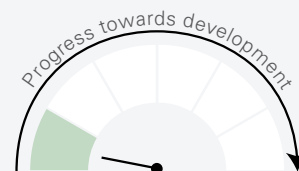
The BREST-OD-300 proposes to use novel, specially developed mixed uranium-plutonium nitride fuel (MNUP). In 2021, the SCC was making test batches of MNUP. To support fuel qualification, samples have been irradiated in the sodium-cooled fast reactor BN600 at the Beloyarsk nuclear power plant in Russia and in the BOR-60 research reactor in Dimitrovgrad. A MNUP fabrication and refabrication module, which complements BREST-OD-300 as part of the Proryv Project, started pilot operations in January 2025. In 2025-2026, construction of an irradiated fuel reprocessing module is planned to start at the same site with commissioning expected in 2030. SCC also signed a contract with CKBM for the manufacturing and supply of equipment for the refuelling complex.

Nuclearis Energy

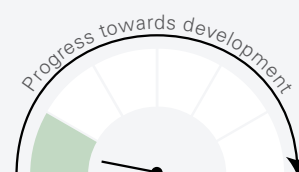
– Nuclearis N1 Nuclear Microreactor (Nuclearis N1)



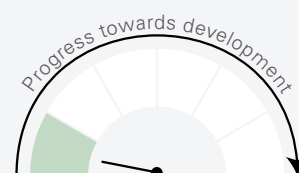
Licensing



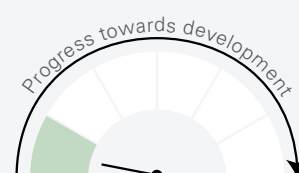
Siting



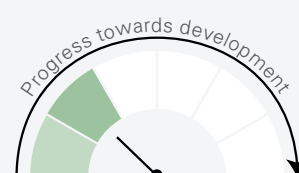
Financing



Supply chain



Engagement



Fuel



Reactor description: Integrated PWR, no refuelling, operating as vertical dry storage for irradiated fuel at the reactor end of life.

Thermal power (MWth)	42
Core outlet temperature (°C)	320
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level < 5%

Assessment of Nuclearis N1’s progress to deployment

Licensing



At the time of the assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



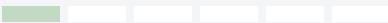
At the time of the assessment, no information related to siting was readily available from any site owners.

Financing



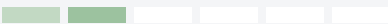
At the time of the assessment, no information about financing was readily available from verifiable public sources.

Supply chain



Nuclearis Energy was established in 2019, as a spin-off of the Argentine company Nuclearis, which obtained the American Society of Mechanical Engineers section III Nuclear Certification in 2022, authorising it to manufacture components for nuclear power plants. At the time of the assessment, no information about supply chain readiness for the Nuclearis N1 design was readily available from verifiable public sources.

Engagement



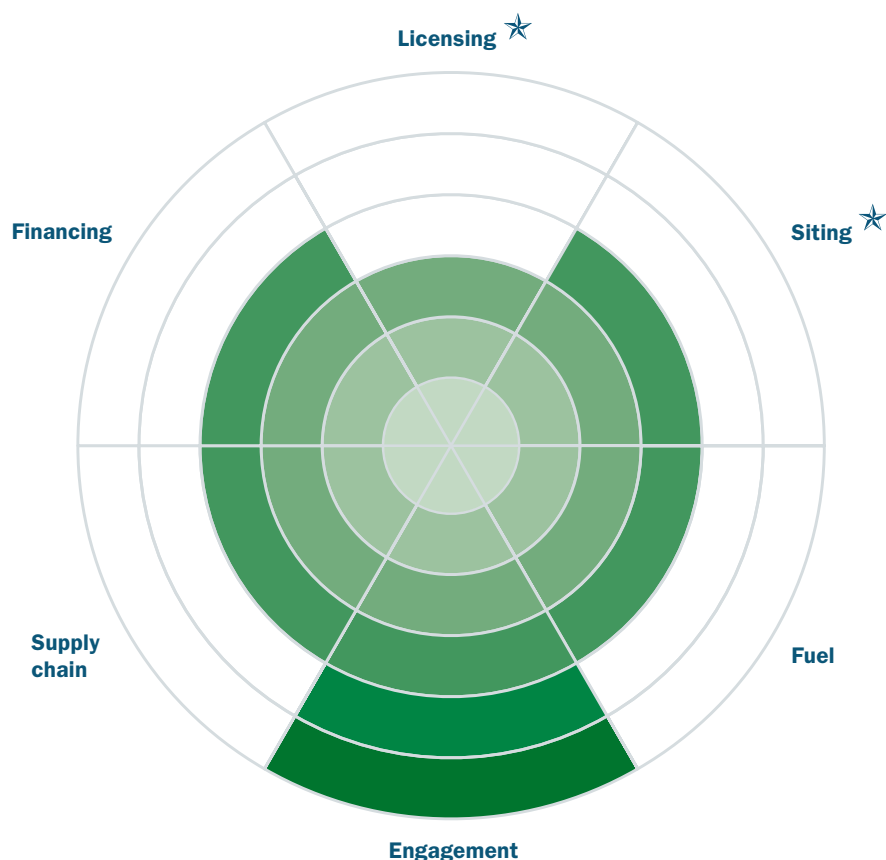
Nuclearis Energy has engaged with the general public with a TEDx Talk delivered by its CEO, Santiago Badran, in 2024. In 2025, former US Congressman Jeff Duncan was appointed to the company’s Board of Directors.

Fuel



The Nuclearis N1 design proposes to use fuel that is the current industry standard for water-cooled reactor technologies and therefore relies on well-established supply chains.

NuScale Power – ENTRA1 Energy Plants™ with NuScale SMRs-inside (ENTRA1 Energy Plants)

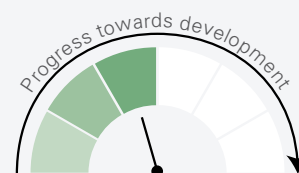


★ Active in multiple jurisdictions or countries.

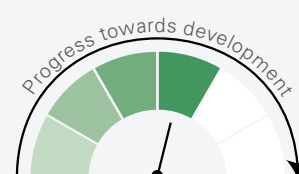
Reactor description: Integral multi-module pressurised water reactor.

Thermal power (MWth)	250
Core outlet temperature (°C)	321
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level: < 5%

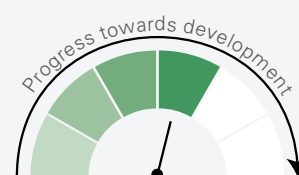
Licensing ★



Siting ★



Financing



Supply chain



Engagement



Fuel



Assessment of ENTRA1's progress to deployment

Licensing ★



ENTRA1 Energy Plants™ with NuScale SMRs-inside (ENTRA1 Energy Plants can be configured with 4, 6 or 12 NuScale Power Modules (NPM) of 77 MWe each, providing 308 MWe (4 NPMs), 462 MWe (6 NPMs), and 924 MWe (12 NPM), respectively. In 2020, the US Nuclear Regulatory Commission (NRC) issued a Standard Design Approval (SDA) for the 50 MWe NPM in a 12-NPM configuration and issued its final rule in February 2023. In July 2023, the NRC docketed for review NuScale's SDA application for the 77 MWe NPM uprate, with a mutually agreed upon 24-month review period. NuScale has also been conducting pre-licensing activities in Poland, Romania and Ukraine.

Siting ★



NuScale continues to progress in Romania, where RoPower, a joint venture of Nuclearelectrica and Nova Power & Gas, contracted NuScale for engineering and site evaluation of an ENTRA1 Energy Plant with a 6-NPM configuration at Doicești. Romania's nuclear regulator approved the project's Licensing Basis Document in 2023 and in 2024, an IAEA mission confirmed the site meets international safety standards. That same year, Nuclearelectrica and RoPower signed a Phase 2 engineering contract with Fluor Corporation. Also in 2024, NuScale advanced in the United Kingdom's Great British Nuclear selection process.

Financing



NuScale has been awarded matching funds from the US Department of Energy (DOE): in 2013, USD 226 million (USD₂₀₂₃ 295 million); and in 2020, USD 1.355 billion (USD₂₀₂₃ 1.595 billion) over ten years. NuScale became publicly traded in 2022 and received USD 380 million (USD₂₀₂₃ 396 million) in the transaction. In 2023, Japan, Korea, the United Arab Emirates and the United States announced up to USD₂₀₂₃ 275 million in new financing. In 2023, export-import banks in Korea and the United States expressed their interest in supporting NuScale deployment. In 2024, NuScale received approximately USD 228 million (USD₂₀₂₃ 221 million) after redemption of all its outstanding warrants.

Supply chain



NuScale has developed a mature supply chain through agreements with Framatome, Ultra Electronics, Sarens, Samsung C&T Corporation, Doosan Enerbility, PaR Systems, and Concurrent Technologies. NuScale has advanced testing, engineering, and review capabilities through agreements including with SIET, the US Reactor Forging Consortium, Oregon State University, Sargent and Lundy, Enercon, and BWXT. Through a strategic alliance with NuScale, ENTRA1 develops, finances, owns and operates ENTRA1 Energy Plants with NuScale SMRs-inside. In 2023, Doosan started forging upper reactor pressure vessels for NuScale. In 2025, Doosan ordered steam generator tubes from Alleima. NuScale was also selected for a Project-Working Group under the European Industrial Alliance on SMRs.

Engagement



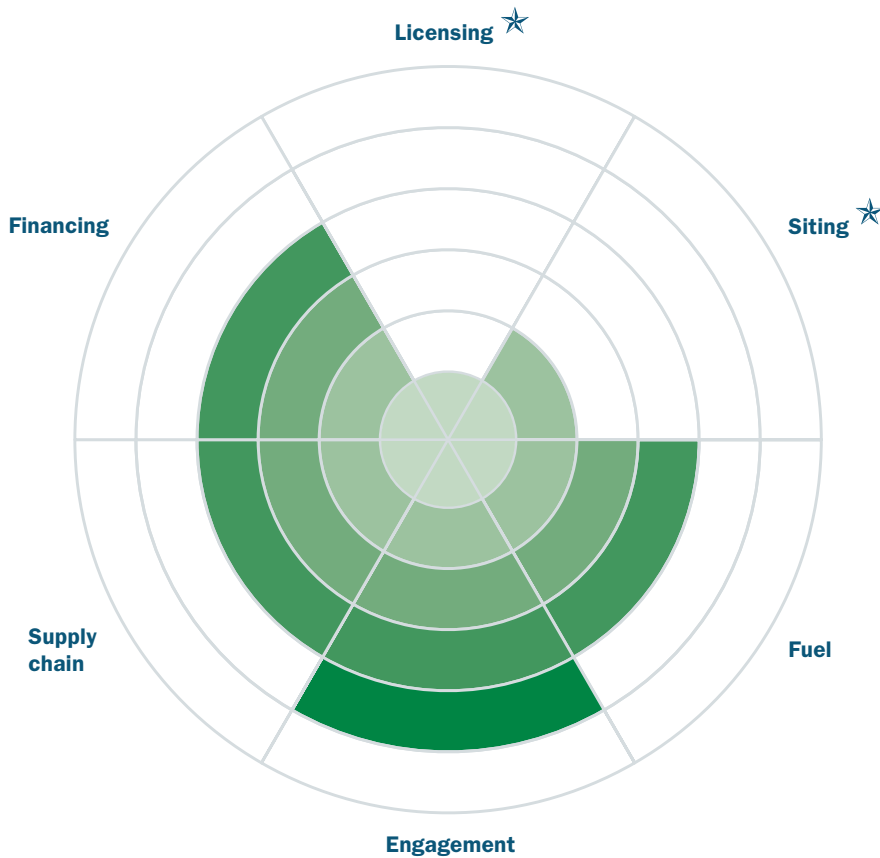
As of early 2025, NuScale had deployed eight Energy Exploration Centers (E2 Centers) in Africa, Asia, Europe, and the United States as educational tools in the use of modular reactors. NuScale has also engaged communities through town hall meetings. In 2023, the Polish Ministry of Climate and Environment issued a Decision-In-Principle to support deployment in Poland of ENTRA1 Energy Plants with NuScale SMRs-inside. NuScale joined the United Nations 24/7 Carbon-free Energy Compact, a group of stakeholders committed to carbon-free energy solutions. In Korea, GS Energy and the Uljin County signed an MoU to conduct feasibility studies of using ENTRA1 Energy Plants for hydrogen production.

Fuel



In 2016, NuScale included its proprietary HTP2 fuel in its application to the NRC for design certification of the 50 MWe NPM in a 12-NPM ENTRA1 Energy Plant configuration. In 2020, the NRC issued an SDA for the 50 MWe NPM in said configuration and issued its final rule in February 2023, approving by incorporation by reference the use of the HTP2 fuel design. HTP2 fuel can be fabricated at Framatome's fabrication facilities in the United States and Europe, and uses proven Framatome components with extensive operating history. HTP2 fuel is similar to fuel used in operating commercial nuclear plants and will not require further development. In 2015, NuScale Power and Areva (now Framatome) signed a contract to manufacture the fuel assemblies for the initial core and reloads.

NUWARD – NUWARD SMR



- ★ Active in multiple jurisdictions or countries.
- Δ Changes since Second Edition (2024).

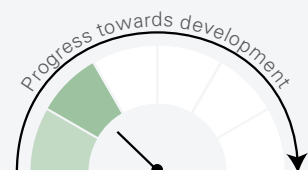
Reactor description: Integrated pressurised water-cooled SMR providing up to 400 MWe.

Thermal power (MWth)	Up to 1 000
Core outlet temperature (°C)	307
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level: < 5%

Licensing ★ Δ



Siting ★



Financing



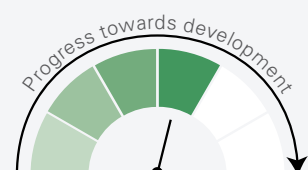
Supply chain



Engagement

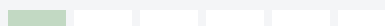


Fuel



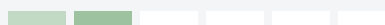
Assessment of NUWARD SMR's progress to deployment

Licensing ★



In 2023, phase two of NUWARD's SMR Joint Early Review (JER) began and is still ongoing in 2025 with the participation of the national nuclear safety regulators from Czechia (SUJB), Finland (STUK), France (ASNR), the Netherlands (ANVS), Poland (PAA), and Sweden (SSM). The goal is for these six European regulators to identify fundamental design requirements for the new NUWARD SMR to meet licensing expectations in their respective countries. In September 2023, the ASNR, STUK, and SUJB released a closure report for phase one of the JER. As of February 2025, the new NUWARD SMR design had not yet started the ASNR pre-licensing process.

Siting ★



EDF owns 18 licensed nuclear sites in France that could be suitable for the NUWARD first-of-a-kind (FOAK) project and has committed to start construction by the early 2030s. In early 2025, the government of France submitted its draft national energy strategy, Programmation Pluriannuelle de l'Énergie (PPE), for public comment, which reaffirms its commitment to supporting construction of a small modular pressurised water reactor under the France 2030 Plan. The government has committed to identify suitable sites and to complete its review of the enabling legislative and regulatory frameworks. In 2023, EDF signed an exclusive co-operation agreement with Respect Energy, a Polish electricity trading company, to conduct siting studies for NUWARD in Poland.

Financing



In 2023, the French government earmarked more than EUR 500 million (USD₂₀₂₃ 575 million) in funding for NUWARD SMR development.

Supply chain



NUWARD is in a partnership with: EDF for engineering; Framatome for the nuclear steam supply system; Edvance for engineering activities on the balance of nuclear plant; and Tractebel for the conventional island and the balance of plant. In 2023, EDF and NUWARD signed a co-operation agreement with Tractebel for nuclear engineering and with Ansaldo Energia and Ansaldo Nucleare to explore the deployment of nuclear in Italy. In 2024, NUWARD was also selected for a Project-Working Group under the European Industrial Alliance on SMRs.

Engagement



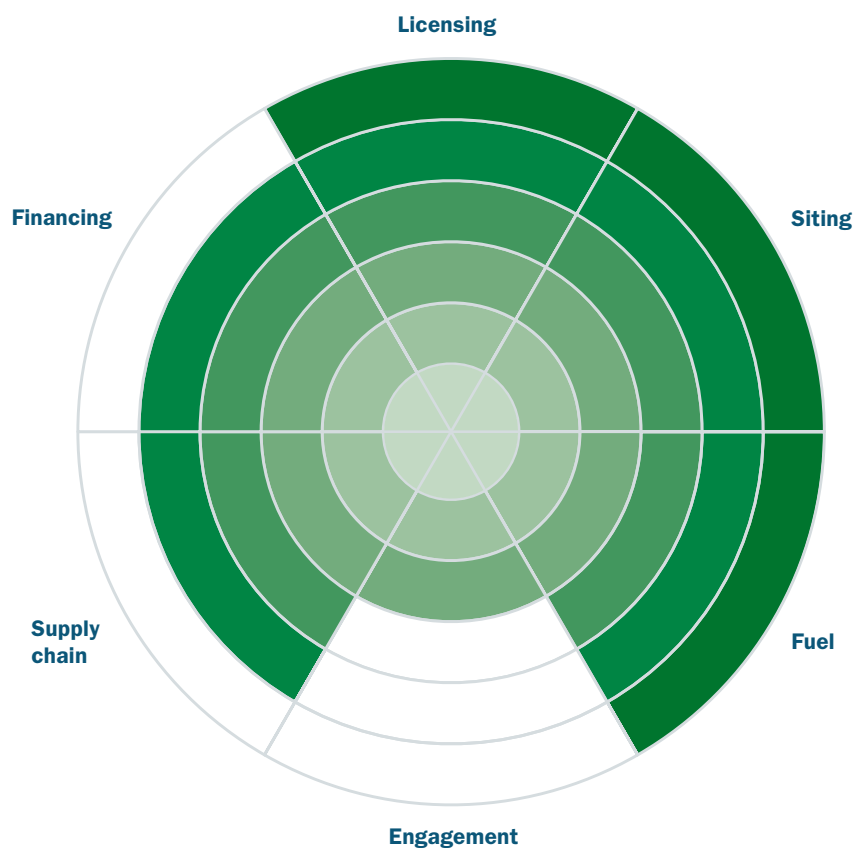
In 2021, the NUWARD International Advisory Board was created to provide advice on subjects including public acceptance. In 2022, the French President stated public support. In 2022 and 2023, EDF advanced various engagements, including through co-operation agreements and letters of intent for feasibility studies and to explore the possible deployment of the NUWARD SMR in Brazil, Czechia, Finland, Italy, Poland, the Slovak Republic and Sweden.

Fuel



The NUWARD SMR uses fuel that is the current industry standard for water-cooled reactor technologies, which are similar in design. There are no barriers expected in the fuel supply chain for this SMR Framatome, a subsidiary of EDF, has been supplying fuel to EDF for its nuclear fleet since the 1960s.

OKBM Afrikantov (OKBM) – KLT-40S



Licensing



Siting



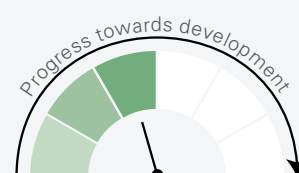
Financing



Supply chain



Engagement



Fuel



Reactor description: Pressurised water reactor for the floating nuclear power unit *Akademik Lomonosov*.

Thermal power (MWth) 300 (plant deployed with two modules of 150 MWth each)

Core outlet temperature (°C) 316

Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic CERMET

Base fissile material U-235, enrichment level: ≥ 10% to < 20%

Assessment of KLT-40S's progress to deployment

Licensing



The *Akademik Lomonosov* floating nuclear power plant powered by two KLT-40S units entered commercial operation in May 2020. Rostekhnadzor issued in 2019 a 10-year licence to Rosenergoatom to operate the Akademik Lomonosov until 2029.

Siting



The *Akademik Lomonosov* is currently docked in the city of Pevek, where it provides electricity and heat to the isolated grid of the Chaun-Bilibino energy centre of Chukotka. The Chukotka Autonomous District administration announced the siting studies to host future floating nuclear power units back in 2010. The reactor has been operating in the city of Pevek since May 2020. In January 2025 the plant had produced 1 billion kWh.

Financing



The *Akademik Lomonosov* has been financed by the Committee of the Russian Federation Budget RF (17%) and JSC Energoatom Concern (83%).

Supply chain



Substantial supply chain capabilities have been developed for the KLT-40S, as it builds on the Russian KLT-40M technology currently used to power icebreakers. Several organisations have contributed to the construction of the Akademik Lomonosov floating nuclear power unit and its KLT-40S reactor units: Rosenergoatom as owner and operator; OKBM as chief designer; the Kurchatov Institute as scientific supervisor of the reactor design; TsKB "Iceberg" as general designer; the Krylov Shipbuilding Research Institute as scientific supervisor; Baltiysky Zavod as builder of the floating nuclear power unit; Atomenergo as the designer of coastal and hydraulic engineering facilities; and Kalouzhsky Turbine Plant as the general designer and supplier of the turbine set.

Engagement



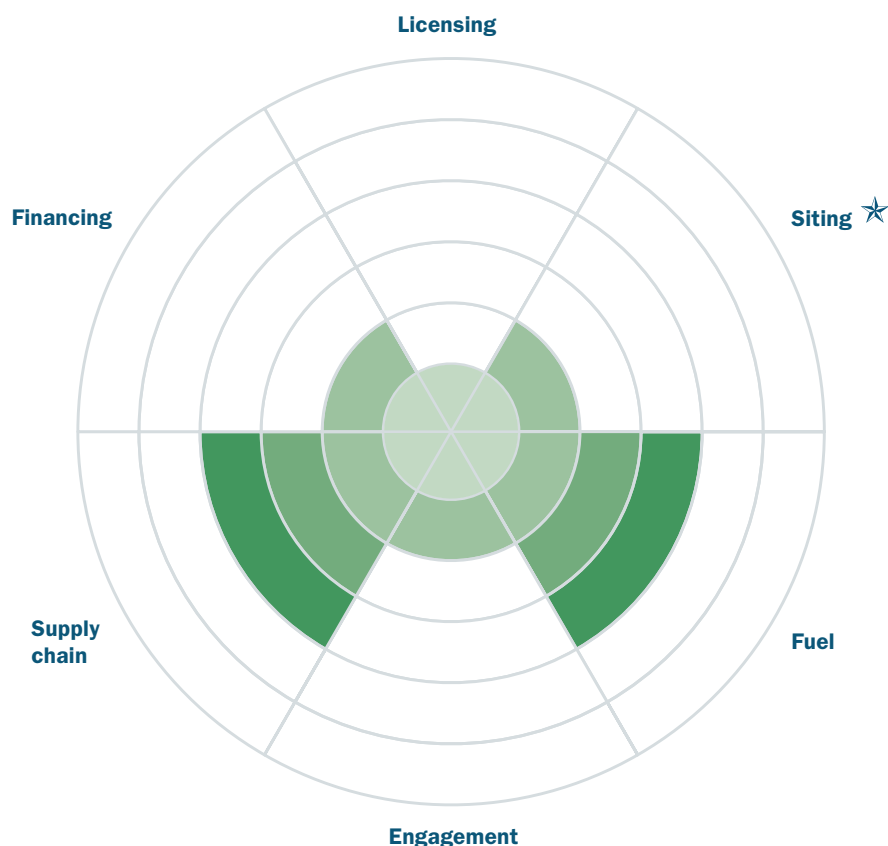
Rosenergoatom has organised public tours on the Akademik Lomonosov barge to showcase the KLT-40S power station and highlight the attractiveness of working in nuclear energy. In 2022, 111 people visited the Akademik Lomonosov of which a significant portion were school-age children. Public hearings were led by Rosenergoatom at Pevek, and Rosenergoatom is engaged with the city of Pevek to leverage the Academic Lomonosov to support the development of social infrastructure within the city. In 2022 Rosenergoatom donated more than RUB 200 million (USD₂₀₂₃ 2.9 million) to support urban infrastructure development within the city of Pevek. The World Association of Nuclear Operators (WANO) conducted a review mission of the plant in 2022 and a follow up in 2024.

Fuel



The KLT-40S design is based on the existing KLT-40M technology, which is currently used to power icebreakers, and therefore relies on well-established fuel supply chains. In particular, the fuel for the Akademik Lomonosov was produced by Mashinostroitelny Zavod (MSZ), a subsidiary of the Russian fuel manufacturer TVEL. First reloading of both reactors happened at the end of 2023 and early 2024 respectively.

OKBM Afrikantov (OKBM) – RITM-200M

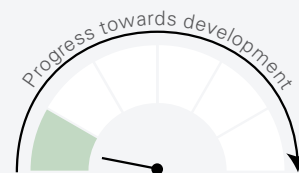


★ Active in multiple jurisdictions or countries.

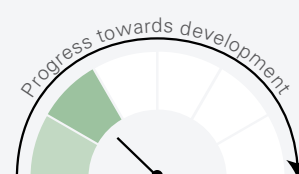
Reactor description: Integral pressurised water reactor for floating nuclear power units for export markets.

Thermal power (MWth)	396 (plant deployed with two modules of 198 MWth each)
Core outlet temperature (°C)	321
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic CERMET
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%

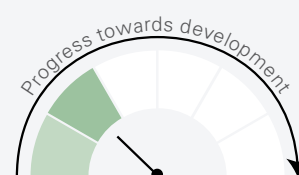
Licensing



Siting ★



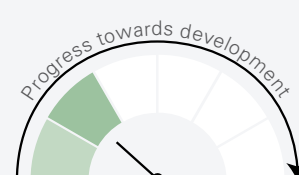
Financing



Supply chain



Engagement

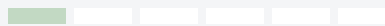


Fuel



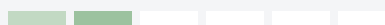
Assessment of RITM-200M's progress to deployment

Licensing



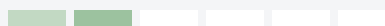
RITM-200M is a floating SMR in the RITM-200 series, designed for export markets by OKBM, a ROSATOM engineering subsidiary. It is a smaller, lighter and higher output evolution of the KLT-40S by OKBM, already deployed on the *Akademik Lomonosov* Floating Nuclear Power Plant. Eight RITM-200 units are already licensed and in operation in pairs on four dual-draft LK-60Ya class icebreakers. Licensing for the floating RITM-200M will benefit from ROSATOM's prior experience, through its subsidiary Atomflot, in licensing and operating the RITM-200 SMRs on icebreakers and the *Akademik Lomonosov* Floating Nuclear Power Plant. At the time of assessment, there was no additional information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting ★



In 2019, ROSATOM signed a Memorandum of Intent with the Ministry of Energy in the Philippines for a preliminary feasibility study on the possible deployment of a RITM-200M. In 2023, ROSATOM indicated it is working with the government of Myanmar on the possible deployment of an RITM-200 series nuclear power plant in Myanmar that could be on land (RITM-200N) or floating (RITM-200M). Russia and Myanmar signed an intergovernmental agreement for co-operation on peaceful nuclear power. In 2024, ROSATOM signed a Memorandum of Understanding with Guinea for the potential deployment of a RITM-200M. In 2023, Energoflot, a joint venture between ROSATOM and Russian company TSS Group, was established to own and operate RITM-200M floating nuclear power units in the Middle East, Southeast Asia and Africa.

Financing



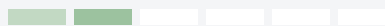
In 2021, the Russian government approved a project cost of RUB 506 billion (USD₂₀₂₃ 7.1 billion) for new nuclear technologies by 2030, which includes the RITM-200M. In 2022, the Russian government announced that the government would invest RUB 100 billion (USD₂₀₂₃ 1.24 billion) in new nuclear technologies, including the RITM-200M.

Supply chain



ROSATOM has significant experience and capabilities in nuclear power development, engineering, procurement and construction, including with the deployment of several RITM-200 units for icebreakers. As RITM-200M is a floating SMR within the RITM-200 series, it is expected that the RITM-200M supply chain will benefit significantly from the work on the other RITM-200 series SMRs. ROSATOM has created a joint venture with TSS Group, a Russian oil and gas construction and engineering company, to construct and export a series of RITM-200M floating power units for markets in the Middle East, Southeast Asia and Africa as early as 2029.

Engagement



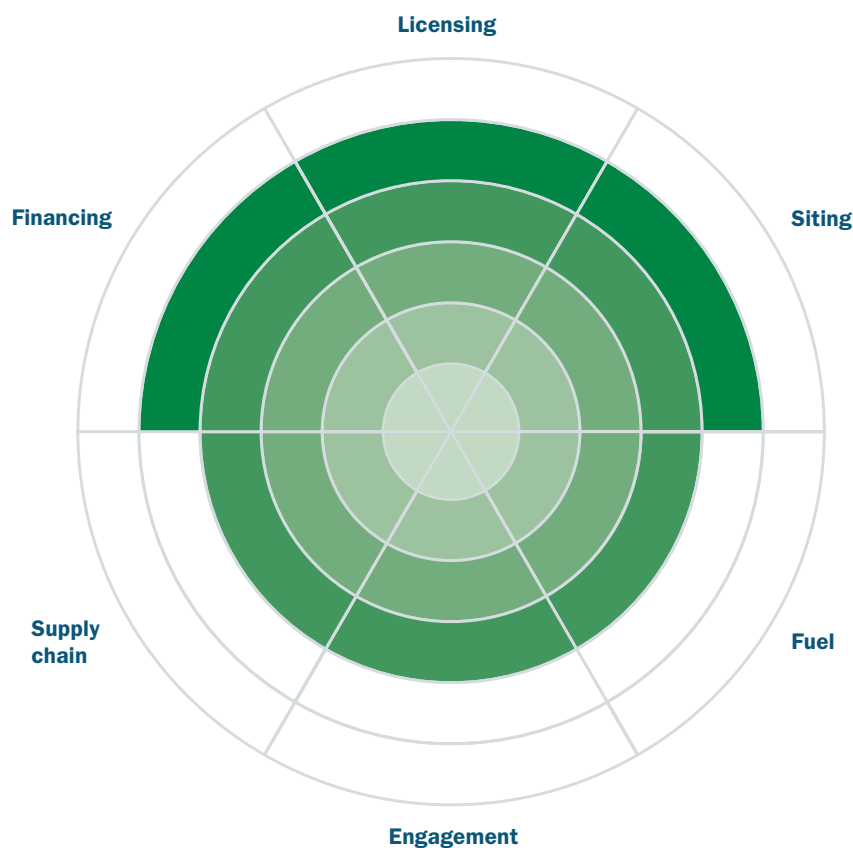
In 2023, ROSATOM participated in the Economic and Humanitarian Russia-Africa Forum and discussed the potential role for ROSATOM to sell electricity from a fleet of RITM-200M floating nuclear power plants in Africa.

Fuel

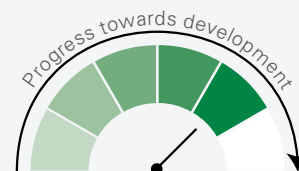


The RITM-200M is designed to use HALEU fuel enriched to less than 20%. Russia has existing capabilities to produce HALEU fuel on a commercial scale. The floating RITM-200M shares aspects of the fuel design for the RITM-200 series SMRs and is expected to leverage the same fuel supply chain. Compared with the RITM-200S for the domestic market, which requires refuelling every five years, the RITM-200M for the export market requires refuelling over a period of up to ten years. Fuel elements for the RITM-200 series reactors have been designed and developed by specialists at the Bochvar Research Institute of Inorganic Materials, part of the Russian fuel manufacturer TVEL, which supplies nuclear fuel to all Russian nuclear power plants, including marine-based reactors.

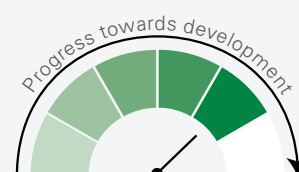
OKBM Afrikantov (OKBM) – RITM-200N



Licensing



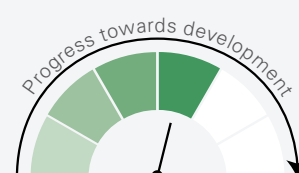
Siting



Financing



Supply chain



Engagement



Fuel



Reactor description: Integral pressurised water reactor for small land-based multi-module nuclear power plants.

Thermal power (MWth) 190

Core outlet temperature (°C) 321

Spectrum (thermal/fast) Thermal

Fuel form Uranium-oxide ceramic CERMET

Base fissile material U-235, enrichment level: ≥ 10% to < 20%

Assessment of RITM-200N's progress to deployment

Licensing



RITM-200N is a land-based SMR in the RITM-200 series, designed by OKBM, a ROSATOM engineering subsidiary. Licensing for RITM-200N will benefit from past experience with RITM-200 SMRs for propulsion on icebreakers. The Federal Environmental, Industrial and Nuclear Supervision Service of Russia (Rostekhnadzor) is drafting amendments to Federal Rules and Regulations to add considerations on the specifics of the RITM-200N, related to design and material standardisation for the deployment of maritime technologies on land. Rostekhnadzor reviewed more than 82 Safety Justification documents for RITM-200N, and in April 2023 issued a licence to construct one RITM-200N in the Ust-Yansky District of the Republic of Sakha (Yakutia) in northern Russia. The project is expected to be completed in 2028.

Siting



In 2019, the Republic of Sakha and ROSATOM signed an agreement of intent to build the first land-based RITM-200N SMR in Yakutia. A formal agreement was reached in 2020 to proceed with the construction. The licence to construct was issued by Rostekhnadzor in 2023, and preparatory work began in 2024. Start-up of the plant is expected in 2028 in a one-unit configuration. In 2024, the government of Russia announced that a second RITM-200N SMR is under consideration be added to the plant by 2031. An agreement between Uzbekistan and Russia was signed in 2024 for construction of a six-unit nuclear power plant using RITM 200N SMRs in the Jizzakh region of Uzbekistan.

Financing



In 2021, the Russian government approved a project cost of RUB 506 billion (USD₂₀₂₃ 7.1 billion) for new nuclear technologies by 2030, including the construction of the land-based RITM-200N in Yakutia. In 2022, the Russian government announced that the government would invest RUB 100 billion (USD₂₀₂₃ 1.2 billion) in new nuclear technologies, including RITM-200N projects. ROSATOM has also secured an off-take agreement for up to 50 MW as well as access to the Far Eastern Concession, a major Russian state investment mechanism, for the Yakutia project. In addition, ROSATOM has an agreement to provide power to the Public Joint Stock Company (PJSC) Seligdar, a Russian mining company, from the land-based RITM-200N project for gold mining operations in Yakutia.

Supply chain



Substantial supply chain capabilities for the RITM-200 SMR series have been developed with the construction of several RITM-200 units for icebreakers. OKBM, an engineering subsidiary of ROSATOM, acts as a designer and a single-source manufacturer for the RITM-200N SMR. For the RITM-200N in Yakutia, Rusatom Overseas is taking an expanded role, acting as developer and builder. Rosenergoatom will be the operator of the Yakutia plant. Preparatory work has been underway since 2024. More than 2000 tonnes of cargo have been delivered to the site, and a temporary village for construction workers has been set to accommodate more than 1000 people. In 2024 forging of long lead items has begun at AEM-Special Steels, a ROSATOM subsidiary.

Engagement



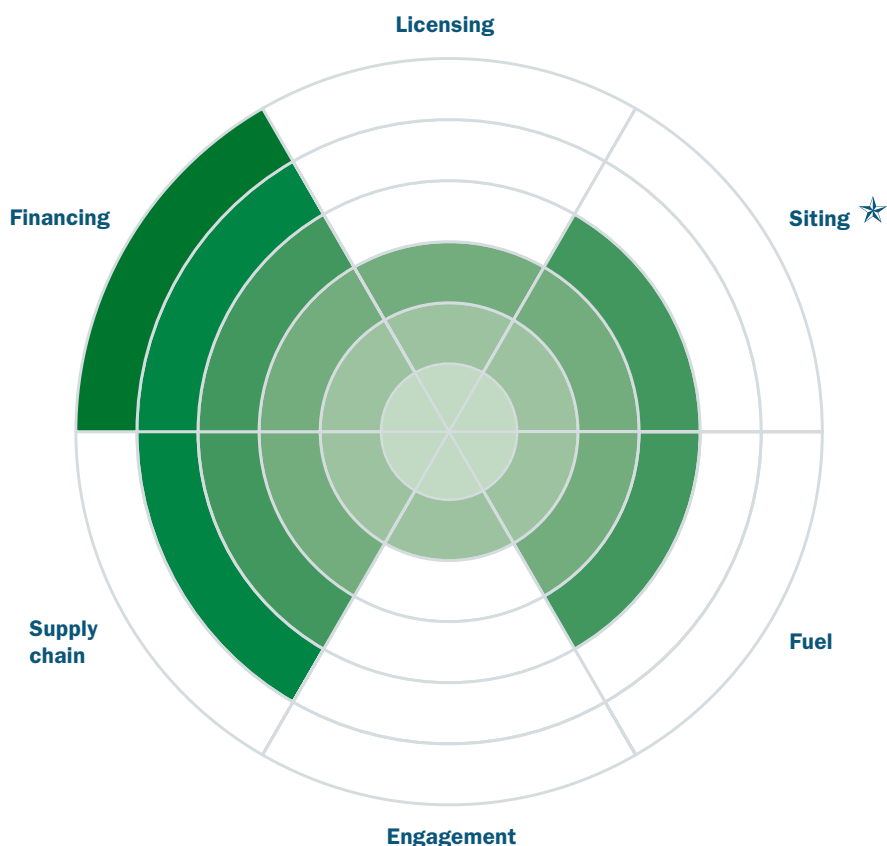
In 2021 and 2023, public discussions were held about the land-based RITM-200N project in Yakutia. The Ministry for the Development of the Russian Far East created a working group to oversee the implementation of the project, with representatives from: Rusatom Overseas; the Russian Ministry of Economic Development; the Russian Ministry of Energy; and Joint Stock Company VEB Infrastructure. ROSATOM is conducting a feasibility study under a 2022 memorandum of co-operation with Kyrgyzstan for the construction of RITM-200N SMRs in Kyrgyzstan. In 2023, Seligdar PJSC signed an agreement of intent with ROSATOM, the Russian Ministry for the Development of the Far East and Arctic, and the Government of the Republic of Sakha (Yakutia) to use the RITM-200N to develop deposits in the Arctic.

Fuel



The land-based RITM-200N SMR shares some aspects of the fuel supply chain of the RITM-200 and KLT-40S designs, which are already operating on four LK-60Ya class icebreakers and the *Akademik Lomonosov* Floating Nuclear Power Plant, respectively. Regarding the RITM-200N fuel design, the Elemash Machine-building plant, a fabrication facility of Russian fuel manufacturer TVEL, produced an experimental nuclear fuel assembly for the land-based RITM-200N to be constructed in Yakutia. Elemash has announced that they will start fabrication of the first fuel assemblies in 2025, with the intent to produce the first core by 2026.

OKBM Afrikantov (OKBM) – RITM-200S



- ★ Active in multiple jurisdictions or countries.
- Δ Changes since Second Edition (2024).

Reactor description: Integral pressurised water reactor for floating nuclear power plants.

Thermal power (MWth)	396 (plant deployed with two modules of 198 MWth each)
Core outlet temperature (°C)	318
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic CERMET
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%

Licensing



Siting ★



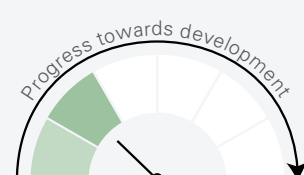
Financing



Supply chain Δ



Engagement



Fuel



Assessment of RITM-200S's progress to deployment

Licensing



RITM-200S is a floating SMR in the RITM-200 series, designed for domestic markets by OKBM, a ROSATOM engineering subsidiary. It is a smaller, lighter and higher output evolution of the KLT-40S by OKBM, already deployed on the *Akademik Lomonosov* Floating Nuclear Power Plant. Eight RITM-200 units are already licensed and in operation in pairs on four dual-draft LK-60Ya class icebreakers. Licensing for the floating RITM-200S will benefit from ROSATOM's prior experience, through its subsidiary Atomflot, in licensing and operating the RITM-200 SMRs on icebreakers and the *Akademik Lomonosov* Floating Nuclear Power Plant. A licence to construct the RITM-200S for deployment in Cape Nagleynyn in north-eastern Russia is under review by Rostekhnadzor, Russia's nuclear safety regulator.

Siting ★



In 2021, the chairman of Baimskaya copper mine and mineral processing facility in Cape Nagleynyn signed an agreement to purchase heat and electricity from four floating nuclear power units, with two RITM 200S each, to be deployed and operated by ROSATOM's subsidiary Atomoflot. The first two units are expected by 2028, the third in 2029, and the fourth in 2031. These timelines are tied to the progress of the mine development. ROSATOMIn 2023, ROSATOM also announced that it is considering up to 15 additional floating nuclear power units in the Arctic by 2042. Kommersant, a Russian news outlet, reported that natural gas company Gazprom is considering RITM-200S units for the Kirinskoye and Yuzhno-Kirinskoye gas condensate fields in Sakhalin region.

Financing



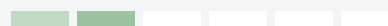
In 2021, the Russian government approved a project cost of RUB 506 billion (USD₂₀₂₃ 7.1 billion) for new nuclear technologies by 2030, including the construction of the marine-based RITM-200S in Cape Nagleynyn. In 2022, the Russian government announced that the government would invest RUB 100 billion (USD₂₀₂₃ 1.2 billion) in new nuclear technologies, including multiple RITM-200S projects. ROSATOM has signed an agreement with Baimskaya for a long-term take-or-pay contract as part of the Cape Nagleynyn floating nuclear power unit project, which is valued at RUB 190.2 billion (USD₂₀₂₃ 2.3 billion). As of 2025, these funds had already been leveraged to start manufacturing hulls in China for the first two RITM-200S floating nuclear power units.

Supply chain



Substantial supply chain capabilities for the RITM-200S SMR have been developed with the construction of several RITM 200 units for icebreakers. In 2022, Atomenergomash, a machinery construction division of ROSATOM, and OKBM started construction of the first RITM-200S floating nuclear power units off-site. Since 2021, ROSATOM has ordered three hulls, with plans announced for a fourth, from Wison Heavy Industry in China for the marine-based RITM-200S in Cape Nagleynyn. In 2022, a keel-laying ceremony was held for the first hull in Nantong, China. In 2023, the Russian government announced over RUB 27.5 billion (USD₂₀₂₃ 322 million) over 4 years for the construction of a marine terminal in the port of Pevek to support a fleet of RITM-200S floating nuclear power units.

Engagement



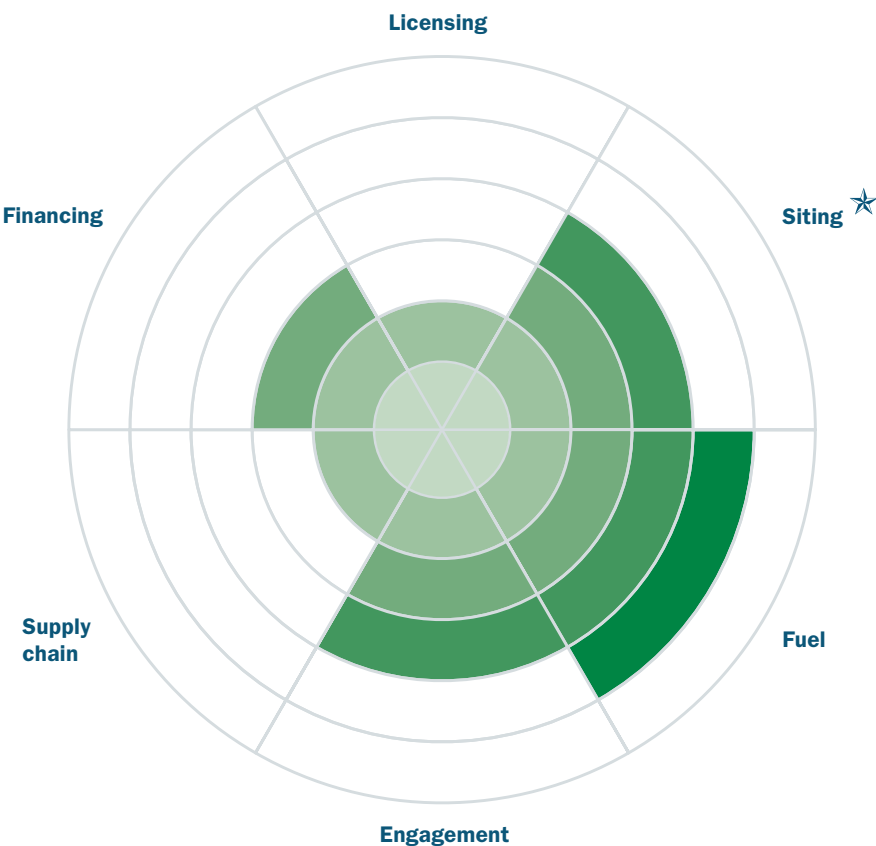
The local district administration of Cape Nagleynyn announced the start of public consultations for the marine-based RITM-200S floating nuclear power unit project in 2022.

Fuel



RITM-200S is designed to use HALEU fuel enriched to less than 20% and requires refuelling every 5 years. Russia has existing capabilities to produce HALEU fuel on a commercial scale. The floating RITM-200S shares aspects of fuel design with the RITM-200 series and is expected to leverage the same fuel supply chain. Fuel elements for the RITM-200 series have been designed and developed by the Bochvar Research Institute of Inorganic Materials (VNIINM), part of the Russian fuel manufacturer TVEL, which supplies nuclear fuel to all Russian nuclear power plants, including marine-based reactors. VNIINM designed the fuel elements, burnable absorber rods, and starting neutron source. The core is being produced at the TVEL machine building plant in Elektrostal, Russia.

Oklo – Aurora Powerhouse



- ★ Active in multiple jurisdictions or countries.
- △ Changes since Second Edition (2024).

Reactor description: Liquid metal-cooled, metal-fuelled fast reactor for heat and cogeneration, using fresh or recycled fuel.

Thermal power (MWth) Up to 205

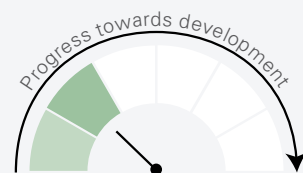
Core outlet temperature (°C) 500

Spectrum (thermal/fast) Fast

Fuel form Metallic U-Zr alloy

Base fissile material U-235, enrichment level:
≥ 10% to < 20%

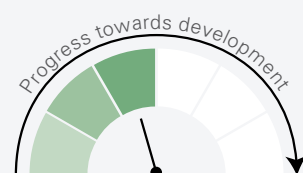
Licensing



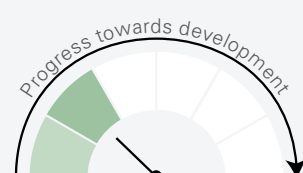
Siting ★



Financing



Supply chain



Engagement △

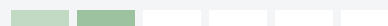


Fuel



Assessment of Aurora Powerhouse's progress to deployment

Licensing



Oklo has been engaged in pre-licensing activities with the US Nuclear Regulatory Commission (NRC) since 2016. In 2020, the NRC approved Oklo's Quality Assurance Program Description and safeguards information protection and handling plan. Since January 2022, Oklo has met with the NRC on a variety of topics, and has docketed multiple reports, licensing project plans and white papers with the NRC on topics related to the design and safety of the Aurora Powerhouse. Oklo plans to submit a Combined License Application (COLA) to the NRC in 2025 and is expected to benefit from previous activities related to the licensing of the Aurora Powerhouse with the NRC, including through the submission of a COLA in March 2020, which was denied without prejudice in 2022 following considerable NRC engagement.

Siting ★



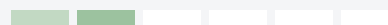
In 2024, Oklo received an Environmental Compliance Permit from the US Department of Energy (DOE) and the Idaho National Laboratory (INL) to construct a first-of-a-kind Aurora Powerhouse at INL. The next step in the project is to begin site characterisation activities. Oklo initially obtained a site use permit from the DOE for this location in 2019. Also in 2024, Oklo entered into a Land Rights Agreement with the Southern Ohio Diversification Initiative (SODI) for right of first refusal to purchase the site for the deployment of two Aurora Powerhouses at the decommissioned Portsmouth uranium enrichment plant in Piketon, Ohio. In 2023, Oklo signed a Memorandum of Understanding (MoU) with Centrus to supply power from these Aurora Powerhouses to the Centrus Piketon HALEU production facility.

Financing



Oklo has benefited from more than five sources of funding, including cost-share awards from the DOE Advanced Research Projects Agency – Energy (ARPA-E) programmes and the Technology Commercialization Fund. In May 2024, Oklo became a publicly traded company on the New York Stock Exchange and received approximately USD 306 million (USD₂₀₂₃ 297 million) of capital in the transaction. In December 2024, Oklo and Switch, a data centre development company based in Las Vegas, signed a collaboration agreement for Oklo to develop, construct, and operate Aurora Powerhouses to provide up to 12 GWe of power to Switch across the United States through a series of to-be-finalised power purchase agreements.

Supply chain



Oklo was granted six GAIN Vouchers to access some US national laboratory capabilities. In 2025, Oklo signed a non-binding agreement with RPower, a provider of natural gas-based power solutions, to propose a phased approach for meeting data centres' power needs – transitioning from natural gas to nuclear energy from the Aurora Powerhouse as it becomes commercially available. In 2024, Oklo selected Siemens Energy as the preferred supplier for the Aurora Powerhouse power conversion system through a non-binding agreement. In 2023, Oklo and Centrus Energy Corp signed an MoU to collaborate on the possible manufacturing of components for the Aurora Powerhouse at the Centrus manufacturing facilities in Piketon, Ohio, and Oak Ridge, Tennessee.

Engagement



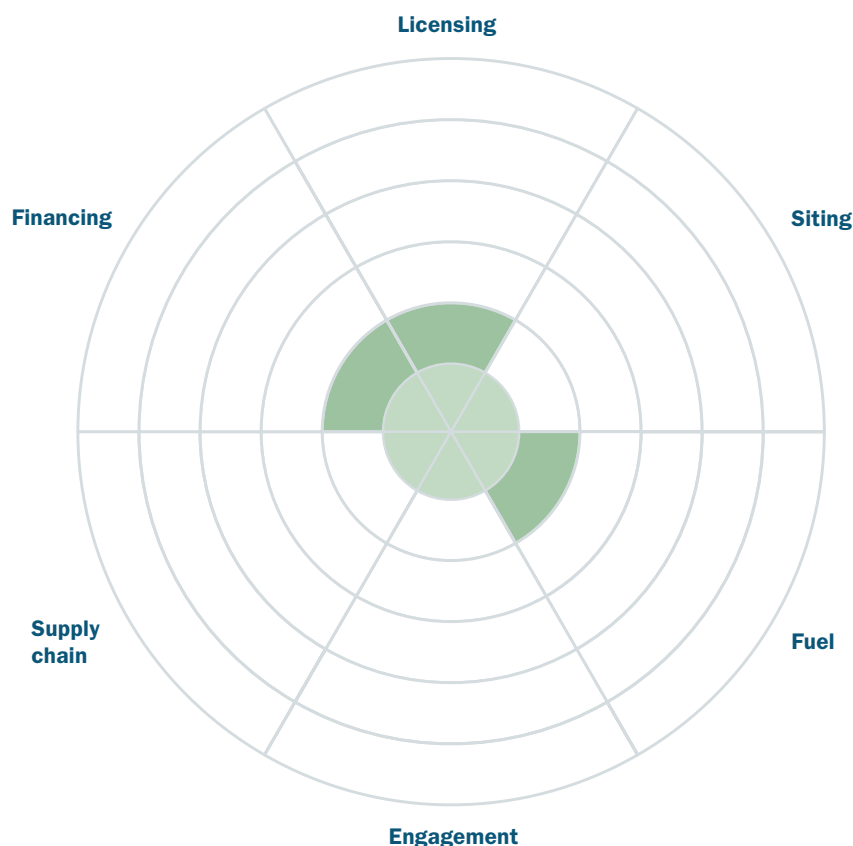
The Aurora Powerhouse team has engaged with community leaders in the region of Piketon, Ohio, including through community events organised in Pike and Scioto counties in 2024, and with SODI in 2023, gathering more than 70 community leaders and officials from federal, state and local governments. In September 2023, Oklo met with officials of the US Air Force and stakeholders in Alaska related to the opportunity to construct a microreactor facility on the Eielson Air Force Base Alaska. The Aurora Powerhouse was also selected as one of TIME's 200 Best Inventions of 2023 in the category of Reuse and Recycle technologies. In 2013, the co-founders of Oklo, both students at MIT, won the energy track of the MIT Entrepreneurship Competition with an early version of the Aurora Powerhouse design.

Fuel



Oklo fabricated and demonstrated metallic fuel prototypes with INL in 2019. In 2020, INL selected Oklo to receive HALEU for its first-of-a-kind Aurora Powerhouse by processing used fuel from the Experimental Breeder Reactor-II. In 2021, Oklo signed a non-binding letter of intent with Centrus to collaborate on deploying a HALEU facility at INL. This partnership expanded in 2023 through an MoU exploring Oklo's potential purchase of HALEU from Centrus. In 2024, the DOE approved Oklo's Conceptual Safety Design Report for the Aurora Fuel Fabrication Facility, planned for deployment at INL. In 2025, Oklo and Lightbridge signed an MoU to study the feasibility of co-locating their fuel fabrication facilities at INL and collaborating on nuclear waste recycling.

Otrera New Energy (Otrera) – Otrera 2x300



△ Changes since Second Edition (2024).

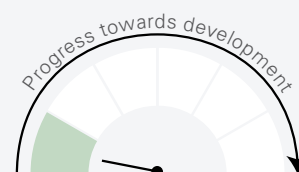
Reactor description: Compact sodium-cooled fast reactor deployed with two 300 MWth modules for cogeneration of electricity and heat.

Thermal power (MWth)	600 (plant deployed with two modules of 300 MWth each)
Core outlet temperature (°C)	550
Spectrum (thermal/fast)	Fast
Fuel form	MOX ceramic or Uranium-oxide ceramic
Base fissile material	- Plutonium, not requiring uranium enrichment (for MOX fuel) - U-235, enrichment level: ≥ 10% to < 20% (for UO₂ fuel)

Licensing



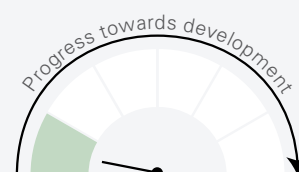
Siting



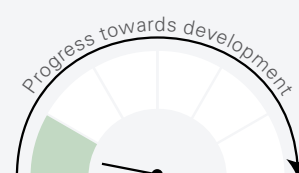
Financing



Supply chain



Engagement

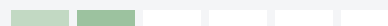


Fuel



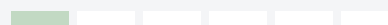
Assessment of Otrera 2x300's progress to deployment

Licensing



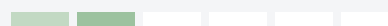
The Otrera 2x300 is designed for cogeneration with two modules, each providing 110 MWe and 180 MWth for process heat. As of early 2025, Otrera New Energy (Otrera) had begun the preparatory review stage of pre-licensing with the French Nuclear Safety and Radiological Authority (ASNR).

Siting



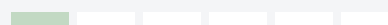
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



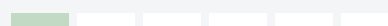
Otrera received EUR 10.0 million (USD₂₀₂₃ 10.8 million) under the France 2030 programme in 2023 and raised EUR 2.5 million (USD₂₀₂₃ 2.6 million) in an equity round in 2024. This equity round involves public and private investors including: Exergon, Invest For The Future, CEA Investissement, RÉEL International, Groupe ADF, EKIUM-Groupe SNEF, AXE CAPITOLE, NGE, and FORTIL.

Supply chain



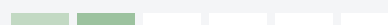
Otrera is one of five SMR projects selected in mid-2023 by the French Atomic Energy and Alternative Energies Commission (CEA) under CEA's French Atomic Sustainable Technologies (FAST) programme, aimed at advancing commercialisation of projects from R&D. As a spin-off from CEA, Otrera is expected to benefit from CEA's history of experience in designing, building and operating sodium-cooled fast reactors. Additionally, Otrera is a member of the Nuclear Valley network, a group of over 400 companies involved in the French nuclear supply chain.

Engagement



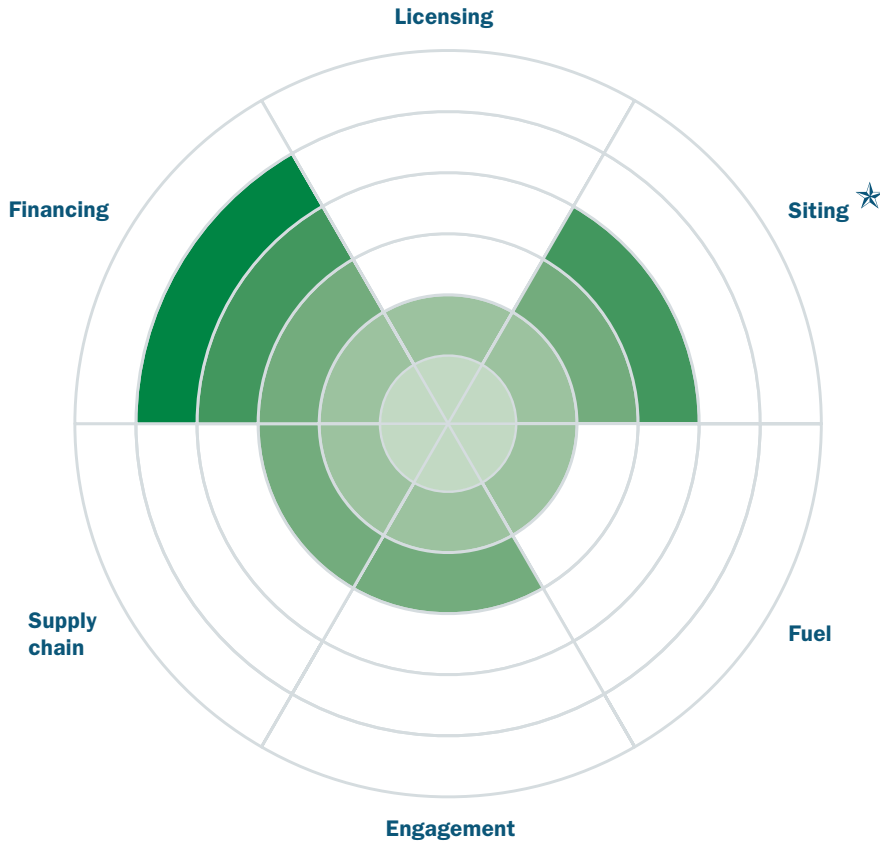
At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

Fuel



Otrera plans to use mixed plutonium-uranium oxide (MOX) fuel with an average plutonium oxide content of 17% or HALEU fuel enriched between 14.5% and 17.0%. In 2024, Orano launched a working group with Otrera and two other French fast-spectrum SMRs to develop a joint strategy for MOX fuel supply and reprocessing in France.

Radiant Industries – Kaleidos

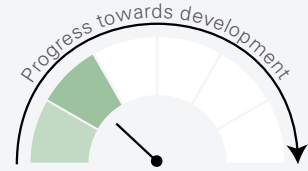


- ★ Active in multiple jurisdictions or countries.
- △ Changes since Second Edition (2024).

Reactor description: Compact, portable high temperature gas microreactor designed to replace diesel generators.

Thermal power (MWth)	3.5
Core outlet temperature (°C)	700
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%

Licensing



Siting ★



Financing △



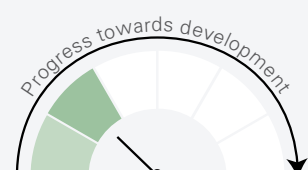
Supply chain △



Engagement △

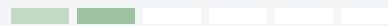


Fuel



Assessment of Kaleidos' progress to deployment

Licensing



Radiant Industries (Radiant) has been engaged in pre-licensing activities with the US Nuclear Regulatory Commission (NRC) for the Kaleidos SMR since October 2022. In October 2023, Radiant submitted a Pre-application Regulatory Engagement Plan for the Kaleidos SMR to the US NRC. The National Reactor Innovation Center (NRIC), a US Department of Energy (DOE) programme led by Idaho National Laboratory (INL), has been supporting regulatory engagement activities with the NRC, the DOE, and Radiant. In June 2024, the DOE announced that it had approved the Kaleidos safety design strategy, which is required before testing the Kaleidos SMR at the Demonstration Of Microreactor Experiments (DOME) facility at INL.

Siting ★



National Reactor Innovation Center (NRIC), a DOE programme led by INL, has selected Kaleidos to be demonstrated at INL's DOME facility in Idaho, United States, where testing in DOME could start as early as mid-2026. Radiant has also signed a Memorandum of Understanding with the mining company Idaho Strategic Resources to jointly study the suitability and licensing process to deploy a Kaleidos SMR at a site owned by Idaho Strategic Resources.

Financing



Radiant raised USD 160 million in venture funding including USD 40 million in Series B fundraising in April 2023 and USD 100 million (USD₂₀₂₃ 97.1 million) in Series C fundraising in November 2024, which it plans to use to test the Kaleidos microreactor at the DOME facility and support siting for a factory that would eventually produce up to 50 microreactors per year. In 2021-2023, Radiant received funding awards from the US Department of Defense to evaluate the integration of microreactors at the Hill Air Force Base. Radiant also received two awards from the NRIC programme: up to USD 1.5 million for front-end engineering and experiment design (FEEED) in 2023; and up to USD 5 million (USD₂₀₂₃ 4.9 million) for detailed engineering and experiment planning (DEEP) in 2024.

Supply chain



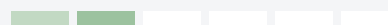
Radiant has placed a purchase order with Amsted Graphite Materials for nuclear-grade graphite for its Kaleidos SMR, and in May 2024 signed a MoU with the company to support domestic supply of graphite. Through the FEEED and DEEP processes, Radiant Industries was added to the qualified supplier list of the INL, and in 2025 will begin to procure major components for the deployment of the Kaleidos SMR at DOME as early as 2026. Radiant has also been awarded two vouchers under the US DOE Gateway for Accelerated Innovation in Nuclear (GAIN) initiative to advance modelling capabilities for their proposed reactor concept in partnership with the INL and Argonne National Laboratory.

Engagement



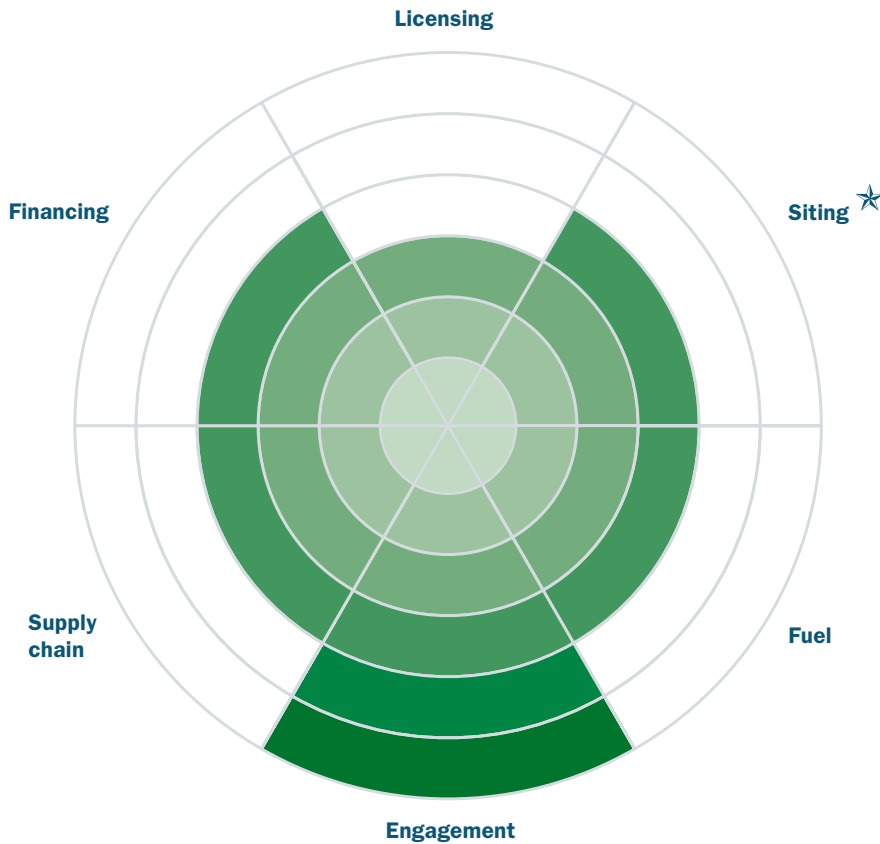
Radiant has engaged the public on the potential benefits and sustainability of the Kaleidos SMR, including through the 2023 Forbes Sustainability Leaders Summit, the 2024 Deep Tech Week in New York in June 2024, and the Atlantic Council's 2024 Frontiers Project where they discussed the role for microreactors to serve off-grid applications such as data centres, military installations, and mining.

Fuel



Radiant plans to use HALEU for its Kaleidos SMR. In 2022, Radiant launched a Request for Proposals for fuel fabricators to produce TRISO fuel, and in 2023 the company entered into an agreement with Centrus Energy to work towards a future supply of HALEU for up to 20 Kaleidos microreactors. In 2023, Radiant also received funding from the US DOE to support activities related to qualification modelling for TRISO fuel.

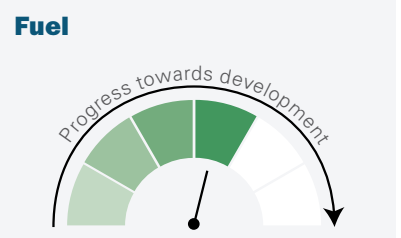
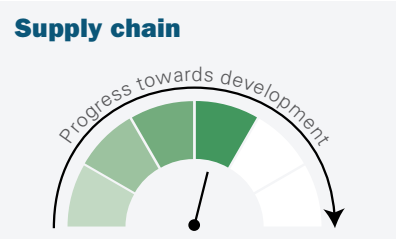
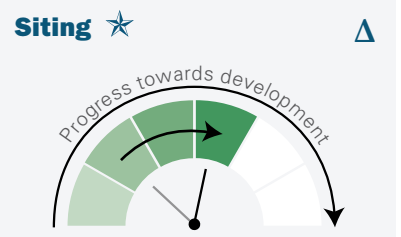
Rolls-Royce SMR – RR SMR



- ★ Active in multiple jurisdictions or countries.
- Δ Changes since Second Edition (2024).

Reactor description: Close-coupled, three-loop pressurised water reactor using low enriched uranium fuel.

Thermal power (MWth)	1 358
Core outlet temperature (°C)	325
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level: < 5%



Assessment of RR SMR's progress to deployment

Licensing



In July 2024, Rolls-Royce SMR (RR SMR) advanced to Step 3 of the UK Office for Nuclear Regulation (ONR) Generic Design Assessment (GDA) process, two years after submitting its design certification application for review. Regulators from Czechia, Finland, Poland, Sweden, and the Netherlands have observed GDA meetings to familiarise themselves with the RR SMR design and the GDA assessment process. This engagement may facilitate future deployment in their respective countries. Additionally, in 2024, the International Atomic Energy Agency conducted a Technical Safety Review of the Rolls-Royce SMR design.

Siting ★



In October 2024, Czech operator CEZ Group chose Rolls-Royce SMR for up to 3 GWe at its Temelín site, with early work planned to start in 2025. That same month, Great British Nuclear (GBN) shortlisted Rolls-Royce SMR among four SMR technologies for the next phase of the United Kingdom's deployment competition, with GBN selecting the sites. Also in 2024, Sweden's Vattenfall shortlisted Rolls-Royce SMR as one of two potential SMR suppliers for its Ringhals site. Poland's Ministry of Climate and Environment approved Industria's plans to deploy Rolls-Royce SMRs for its Central Hydrogen Cluster. UK-based Chiltern Vital Group (CVG) partnered with Rolls-Royce SMR for potential deployment at the Gloucestershire Science and Technology Park, which CVG acquired in 2024.

Financing



Rolls-Royce SMR has attracted more than GBP 490 million (USD₂₀₂₃ 713 million) in combined public and private sector funding to support its development and deployment. The UK government has supported the RR SMR, including with GBP 210 million (USD₂₀₂₃ 305 million) in funding through the UK Research and Innovation funding. In the private sector, partners in the Rolls-Royce SMR business have made equity investments. Specifically, Rolls-Royce Group, BNF Resources, Constellation, and the Qatar Investment Authority have raised nearly GBP 280 million (USD₂₀₂₃ 407 million) in combined funding for the RR SMR. In October 2024, ČEZ Group acquired a 20% equity position in Rolls-Royce SMR.

Supply chain



In 2024, Rolls-Royce SMR signed contracts with GSE Solutions for a power station simulator and the University of Sheffield to establish Module Development Facility concepts at the University Advanced Manufacturing Research Centre. The facility hosted the first supplier conference for the RR SMR in June 2024. Rolls-Royce SMR has secured multiple agreements for design and engineering services with: AtkinsRéalis, BAM Nuttall, Laing O'Rourke, Jacobs, Nuclear AMRC, Assystem, The Welding Institute, and the UK National Nuclear Laboratory. It also has Memoranda of Understandings (MoUs) with: Sheffield Forgemasters, and Škoda JS. In 2024, Rolls-Royce SMR was selected to form a Project-Working Group under the European Industrial Alliance on SMRs.

Engagement



Since 2021, Rolls-Royce SMR has received endorsements from UK government officials, including former Business and Energy Secretary Kwasi Kwarteng, and has been working with local stakeholders in six communities to identify potential sites for an SMR module factory. Additionally, Solway Community Power Company, an SMR development facilitator, chose Rolls-Royce SMR for deployment in West Cumbria, United Kingdom. Rolls-Royce SMR has also signed MoUs with: ULC-Energy in the Netherlands, the Jordan Atomic Energy Commission in Jordan, Constellation in the United States, and Energoatom in Ukraine.

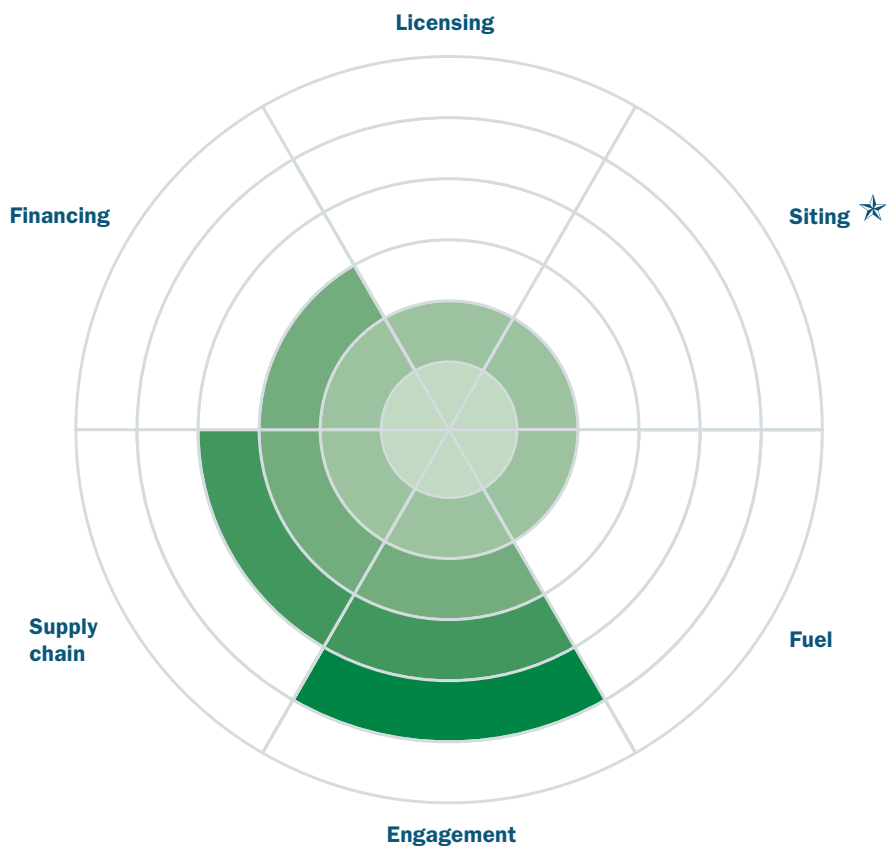
Fuel



The RR SMR utilises fuel that is the current industry standard for water-cooled reactor technologies that are similar in design. Given this, there are no barriers expected in the fuel supply chain for this SMR. In 2023, Rolls-Royce SMR signed an agreement with Westinghouse Electric Company to develop and advance work on the design of the RR SMR nuclear fuel.

Saltfoss Energy (Saltfoss)

– Compact Molten Salt Reactor (CMSR)



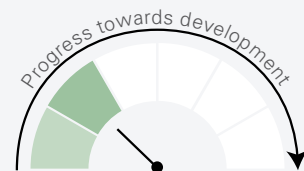
★ Active in multiple jurisdictions or countries.

Δ Changes since Second Edition (2024).

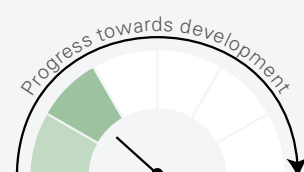
Reactor description: Multi-module, graphite moderated, liquid fluoride molten salt fuelled reactor on a power barge.

Thermal power (MWth)	250
Core outlet temperature (°C)	650
Spectrum (thermal/fast)	Thermal
Fuel form	Molten fluorides
Base fissile material	U-235, enrichment level: < 5%

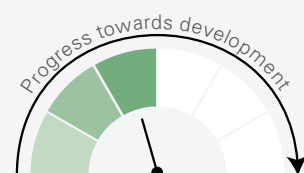
Licensing



Siting ★



Financing



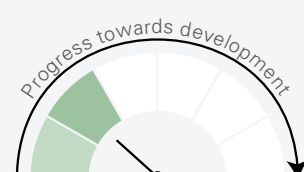
Supply chain



Engagement Δ

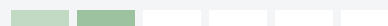


Fuel



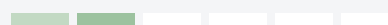
Assessment of CMSR's progress to deployment

Licensing



Saltfoss Energy (Saltfoss), formerly Seaborg, has been advancing qualification activities with respect to the international maritime legal framework. The American Bureau of Shipping (ABS) evaluated the CMSR through its New Technology Qualification process and in 2020 found that it satisfied the Feasibility Stage, the first of five phases. In 2022, ABS issued an Approval in Principle to South Korean shipbuilder Samsung Heavy Industries for the use of the CMSR on a power barge. The CMSR will require separate licensing from a nuclear safety regulator. At the time of assessment, there was no information readily available from verifiable public sources related to engagement with nuclear safety regulators.

Siting ★



Saltfoss signed Memoranda of Understanding (MoU) with Thai utility GPSC for a feasibility study in 2024 and Indonesian power company Pertamina NRE in 2023. In 2023, Saltfoss signed a letter of intent with Norwegian nuclear power project developer Norsk Kjernekraft to investigate the possible deployment of CMSR in Indonesia and in Norway. In 2022, Saltfoss conducted a joint study with Siemens Energy and Vietnamese power engineering consultancy PECC2 exploring the requirements for deploying floating nuclear power plants in Viet Nam, which included the preparation of a Strategic Environmental Assessment and the identification of potential sites for a CMSR power barge.

Financing



Saltfoss has received private and public sector funding to advance its CMSR and related technologies. In 2018, Saltfoss raised DKK 11.5 million (USD₂₀₂₃ 1.91 million) in venture financing, including from PreSeed Ventures, whose shares were later acquired by Heartland. In 2020, the company secured two loans from the Danish state investment fund Vaekstfonden and announced that they had secured USD 24 million (USD₂₀₂₃ 31 million) in private funding from a range of investors to accelerate the deployment of its CMSR technology. In 2022, the European Innovation Council (EIC) provided a EUR 2.5 million (USD₂₀₂₃ 2.3 million) grant and committed EUR 15 million (USD₂₀₂₃ 17.2 million) in equity through the 2022 EIC Accelerator programme.

Supply chain



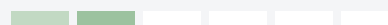
In 2024, Saltfoss signed MoUs with the South Gyeongsang Province in Korea and Korean Atomic Energy Research Institute on R&D and design of Molten Salt Reactors. In 2023, Saltfoss joined a consortium with Korea Hydro & Nuclear Power (KHNP) and Samsung Heavy Industries (SHI). SHI completed the conceptual design for the CMSR Power Barge. In 2022, Saltfoss signed an MoU with Best Engineering in Energy Solutions for regulatory support services. Saltfoss has been advancing research with the Technical University of Denmark (DTU), the ISIS Neutron and Muon Source in the United Kingdom and the European Spallation Source research centre in Sweden. Via the European Commission ERA-LEARN programme, Saltfoss is working with partners on development of multi-physics code.

Engagement



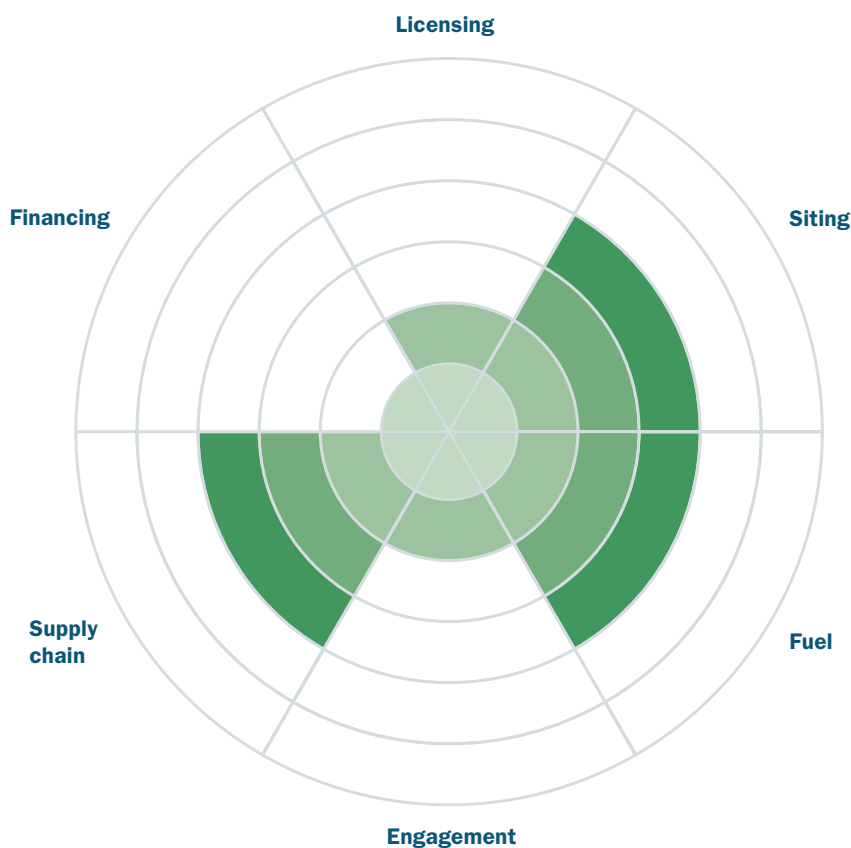
Saltfoss has engaged in outreach beyond the nuclear sector through: a 2025 DTU-Korean embassy seminar on SMRs in Denmark; articles in Danish news outlets; a panel on Danish public radio DR; an investment-focused webinar in 2025; a Harvard Business Review podcast in 2022; and discussions on Korea's Arirang News in 2022. In 2023, Saltfoss participated in Copenhagen Culture Night, hosting an open-door event for the public to meet Saltfoss scientists and learn about the CMSR. Further engagement includes Denmark's Folkemødet democracy festival in 2024. Saltfoss joined the EU SMR Alliance and the coal-to-nuclear REPOWER initiative led by NGO TerraPraxis.

Fuel



CMSRs are designed to use LEU in a molten salt blend of uranium, sodium and potassium fluorides. In 2024, KEPCO Nuclear Fuel, GS Engineering & Construction, and Saltfoss launched a joint feasibility study on LEU fuel salt production, building on their 2023 MoU to explore developing a fuel salt production facility. In 2023, Saltfoss chose to keep fuel enrichment below 5% for first units due to HALEU supply uncertainties. This led to choosing graphite as a moderator. Saltfoss also announced continued collaboration with Urenco on LEU for first units and HALEU for future CMSRs. Also in 2023, Indonesia's National Research and Innovation Agency (BRIN) discussed potential collaboration with Saltfoss on nuclear fuel and waste management R&D.

State Power Investment Corporation (SPIC) – HAPPY200



△ Changes since Second Edition (2024).

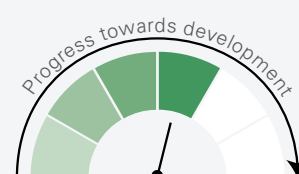
Reactor description: Low-temperature, low-pressure light water reactor.

Thermal power (MWth)	200
Core outlet temperature (°C)	120
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level: < 5%

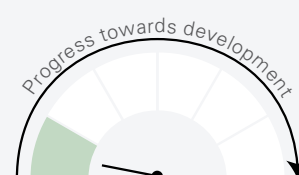
Licensing



Siting



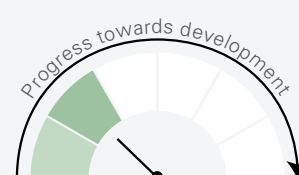
Financing



Supply chain



Engagement

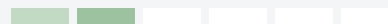


Fuel



Assessment of HAPPY200's progress to deployment

Licensing



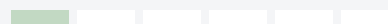
In September 2024, China's National Nuclear Safety Administration (NNSA) approved the Site Selection Safety Analysis Report for the deployment of HAPPY200 as part of the Shandong Haiyang Integrated Small Reactor Demonstration Project, allowing China's State Power Investment Corporation (SPIC) to proceed with design work based on the reported detailed site parameters.

Siting



In 2018, the municipal government of Baishan in Jilin province of the People's Republic of China (China) selected the site for the Baishan Nuclear Energy Heating Project. In 2019, following on a co-operation agreement signed in 2019 between SPIC and the municipal government of Baishan, the HAPPY200 SMR was selected for the Baishan Nuclear Energy Heating Project to provide district heating to an area over 80 square kilometres. The Site Selection Safety Analysis Report for HAPPY200 in Baishan was approved by the NNSA in September 2024.

Financing



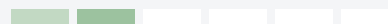
At the time of assessment, no information about financing was readily available from verifiable public sources.

Supply chain



In 2015, China Power Investment Corporation and the State Nuclear Power Technology Corporation merged to form SPIC. SPIC is an integrated energy organisation and performs nuclear supply chain roles associated with the development of the HAPPY200 SMR. This includes research, design and manufacturing of nuclear power components and equipment, as well as construction, operation and management activities. SPIC signed a general contracting framework agreement with the Shanghai Nuclear Engineering Institute to support the Nuclear Energy Heating Demonstration Project in Baishan.

Engagement



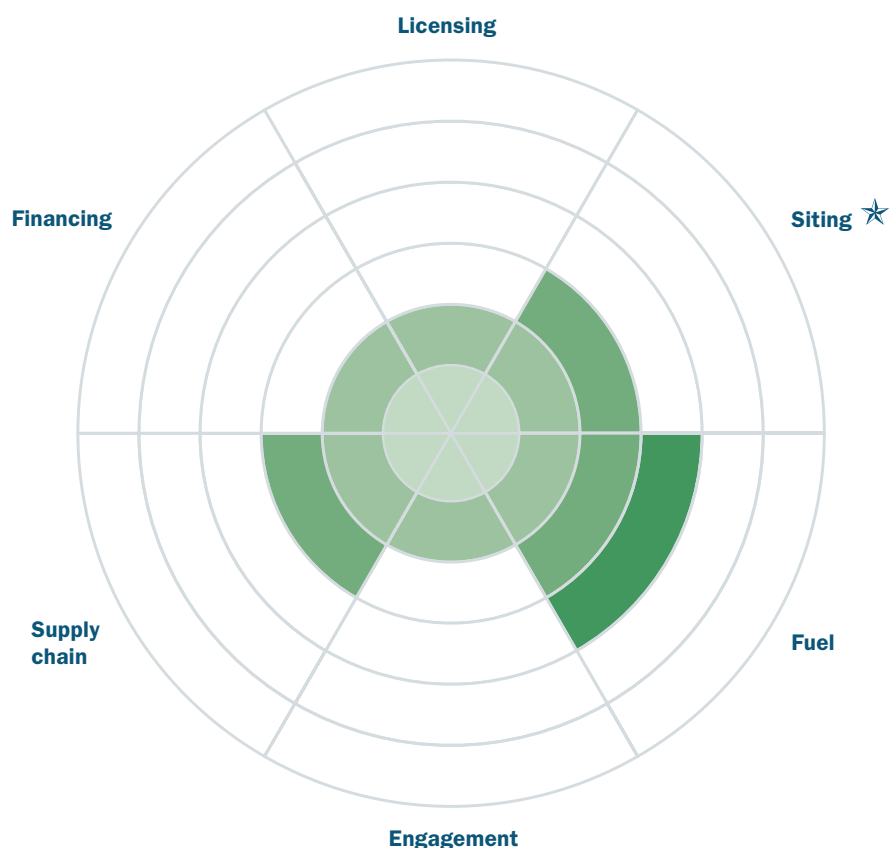
In June 2019, Zhang Zhijun, secretary of the Baishan Municipal Party Committee of Jilin Province, affirmed support from the city of Baishan for the demonstration of HAPPY200.

Fuel



The HAPPY200 utilises fuel that is the current industry standard for water-cooled reactor technologies and therefore relies on well-established supply chains.

Steady Energy – LDR-50



★ Active in multiple jurisdictions or countries.

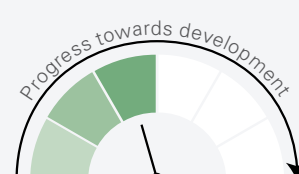
Reactor description: Simplified light water PWR that is specifically designed to produce low temperature heat (<150°C).

Thermal power (MWth)	50
Core outlet temperature (°C)	150
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level < 5%

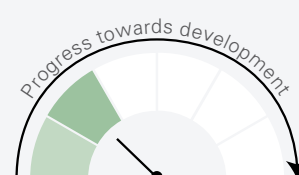
Licensing



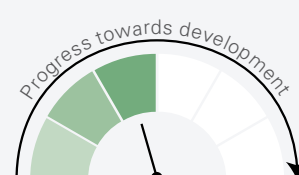
Siting ★



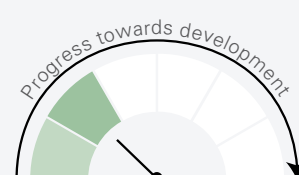
Financing



Supply chain



Engagement



Fuel



Assessment of LDR-50's progress to deployment

Licensing



In August 2024, Finland's Radiation and Nuclear Safety Authority (STUK) received documentation from Steady Energy to conduct a preliminary safety review of the LDR-50 design. This review may help inform the Finnish government's Decision-in-Principle, a required step before formal licensing in Finland.

Siting ★



In 2023 and 2024, Steady Energy signed letters of intent and agreements with Finnish utilities Helen, Kuopion Energia, and Keravan Energia about the potential deployment of LDR-50 SMRs for district heating in the Helsinki, Kuopio, and Kerava regions, respectively. In July 2024, Kuopion Energia initiated an environmental impact assessment for various potential sites in its region.

Financing



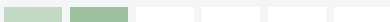
Steady Energy is a spin-off of the VTT Technical Research Centre where development of the LDR-50 first began during 2020-2023. In 2023, VTT announced a EUR 5 million (USD₂₀₂₃ 5.4 million) investment over two years to support the development of the LDR-50 design. In 2024, Steady Energy secured EUR 10 million (USD₂₀₂₃ 10.5 million) in seed funding, with backing from Yes VC and Lifeline Ventures.

Supply chain



In 2024, Steady Energy signed a contract with Tractebel for engineering services, including reactor design, safety assessment, and accident management. That same year, Steady Energy was selected to form a joint Project-Working Group, alongside another SMR developer, under the European Industrial Alliance on Small Modular Reactors. The company also announced plans to begin construction of an electrical pilot plant for the LDR-50 reactor in 2025.

Engagement



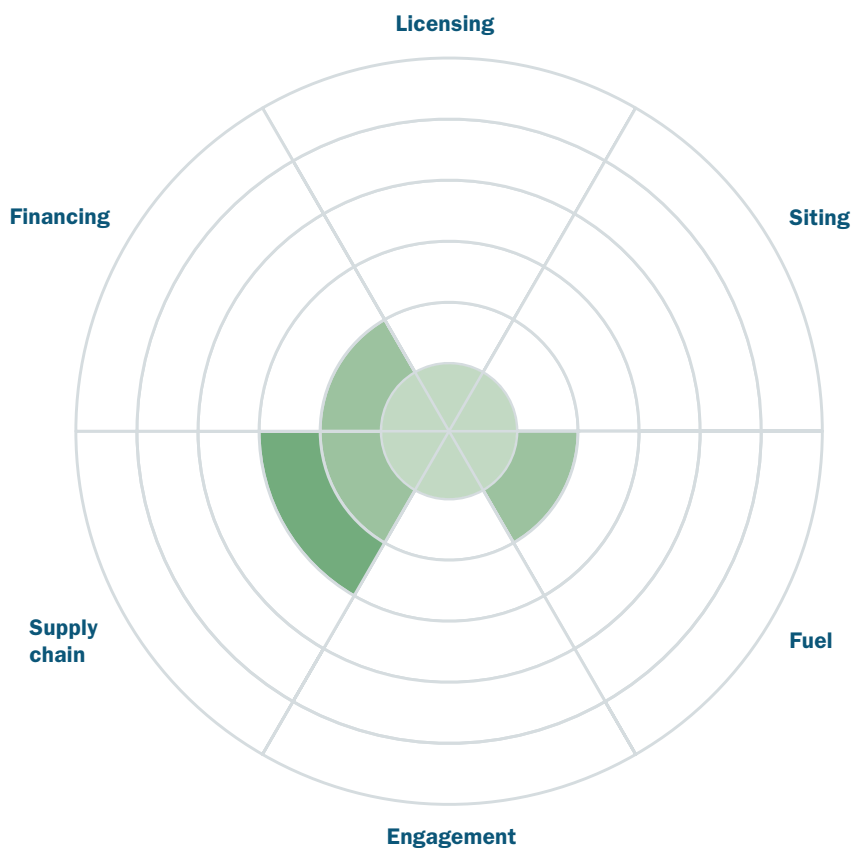
In September 2024, Last Energy participated in a public hearing organised by the City of Kuopio to discuss the utilisation of small modular reactors for district heating.

Fuel



The LDR-50 utilises fuel that is the current industry standard for water-cooled reactor technologies and therefore relies on well-established supply chains. At the time of the assessment, no additional information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel for the LDR-50.

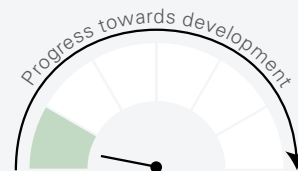
Stellaria – Stellarium



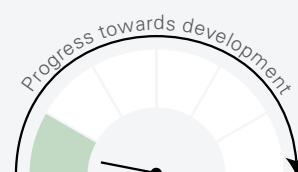
Reactor description: Multi-module, land-based, fast spectrum, molten salt breeder reactor.

Thermal power (MWth)	1 080 (plant deployed with two modules of 540 MWth each)
Core outlet temperature (°C)	750
Spectrum (thermal/fast)	Fast
Fuel form	Molten chloride
Base fissile material	- U-235, enrichment level: $\geq 10\%$ to $< 20\%$ - Plutonium, not requiring uranium enrichment

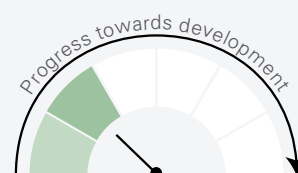
Licensing



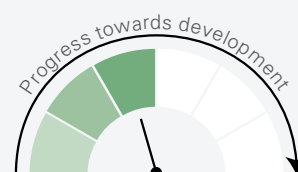
Siting



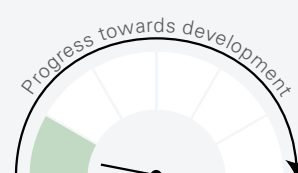
Financing



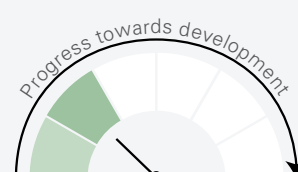
Supply chain



Engagement

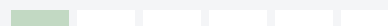


Fuel



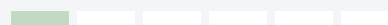
Assessment of Stellarium's progress to deployment

Licensing



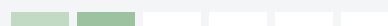
Stellaria plans to implement its Stellarium design through a three-phase strategy. The first phase involves developing the experimental ALVIN reactor, which will produce less than 1 MWth. This will be followed by the prototype reactor, MegaAlvin, and then the construction of the Stellarium for commercial use. Stellaria has initiated the pre-licensing process for the experimental ALVIN reactor with France's Nuclear Safety and Radiation Protection Authority (ASNR). At the time of the assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities for the Stellarium commercial SMR.

Siting



At the time of the assessment, no information related to siting was readily available from any site owners.

Financing



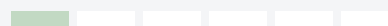
In 2024, the French government awarded EUR 8 million (USD₂₀₂₃ 8.6 million) to Stellaria for its project under the France 2030 programme.

Supply chain



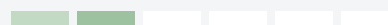
Stellaria has partnered with Unitel Technologies to deploy an Artificial Intelligence for High-Performance Computing (AI/HPC) cluster. This infrastructure will be used to perform advanced simulations and optimise reactor design. As part of its France 2030 project, Stellaria is working with engineering firms and research institutions to develop an initial reactor design with partners such as Technip Energies, Schneider Electric, and the French Atomic Energy and Alternative Energies Commission (CEA). Stellaria was one of two spin-offs from the CEA announced in March 2023.

Engagement



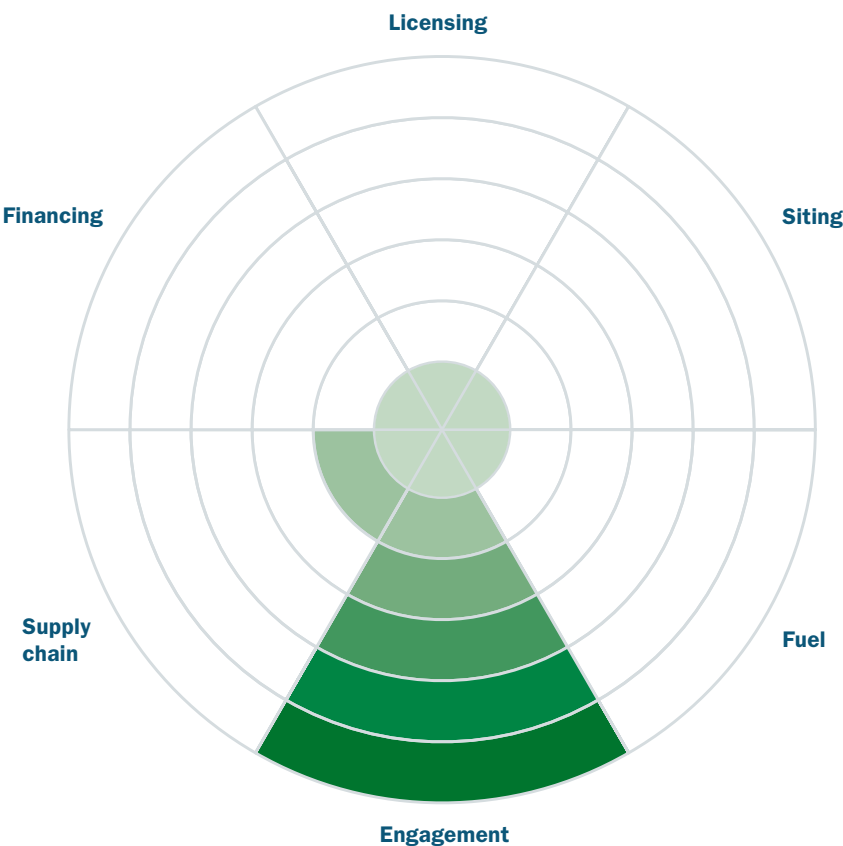
At the time of the assessment, no information was readily available from verifiable public sources related to engagement activities.

Fuel



Through its France 2030 project, Stellaria is partnering with Orano to advance the development of liquid molten-salt fuel. Stellaria is also part of a working group with Orano and two other fast spectrum molten-salt fuelled SMRs to develop a joint strategy for liquid molten-salt fuel supply, logistic and processing in France.

Stratek Global – HTMR-100



△ Changes since Second Edition (2024).

Reactor description: High-temperature gas-cooled TRISO pebble bed reactor.

Thermal power (MWth) 100

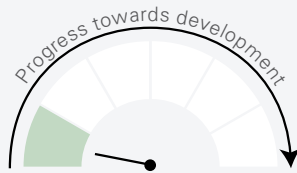
Core outlet temperature (°C) 750

Spectrum (thermal/fast) Thermal

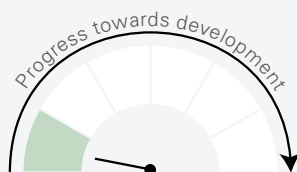
Fuel form Uranium-oxide ceramic TRISO pebble

Base fissile material U-235, enrichment level: ≥ 10% to < 20%

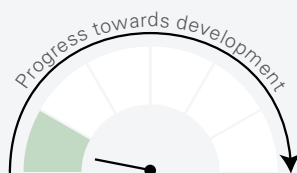
Licensing



Siting



Financing



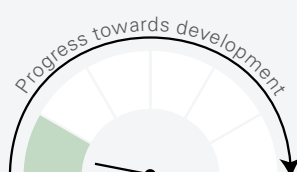
Supply chain



Engagement

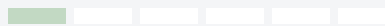


Fuel



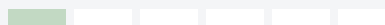
Assessment of HTMR-100's progress to deployment

Licensing



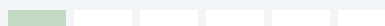
Stratek Global (Stratek) was formed in 2022 to commercialise the HTRM-100, which is based on the South African Pebble Bed Modular Reactor (PBMR) project, which was led by a consortium with Eskom, South Africa's largest utility, during the 1990s and 2000s. The PBMR project has been in a state of "care and maintenance" since 2010, when the government decided to cease investments. In November 2024, the South African Department for Mineral Resources and Energy announced its intention to recommend that the Cabinet approve lifting the care and maintenance status on the PBMR programme. At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



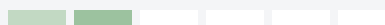
In 2023, Moneyweb, a South African financial news outlet, reported that a consortium led by C5 Capital, a US-based venture capital firm, is conducting a feasibility study with plans to construct the first in a series of HTMR-100 SMRs in the Western Cape, South Africa. At the time of assessment, no information related to siting was readily available from any site owners.

Financing



In 2024, Stratek and Koya Capital, a South African technology consultancy and financing firm, signed a partnership agreement under which Koya Capital will support efforts to secure ZAR 9.0 billion (USD₂₀₂₃ 467.2 million) in financing for the construction of the first-of-a-kind HTMR-100 reactor in South Africa. At the time of assessment, no additional information about financing was readily available from verifiable public sources.

Supply chain



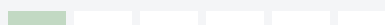
Stratek is working with: Dura-Power Consultants, a South African consulting company; ARINT, a South African engineering services company; and SMR Global Energy, an Australian company in which Stratek holds an equity share, to support the development of the HTMR-100. Stratek is also working with JK Design Architects, a South African architectural firm, to develop architectural renderings for nuclear power plants based on HTMR-100 technology. In 2025, Stratek signed an agreement with Omega Pinnacle Investments, a South African engineering and consulting services company, for the potential application of the HTMR-100 in the mining sector.

Engagement



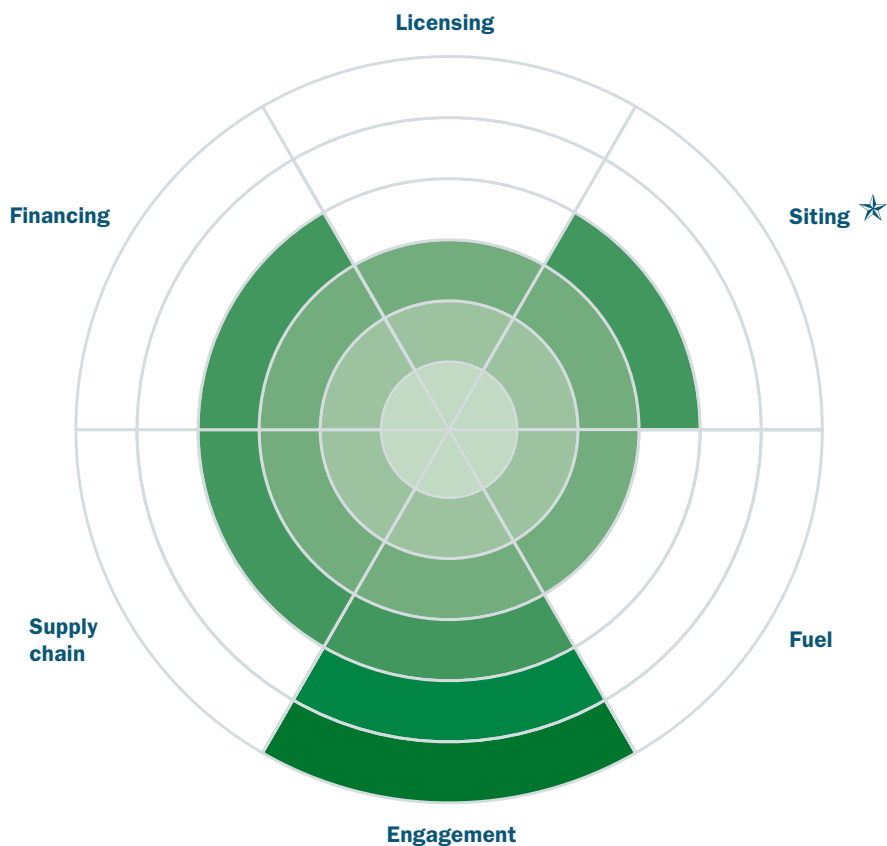
Stratek has taken steps to engage the public through contributions and interviews in online media, journals, news outlets, podcasts, radio, and YouTube, among others. Stratek has also engaged with military officials, non-profit organizations, universities, and farmers. In 2023, Stratek met with Limpopo Economic Development Agency to discuss the potential deployment of the HTMR-100 in the Limpopo province, South Africa. In 2024, the Southern African Agri Initiative, an agricultural network advocating for farmers' rights, and C5 Capital formed a strategic partnership to explore the use of HTMR-100 nuclear technology in agriculture. Since 2024, Stratek has been publishing a monthly magazine to raise awareness of nuclear technology, including the HTMR-100.

Fuel



In the 2000s, as part of the PBMR project, TRISO fuel pebbles were produced in South Africa and later sent to Idaho National Laboratory, United States, for irradiation tests. The South African Nuclear Energy Corporation (NECSA) has a TRISO pebble fuel manufacturing laboratory in Pretoria, South Africa, which has been in care and maintenance since 2010. A TRISO consortium led by Koya Nuclear Energy was created to accelerate the deployment of TRISO nuclear fuel for the HTRM-100. At the time of assessment, no additional information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

TerraPower – Sodium Reactor Plant

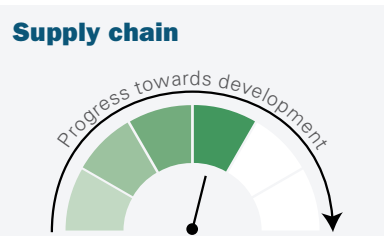
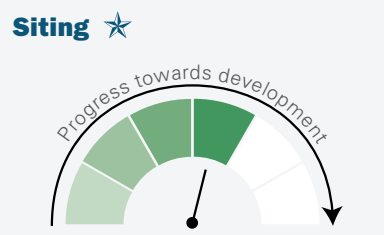


★ Active in multiple jurisdictions or countries.

Δ Changes since Second Edition (2024).

Reactor description: Sodium fast reactor coupled with a molten salt energy storage system designed to provide dispatchable power.

Thermal power (MWth)	840
Core outlet temperature (°C)	500
Spectrum (thermal/fast)	Fast
Fuel form	Metallic U-Zr alloy
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%



Assessment of Natrium's progress to deployment

Licensing



In 2024, the US Nuclear Regulatory Commission (NRC) registered the application submitted by TerraPower for a construction permit for its Natrium SMR at the Kemmerer Power Station Unit 1 site in Kemmerer, Wyoming, United States. The US NRC is reviewing the associated technical documentation and conducting Safety and Environmental Reviews.

Siting ★



PacifiCorp has selected the Natrium Reactor Plant to replace the retiring coal-fired Naughton Power Plant in Kemmerer, Wyoming. PacifiCorp has also launched feasibility studies to potentially deploy up to five additional Natrium Reactor Plants at multiple sites in its service territory by 2035. In 2023, TerraPower announced its purchase of land in Kemmerer for the Natrium demonstration project. In 2024, TerraPower began site preparation in Kemmerer, and, in 2025, the Wyoming Industrial Siting Council granted a construction permit for TerraPower to proceed with construction of non-nuclear portions of the Kemmerer Power Station Unit 1.

Financing



In 2020, TerraPower was selected as one of the two awardees of the US Department of Energy Advanced Reactor Demonstration Program (ARDP) to receive USD 1.98 billion (USD₂₀₂₃ 2.3 billion) from the US DOE in a matching-grant over seven years for the Natrium demonstration project. In 2021, USD 1.23 billion (USD₂₀₂₃ 1.4 billion) of this ARDP award was appropriated through the Infrastructure Investment and Job Act. TerraPower has also raised over USD 830 million (USD₂₀₂₃ 864 million) through one of the largest private capital raises in the advanced nuclear industry.

Supply chain



TerraPower has been developing Natrium with GE Hitachi since 2020 and has partnerships with Bechtel and Energy Northwest. TerraPower has agreements with Japan Atomic Energy Agency, Mitsubishi Heavy Industries and Mitsubishi FBR Systems, South Korea SK Group, and KHNP. It has selected Western Service Corporation, James Fisher Technologies, BWXT Canada, and Curtiss-Wright Flow Control Service for engineering services, and has partnerships with universities and Idaho and Argonne National Laboratories. In 2024, Bechtel and TerraPower began site preparation in Kemmerer, and placed orders for long-lead items with Teledyne Brown Engineering, ENSA, Doosan Enerbility, HD Hyundai, and Marmen. TerraPower has completed the procurement selection for all long lead items for its Natrium Reactor Plant.

Engagement



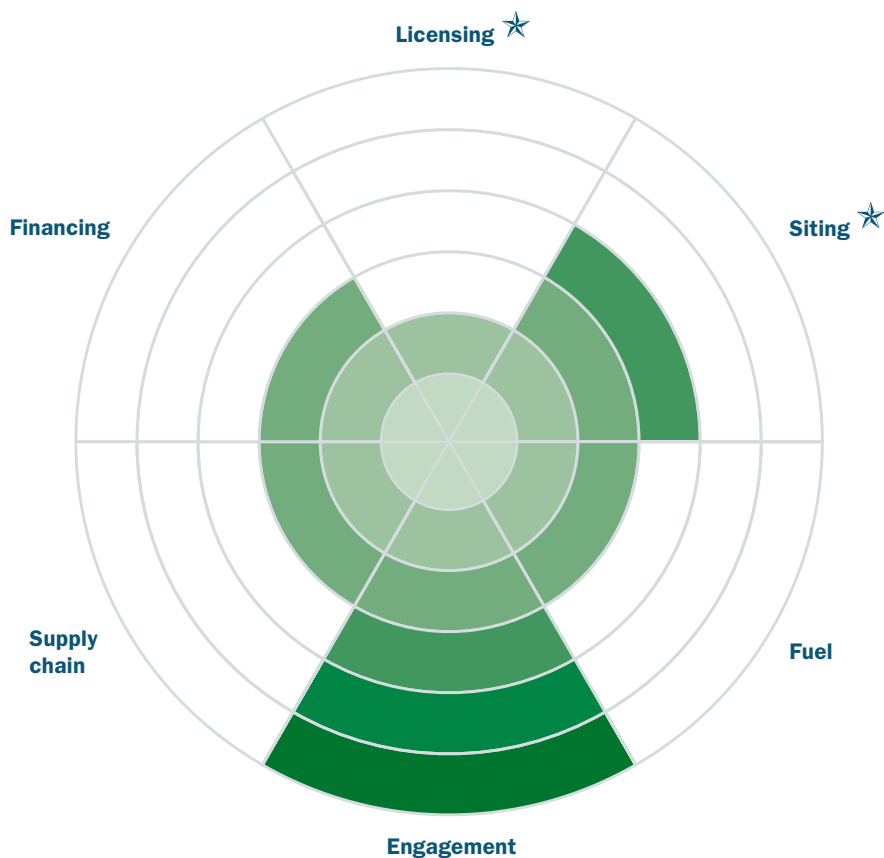
TerraPower has received endorsements from the Wyoming Governor Mark Gordon, Wyoming President of the Senate Dan Dockstader and Bill Thek, the Mayor of Kemmerer and has engaged with local communities in Kemmerer, including through public webinars, town hall meetings, and public hearings. In 2024, TerraPower also engaged with Sabey Data Centers to explore meeting their energy needs with Natrium reactors, and the general public through media and podcasts.

Fuel



TerraPower plans to use HALEU fuel for its Natrium Reactor Plant design. Global Nuclear Fuel-Americas (GNF-A), a GE-led joint venture, and TerraPower have announced a binding agreement to build a fuel fabrication facility at the site of GNF A's existing plant near Wilmington, North Carolina, United States. In 2023, TerraPower signed an MoU with Centrus Energy to work towards securing HALEU supply for the Natrium demonstration project. In 2024, TerraPower entered into partnerships with Framatome North America to support the development of a pilot facility for HALEU metallisation, and with ASP Isotopes for the production of fuel for the Natrium demonstration project and TerraPower's commercial multi-unit build programme.

Terrestrial Energy – Integral Molten Salt Reactor (IMSR)



★ Active in multiple jurisdictions or countries.

△ Changes since Second Edition (2024).

Reactor description: Fluoride molten salt fuelled, graphite moderated reactor using low-enriched uranium.

Thermal power (MWth)	884 (plant deployed with two modules of 442 MWth each)
Core outlet temperature (°C)	700
Spectrum (thermal/fast)	Thermal
Fuel form	Molten fluorides
Base fissile material	U-235, enrichment level: < 5%

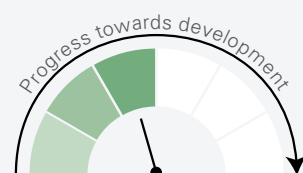
Licensing ★



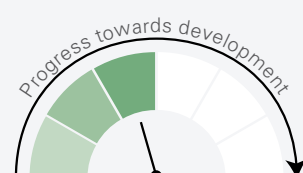
Siting ★



Financing



Supply chain



Engagement

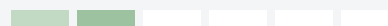


Fuel



Assessment of IMSR's progress to deployment

Licensing ★



Terrestrial Energy (TE) has completed Phase 2 of the Canadian Nuclear Safety Commission (CNSC) vendor design review (VDR) pre-licensing process for its Integral Molten Salt Reactor (IMSR) and is also engaged in pre-application activities with the US Nuclear Regulatory Commission (NRC). TE has submitted 10 topical reports and white papers to the US NRC, of which 9 have already been reviewed and received feedback. The CNSC and NRC are collaboratively reviewing the IMSR technology and aim to issue a common report providing feedback on the IMSR design. In its VDR report of the IMSR the CNSC did not identify any fundamental barriers to licensing.

Siting ★



In February 2025, John Sharp, Chancellor of the Texas A&M University System – a network of universities and state agencies in Texas – announced that Texas A&M University System had selected TE's IMSR for deployment at its RELLIS campus. This decision is part of "The Energy Proving Ground" project, which also includes the deployment of reactors from Kairos Power, Natura Resources, and Aalo Atomics at the RELLIS campus. In 2024, TE signed a Memoranda of Understanding (MoU) with EnergySolutions, driven by the rising demand for power for data centres. Under the MoU, TE and EnergySolutions are exploring the potential deployment of IMSR technology at sites owned by EnergySolutions, including the decommissioned Kewaunee Nuclear Power Station in Carlton, Wisconsin, United States.

Financing



TE has raised more than USD₂₀₂₃ 43 million in public and private sector financing in support of commercialising the IMSR. This includes more than CAD 15 million (USD₂₀₂₃ 13.9 million) in Series A financing, support from the US federal government for advancing R&D and pre-licensing activities with the NRC, and CAD 25.7 million (USD₂₀₂₃ 22.2 million) in Canadian federal funding through Sustainable Development Technology Canada and Canada's Strategic Innovation Fund.

Supply chain



TE has contracts with Aecon Group to confirm construction costs and timelines, Hatch for engineering, procurement, and project management, and Siemens Energy Canada to supply steam turbines and other equipment. TE also has contracts with BWXT, L3Harris Technologies, and KSB, as well as an MoU with Energy Northwest, Schneider Electric USA, and Numerical Advisory Solutions, a Zachry Group company. R&D is being advanced with universities, CNL, Oak Ridge National Laboratory, and the Nuclear Research and Consultancy Group in the Netherlands. TE has received three vouchers under the US Gateway for Accelerated Innovation in Nuclear initiative. Frazer-Nash, an engineering consultancy in the United Kingdom, has been contracted for graphite moderator fabrication engineering services.

Engagement



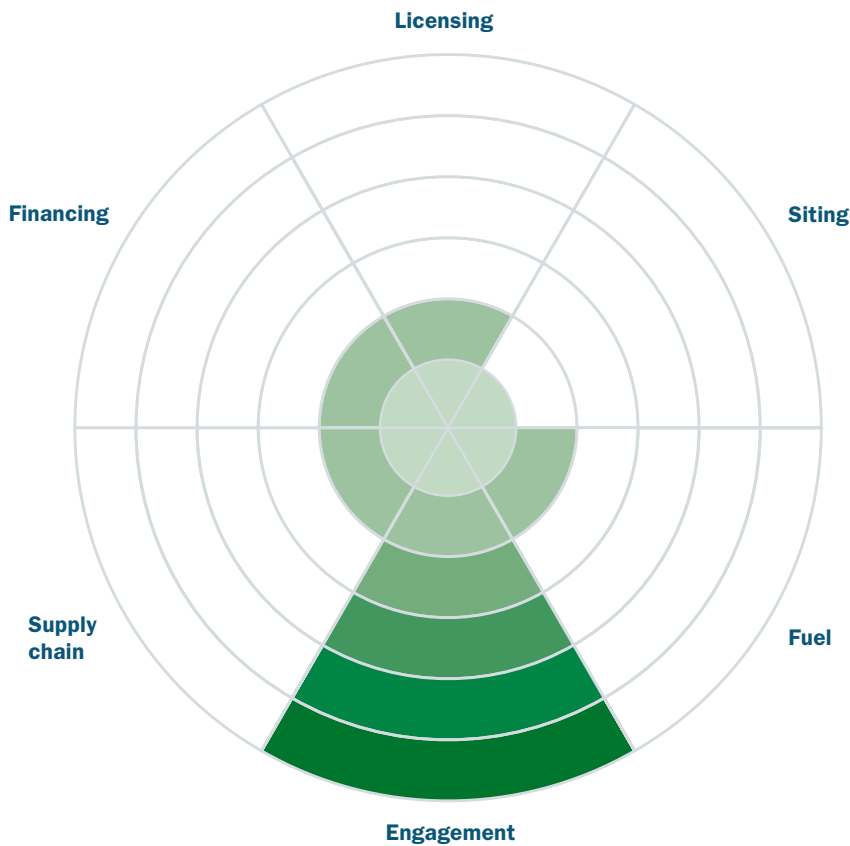
TE has signed a Letter of Intent with NGO TerraPraxis and joined the Fuel Cell & Hydrogen Energy Association. TE is a member of the First Nations Power Authority and has signed an MoU with Invest Alberta, Canada. In 2023 and 2024, TE signed MoUs with Emirates Nuclear Energy Corporation in the United Arab Emirates and Viaro Energy in the United Kingdom to explore the potential development of the IMSR, including applications in electricity generation and industrial heat. TE's advisors include senior experts in technical, industrial, financial, policy, regulatory, and environmental fields, including former government officials and former CEOs from industry and banking. The TE President has advanced public outreach since 2017 through such media as the New York Times, Mother Jones and others.

Fuel



TE plans to use low-enriched uranium in molten salt in a once-through fuel cycle. TE is advancing research to develop molten salt for the IMSR, including with: ENGIE Laborelec in Belgium; Canadian Nuclear Laboratories; the Joint Research Centre in Germany; and Argonne National Laboratory and Pacific Northwest National Laboratory in the United States. TE has been working to transition from laboratory to commercial scale, including through contracts with Orano and MoUs with Centrus Energy and Cameco for fuel and uranium supply, respectively. In 2023, TE also signed a contract with Springfields Fuels, a subsidiary of Westinghouse, to design and construct a pilot fuel plant for the IMSR in the United Kingdom.

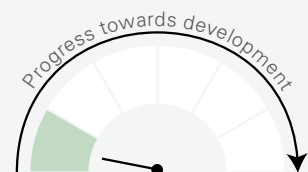
ThorCon International – ThorCon 500



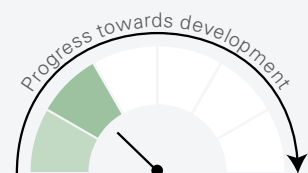
Licensing



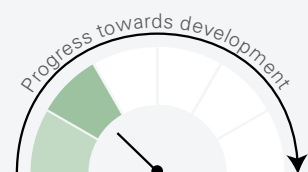
Siting



Financing



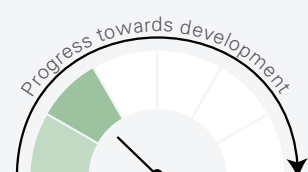
Supply chain



Engagement



Fuel

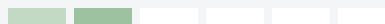


Reactor description: Molten salt reactor on a floating hull, designed for two power modules, ballasted to the seabed or shore.

Thermal power (MWth)	1 114 (plant deployed with two modules of 557 MWth each)
Core outlet temperature (°C)	704
Spectrum (thermal/fast)	Thermal
Fuel form	Molten fluorides
Base fissile material	U-235, enrichment level: < 5%

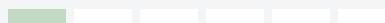
Assessment of ThorCon 500's progress to deployment

Licensing



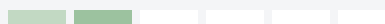
In March 2023, ThorCon submitted a consultation paper for its ThorCon 500 to Indonesia's Nuclear Energy Regulatory Agency (BAPETEN), to initiate the pre-licensing process in Indonesia.

Siting



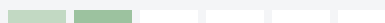
ThorCon has reported their interest to demonstrate and site ThorCon's ThorCon 500 on Gelasa (Kelasa) Island in Bangka Belitung (Babel) Islands Province. At the time of assessment, no information related to siting was readily available from any site owners.

Financing



ThorCon has created a Singapore-based special purpose company, supported by private investors, for the purpose of financing the ThorCon 500 project in Indonesia. ThorCon has prepared an investment fund of RP 17.0 trillion (USD₂₀₂₃ 1.11 billion) for the construction of the ThorCon 500 on Gelasa Island. ThorCon has indicated that the funds will not be dispersed until the project is approved.

Supply chain



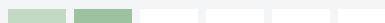
In 2017, ThorCon began feasibility studies with the Indonesia Thorium Consortium of state-owned companies. ThorCon is also collaborating with the Indonesian National Research and Innovation Agency (BRIN), University of Sebelas Maret (UNS), University of Bangka Belitung (UBB), Bandung Institute of Technology (ITB), and University of California Berkeley. In 2019, ThorCon signed Memoranda of Understanding (MoUs) with Daewoo Shipyard & Marine Engineering in South Korea and PT PAL Indonesia. In 2022, ThorCon signed collaboration agreements with the Spanish firm Empresarios Agrupados and with the University of Gadjah Mada for safety assessments of the ThorCon 500 design. ThorCon has also signed an agreement with Bureau Veritas, a conformity assessment body, to work on project certifications.

Engagement



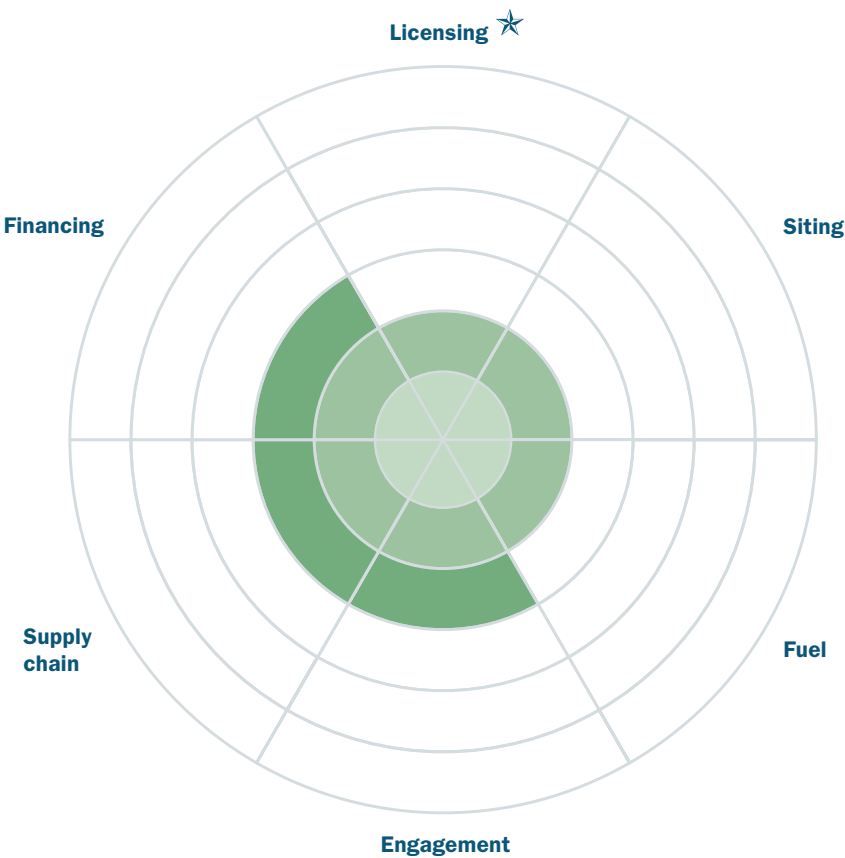
In the fall of 2023, engaging at the national level, ThorCon presented its latest progress and met with the National Energy Council of Indonesia. At the local level, ThorCon participated in two meetings organised by BAPETEN. ThorCon met with communities and representatives of the Central Bangka Regency, a province of the Bangka Belitung Islands. ThorCon also met with the acting governor of the Bangka Belitung Islands. Under ThorCon's three MoUs with UNS, UBB, and ITB universities, it participates in research and education programmes on nuclear energy in Indonesia. In addition, ThorCon organised one seminar and one webinar to raise awareness of the impacts of the deployment of its ThorCon 500 SMR.

Fuel



ThorCon intends to use low-enriched uranium (< 5% LEU) as a source of fuel for its ThorCon 500 design. In 2023, PT ThorCon Power Indonesia, opened the ThorCon-ITB Molten Salt Lab, developed in collaboration with ITB, to advance R&D on its molten salt fuel. Additionally, ThorCon has been awarded a voucher under the United States Department of Energy Gateway for Accelerated Innovation in Nuclear (GAIN) initiative in partnership with Argonne National Laboratory (ANL) to work on sodium fluoride and beryllium fluoride salt properties for liquid fuelled fluoride molten salt reactors.

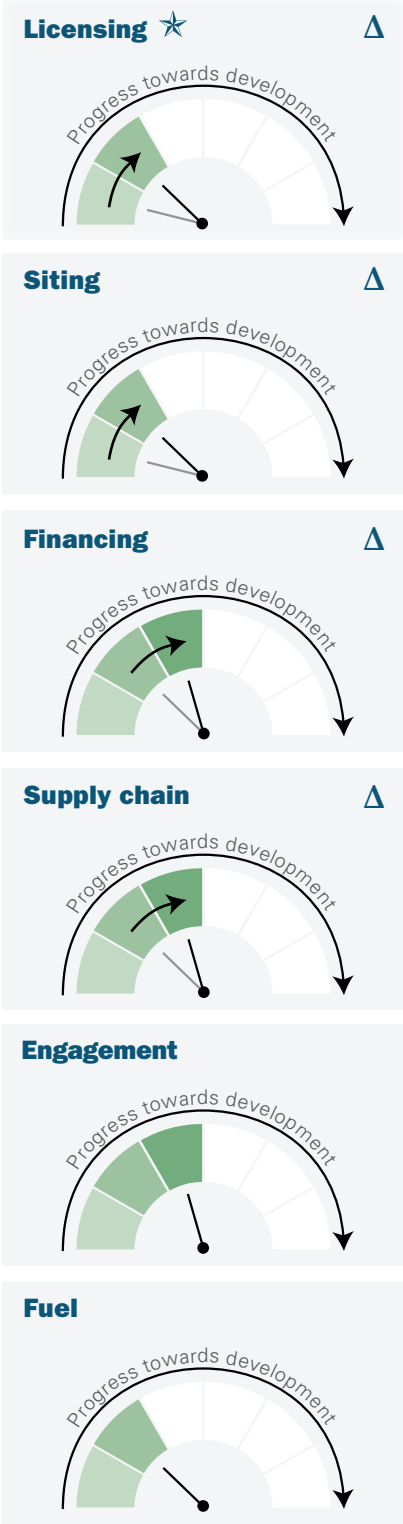
Thorizon – Thorizon One



- ★ Active in multiple jurisdictions or countries.
- △ Changes since Second Edition (2024).

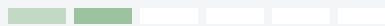
Reactor description: Cartridge core molten salt reactor using spent fuel and thorium with option for thermal or fast spectrum.

Thermal power (MWth)	250
Core outlet temperature (°C)	550
Spectrum (thermal/fast)	Both thermal and fast
Fuel form	Molten chlorides
Base fissile material	Plutonium, not requiring uranium enrichment



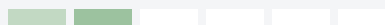
Assessment of Thorizon One's progress to deployment

Licensing



Thorizon is undergoing a joint preparatory review with the French Nuclear Safety and Radiation Protection Authority (ASNR) and the Dutch Authority for Nuclear Safety and Radiation Protection (ANVS). This review focuses on evaluating the reactor's design and safety features to streamline future licensing processes in both countries. This joint preparatory review is progressing under a Memorandum of Understanding signed in September 2023 between ASNR and ANVS.

Siting



Thorizon has announced it is working with EPZ, the Dutch nuclear power plant operator, to evaluate the feasibility of deploying a demonstration reactor at the Borssele nuclear power plant site in Zeeland, Netherlands. EPZ is collaborating with industry to prepare for SMR development at industrial sites in the Netherlands and is engaged with Thorizon as well as a shortlist of other SMRs for this purpose.

Financing



In 2020, Thorizon received EUR 350 000 (USD₂₀₂₃ 449 000) from the Dutch Government in loans. In 2021, Positron Ventures led a EUR 750 000 (USD₂₀₂₃ 938 000) pre-seed round for Thorizon. In 2022, Thorizon raised EUR 12.5 million (USD₂₀₂₃ 14.4 million) from private investors, including Positron Ventures and Huisman, and public investors, including Invest-NL, the Participation Fund Sustainable Economy North Holland (PDENH), and Impuls Zeeland. In 2024, the Thorizon-led consortium PROXIMA was awarded EUR 10 million (USD₂₀₂₃ 10.8 million) from the France 2030 programme. Also in 2024, Thorizon formed a consortium with Demcon and VDL Group to establish an advanced testing facility for molten salt reactor technologies. The consortium submitted a matching grant application to the Dutch Province of Noord-Brabant in October 2024.

Supply chain



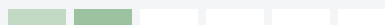
Thorizon, a spin-off of the Netherlands' national nuclear research institute NRG, is working with Tractebel for engineering services. Thorizon is part of two consortia: with Oakridge, Ecole Centrale Lille, Lille University, Orano, and France's Alternative Energies and Atomic Energy Commission (CEA) in France for research and development; and with Demcon and VDL Group in the Netherlands for manufacturability and safety analysis. Thorizon has also contracted Electricité de France (EDF) for neutronic core calculations and nuclear fuel cycle scenario analyses, and the Dutch Institute for Fundamental Energy Research (DIFFER) for corrosion studies. Thorizon was selected for a Project-Working Group under the European Industrial Alliance on SMRs.

Engagement



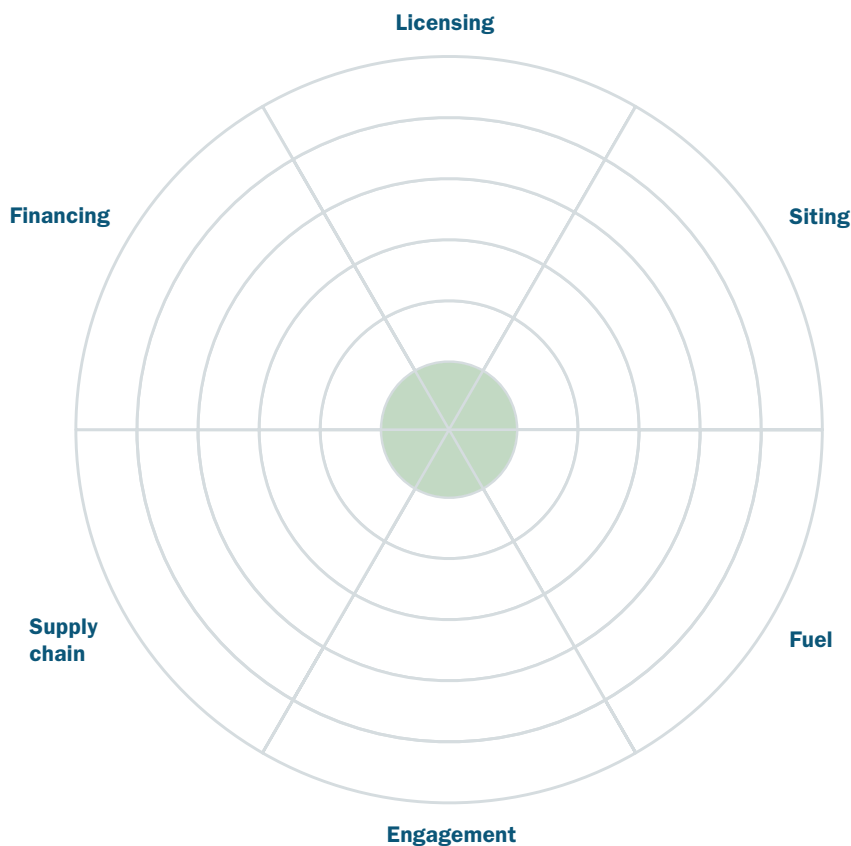
Thorizon has initiated engagement and outreach activities, including interviews with the Dutch news radio channel BNR, a video interview with the Energy Impact Center, and podcasts with Studio Energy. In 2023, Sander de Groot, Thorizon's co-founder and CTO, presented his project and vision to the Royal Netherlands Institute of Engineers.

Fuel

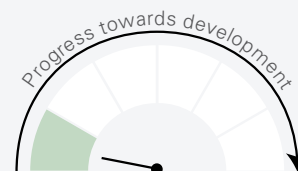


Thorizon plans to use a chloride molten fuel salt mixture of uranium, plutonium from spent nuclear fuel, and thorium. The fuel is contained in cartridges, separated from the molten salt coolant. Under the PROXIMA project, Thorizon is working with Orano to develop liquid molten-salt fuel and is part of a working group with Orano and two other fast spectrum molten-salt fuelled SMRs to develop a joint strategy for liquid molten-salt fuel supply, logistic and processing in France. Thorizon is a member of MIMOSA (Multi-recycling strategies of LWR SNF focusing on MOlten SAIt technology), an Orano-led consortium of 14 parties including EDF, Delft University and NRG, with a budget of EUR 5.75 million (USD₂₀₂₃ 6.6 million) from the Euratom HORIZON fund to work on recycling spent nuclear fuel.

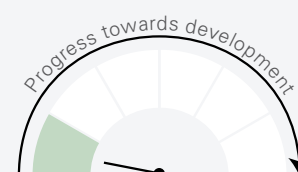
Toshiba Energy Systems & Solutions Corporation – MoveLuX



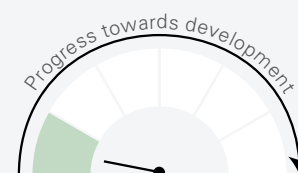
Licensing



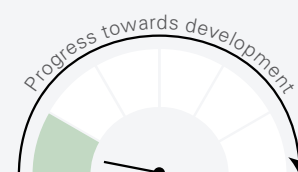
Siting



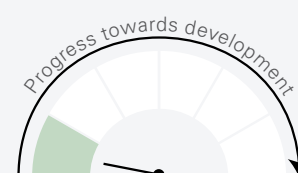
Financing



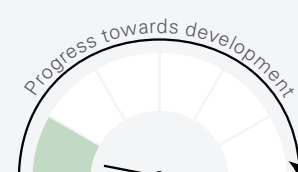
Supply chain



Engagement



Fuel

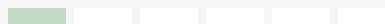


Reactor description: Heat-pipe-cooled calcium-hydride moderated microreactor.

Thermal power (MWth)	10
Core outlet temperature (°C)	685
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium silicide ceramic
Base fissile material	U-235, enrichment level: < 5%

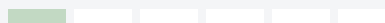
Assessment of MoveluX's progress to deployment

Licensing



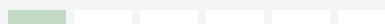
The MoveluX is in the conceptual design phase. At the time of assessment, there was no information readily available from verifiable public sources related to licensing and pre-licensing activities.

Siting



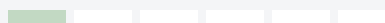
At the time of assessment, no information related to siting was readily available from any site owners.

Financing



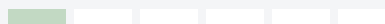
At the time of assessment, no information about financing was readily available from verifiable public sources.

Supply chain



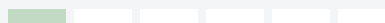
At the time of assessment, no information about supply chain readiness was readily available from verifiable public sources.

Engagement



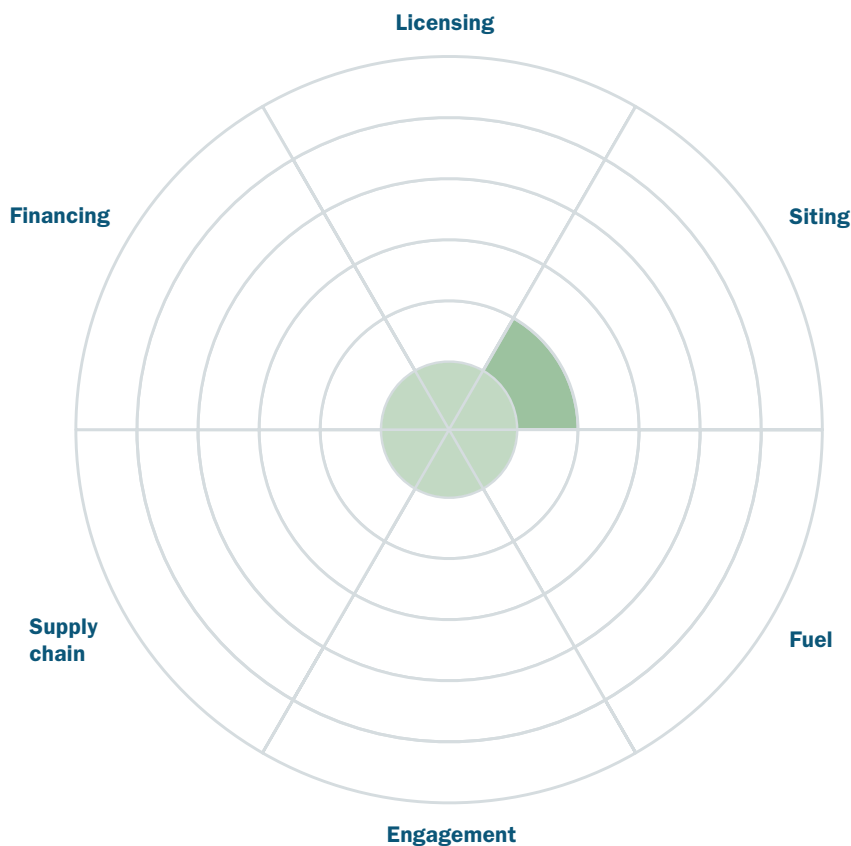
At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

Fuel

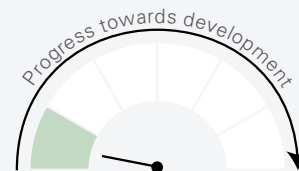


The MoveluX proposes to use a hexagonal uranium silicide (U_3Si_2) fuel assembly with a uranium enrichment of less than 5%. At the time of assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

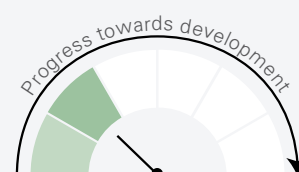
Toshiba Energy Systems & Solutions Corporation – 4S



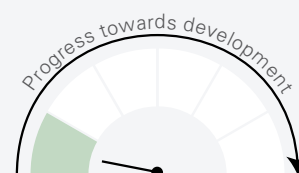
Licensing



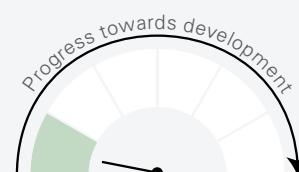
Siting



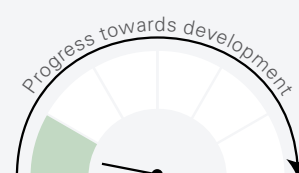
Financing



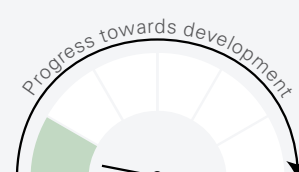
Supply chain



Engagement



Fuel



Reactor description: Sodium-cooled fast reactor intended for remote areas.

Thermal power (MWth) 30

Core outlet temperature (°C) 510

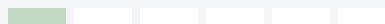
Spectrum (thermal/fast) Fast

Fuel form Metallic U-Zr alloy

Base fissile material U-235, enrichment level:
≥ 10% to < 20%

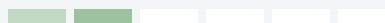
Assessment of 4S's progress to deployment

Licensing



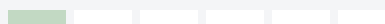
In 2007 Toshiba first began engagement with the US Nuclear Regulatory Commission (NRC) through a pre-application review for design approval of the 4S. In 2012, Toshiba submitted Technical Reports to the NRC related to safety design criteria for the 4S to support this pre-application review. These materials have since been archived by the NRC. At the time of assessment, there was no additional information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



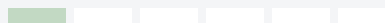
In 2007, the Alaska Electric Light and Power Company and the City of Nome both wrote to the NRC indicating their interest in the 4S SMR for replacing diesel generators in remote areas of Alaska.

Financing



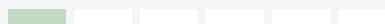
At the time of assessment, no information about financing was readily available from verifiable public sources.

Supply chain



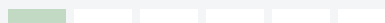
At the time of assessment, no information about supply chain readiness was readily available from verifiable public sources.

Engagement



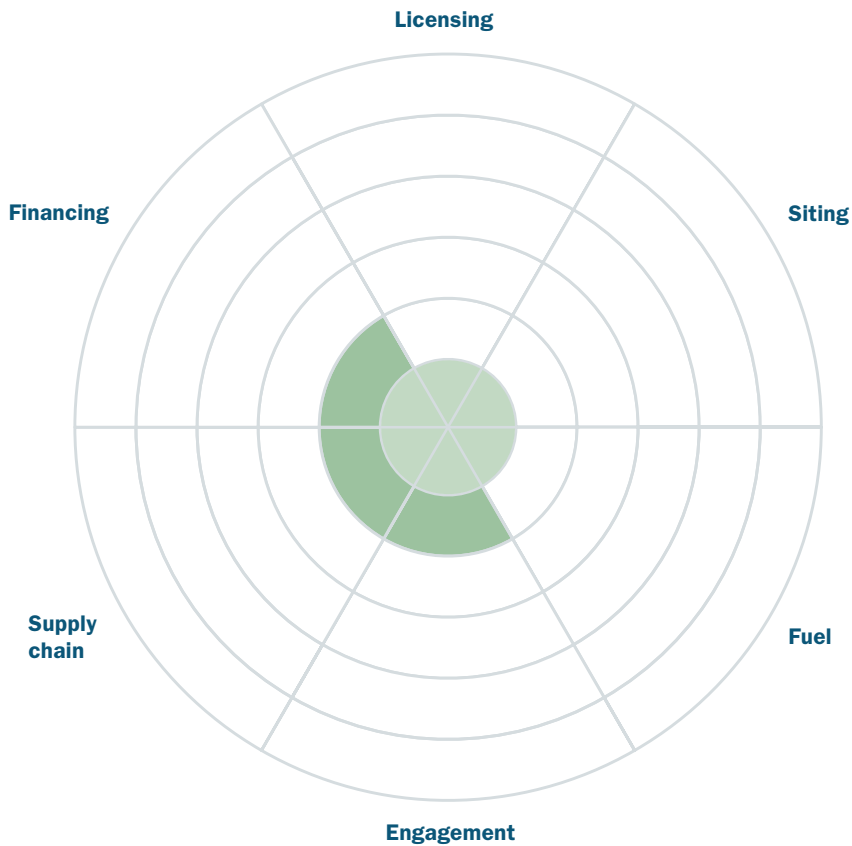
At the time of assessment, no recent information was readily available from verifiable public sources related to engagement activities.

Fuel

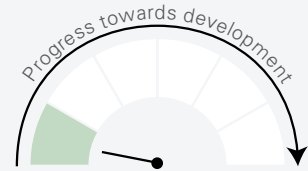


The 4S proposes to use HALEU fuel. At the time of assessment, no information was readily available from verifiable public sources to assess progress towards the commercial supply of qualified fuel.

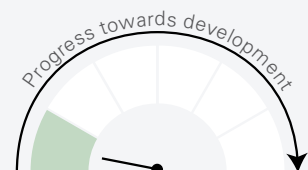
Transmutex – START



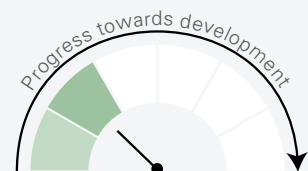
Licensing



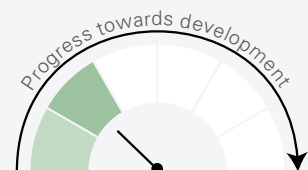
Siting



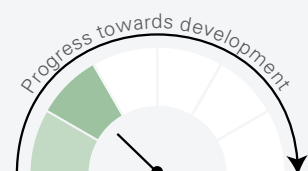
Financing



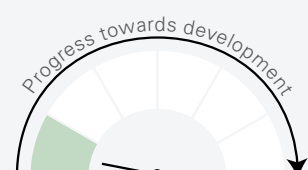
Supply chain



Engagement



Fuel



Reactor description: A land-based subcritical accelerator-powered lead-cooled reactor.

Thermal power (MWth)	600
Core outlet temperature (°C)	450
Spectrum (thermal/fast)	Fast
Fuel form	Other (metallic thorium fuel)
Base fissile material	Plutonium, not requiring uranium enrichment

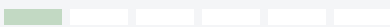
Assessment of START's progress to deployment

Licensing



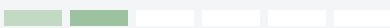
START combines a nuclear reactor with an external accelerator that provides fast neutrons to affect a sub-critical fission reaction that can transmute long-lived radioactive waste. A study funded by Germany's Federal Agency for Disruptive Innovations, completed in early 2025, considered technical readiness, safety compliance, and legal frameworks for START. The study was independently validated by TÜV Nord, the Technical University of Munich, and the law firm PSWP, and found that the plant design complies with German safety requirements and is consistent with Germany's legal framework excepting the reprocessing of spent fuel which is currently prohibited in Germany. At the time of the assessment, no additional information about licensing was readily available from verifiable public sources.

Siting



At the time of the assessment, no information about siting was readily available from verifiable public sources.

Financing



As of January 2025, Transmutex had joined the NEWTON (Nuclear Energy Waste Transmutation Optimized Now) programme, sponsored by the Advanced Research Projects Agency-Energy (ARPA-E) under the US Department of Energy (DOE), and had been awarded USD 4.3 million (USD₂₀₂₃ 4.2 million). In January 2024, Transmutex raised USD 23.1 million (USD₂₀₂₃ 22.4 million) in a Series A private investment round led by Union Square Ventures, Steel Atlas, and eight other investors. Earlier funding rounds include USD 5.78 million (USD₂₀₂₃ 6.50 million) in July 2021 from Union Square Ventures and AlleyCorp, and a USD 102 000 (USD₂₀₂₃ 106 198) investment from Fondation Genevoise pour l'Innovation Technologique (FONGIT) in July 2022.

Supply chain



In 2022, the CNRS (French National Centre for Scientific Research) signed a five-year research partnership with Transmutex.

Engagement



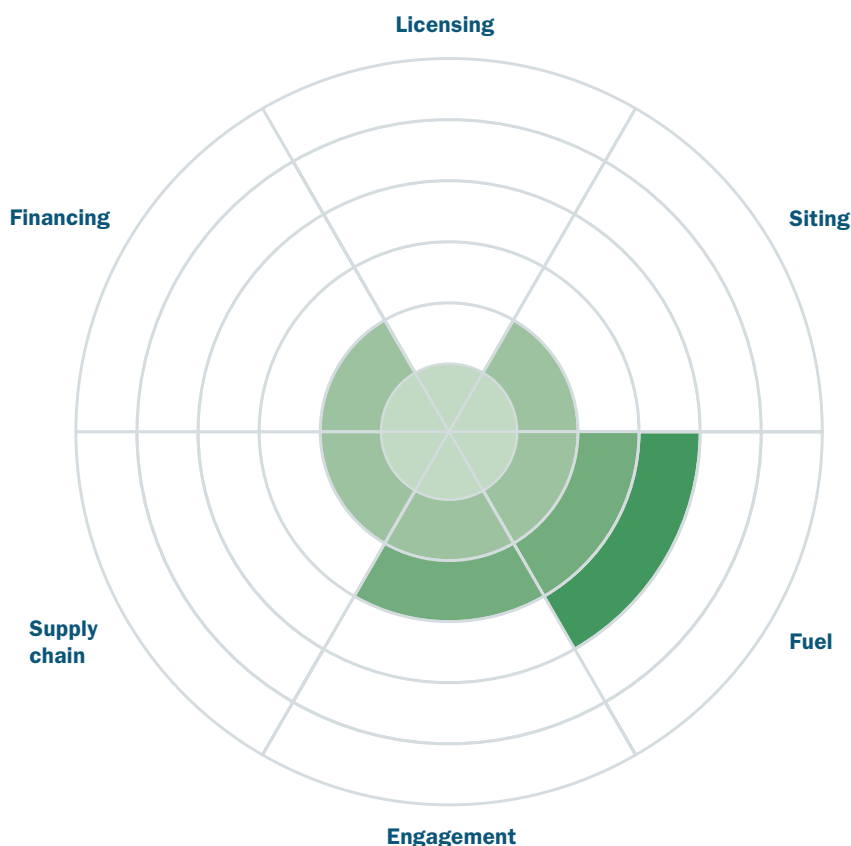
In February 2025, Transmutex CEO, Franklin Servan-Schreiber, presented the START and its waste recycling features to the World Governments Summit, an annual event that convenes governments, international organisations, and industry, in Dubai.

Fuel

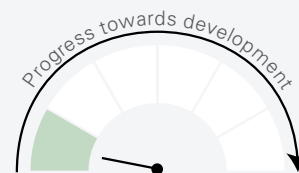


Transmutex aims to close the thorium fuel cycle and reduce long-lived radioactive waste streams with its transmutation reactor. The START SMR is designed to use thorium metallic fuel containing a high percentage of transuranics from spent nuclear fuel. At the time of the assessment, no information about fuel was readily available from verifiable public sources.

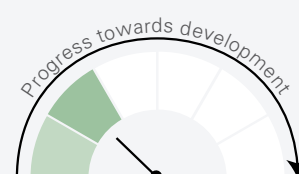
University of West Bohemia and the Czech Technical University in Prague (UWB and CIIRC CTU) – TEPLATOR



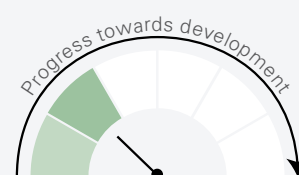
Licensing



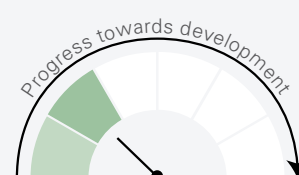
Siting



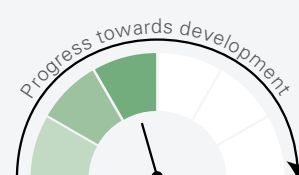
Financing



Supply chain



Engagement



Fuel

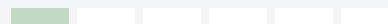


Reactor description: Low temperature, low pressure and low power density heavy water reactor for district heating.

Thermal power (MWth)	50 to 170
Core outlet temperature (°C)	180
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, not requiring enrichment

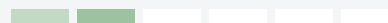
Assessment of TEPLATOR's progress to deployment

Licensing



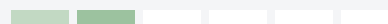
At the time of assessment, there was no information readily available from verifiable public sources related to licensing or pre-licensing activities.

Siting



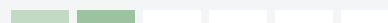
In 2023, a tripartite agreement was signed between World ThermoExport, a company created in early 2023 to commercialise the TEPLATOR reactor, Teploatom Service, the company that supports the development of TEPLATOR in Ukraine, and the Ukrainian city of Slavutych, to explore the possible deployment of a TEPLATOR reactor in Slavutych, in northern Ukraine.

Financing



Invest & Property Consulting (IPC), based in Czechia, is the primary investor in TEPLATOR. In 2021, the Czech engineering company VVV MOST and CIIRC CTU received a matching grant of CZK 45.9 million (USD₂₀₂₃ 2.4 million) from the Ministry of Industry and Trade of the Czech Republic to support the research and development of the reactivity control system for the TEPLATOR design.

Supply chain



The TEPLATOR is being advanced in a partnership with the Czech Technical University in Prague and the University of West Bohemia in Pilsen, Czechia. Dr. Radek Škoda, a senior nuclear scientist at the Czech Technical University involved in the TEPLATOR project, served as the principal investigator for the matching grant awarded by the Ministry of Industry and Trade of the Czech Republic to VVV MOST and CIIRC CTU.

Engagement



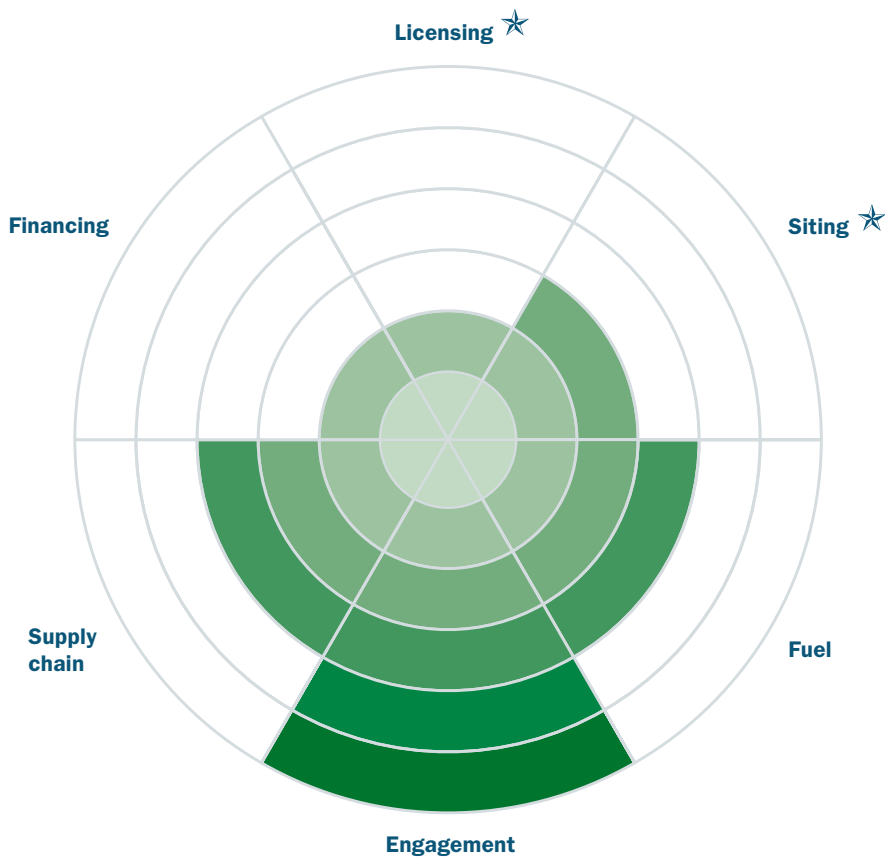
Development of the TEPLATOR design is taking place in partnership with the Czech Technical University in Prague and the University of West Bohemia in Pilsen, which creates opportunities for engagement with students. In June 2023, the ABF Foundation presented TEPLATOR with the Urban Project of the Year award for architect Michal Postránecký's artwork of the TEPLATOR facility. In September of that year, Teploatom participated in the Spent Nuclear Fuel Management Workshop organised by the International Conference on Nuclear Decommissioning and Environmental Recovery in the Slavutych City Council meeting hall to present the TEPLATOR technology in the presence of the mayor of Slavutych.

Fuel



TEPLATOR proposes to use spent nuclear fuel assemblies from VVER, BWR or PWR reactors or natural uranium as fuel. While contracts have not been signed to procure the spent fuel assemblies, the current fleets of operating reactors could provide the supply of fuel required for the TEPLATOR.

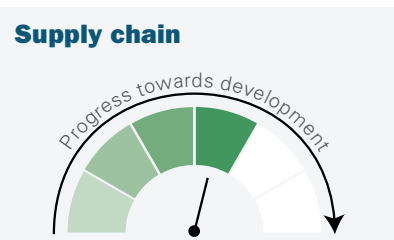
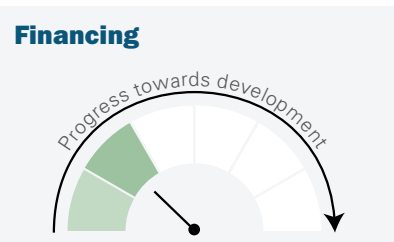
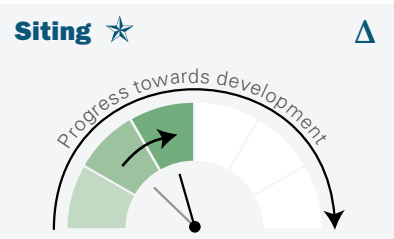
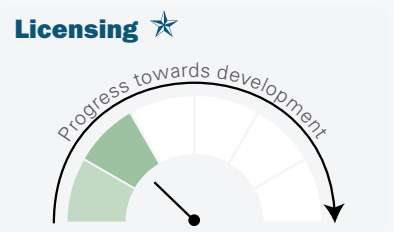
Westinghouse Electric Company – AP300 SMR



- ★ Active in multiple jurisdictions or countries.
- △ Changes since Second Edition (2024).

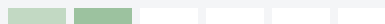
Reactor description: One-loop pressurised light water reactor based on the proven AP1000 design.

Thermal power (MWth)	990
Core outlet temperature (°C)	285
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxide ceramic
Base fissile material	U-235, enrichment level: < 5%



Assessment of AP300 SMR's progress to deployment

Licensing ★



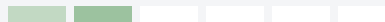
In 2023, Westinghouse Electric Company (Westinghouse) submitted a Pre-Application Regulatory Engagement Plan (REP) to the US Nuclear Regulatory Commission (NRC) to prepare pre-licensing activities in the United States. This REP describes the anticipated pre-application interactions Westinghouse intends to have with the US NRC to support the future licensing of the AP300. As of February 2025, Westinghouse had submitted twelve white papers to the US NRC and received feedback for seven of them. In April 2024, Westinghouse also submitted an application to the UK Department for Energy Security and Net Zero (DESNEZ) seeking approval to enter UK Office for Nuclear Regulation's Generic Design Assessment (GDA) pre-licensing process for its AP300 SMR, which was approved by DESNEZ in August 2024.

Siting ★



In February 2024, Westinghouse signed a Memorandum of Understanding (MoU) with Community Nuclear Power (CNP), an SMR development facilitator, for the construction of four AP300 SMRs at the North Tees Group Estate site in North Tees, United Kingdom. The site is owned by Green Lizard Group, a UK-based estate management company. CNP is working with partners, including Jacobs UK and Interpath Advisory, to develop a fully licensed site by 2027. In September 2024, Great British Nuclear (GBN) shortlisted AP300 SMR among four SMR technologies for the next phase of the United Kingdom's deployment competition, with GBN selecting the sites.

Financing



The AP300 SMR is based on the licensed AP1000 pressurised light water technology. Past US Department of Energy (DOE) funding for the AP1000 programme contributed to the development of the advanced passive safety technology used in the AP300 SMR. In 2024, CNP announced that the project at North Tees Group Estate site in North Tees, United Kingdom, which aims to deploy four AP300 SMRs, will be the first privately funded SMR project in Europe.

Supply chain



Westinghouse proposes to use components identical to the AP1000, currently deployed and operating in China and the United States, including major equipment, structural elements, safety systems and fuel. The AP300 system is expected to benefit from the existing and future supply chain capabilities of the AP1000. In 2024, Westinghouse signed a series of MoUs with key reactor component suppliers to support nuclear new-build projects with Westinghouse's reactor technologies, including the AP300, in Canada and globally. These suppliers include Seaspan ULC, a Canadian shipbuilding company, BWXT Canada, and Curtiss-Wright Nuclear Canada. Westinghouse also opened a new global engineering hub in Kitchener, Ontario, to support its nuclear business, including the development of the AP300 technology.

Engagement



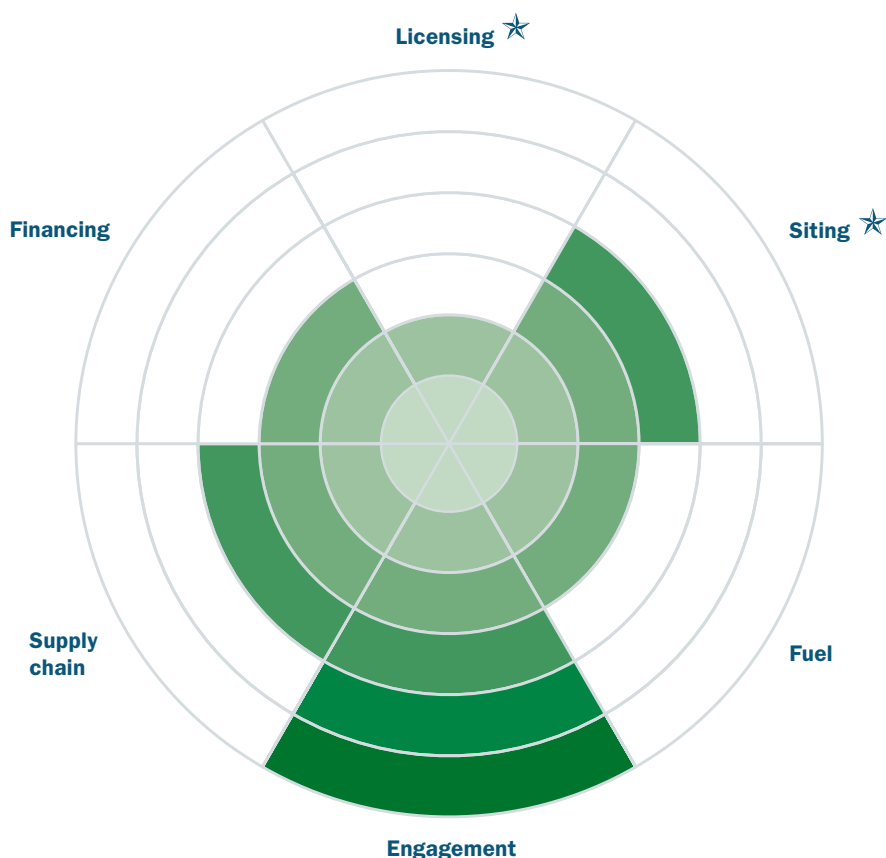
Westinghouse has conducted outreach and engagement activities about the AP300, including: meeting with elected officials, professors and students in Australia; hosting an industry night at the University of Waterloo in Canada; discussions on nuclear energy, including the AP300, with the League of Women Voters of New Mexico, United States; presentations to US elected officials; and media interviews, YouTube videos, and podcasts for public audiences as well as potential customers for non-electric applications. Westinghouse also advances customer engagement through MoUs, including with: Fortum in Finland and Sweden; JAVYS in the Slovak Republic; Energoatom in Ukraine; the North Shore Mi'kmaq Tribal Council; and Saskatchewan Power Corporation in Canada.

Fuel



The AP300 reactor uses proven light water reactor fuel. In July 2023, Westinghouse was awarded GBP 10.5 million (USD₂₀₂₃ 13.0 million) in matching-grants from the UK government's Nuclear Fuel Fund to expand Westinghouse's Springfield fuel fabrication facility in Lancashire, United Kingdom. Part of the funds will be allocated to develop fuel for the AP300 as well as potential production of HALEU. Westinghouse has additional fabrication capabilities in Columbia, South Carolina (United States) and in Vasteras, Sweden.

Westinghouse Electric Company – eVinci microreactor



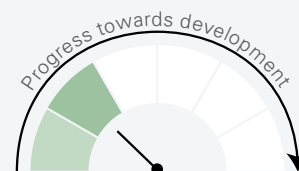
★ Active in multiple jurisdictions or countries.

△ Changes since Second Edition (2024).

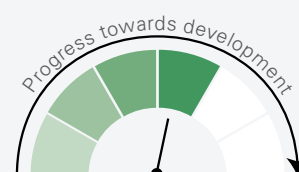
Reactor description: High temperature sodium heat pipe microreactor using TRISO fuel with a minimum eight-year refuelling period.

Thermal power (MWth)	15
Core outlet temperature (°C)	750
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-oxycarbide ceramic TRISO prismatic
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%

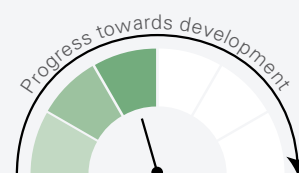
Licensing ★



Siting ★



Financing △



Supply chain △



Engagement △

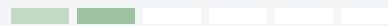


Fuel △



Assessment of eVinci's progress to deployment

Licensing ★



Westinghouse Electric Company (WEC) is engaged in pre-licensing activities with the US Nuclear Regulatory Commission (NRC) and the Canadian Nuclear Safety Commission's (CNSC) individually, and through a joint review of the eVinci. The National Reactor Innovation Center (NRIC), a US Department of Energy (DOE) programme led by Idaho National Laboratory (INL), has been supporting regulatory engagement activities with the NRC to determine an appropriate regulatory strategy for the deployment and testing of microreactors, including the eVinci, at the Demonstration Of Microreactor Experiments (DOME) facility at INL. WEC has also engaged the NRC on the potential deployment of the eVinci at Pennsylvania State University for research applications.

Siting ★



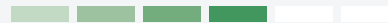
NRIC has selected the eVinci microreactor to be demonstrated at one of the sites in INL's DOME facility in Idaho, United States, where testing in DOME could start as early as 2027. WEC is also working with the Saskatchewan Research Council (SRC) to develop a project to deploy the eVinci microreactor in Saskatchewan, Canada, where it would provide heat and electricity and support the testing of industrial, research and energy use applications. In May 2022, WEC signed a Memorandum of Understanding (MoU) with Pennsylvania State University (PSU) for discussions about possible deployment at PSU's University Park campus.

Financing



WEC received funding from US Industry Opportunities for Advanced Nuclear Technology Development, Advanced Research Project Agency–Energy, Advanced Research Development Program, and Modeling Enhanced Innovations Trailblazing Nuclear Energy Reinvigoration. WEC received two awards from NRIC: up to USD 1.5 million in 2023; and up to USD 5.0 million (USD₂₀₂₃ 4.7 million) in 2024. For the space version of the eVinci, WEC received approximately USD 17 million under the Space Force JETSON project, and USD 5 million (USD₂₀₂₃ 5.2 million) from INL for its participation in the NASA Artemis programme. In Canada, WEC received CAD 27 million (USD₂₀₂₃ 20.9 million) under Canada's Strategic Innovation Fund and SRC received CAD 80 million (USD₂₀₂₃ 59.3 million) from the Government of Saskatchewan to deploy eVinci.

Supply chain



WEC intends to build the eVinci at its facilities in Etna, Pennsylvania, United States. WEC has demonstrated its manufacturing capabilities with: a 12-foot prototype heat pipe in 2023; and a Manufacturing Demonstration Unit of the microreactor in 2024. In 2023, WEC invested over USD 18 million for an eVinci microreactor accelerator hub for engineering, licensing, operations, testing and prototype trials in Etna, Pennsylvania. WEC is advancing collaborations with INL, Penn State University, McMaster University, Southern Company Services, Canadian Nuclear Laboratories, and Hatch. WEC also has agreements with Core Power and Prodigy Clean Energy to design a marine-based eVinci, and with Astrobotic to develop a space version of eVinci.

Engagement



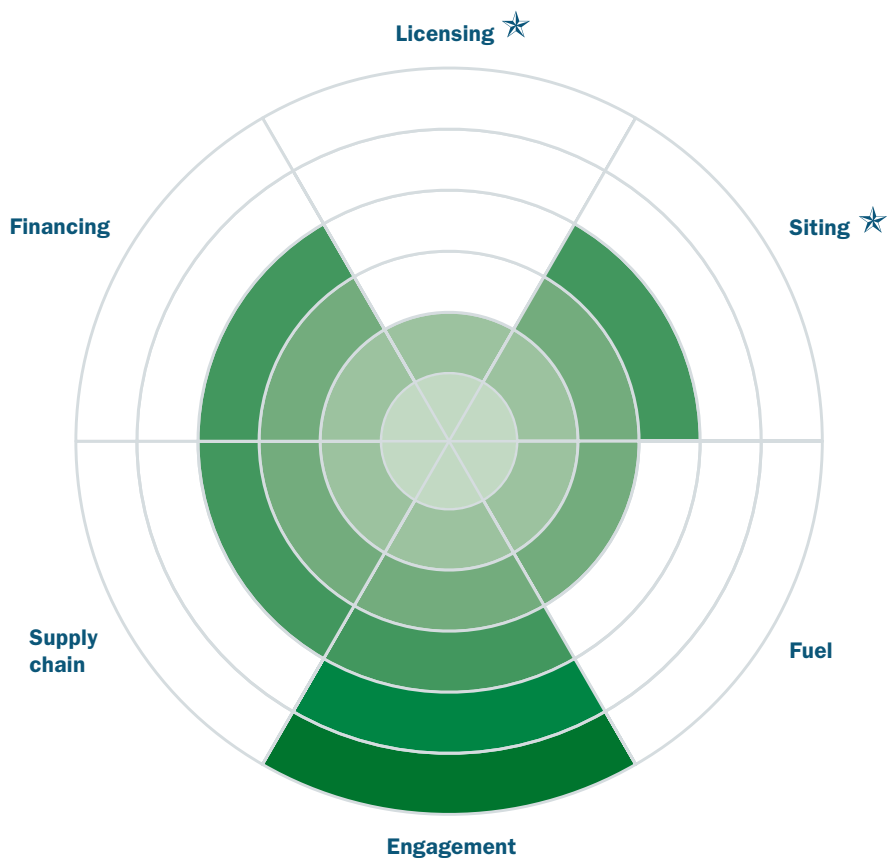
Since 2020, WEC has advanced Indigenous engagement in Canada including with North Shore Mi'kmaq Tribal Council and Saskatchewan First Nation Natural Resource Centre of Excellence, and community engagement in the United States including through the Energy for All Alaska Task Force. WEC sponsors Women in STEM scholarships at Ontario Tech University and has partnered with Pennsylvania State University and Carnegie Mellon University to engage students on SMRs. WEC has agreements with Bruce Power, Ontario Power Generation, and Emirates Nuclear Energy Corporation to explore deployment opportunities. WEC is engaging potential end-users through the Post-Industrial Midwest and Appalachia Nuclear Alliance and their membership in the Nuclear Energy Maritime Organization.

Fuel



In October 2024, WEC was one of six companies awarded a US DOE contract for a minimum of USD 2.0 million to perform HALEU deconversion in support of advanced reactors in the United States. Through the UK Nuclear Fuel Fund, WEC was awarded a GBP 10.5 million (USD₂₀₂₃ 12.9 million) matching-grant in 2023 to upgrade and expand its Springfields fuel fabrication facility to support the potential production of HALEU fuels. WEC is also collaborating with Urenco on a Pre-Front End Engineering Design study about TRISO fuel.

X-energy – Xe-100



★ Active in multiple jurisdictions or countries.

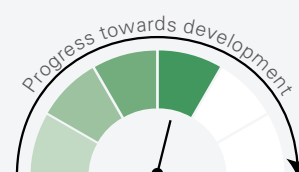
Reactor description: High temperature gas-cooled reactor designed for online refuelling using proprietary TRISO-X pebble fuel.

Thermal power (MWth)	200
Core outlet temperature (°C)	750
Spectrum (thermal/fast)	Thermal
Fuel form	Uranium-carbide ceramic TRISO-X pebble
Base fissile material	U-235, enrichment level: ≥ 10% to < 20%

Licensing ★



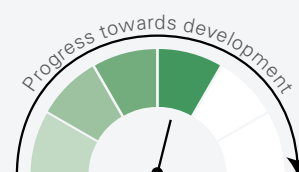
Siting ★



Financing



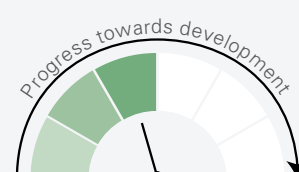
Supply chain



Engagement

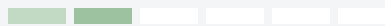


Fuel



Assessment of Xe-100's progress to deployment

Licensing ★



The Xe-100 reactor is undergoing pre-licensing review by the US Nuclear Regulatory Commission (NRC). X-energy has submitted 28 topical reports and white papers to the US NRC, of which 22 have already been reviewed. In 2024, X-energy completed the combined phase 1 and 2 vendor design review with the Canadian Nuclear Safety Commission (CNSC). No fundamental barriers to licensing were identified in this pre-licensing review by CNSC. The NRC and CNSC co-ordinated a joint review of X-energy's reactor pressure vessel construction code assessment white paper in 2021, under the 2019 NRC-CNSC Memorandum of Cooperation.

Siting ★



In the United States: Dow Chemical has selected X-energy to provide power and steam using four Xe-100 reactors to its manufacturing facility in Seadrift, Texas; X-energy and Energy Northwest signed a Joint Development Agreement in 2023 for deployment of Xe-100 SMRs at the Columbia site in Richland, Washington; the Maryland Energy Administration partnered with X-energy to investigate the feasibility of repowering a current Maryland coal site with a Xe-100 SMR; and Grant County Public Utility District (PUD) is currently evaluating the potential of adding Xe-100 SMRs to its portfolio. In Canada, Ontario Power Generation (OPG) has signed a Framework Agreement with X-energy to explore potential deployment of Xe-100 in Canada at remote and industrial sites.

Financing ★



In 2025, X-energy closed a USD 700 million (USD₂₀₂₃ 680 million) C-1 capital raise led by Amazon. In 2024, X-energy matched a UK government grant of GBP 3.34 million (USD₂₀₂₃ 4.05 million) directed to its local partner Cavendish Nuclear. X-energy was one of two awardees of the US Department of Energy (DOE) Advanced Reactor Demonstration Program (ARDP) in 2020 and received USD 80 million (USD₂₀₂₃ 94.2 million). ARDP authorised USD 1.23 billion (USD₂₀₂₃ 1.4 billion) across seven years for X-energy to develop and construct a plant with four Xe-100 units and the X-energy fuel fabrication facility. In 2023, X-energy closed series C financing round of USD₂₀₂₃ 235 million with investors including OPG, Doosan Enerbility, DL E&C and Ares Management.

Supply chain



X-energy has selected: Zachry Group, Day & Zimmerman, and Burns & McDonnell as constructors; Doosan for major components; Curtiss-Wright for primary components; and Howden for helium circulators. X-energy has agreements with OPG, Cavendish Nuclear, Sheffield Forgemasters, Hatch, and Kinectrics, and, in 2023, contracted Kinectrics for a Helium Test Facility. It also has agreements with Amsted Graphite Materials and with SIMSA for supply chain development in Saskatchewan. In 2023, X-energy signed a contract with Mirion Technologies for a detailed design of its Burn-Up Measurement System. In 2024, X-energy inaugurated an operator training centre in Maryland, United-States. Also in 2024, Energy Northwest selected AtkinsRealis as Owner's Engineer for the Xe-100 project in Columbia, Washington.

Engagement



In 2024, X-energy has advanced engagement in Seadrift, Texas, for example through community town halls. In 2023, X-energy held public meetings to discuss its fuel fabrication facility in Oak Ridge as well as the potential Xe-100 deployment at Dow's site. In Canada, X-energy has Memoranda of Understanding (MoU) with First Nations Power Authority, Millwright Regional Council of Ontario, Building Trades of Alberta, Invest Alberta Corporation, and OPG. In the Middle East, X-energy has an MoU with the Jordan Atomic Energy Commission and one with the Emirates Nuclear Energy Company. X-energy's CEO has participated in interviews and events such as the CERA week.

Fuel



Xe-100 plans to use TRISO-X pebble fuel. X-energy is working with MIT, Sargent & Lundy, Centrus and others on design, production and licensing. In 2022, X-energy started construction of a TRISO-X Fuel Fabrication Facility (TX1) in Oak Ridge, Tennessee, with startup planned in 2027, building on the TRISO pilot line at Oak Ridge National Laboratory that X-energy has owned and operated since 2017. The US NRC is reviewing an Xe-100 topical report on TRISO-X qualification methodology and the licence application, submitted in 2022, for the fuel fabrication facility. In 2024, X-energy received USD 148 million (USD₂₀₂₃ 144 million) in tax credits for construction of this facility. In 2024, X-energy awarded Geiger Brothers a USD 40.8 million (USD₂₀₂₃ 39.6 million) site development contract for TX1.

Annex 1. The NEA SMR Dashboard assessment criteria definitions

The first edition of the *NEA SMR Dashboard* (NEA, 2023) introduced and defined six assessment criteria to track the progress toward commercialisation and deployment of respective SMR designs. The second and third editions apply the same criteria. Table 16 below summarises the definitions of the criteria applied to assess progress in the areas of: licensing, siting, financing, supply chain, engagement and fuel.

Table 16. *The NEA SMR Dashboard assessment criteria definitions*

Licensing	No information	Pre-licensing	Licence/construction/design certification application submitted	Design approved	Licence to construct approved	Licence to operate approved
	* Bonus for multiple jurisdictions					
Siting	No information	Non-binding agreements/MoUs/non-binding announcements	Site owner has shortlisted the technology	Site owner has selected the technology	Received permit(s) and/or licence(s) for construction on the site	Construction has started/ been completed on the site
	* Bonus for multiple sites					
Financing ⁽¹⁾	No information	At least one announcement	Five or more announcements or USD 100 million	Ten or more announcements or USD 500 million	FOAK is fully financed	FOAK financed + progress for NOAK finance
Supply chain ⁽²⁾	No information	Supplier days/events/workshops/trade shows/non-binding agreements/MoUs/non-binding announcements	Binding contracts for services & materials	Partnerships/joint ventures/consortia/in-house capabilities for EPC services	FOAK construction ongoing/complete	NOAK construction ongoing
Engagement ^(3,4)	No information	One or more engagements	Three or more engagements	Five or more engagements	Seven or more engagements	Ten or more engagements
Fuel	No information	Non-binding agreements & studies with national labs for RDD/Lab-scale production of fuel	Contracts/agreements with fuel supply chain (uranium/conversion/enrichment/fabrication)	Operating fabrication facility producing fuel, or uses same fuel as existing/generation-III commercial reactors	Contracts for fuel for FOAK	Fuel loading has begun/ been completed

- (1) **Types of financing announcements in scope:** Funding from private investors; government grants; loans; loan guarantees; cost sharing agreements; public-private partnerships; equity partnerships; announcements of becoming publicly traded; announcements of sizeable investments; power purchase agreements; financing approval from rate payers; export credit financing; bank financing; multilateral development bank financing.
- (2) **Types of suppliers of interest in scope:** Engineering, procurement and construction organisations; universities, labs and research institutions when they are supplying research and development services to an SMR project.
- (3) **Types of stakeholders for “engagement” in scope:** Subnational governments; Indigenous governments; labour unions; non-governmental organisations; civil society organisations; community organisations; universities; end users and customers; advisory boards.
- (4) **Types of announcements for “engagement” in scope:** Memorandums of understanding; endorsements; town hall meetings; benefit-sharing agreements.

References

The references used to compile the SMR assessments (pages 102 to 249) are available at: www.oecd-nea.org/dashboard-edition3-ref.

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The NEA Small Modular Reactor Dashboard: Third Edition

All low-carbon solutions will be required to achieve the world's net zero targets. Nuclear energy has a role to play in meeting this need. A wave of innovation in small modular reactors (SMRs) is advancing quickly with the potential to help decarbonise hard-to-abate sectors. Progress is real and is positioned to accelerate pathways to net zero. SMRs could replace coal on-grid, fossil fuel cogeneration of heat and power for heavy industry, diesel at off-grid mines, as well as producing hydrogen and synthetic fuels.

Looking beyond technical feasibility, The *NEA SMR Dashboard* defines criteria for assessing real progress in six dimensions of readiness: licensing, siting, financing, supply chain, engagement, and fuel. In the *NEA SMR Dashboard: Second Edition*, the NEA presented assessments of the progress of 56 SMRs. In 2023, the NEA presented assessments of the progress of 42 SMRs in two volumes.

For this third edition of the *NEA SMR Dashboard*, the NEA's comprehensive global review identified 127 SMR technologies around the world. Seventy-four SMRs are included in this edition of the *NEA SMR Dashboard*; these are the SMRs for which the requisite publicly available information was assessable and for which the relevant designers were willing to participate. The other 53 include approximately 25 that are under development but requested not to be included in the *SMR Dashboard* at this time but may be included in the future; the remainder include SMR technologies that are not under active development, may be without human or financial resources, or have been cancelled or paused indefinitely. The assessments in this edition of the *SMR Dashboard* are based on progress up to a cutoff date of 14 February 2025.